

Overlap-Aware Calculus for Polynomial Random Matrices

Exact descriptors, two-sided certificates,
and the Walsh sampling frontier

This report unifies the research around one requirement: a retained state must preserve the actual random dependencies and ordered coefficient information needed by every allowed future interaction. It develops exact Gram algebras, extension-stable signed cores, same-input lower certificates, and explicit confidence consequences in a common framework. Results developed in this project include an exact aggregate-moment calculus that avoids enumerating an exponentially large aggregate law, a common two-sided theorem for signed quadratic insertions in fixed-Sidon Walsh rows, independent Rademacher rows, and actual SparseStack layers in the stated oversampled fixed-input regime, and a simultaneous finite-witness theorem for the strict tail counts of every Sidon Walsh profile on one original sample. Sparse-block and dense permutation-sector theorems give reusable norm and strict-quantile calculi; cyclic seed spaces and joint-Gram Clifford states enlarge the dense classes. Peak-orientation and cross-cell examples developed here delimit two tempting pooling reductions.

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13 September 2026

Updated through the finite-witness and coherence checkpoint of 12 September 2026. Proofs, finite-count certificates, assumptions and remaining gaps are distinguished. Full FR-10 is not claimed solved.

Contents

Flagship results and applications	1
1 Purpose, scope, and the common method	7
1.1 A descriptor has five obligations	7
1.2 Conventions	8
1.3 Principal quantitative conclusions	8
2 Representation before estimation	11
2.1 Random-word Grams and finite vacuum models	11
2.2 Vacuum renewal and covariance-sensitive bounds	12
3 Overlap identities and legitimate state reduction	13
3.1 The actual collision Gram	13
3.2 One block-partition formula contains the low-order corrections	14
3.3 Innovation weights must come from the Gram	15
3.4 A frame contraction that really survives repeated visits	15
3.5 The exact acceptance test for a proposed compression	16
4 A signed Gram calculus with exact and approximate law interfaces	17
4.1 Model and deterministic algebra	17
4.2 The probability interface: retain a joint feature law	19
4.3 One same-input error and confidence transfer principle	25
4.4 What the calculus establishes and what remains open	28
5 From exact bucket geometry to two-sided bounds	28
5.1 The conditional descriptor and its upper contraction	29
5.2 Lower certificates for the same matrix	30
5.3 Integrating the random geometry: the diffuse theorem	32
5.4 A second route to lower bounds: invariant sign projections	34
5.5 An anisotropic interface for a retained heavy part	35
6 Rare collisions and the sharp confidence exponent	36
7 From an exact law to exact constants	40
7.1 An explicit two-dimensional calculation	40
8 Exact contraction with a controlled live boundary	41
8.1 Ordered parity messages and norm moments	42
8.2 Even common full rank does not replace named constraints	43
8.3 Coherent cell sums: a small closure with an explicit boundary	44
9 Computable calculi: symmetric lifts, tensor factors, and interacting chains	46
9.1 A common schedule and the distinction between moments and quantiles	46
9.2 Symmetric powers remove the exponential moment cost at bounded size	47
9.3 Tensor factors and growing irreducible coefficient algebras	48
9.4 Matrix-product operators retain coherent sums and physical interactions	50
9.5 An actual continuation disproves full replica symmetrization	52
10 A complete norm and strict-quantile calculus at controlled block sparsity	53
10.1 The retained class and a constructive pointwise certificate	53
10.2 An exact envelope law and matching strict quantiles	54
10.3 Useful arithmetic, bit costs and error control	56
10.4 Full operations and final structural losses	56
10.5 Banded polynomials and collision thresholds	57
11 A dense calculus from complete permutation sectors	58
11.1 The supplied matrix class and its original random inputs	58
11.2 All spin sectors, their metric, and their positive multiplicities	59

11.3	The coefficient algebra is the full group-permutation centralizer	60
11.4	A same-input moment and strict-quantile theorem	61
11.5	Exact transfer of the overlapping score law	62
11.6	Arithmetic, bit costs, and useful confidence orders	62
11.7	Extension, symmetry breaking, and common refinement	64
11.8	Dense full-rank geometry and exact verification examples	64
12	Two complementary dense calculi: rank and Clifford geometry	65
12.1	Dense low-rank matrix-product programs	66
12.2	Rank updates, computational cost, and confidence scope	67
12.3	An even-copy homogeneous Clifford class	69
12.4	Graded closure and consistent pencil names	71
12.5	Examples and the mixed-degree obstruction	72
13	Walsh sampling: a complete contraction chain and its uniformity boundary	73
13.1	Model, target, and the retained state	73
13.2	An exact extension and a closed upper potential	74
13.3	A density filtration that keeps quotient multiplicities	75
13.4	The scalar input and additive Schur assembly	76
13.5	From the same contraction to a uniform finite sample bound	78
13.6	Confidence is a separate, cheaper operation	80
13.7	An overlap certificate on the original rows	80
13.8	A matching theorem from exact core-quotient geometry	82
13.9	Why completing every support to core cosets is insufficient	84
13.10	What is proved, and the remaining global question	85
14	Larger geometric classes and one-sided uniform control	86
14.1	Unrelated affine-flat directions	86
14.2	All-Sidon lower isometry and the fourth-moment input	87
15	Ordered moment corrections and the cost of summing them	87
15.1	The actual sample-count polynomial	88
15.2	A finite hierarchy stable under further paired names	88
15.3	The first two corrections retain joint relations	89
15.4	Ordered integrations that preserve second-moment gains	90
15.5	The complete second correction at logarithmic order	91
15.6	Why a fixed number of corrections cannot finish the norm theorem	93
16	Complete moment transfer with general Parseval geometry	93
16.1	Hard and relative relation horizons	94
16.2	A boundary certificate stable through the whole moment	95
16.3	Two-sided complete norm moments and a separate confidence bound	96
16.4	Explicit low-girth supports and signed obstructions	98
16.5	What is tractable, and what remains model-specific	99
17	Conditional registers and a relation calculus without depth loss	100
17.1	All independent names may interleave	100
17.2	Innovation in the actual affine constraint Gram	101
17.3	Covariance and pairwise laws do not survive all continuations	103
17.4	Retained subcodes and relative horizons	104
17.5	An exact tree of affine innovations	104
17.6	Two-sided covariance and positive-polynomial consequences	106
17.7	Updates, useful costs, and growing noncommuting examples	108
17.8	Why bounded overlap alone is insufficient	109
18	Summing the baseline before estimating relations	110
18.1	A deterministic recurrence on the original sign matrix	110
18.2	A complete independent baseline at logarithmic order	111
18.3	A positive certificate for the full spectral edge	112

19 Sidon confidence and shared cubic coefficient geometry	113
19.1 A probability baseline can be amplified without changing the matrix	113
19.2 An exact cubic law and a same-input lower certificate	115
19.3 Shared original cells remove an ambient-rank loss for modulations	116
19.4 Arbitrary coefficients: a shared near-peak theorem and its limit	117
19.5 Even every orbit modulation is not a deterministic norm certificate	118
19.6 A direct common-event theorem for growing coefficient spaces	119
20 Closing cubic peaks and their coefficient interactions	121
20.1 A reused direction that diagonal Gram data miss	121
20.2 Signed isotropic channels at almost linear dimension	122
20.3 A matching common-event sampling theorem	123
20.4 A common algebra for noncommuting repaired projections	125
20.5 Polynomial programs and the unchanged Walsh sampling event	127
21 Actual Walsh couplings beyond the cubic block algebra	128
21.1 Original cross covariance and whole-sample contractions	128
21.2 A constant-probability obstruction to adaptive low rank	130
21.3 An exact flat cross coupling in the original sample law	130
21.4 Consequences for the cubic descriptor and explicit costs	131
22 Dense closure after symmetry breaking and joint Grams	132
22.1 Cyclic seed spaces are stable under every later product	132
22.2 Complete attained Grams close mixed-degree Clifford words	133
22.3 What these dense extensions establish	135
23 Extension-stable signed conditional cores	135
23.1 The conditional projector and its exact algebra	135
23.2 High moments and continuation through a circuit	136
23.3 A noncommuting Parseval benchmark with a constant-size core	138
23.4 Costs and exact scope	139
24 Finite signed-polynomial transfer and a two-sided quadratic calculus	139
24.1 Exact ordered linearization and finite spectral transfer	140
24.2 The exact quadratic energy identity	142
24.3 A tractable two-sided theorem for HDH	142
24.4 Three distinct input laws covered by the same theorem	144
24.5 Limitations and the remaining bridge	145
25 Exact aggregate-moment boundary calculus	145
25.1 The model and its live boundary	145
25.2 An exact truncated exponential transfer	146
25.3 Two-sided norm moments and a separate strict-tail certificate	147
25.4 Reusable operations and explicit costs	147
25.5 Applications, verification, and the boundary of the theorem	148
26 Signed Walsh pooling: what is closed and what remains open	149
26.1 The common process and a complete clipped contraction	149
26.2 Finite witnesses on the original Sidon sample	151
26.3 The noise floor of a positive witness	152
26.4 Exact signed large-observation families	153
26.5 Pooling that is useful but does not cover all supports	155
26.6 Earlier reductions that the exact examples rule out	155
26.7 Peak orientation defeats an all-sample smooth contraction	157
26.8 Exact cell aggregation still loses cross-cell coherence	158
26.9 The exact remaining routes	159
27 What a unified descriptor cannot discard	160
27.1 Coherent cancellation is not permission to retire a live label	160

27.2	Even all pairwise positive data can miss a signed cycle	160
27.3	Positive return kernels do not give a Loewner recurrence	161
28	The unified result and the remaining frontier	162
28.1	A concrete general target	162
28.2	The current calculus frontier and the full FR-10 pooling problem	165
28.3	Provenance and verification	165
References		167

Flagship results and applications

This section records the strongest conclusions of the project as of 13 September 2026. The project now contains several complete calculi for explicit structural classes: each retains the random identifications and ordered coefficient geometry needed by its allowed updates, has an explicit state and arithmetic cost, and gives an upper estimate together with a same-input lower certificate. It also contains a common two-sided moment theorem for a fixed quadratic continuation in three actual random-matrix models. These results meet the general target on their stated classes; they do not give one economical descriptor for every polynomial random matrix. The arbitrary-support Walsh problem FR-10 and the all-Sidon excess estimate in Flagship 7 remain open. The exact suites and independent internal reviews recorded in the project are evidence for the proofs, not external peer review.

Here “flagship” denotes a result that best exposes the project’s current method, not a claim of literature priority. Flagships 1–3 specialize classical Schur–Weyl and commutant decompositions, invariant cyclic spaces with matrix-product contraction, and Clifford universal-property and exterior-algebra representations [23, 20, 22]. Flagship 4 uses subgroup Fourier projection and Bonami–Beckner hypercontractivity [8, 9]. Flagship 5 specializes the Brailovskaya–van Handel comparison and is adjacent to the oblivious sparse norm-approximating projection (OSNAP) sandwich recursion of Chenakkod, Dereziński, and Dong [12, 10]. Flagship 6 combines truncated exponential coefficient extraction with bounded-width variable elimination [21]. The report claims the displayed within-project statements and proofs; it does not claim historical priority for these ingredients or their combinations. An exhaustive novelty audit has not been completed.

Flagship 1: complete grouped sectors give a dense anchor calculus

Let the physical space be $(\mathbb{C}^2)^{\otimes L}$. Mark t supplied sites and partition the other sites into g permutation-invariant groups of sizes $M_1, \dots, M_g$. Allow arbitrary matrices on the marked factor and arbitrary ordered polynomials in the collective generators H_j, E_j, F_j of every group. Let the scalar coefficients depend on the actual joint score

$$S(z) = s_0 + \sum_{v=1}^m \phi_v(z_{V_v}) \in \mathbb{Z}^c, \quad |V_v| \leq a, \quad \phi_v \in \{-K, \dots, K\}^c,$$

with live primitive boundary at most w . Complete spin sectors reduce the same physical matrix and retain total dimension

$$D_{\text{eff}} = 2^t \prod_{j=1}^g \left\lfloor \frac{(M_j + 2)^2}{4} \right\rfloor.$$

For the positive sector readout $G_h(S)$ and every integer $h \geq 1$,

$$D_{\text{eff}}^{-1/(2h)} G_h(S(z)) \leq \|Z(z)\| \leq G_h(S(z)).$$

The same inequalities hold after taking every positive norm moment and, separately, every strict upper-tail quantile of the same Z . Thus $h \geq \log D_{\text{eff}}/[2 \log(1 + \epsilon)]$ gives ratio at most $1 + \epsilon$. With

$$U = (2mK + 1)^c, \quad C_{\text{sec}} = 2^{3t} \prod_j (M_j + 1)^4,$$

an explicit T -term input of word length at most R and scalar degree at most q has the conservative arithmetic cost

$$O\left(mc2^{w+a}U + U[(T(R+1) + \log(h+1))C_{\text{sec}} + Tc \log(q+2)]\right),$$

plus local-table and input-leaf formation and the stated rational bit costs. This is an exact dense calculus for collective-spin polynomials, arbitrary marked-site interactions, and overlapping

additive score laws. Its lower witness is a maximizing vector in a sector embedded in the original physical space. The input must supply the partition and an expression in its centralizer algebra; the result does not recognize or compress an arbitrary 2^L by 2^L array, and large g, t, c, w retain their displayed costs.

Flagship 2: cyclic seed spaces survive dense symmetry breaking

Suppose a supplied coefficient algebra has the exact representation

$$\mathcal{A} = \bigoplus_{\lambda} M_{d_{\lambda}} \otimes I_{\mu_{\lambda}}, \quad N_{\mathcal{A}} = \sum_{\lambda} d_{\lambda}^2.$$

For r supplied seed vectors, allow all outer products $v_a v_b^*$ and every subsequent sum, ordered product, and adjoint with the old algebra. In sector λ , let M_{λ} be the multiplicity span generated by the seed coordinates and put $t_{\lambda} = \dim M_{\lambda}$. The exact active space is

$$W = \mathcal{A} \text{span}\{v_a\} = \bigoplus_{\lambda} \mathbb{C}^{d_{\lambda}} \otimes M_{\lambda}, \quad t_{\lambda} \leq \min(\mu_{\lambda}, r d_{\lambda}), \quad \dim W \leq r N_{\mathcal{A}},$$

Define

$$D_{\text{keep}} = N_W + \sum_{\lambda: \mu_{\lambda} > t_{\lambda}} d_{\lambda}, \quad N_W = \dim W.$$

Thus D_{keep} counts the active block together with one copy of every nonempty untouched sector. The space W reduces every old coefficient and every seed operator. All seed-induced cross-sector couplings form one fully coupled matrix on W ; on $W^{\perp}$ the old sectors remain with their residual multiplicities. A positive trace readout over these retained blocks gives pointwise, moment, and strict-quantile comparisons within $D_{\text{keep}}^{1/(2h)}$, with no accumulation of this factor over later operations.

For one collective-spin group, $N_{\mathcal{A}} = \binom{L+3}{3}$. If the seeds are matrix-product states of bond at most χ , their complete coefficient Grams and active coordinates have the reusable preprocessing bound

$$O\left(r^2 L \chi^4 N_{\mathcal{A}} + N_{\mathcal{A}} L^2 + N_{\mathcal{A}}^3 + r^2 N_{\mathcal{A}}^2 + \sum_{\lambda} (r d_{\lambda})^3\right).$$

This permits genuinely irreducible, noncommuting dynamics on the active space while the ambient multiplicities may be exponential. It applies to low-rank perturbations by supplying range and adjoint-range seeds. The restriction is substantive: the base finite-sector representation, all cross-seed coefficient Grams, a tractable seed representation, and a tractable shared-score law must be supplied. Arbitrary high-rank symmetry breaking or uncontrolled MPS bond growth is outside this calculus.

Flagship 3: the actual joint Gram closes mixed Clifford words

Let $v_1(z), \dots, v_q(z)$ be named real vectors in one even-dimensional Clifford coordinate space, let A_i be their faithful Clifford pencils, and retain the complete joint Gram

$$Q_{ij}(z) = v_i(z)^T v_j(z), \quad A_i A_j + A_j A_i = 2Q_{ij}(z)I.$$

For arbitrary mixed-degree ordered words and arbitrary signed noncommuting coefficients on an external b -dimensional factor, an attained Gram of rank r has an exact exterior representation of dimension $b2^r$, with rational Clifford generators and metric when the attained Gram is rational. It preserves the largest singular value of the same matrix exactly. If the full Gram is determined by an additive local score with parameters (m, c, K, a, w) , and the input is a T -term expression of maximum word length R , and if r_{max} is the largest attained Gram rank, put $U_s = (2mK + 1)^c$

and $D_{\text{eff}} = b2^{r_{\text{max}}}$. The positive readout gives moment and strict quantile comparisons within $D_{\text{eff}}^{1/(2h)}$ and has conservative cost

$$O\left(mc2^{w+a}U_s + U_s \left[q^3 4^q + (T(R+1) + \log(h+1))(b2^q)^3\right]\right),$$

apart from local-program, Gram-program, and input formation.

As a concrete application, one sign vector and its first two cyclic shifts have a joint Gram determined by the two overlapping correlations $\sum_i z_i z_{i+1}$ and $\sum_i z_i z_{i+2}$, with local arity three and live boundary at most four. The Gram changes rank, but the calculus gives the exact finite norm law for $I + A_1 + (A_2 A_3 - A_3 A_2)/4$. In the 32-sign audit there are 130 attained score pairs although the physical dimension is 65,536; with $2h = 40$ the general factor is $8^{1/40} < 1.1$, and the special expression has strict 2^{-20} quantile $\sqrt{285 + \sqrt{19840}}$. Every new pencil must contribute its cross Gram with every retained pencil. The exponential dependence on q is explicit, and an irreducible Pauli representation of an odd-dimensional Clifford coordinate space can lose a central orientation sign, so Gram sufficiency is not asserted there.

Flagship 4: signed cores have a continuation-safe remainder rule

On a nonempty actual affine event $\mathcal{E} = \{y \in \mathbb{F}_2^D : Cy = b\}$, let $F : \mathcal{E} \rightarrow \mathbb{C}^{d \times d}$ and first combine Fourier coefficients with their affine phases. Let H be a rank- h subgroup of the resulting character quotient, let P_H denote Fourier projection onto H , and put $A = P_H F$ and $R = F - A$. The core contains at most 2^h signed matrices. It is an exact bimodule for H -supported insertions, while a general product obeys the diagnostic identity

$$P_H(FG) = (P_H F)(P_H G) + P_H[(F - P_H F)(G - P_H G)].$$

The last term shows exactly how two discarded modes can return. When it is not recomputed from source coefficients, it is charged to a certified error. For a circuit product the extension-stable recursion is

$$e_{FG} \leq e_F n_G + a_F e_G,$$

where e_F, e_G are the certified remainder errors, a_F bounds the core, and n_G bounds the full operand at the required Hölder moment. For $p \geq 2$, a discarded free-coordinate Fourier polynomial of degree at most s satisfies

$$(\mathbb{E}_{\mathcal{E}} \|R\|_F^p)^{1/p} \leq (p-1)^{s/2} (\mathbb{E}_{\mathcal{E}} \|R\|_F^2)^{1/2},$$

and a further consistent affine restriction of new rank r_0 costs at most the visible factor $2^{r_0/p}$; reused constraints have $r_0 = 0$. Direct core gates cost $O(2^h d^3)$ arithmetic, direct convolution costs $O(4^h d^3)$, and core storage is $O(2^h d^2 + Dh + D^2)$ fields, in addition to source and error certificates.

The diagnostic application is a noncommuting Parseval-compressed Walsh cross block with arbitrarily many diffuse tail rows and only 32 retained core characters. With tail-to-core ratio λ , for every moment index $P \geq 2$,

$$e_P^2 = 12(P-1)\lambda^2 + 4(P-1)^2\lambda^4, \quad \|\mathcal{Q}(X) - \mathcal{Q}(A)\|_{L^p} \leq 108e_{3p}(3 + e_{3p})^2.$$

The same signed cubic continuation distinguishes exact core values zero and two. Its plus-phase upper certificate is $O(\lambda\sqrt{p})$ when $\lambda\sqrt{p} \leq 1$, whereas the corresponding wordwise triangle envelope is at least one, giving an unbounded certificate improvement as $\lambda \rightarrow 0$. This theorem does not discover a small accurate core for an arbitrary input. Large discarded energy, large continuation degree, or adversarial new constraints can make its explicit error useless.

Flagship 5: a specialized quadratic theorem for three models

Let $U \in \mathbb{R}^{k \times n}$ satisfy $U^T U = I_n$ with $n \geq 64$. Let $Y_1, \dots, Y_m$ be iid centered real symmetric matrices with

$$\|Y_a\| \leq K \quad \text{almost surely,} \quad K \geq n, \quad \text{Cov}(Y_a) = \nu \text{Cov}(G_U), \quad 0 < \nu \leq 1,$$

where the covariance is the complete real entry covariance and G_U is the Gaussian matrix matching the covariance of the off-diagonal feature Gram error. Put $H = m^{-1} \sum_a Y_a$. For every fixed real symmetric D , with no rank, definiteness, or commutation condition, and every integer

$$\max\{2, \lceil \log n \rceil\} \leq p \leq n, \quad m \geq \max \left\{ \frac{24K^2}{\nu n}, \frac{4C_0^2 K^2 p^4}{\nu(n-1)} \right\},$$

the proved same-input bounds are

$$\boxed{\frac{\nu n}{4m} \|D\| \leq (\mathbb{E} \|H D H\|^p)^{1/p} \leq 576 \frac{\nu n}{m} \|D\|}.$$

Here C_0 is the identified but nonnumerical universal constant in the imported Brailovskaya–van Handel moment-universality inequality. With $X = (m\nu n)^{-1/2} \sum_a Y_a = \sqrt{m/(\nu n)} H$, the lower certificate tests the actual output on the random vector Xv , where v is a fixed extreme eigenvector of D ; it does not use a different saturating model.

For fixed-Sidon Walsh rows and for independent Rademacher feature rows, $\nu = 1$ and $K = k$, so the scale is $(n/m)\|D\|$. For an actual SparseStack layer S_a with b buckets, $Y_a = U^T S_a^T S_a U - I$, one has $\nu = 1/b$ and $K = k$. With $S = m^{-1/2} [S_1^T \dots S_m^T]^T$, stacking m layers gives mb sketch rows and

$$\frac{n}{4mb} \|D\| \leq (\mathbb{E} \|(U^T S^T S U - I) D (U^T S^T S U - I)\|^p)^{1/p} \leq 576 \frac{n}{mb} \|D\|,$$

provided

$$m \geq \max \left\{ \frac{24bk^2}{n}, \frac{4C_0^2 bk^2 p^4}{n-1} \right\}.$$

The covariance calculation uses the actual signed edge Gram: a reused triangle collision has probability b^{-2} , not b^{-3} . The theorem is for fixed predetermined U, D in an explicitly oversampled regime. It gives matching norm moments, not matching extreme quantiles, an optimal bucket-sparsity tradeoff, or simultaneous postselection over Walsh supports. A directly related OSNAP calculation is more specific. In Lemma 5.1 of Chenakkod, Dereziński, and Dong, $X_i = S_i U$ and $Y_i = X_{2i-1}^T X_{2i}$ are formed from independent copies of a fully independent unscaled OSNAP, and the lemma bounds $\|Y_1^T V^T V Y_1\|_{2q}$ from above by explicit diagonal terms and an independent-copy successor [10]. It is a recursive upper estimate with the positive insertion $V^T V$. Their complete argument attains near-optimal embedding dimension and sparsity. The theorem here instead concerns a fixed empirical-Gram sum H , allows arbitrary symmetric D , and supplies a lower certificate in the same realization. Its oversampled fixed-input regime does not compete with their embedding guarantee, and historical novelty of this combination is not claimed.

Flagship 6: exact aggregate moments avoid enumerating the aggregate law

Let independent primitive signs feed local rational statistics

$$T = \sum_{j=1}^L s_j(\xi_{V_j}) \in \mathbb{Q}^w, \quad Z = P(T) \in M_d(\mathbb{Q}(i)), \quad \deg P \leq R,$$

with local arity at most a and live primitive boundary at most b . Actual zero-one constraints and their conditioning factors may be included; let $\mathcal{E}$ be the resulting named event with $\mathbb{P}(\mathcal{E}) > 0$, taking $\mathcal{E}$ to be the whole space when no constraint is imposed. At even order $2p$, set

$$D_0 = 2Rp, \quad H_{D_0} = \binom{w + D_0}{w}.$$

A truncated exponential message has exactly H_{D_0} scalar coefficients per boundary assignment and computes

$$M_p = \mathbb{E}[\text{tr}((Z^*Z)^p) \mid \mathcal{E}], \quad d^{-1/(2p)} M_p^{1/(2p)} \leq (\mathbb{E}[\|Z\|^{2p} \mid \mathcal{E}])^{1/(2p)} \leq M_p^{1/(2p)}.$$

The ratio is at most $\sqrt{e}$ for $p \geq \log d$. The lower endpoint is the spectral mass of the same matrix. Markov gives an upper tail, while an independently computed Paley–Zygmund expression gives a strict lower-tail certificate. These are kept separate because moment comparison alone does not imply matching quantiles.

The conservative arithmetic count is

$$O\left([L2^{b+a}(w+1) + d^3 \log(p+1)] H_{D_0}^2\right),$$

plus supplied-polynomial preprocessing and explicit rational bit costs. Products, adjoints, word extension, aggregate-list combination, and actual constraint identification have exact update rules; mathematical approximation error is zero. Applications include independent rationally weighted signs, overlapping path statistics, bounded-width sparse collision statistics with the correct triangle probability B^{-2} for bucket count B , and dense irreducible noncommuting coefficient algebras. One audit uses 80 signs whose aggregate has 2^{80} distinct values but needs only 13 message coefficients at degree horizon 12. The method requires small supplied (w, a, b) at the desired moment horizon. It does not produce such a representation for an arbitrary compressed SparseStack matrix or for uniform selection over all Walsh supports.

Flagship 7: Walsh pooling, strict tail counts, and the coherence frontier

Let $G = \mathbb{F}_2^d$, $N = 2^d$, and $\chi_v(y) = (-1)^{v \cdot y}$. For a deterministic Walsh row list $Q = (y_1, \dots, y_m)$, let $\mathcal{C}_k$ contain every real unit vector on every Sidon support of size at most k , and define

$$F_c(y) = \sum_v c_v \chi_v(y), \quad f_c = F_c^2, \quad R_Q^2 = \sup_{c \in \mathcal{C}_k} \sum_i f_c(y_i).$$

For any family $g = (g_c)$, write

$$\mathcal{W}_Q(g) = \mathbb{E}_\varepsilon \sup_{c \in \mathcal{C}_k} \left| \sum_i \varepsilon_i g_c(y_i) \right|.$$

For probabilistic statements below, take $Q = (Y_1, \dots, Y_m)$ to be one iid uniform Walsh row list, write $y_i = Y_i$, and let $\mathbb{E}_Q$ average over that same list. For every $L > 0$, every $1 \leq k \leq N$, every row list including repetitions, and even without the Sidon restriction, the pooled clipped process satisfies

$$\boxed{\mathbb{E}_\varepsilon \sup_c \left| \sum_i \varepsilon_i \min(f_c(y_i), L) \right| \leq 4\sqrt{2Lmk \log(2eN/k)} \leq 4\sqrt{2} R_Q \sqrt{Lk \log(2eN/k)}}.$$

At $L = \log(2k)$ this has the desired FR-10 scale simultaneously over all supports and amplitudes. Thus the signed-width route reduces the all-Sidon stage to the original-sample estimate

$$\mathbb{E}_Q \mathbb{E}_\varepsilon \sup_{c \in \mathcal{C}_k} \left| \sum_i \varepsilon_i (f_c(y_i) - \log(2k))_+ \right| \stackrel{?}{\leq} C \sqrt{k \log(2k) \log(2N/k)} \mathbb{E}_Q R_Q.$$

This displayed excess estimate is open. It keeps the original iid Walsh rows; replacing their high-observation events by independent edge weights would change the problem.

A second route now gives a stronger proved statement on those same original rows. Put

$$n_c(t) = \#\{i : F_c(Y_i)^2 > t\}, \quad A_w = 2N + 1, \quad L_t = \left\lceil \frac{32k}{t} \log(2t^2) \right\rceil.$$

For every $2 \leq t < k$ and $u \geq 0$, with probability at least $1 - e^{-u}$, simultaneously for every Sidon support of size at most k and every real unit amplitude vector,

$$\sup_c n_c(t) \leq \frac{131m}{t^2} + 3L_t \log A_w + 3u, \quad \mathbb{E}_Q \sup_c n_c(t) \leq \frac{131m}{t^2} + 3L_t \log A_w + 3.$$

The thresholds are strict and the sample is shared by the whole supremum. Writing $D_0 = k \log(2eN/k)$, the complexity term is $O(D_0 \log(2t)/t)$ whenever a size- k Sidon support exists. Under that same existence hypothesis, dyadic integration gives a verified fixed-confidence all-Sidon half-isometry at sample order

$$O\left(k \log^2(2k) \log(2N/k)\right),$$

one sparsity logarithm above the target. The proof uses a finite alphabet of A_w functions, but its $A_w^{L_t}$ possible words are a probabilistic certificate rather than an efficient supremum algorithm.

The newest obstructions explain what the remaining logarithm cannot be removed by. An absolute-constant smooth contraction normalized only by the ramp mass is false when required at every sample size: actual cubic Sidon peaks with $2^{\Theta((\log k)^2)}$ orientations force a $\sqrt{\log k}$ divergence. Exact signed aggregation inside every high-rank agreement cell also fails after taking a triangle bound across cells by an unbounded $\sqrt{\log k}$ factor, although the true coherent residual on the same input has the target-scale upper bound. These are strongly oversampled counterexamples to the stated proof reductions; neither disproves the open excess estimate, a target-sample-only contraction, or FR-10. Section 26 records a summable orientation correction whose implication is proved but whose contraction hypothesis remains open.

There is also a complete full-amplitude application for deterministic lists. For $1 \leq k \leq N/2$, if $n_{\max}$ is the largest row multiplicity and $Nn_{\max} \leq BR_Q^2 \log(2k)$, then

$$\mathcal{W}_Q(f) \leq 2\sqrt{6Nn_{\max}\{\log 2 + k \log(9eN/k)\}},$$

which has the target scale with a universal loss $O(\sqrt{B})$. Equal positive row multiplicities give the stronger $O(\sqrt{mk \log(2N/k)})$ bound. This application has a dense-sample condition and does not cover sparse iid FR-10. There is positive evidence: on the cubic Sidon graph with $N = k^2$, flat profiles have exact values $f \in \{0, 2, k\}$, and for $2 < t < k$ their translated excess has averaged signed width between $(k - t)/2$ and $4(k - t)$ for $1 \leq m \leq k^{4/3}$. A further theorem handles an amplitude vector α and excess threshold τ satisfying $\|\alpha\|_2 = 1$ and $4\|\alpha\|_1^2 \leq \tau < k$ through common affine-cell maxima.

The remaining difficulty is joint geometry and the last height logarithm, rather than an unavailable original-sample tail estimate: empirical Gram eigenvalues without their orientation can have the same energy radius and unboundedly different signed widths. For one named Sidon support the ordered pair lift has at most $1 + \binom{k}{2}$ coefficients and direct empirical-Gram formation costs $O(mk^4)$ arithmetic, but no economical extension-stable pooling of these oriented states over every support is proved. Even the all-Sidon excess bound would still require a further passage to arbitrary supports to finish full FR-10.

1 Purpose, scope, and the common method

The objective is to estimate the norm of a polynomial random matrix without discarding either the geometry of its coefficients or dependencies created by reusing the same random input. This report consolidates the project’s results into a small number of mathematical mechanisms. The organizing distinction is between an *exact representation*, a *tractable retained state*, and a *quantitative certificate*. An identity at the first level is not, by itself, a sharp bound at the third.

The principal conclusions are as follows. Ordered polynomial circuits generated by specified collision sums admit an exact Gram algebra, and exact live-boundary transfer evaluates programs whose primitive interaction graph has controlled width. A new aggregate-moment transfer goes further when a large random input enters through a few polynomial statistics: it computes the required trace moment from a truncated exponential message, without enumerating the aggregate-value law. At logarithmic moment order its dimension loss is at most $\sqrt{e}$, and its arithmetic and bit costs are explicit.

Two newer analytic mechanisms preserve signed information through later interactions. Conditional Fourier projection onto a supplied character core is an exact bimodule, while the product of two discarded remainders returns as one explicitly charged error channel. Finite nilpotent linearization retains the full order and random-name reuse of an arbitrary written polynomial. In the quadratic case this combines with an exact fourth-order correction and a same-input random-vector witness to give a two-sided *HDH* theorem for every fixed symmetric insertion, including actual SparseStack layers, fixed-Sidon Walsh rows, and independent Rademacher rows.

The dense checkpoint is also extended. Cyclic spaces generated by finitely many supplied seed vectors absorb arbitrary low-rank symmetry-breaking interactions into one exact active block of dimension at most the seed count times the base-algebra dimension. Complete attained joint Grams give an exact mixed-degree Clifford calculus, including rank changes and all cross-pencil terms. These results extend the dense classes while retaining direct strict-quantile brackets whenever their finite score laws are tractable.

For Walsh pooling, the clipped quadratic process is controlled uniformly over all sparse supports, and the remaining signed excess is solved for a nontrivial class of embedded cubic supports and arbitrary profiles satisfying an explicit ℓ_1 -spread condition. The finite-witness theorem proved here bounds the strict large-observation counts of every Sidon profile on the same original sample and yields a fixed-confidence all-Sidon result with one extra sparsity logarithm. Flat translated peaks still pool at the target scale. The target all-Sidon signed-excess estimate and the passage to arbitrary supports remain open, so full FR-10 is not claimed. Exact affine-phase, peak-orientation, and cross-cell examples identify data that covariance, even tensors, singular-value laws, Gram eigenvalues, or a cellwise triangle step cannot safely discard.

Definitions and central project-specific arguments are given in the text; where a lengthy finite enumeration is compressed, its counting rule and verification scope are identified explicitly. Long counting arguments with omitted details are labeled as proof outlines, with the companion proof identified in the source map. Standard concentration and matroid results used as inputs are stated with their hypotheses and cited. The report is a synthesis of proved research notes, not a claim of literature priority or external peer review. In particular, the unrestricted intermediate-sparsity part of FR-10 remains open.

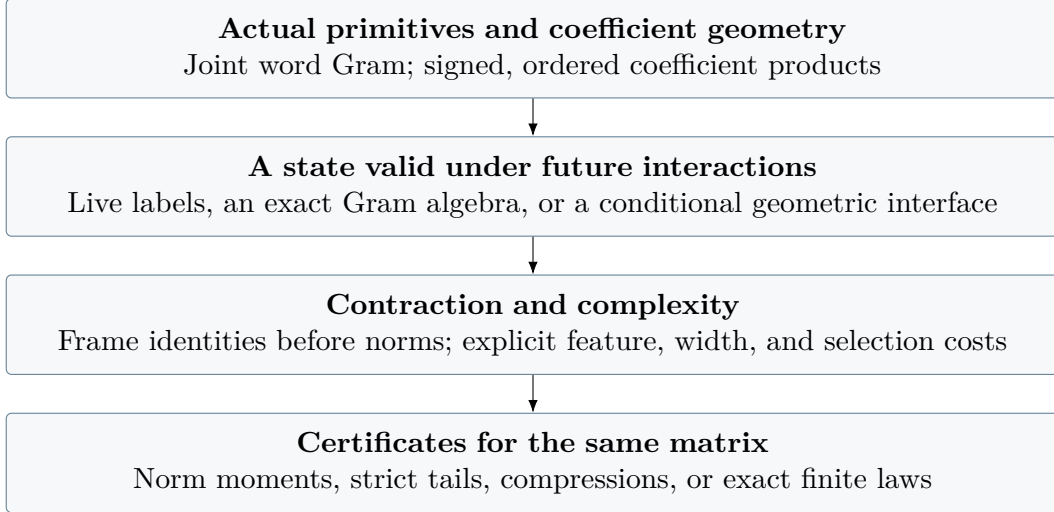
1.1 A descriptor has five obligations

For a model $Z(\omega) = \sum_w \phi_w(\omega)A_w$, a proposed calculus must specify:

- (i) the actual joint law of the scalar words ϕ_w , including reuse of primitive variables;
- (ii) which ordered coefficient information survives the encoding;

- (iii) why every allowed subsequent interaction acts on that encoding;
- (iv) the cost of enumerating, integrating, or bounding the retained states;
- (v) an upper certificate and, separately, a lower witness for the same input matrix, together with the probability convention used to compare them.

The results cover explicit structural classes; counterexamples test simpler states.



1.2 Conventions

All logarithms are natural unless a base is displayed. Matrices in the SparseStack model are real; $*$ denotes transpose there and conjugate transpose in complex statements. The norm $\|A\|$ is the operator norm and $\|A\|_F^2 = \text{tr}(A^*A)$. For a nonnegative random variable Y and $p > 0$, write $\|Y\|_{L^p} = (\mathbb{E}Y^p)^{1/p}$. A strict upper-tail quantile is

$$Q_Y(\delta) = \inf\{t \geq 0 : \mathbb{P}(Y > t) \leq \delta\}, \quad 0 < \delta < 1. \quad (1)$$

The distinction between $>$ and $\geq$ is material for finite collision laws. An asymptotic assertion at fixed confidence does not automatically extend to confidence tending to zero. Arithmetic operation counts are not bit-complexity bounds unless bit sizes are specified.

For SparseStack, fix $U \in \mathbb{R}^{n \times d}$ with $U^*U = I_d$, and write its rows as u_i^* . Let s be the number of blocks and b their common bucket count, so the sketch has $m = sb$ rows. For every pair (g, i) , independently choose h_{gi} uniformly in $[b]$ and a Rademacher sign σ_{gi} . Each input column has entry $\sigma_{gi}/\sqrt{s}$ in its selected row of block g . With S the sketch and $\ell_i = \|u_i\|^2$, define

$$X = U^*S^*SU - I_d = \frac{1}{s} \sum_{g=1}^s \sum_{i < j} z_{g,ij} A_{ij}, \quad (2)$$

$$z_{g,ij} = \sigma_{gi}\sigma_{gj} \mathbf{1}_{\{h_{gi}=h_{gj}\}}, \quad A_{ij} = u_i u_j^* + u_j u_i^*, \quad \mu = \max_i \ell_i.$$

Thus $\sum_i u_i u_i^* = I_d$, $0 \leq \ell_i \leq 1$, and $\sum_i \ell_i = d$. Statements about a fixed U are not simultaneous over all Parseval frames unless explicitly stated. The Walsh application will use $N = 2^d$ for the dictionary size; its d is a Boolean ambient dimension, distinct from the matrix dimension in the preceding model.

1.3 Principal quantitative conclusions

The results can be located through the following guarantees, each with a different structural assumption.

1. **Exact polynomial evaluation.** A retained list of q coefficient vectors supports every allowed ordered circuit of size J in $O(J(q+1)^3)$ arithmetic operations per law atom, independent of its formal degree except through the circuit. For K repeated types and a fixed list of bounded-degree bucket features, the full joint law has polynomial arithmetic cost in the explicit population size, with the K -dependent exponent stated in Theorem 4.2. This gives exact moments and strict tails for the specified generators.

2. **A uniform diffuse scale.** For $b \geq 2$, $L = \log(4dm)$ and $p \geq 1$, if $\mu \leq [16b(L+p)]^{-1}$, then

$$\frac{F_p}{460} \leq (\mathbb{E}\|X\|^p)^{1/p} \leq 24F_p, \quad F_p = \sqrt{(d+p)/m} + (d+p)/m.$$

Theorem 5.6 also supplies two-sided strict quantiles under its stated confidence-dependent leverage condition.

3. **The rare-collision exponent.** With d, s, ε fixed, $0 < \varepsilon \leq 1$, the supremum over all finite Parseval frames of $\mathbb{P}(\|X\| > \varepsilon)$ has order $b^{-\lceil s\varepsilon \rceil}$ as $b \rightarrow \infty$. Theorem 6.1 states the finite upper bound and explains why fixing the finite row population first changes an integer endpoint. This is not a uniform joint-parameter sampling formula.
4. **A matching Walsh support class.** Write $N = 2^d$, $a = 2^t$, $D = d - t$ and $k = aK \geq 2$. Over all supports contained in at most K cosets of any t -dimensional core, with $K = 1$ or $2 \leq K \leq \min(2D, 2^D)$, Theorem 13.12 gives, for $0 < \eta \leq .1$,

$$\frac{\mathcal{R}}{400000} < m_*^{\text{class}} \leq \lceil 2^{32}\mathcal{R} \rceil, \quad \mathcal{R} = k \log(2k) \log(2N/k) + k \log(1/\eta).$$

The coefficients and core both vary, the observations are the original independent sampled rows, and the lower witness lies in the same class.

5. **Exact extension at controlled width.** Theorem 8.1 computes $\mathbb{E} \text{tr}((R^*R)^p)$ for a length- L product program in $O(L2^w t^{2p} n^{6p})$ arithmetic operations, with norm-moment comparison factor $d^{1/(2p)}$. Primitive boundary width, local term count, and internal matrix dimension are separate costs. The lower event in Proposition 8.2 is computed from the same program.
6. **Complete compressed moments with localized relations.** For $g_B > 4p$ and $\max(2, \lceil \log n \rceil) \leq p \leq n$, Theorem 16.3 gives

$$\frac{1}{4}(\sqrt{n/m} + n/m) \leq (\mathbb{E}\|H_{S,U}\|^{2p})^{1/(2p)} \leq 160\kappa(\sqrt{n/m} + n/m).$$

The explicit κ depends on local code geometry and frame weight; one localized six-relation gives $\kappa \leq 2^{1/3}$ for every fixed Parseval compression. A separate lower event and upper confidence bound are stated with their horizons.

7. **Growing coefficient spheres and a full spectral certificate.** For odd $r \geq 11$, $k = 2^r$, and $d = 2r$, Theorem 19.9 gives matching sample scale $k + (k/\log k) \log(1/\eta)$ simultaneously over every cubic orbit space spanned by at most $\lfloor \sqrt{k/r} \rfloor$ original row modulations, with all coefficients on each span. Separately, Theorem 18.3 bounds the complete fixed-support spectrum through an explicit nonnegative relation remainder. The improved first and second moment corrections alone do not control that remainder.
8. **A complete restricted moment and quantile calculus.** For maximum block size b , block-row/column degrees Δ_r, Δ_c , and the exact shared-primitive envelope law in Theorem 10.2,

$$b^{-1/2}\mathcal{F}_p \leq (\mathbb{E}\|Z\|^p)^{1/p} \leq \sqrt{\Delta_r \Delta_c} \mathcal{F}_p \quad (p > 0).$$

The same factors bracket every strict quantile, including zero and atom endpoints. Full sums, ordered products, adjoints and primitive identifications have exact updates. Fixed local parameters and logarithmic random boundary give polynomial work in the input and confidence bit length.

9. **All-name contraction and deep relation trees.** Section 17 proves constant-one contraction for arbitrarily interleaved mutually independent names, and gives actual conditional Gram updates for fresh and reused affine constraints. Even private innovations on a supplied junction tree give a real Parseval covariance proxy with no tree-depth factor. A width-four chain has growing coefficient dimension, complete original relation code and proxy norm at most two.
10. **Closed cubic interactions at larger coefficient rank.** Section 20 enlarges the structured coefficient spaces to dimension of order $k/(\log k)^2$ at the same matching sample scale. Different isotropic projections need not commute, yet their common block size is at most 2^{h+1} , where h is the dimension of their sum minus that of their intersection. Ordered sums and products stay in this algebra.
11. **Dense matrices with nearly exact moments and strict quantiles.** For t marked sites and g permutation groups of sizes M_j , Theorem 11.3 keeps all physical coefficient sectors, of total effective dimension

$$D_{\text{eff}} = 2^t \prod_{j=1}^g \left\lfloor \frac{(M_j + 2)^2}{4} \right\rfloor.$$

Its positive readout G_h satisfies $D_{\text{eff}}^{-1/(2h)} G_h \leq \|Z\| \leq G_h$ pointwise. The exact joint score law therefore gives both every positive norm moment and every strict quantile, with ratio at most $1 + \epsilon$ when $2h \log(1 + \epsilon) \geq \log D_{\text{eff}}$. Full signed sums, products, physical adjoints, primitive identifications, marking and partition refinement have exact rules, with all resulting sizes charged.

12. **Other dense geometries and a Walsh obstruction.** Section 12 supplies compact moment calculi for low-rank matrix-product operators, scalar identity shifts with residual rank strictly below the physical dimension, and homogeneous Clifford expressions. Their lower witnesses belong to the same matrix; general matching extreme quantiles require more information. For a balanced Sidon cross block C_m of size $q \geq 2$ and $m \geq k^2/(4q)$, Theorem 21.2 shows, with probability at least $1/16$, that every adaptive rank- R approximation with $R \leq q/1024$ has operator error greater than $\frac{1}{2}\sqrt{q/m}$. The statement has nonzero-rank content at $q \geq 1024$ and concerns the original Walsh observations.
13. **Continuation-safe signed cores.** Section 23 retains a conditional Fourier subgroup H and propagates one certified remainder norm. The exact product rule

$$P_H(FG) = (P_H F)(P_H G) + P_H[(F - P_H F)(G - P_H G)]$$

keeps the returning diffuse channel visible; a fresh affine restriction of rank s costs $2^{s/p}$ at order p , while a reused restriction costs no new-rank factor. A noncommuting Parseval continuation with only 32 core characters improves the wordwise envelope by an unbounded factor as its diffuse scale tends to zero.

14. **A common two-sided quadratic theorem.** Section 24 proves, under its explicit covariance, cap, and sample hypotheses, that every fixed real symmetric D satisfies

$$\frac{\nu n}{4m} \|D\| \leq (\mathbb{E} \|H D H\|^p)^{1/p} \leq 576 \frac{\nu n}{m} \|D\|.$$

The same realized H supplies the lower witness. Fixed-Sidon Walsh and independent-sign rows have $\nu = 1$; actual SparseStack layers have $\nu = 1/b$, giving scale $n/(mb)$ while retaining the true b^{-2} triangle collision probability.

15. **Exact aggregate moments without the aggregate law.** For w rational aggregate coordinates, degree R , even order $2p$, and a named positive-probability constraint event $\mathcal{E}$ (possibly the whole space), Section 25 uses $H_D = \binom{w+2Rp}{w}$ truncated coefficients and computes $M_p = \mathbb{E}[\text{tr}((Z^*Z)^p) \mid \mathcal{E}]$ exactly. It brackets the conditional norm moment within $d^{1/(2p)}$ and costs

$$O\left(\left[L2^{b+a}(w+1) + d^3 \log(p+1)\right] H_D^2\right).$$

The update rules cover products, adjoints, aggregate extension, and actual constraint identification. Matching strict quantiles still need a separate finite-law or lower-tail argument.

16. **Signed pooling progress and its exact frontier.** Section 26 proves the clipped sparse Walsh width for every support, a target-scale excess theorem for a broad embedded cubic profile class, and a simultaneous strict tail-count theorem for every Sidon profile on the actual iid sample. The latter yields fixed-confidence sampling at $O(k \log^2(2k) \log(2N/k))$ and leaves one height logarithm. Pair-lift, positive-witness, peak-orientation, and cross-cell counterexamples show why empirical spectra, positive surrogates, an all-sample log-free contraction, or cellwise triangle aggregation cannot supply the missing sharp bridge.
17. **Dense closure after symmetry breaking.** Section 22 gives an exact active-space update for arbitrary supplied low-rank seed interactions, with active dimension at most $r \sum_{\lambda} d_{\lambda}^2$, and an exact mixed-degree Clifford calculus from the complete attained joint Gram. Their comparison factors and score, rank, arithmetic, and bit costs are stated separately.

These guarantees state the main achievements without equating their input classes or suppressing the remaining complexity costs.

2 Representation before estimation

2.1 Random-word Grams and finite vacuum models

Let $Z = Z^*$ be a matrix-valued function on a probability space. For finite spaces, multiplication by Z is a self-adjoint operator $\mathcal{M}_Z$ on $\mathbb{C}^d \otimes L^2(\Omega)$. Let P project onto constant functions. Then

$$P\mathcal{M}_Z^r P = \mathbb{E}Z^r, \quad \mathbb{E} \text{tr} Z^{2q} = \|\mathcal{M}_Z^q P\|_{\text{HS}}^2. \quad (3)$$

For polynomial models on infinite spaces, the same formulas hold on the finite-dimensional polynomial subspaces needed for the stated finite moments; one need not assume that the full multiplication operator is bounded.

Lemma 2.1 (Finite cutoff). *Suppose a finite self-adjoint compression M_q satisfies $M_q^j P = \mathcal{M}_Z^j P$ for $0 \leq j \leq q$, with vectors identified in the full space. Then $P\mathcal{M}_Z^r P = \mathbb{E}Z^r$ for every $0 \leq r \leq 2q$.*

Proof. Write $r = a + c$ with $a, c \leq q$. Self-adjointness gives $P\mathcal{M}_Z^r P = (\mathcal{M}_Z^a P)^*(\mathcal{M}_Z^c P) = (M_q^a P)^*(M_q^c P) = P\mathcal{M}_Z^r P$. $\square$

If $Z^q = \sum_w \phi_w A_w$, its exact moment is instead written

$$\mathbb{E} \text{tr} Z^{2q} = \sum_{w, w'} \Gamma_{ww'} \text{tr}(A_w^* A_{w'}), \quad \Gamma_{ww'} = \mathbb{E}(\overline{\phi_w} \phi_{w'}). \quad (4)$$

The Gram Γ is positive semidefinite. Individual coefficient pairings need not be positive. A Loewner comparison $\Gamma \preceq \tilde{\Gamma}$ therefore yields a valid upper bound by applying it to each matrix coordinate; entrywise domination alone does not. Formula (4) and the vacuum formula are two realizations of the same squared Hilbert-space norm.

The multiplication rule depends on the primitive law. For Walsh characters, $\chi_A \chi_B = \chi_{A \triangle B}$; hence $\chi_A^2 = 1$. For normalized Hermite polynomials $h_j = H_j / \sqrt{j!}$ under a standard Gaussian,

$$h_a h_b = \sum_{r=0}^{\min(a,b)} r! \binom{a}{r} \binom{b}{r} \sqrt{\frac{(a+b-2r)!}{a!b!}} h_{a+b-2r}.$$

This follows by multiplying the generating functions $e^{tx-t^2/2} e^{ux-u^2/2} = e^{tu} e^{(t+u)x-(t+u)^2/2}$ and comparing coefficients. In particular, no universal penalty for a repeated primitive can replace these distinct exact multiplication laws.

2.2 Vacuum renewal and covariance-sensitive bounds

Assume $\mathbb{E}Z = 0$ and write a finite multiplication model relative to P as

$$M = \begin{pmatrix} 0 & B^* \\ B & C \end{pmatrix}, \quad V = B^* B, \quad K_r = P M^r P, \quad F_j = B^* C^{j-2} B \quad (j \geq 2).$$

The F_j are ordered first-return operators. They include both the cost of leaving the constant sector and the dynamics before returning.

Theorem 2.2 (Ordered renewal and covariance-spectrum comparison). *For $r \geq 2$,*

$$K_r = \sum_{j=2}^r F_j K_{r-j}, \quad K_0 = I, \quad K_1 = 0, \quad (5)$$

$$\sum_{r \geq 0} K_r z^r = \left[I - z^2 B^* (I - zC)^{-1} B \right]^{-1}. \quad (6)$$

These identities are formal power-series identities. Write the polar decomposition $B = W V^{1/2}$, and let $h_j \geq \|W^ C^j W\|$, $h_j \geq 0$. With $h(z) = \sum_{j \geq 0} h_j z^j$,*

$$|\operatorname{tr} K_r| \leq [z^r] \operatorname{tr} \left(I - z^2 h(z) V \right)^{-1}. \quad (7)$$

All statements may be restricted to a finite range of orders. If $V = 0$, every positive-order vacuum moment is zero.

Proof. Set $L_r = (I - P) M^r P$. Block multiplication gives $K_r = B^* L_{r-1}$ and $L_j = B K_{j-1} + C L_{j-1}$, with $L_0 = 0$. Iterating gives (5); summing yields (6). Every term in the expanded recurrence is an ordered product $F_{j_1} \cdots F_{j_t}$ with $j_i \geq 2$ and $\sum j_i = r$. Put $D_i = W^* C^{j_i-2} W$. Cyclicity and Schatten Hölder give

$$|\operatorname{tr}(F_{j_1} \cdots F_{j_t})| = |\operatorname{tr}(V D_1 \cdots V D_t)| \leq \prod_i \|V D_i\|_{S_t} \leq \operatorname{tr}(V^t) \prod_i h_{j_i-2}.$$

For $t = 1$ this is the trace-norm inequality. Summing the nonnegative majorants proves (7). Finally $V = 0$ implies $B = 0$. $\square$

Corollary 2.3 (Graded rest space). *Suppose the rest space has orthogonal sectors $\mathcal{H}_1, \dots, \mathcal{H}_a$, $\operatorname{range} B \subseteq \mathcal{H}_1$, and a symmetric nonnegative matrix G satisfies $\|P_i C P_j\| \leq G_{ij}$. Put*

$$T(v) = \begin{pmatrix} 0 & \sqrt{v} e_1^* \\ \sqrt{v} e_1 & G \end{pmatrix}.$$

If $\lambda_1, \dots, \lambda_d$ are the eigenvalues of V , then for $q \geq 1$

$$\operatorname{tr} K_{2q} \leq \sum_{j=1}^d (T(\lambda_j)^{2q})_{00} \leq \frac{\operatorname{tr} V}{v} (T(v)^{2q})_{00}, \quad v \geq \|V\|, \quad v > 0.$$

Proof. Block-norm induction gives $\|P_1 C^j P_1\| \leq (G^j)_{11}$. Use these h_j in Theorem 2.2 and apply the scalar renewal identity for $T(\lambda)$. Each positive-order coefficient is a polynomial in λ with nonnegative coefficients and zero constant term; thus $\lambda^a \leq \lambda v^{a-1}$ proves the second inequality. $\square$

This comparison is best possible from V and G alone. Indeed, on $\mathbb{R}^a \otimes \mathbb{R}^d$ take $Bx = e_1 \otimes V^{1/2}x$ and $C = G \otimes I_d$. After diagonalizing V , the full matrix is a direct sum of the scalar $T(\lambda_j)$, giving equality in the covariance-spectrum bound (the later $\text{tr } V/v$ relaxation need not be an equality). Further improvements must retain more than covariance and scalar sector norms.

Proposition 2.4 (Exact SparseStack covariance). *For (2),*

$$V = \mathbb{E}X^2 = \frac{1}{m}V_0, \quad V_0 = \sum_{i < j} A_{ij}^2 = (d+1)I_d - 2 \sum_i \ell_i u_i u_i^*. \quad (8)$$

In particular $0 \preceq V_0 \preceq (d+1)I_d$ and $\text{tr } V_0 = d(d+1) - 2 \sum_i \ell_i^2$.

Proof. Distinct unordered edges have zero scalar covariance because an independent sign occurs oddly; different centered blocks also have zero covariance. The remaining factor is $1/(sb)$. In the sum of A_{ij}^2 , terms involving $\langle u_i, u_j \rangle$ total $I - \sum_i \ell_i u_i u_i^*$, while the squared-length terms total $dI - \sum_i \ell_i u_i u_i^*$. Both equalities follow from $\sum_i u_i u_i^* = I$ and $\sum_i \ell_i = d$. $\square$

A useful test is the scalar two-row frame $u_1 = u_2 = 1/\sqrt{2}$ with $s = 1$. Its error is $\sigma_1 \sigma_2 \mathbf{1}_{h_1=h_2}$, so every positive even moment equals $1/b$. An exact three-state vacuum model has successive off-diagonal entries $a = 1/\sqrt{b}$ and $c = \sqrt{1-1/b}$ and zero diagonal. Its first returns satisfy $F_{2j} = a^2 c^{2j-2}$ and $F_{2j+1} = 0$; renewal gives $K_{2q} = a^2(a^2 + c^2)^{q-1} = 1/b$. Charging a new collision at every return would fail this test. Odd returns cannot generally be suppressed: for three scalar rows $1/\sqrt{3}$, the triangle gives $\mathbb{E}X^3 = 16/(9b^2) > 0$ when $s = 1$.

3 Overlap identities and legitimate state reduction

3.1 The actual collision Gram

For a word w of signed-hash edges, form in each block the multigraph of its incident row labels. Let $r_g(w) = v_g(w) - c_g(w)$ be its equality rank, where v_g counts incident vertices and c_g connected components; empty graphs have rank zero. Let $r(w) = \sum_g r_g(w)$.

Proposition 3.1 (Word weights and positive half-word decomposition). *The expectation of the scalar word is zero unless every blockwise vertex degree is even, and otherwise equals $b^{-r(w)}$. For length- q half words,*

$$K(w, w') = \mathbf{1}_{\{\text{matching blockwise sign parity}\}} b^{-r(w \cup w')}. \quad (9)$$

This is the actual Gram kernel in (4). If $\Pi = (\pi_g)_{g=1}^s$ specifies the actual equal-hash partition in each block and $a_g = |\pi_g|$, then

$$\mathbb{E} \text{tr } X^{2q} = s^{-2q} \sum_{\Pi} \left(\prod_g \frac{(b)_{a_g}}{b^n} \right) \sum_{\alpha} \|C_{\Pi, \alpha}\|_F^2. \quad (10)$$

Here $(b)_a = b(b-1) \cdots (b-a+1)$, zero if $a > b$, and $C_{\Pi, \alpha}$ is the sum of ordered coefficient products of all compatible length- q words with sign parity α .

Proof. Each sign has expectation zero in odd degree. A connected hash constraint on v labels has probability b^{1-v} . Different required components may accidentally receive the same colour; there

is no inequality condition between them. This proves the word weight and (9). In contrast, a specified actual partition with a classes requires distinct colours for its classes and has probability $(b)_a/b^n$. Conditional on these partitions, Fourier Parseval in the independent signs, entry by entry in X^q , proves (10). $\square$

The distinction between required constraints and actual partitions is essential: using $(b)_c$ in a required-constraint weight would discard legitimate coincidences. Positivity in (10) occurs after coherent coefficient summation. For example, the real Parseval frame with $UU^* = I_3 - \mathbf{1}\mathbf{1}^*/3$, expanded with the oriented rank-one coefficients $u_i u_j^*$ rather than the symmetric A_{ij} , has the surviving word $(1, 2), (1, 3), (2, 1), (3, 1)$ with trace coefficient $-2/81$ and hash weight b^{-2} . Thus even fourth-order surviving words need not be positive.

3.2 One block-partition formula contains the low-order corrections

For one block let $Y = \sum_e z_e A_e$. In expanding $s^{-r} \mathbb{E} \operatorname{tr}(Y_1 + \dots + Y_s)^r$, partition the positions $[r]$ according to their block label. For a partition π let $H_\pi(b)$ be the ordered coefficient trace sum whose scalar weight is obtained by applying Proposition 3.1 separately in each class of π . Then, exactly,

$$\mathbb{E} \operatorname{tr} X^r = s^{-r} \sum_{\substack{\pi \text{ partition of } [r] \\ \text{no singleton classes}}} (s)_{|\pi|} H_\pi(b). \quad (11)$$

The proof is grouping the expanded block assignments; a singleton block is centered, and $(s)_{|\pi|}$ counts injective assignments of distinct blocks. This formula retains multiplication order even when occurrences of different blocks interlace.

For clarity, its fourth-order specialization can be contracted completely. Put

$$R_1 = \sum_e \operatorname{tr} A_e^4, \quad T = \sum_{e,f} \operatorname{tr}(A_e A_f A_e A_f), \quad W = 2 \operatorname{tr} V_0^2 + T,$$

and let R_3 sum $\operatorname{tr}(A_{e_1} A_{e_2} A_{e_3} A_{e_4})$ over all 24 edge orders of every undirected simple four-cycle. Then

$$\mathbb{E} \operatorname{tr} X^4 = \frac{R_1}{ms^2} + \frac{W - 3R_1/s}{m^2} + \frac{R_3}{m^3}. \quad (12)$$

Indeed the one-block edge multisets are one edge four times, two distinct edges twice each, or a four-cycle. Their rank weights are b^{-1}, b^{-2}, b^{-3} ; the middle coefficient is $W - 3R_1$. The two-block contribution has coefficient W , and (11) gives (12). The subtraction $-3R_1/s$ removes the repeated-edge overcount; it cannot be omitted from an exact identity.

At sixth order, define Q_j as the trace sum of sign-surviving six-edge one-block words of rank j ($1 \leq j \leq 5$). Let B_j sum all 15 choices of a two-position block carrying a repeated edge and a four-position block of rank j ($1 \leq j \leq 3$). Let C_Δ sum the ten partitions into two triples, each triple carrying an ordered triangle. Finally, P_6 sums the 15 pair partitions, each pair carrying an arbitrary repeated edge. These definitions specify finite ordered coefficient sums without an independence assumption on edges inside a block. Formula (11) becomes

$$\begin{aligned} \mathbb{E} \operatorname{tr} X^6 = & \sum_{j=1}^5 \frac{Q_j}{m^j s^{5-j}} \\ & + \left(1 - \frac{1}{s}\right) \left\{ \frac{B_1}{m^2 s^2} + \frac{B_2}{m^3 s} + \frac{B_3 + C_\Delta}{m^4} \right\} \\ & + \left(1 - \frac{1}{s}\right) \left(1 - \frac{2}{s}\right) \frac{P_6}{m^3}. \end{aligned} \quad (13)$$

The possible block sizes are 6, $4 + 2$, $3 + 3$, and $2 + 2 + 2$; their total hash ranks explain every denominator. In particular the triangle-pair term survives. The coefficient P_6 is the sixth

trace moment of $\sum_e \gamma_e A_e$ for independent standard Gaussians, by the Gaussian pairing formula. Equations (12)–(13) are consistency conditions on an all-order rule, not separate recipes to be extrapolated.

3.3 Innovation weights must come from the Gram

In an inner-product space, let columns of a map R be retained functions, with Gram $G = R^*R$. For a new function f , write $g = R^*f$ and $\nu = \|f\|^2$. The orthogonal projection is $RG^\dagger g$, where $G^\dagger$ is the Moore–Penrose inverse, and its residual variance is

$$\|f - RG^\dagger g\|^2 = \nu - g^*G^\dagger g \geq 0. \quad (14)$$

To check the singular case, g lies in range G because it annihilates $\ker R$; expansion of the squared norm then proves the identity. Zero residual variance means that no new normalized channel exists.

For an equality indicator δ with probability $p \in (0, 1)$, the basis $1, (\delta - p)/\sqrt{p(1-p)}$ gives the multiplication matrix

$$J_p = \begin{pmatrix} p & \sqrt{p(1-p)} \\ \sqrt{p(1-p)} & 1-p \end{pmatrix}, \quad J_p^2 = J_p. \quad (15)$$

Hence $(J_p^r)_{00} = p$ for every $r \geq 1$. Reprojecting onto constants after each multiplication would give p^r . The off-diagonal amplitude is the square root of the actual residual variance, not a guessed probability penalty.

There is a similarly explicit first cycle. With independent uniform hashes on $Q \geq 2$ colours, set $p = 1/Q$ and $\eta_{ij} = \delta_{ij} - p$ for three distinct labels. Distinct edge residuals are orthogonal, but $\mathbb{E}\eta_{12}\eta_{13}\eta_{23} = p^2(1-p)$. Thus

$$\begin{aligned} \zeta &= \eta_{12}\eta_{13} - p\eta_{23} = \mathbf{1}_{h_1=h_2=h_3} - p(\delta_{12} + \delta_{13} + \delta_{23}) + 2p^2, \\ \mathbb{E}\zeta^2 &= p^2(1-p)(1-2p). \end{aligned} \quad (16)$$

It is orthogonal to constants and to all three edge residuals. To verify this, use pairwise independence of distinct equalities, their triple probability p^2 , and subtract the projection coefficient $p^2(1-p)/(p(1-p)) = p$. The residual variance is $p^2(1-p)^2 - p^3(1-p)$. At $Q = 2$ the residual vanishes; at $Q \geq 3$ the constant, three edges, and triangle residual span the five actual partition states. Degenerate Gram ranks have to be removed rather than normalized.

The same calculation explains the surviving coefficient join in the research question. Take four distinct terminal labels a, b, c, d and an internal label set disjoint from them. Conditional on $h_a = h_b$ and $h_c = h_d$, put $x_v = \mathbf{1}_{h_v=h_a}$ and $y_v = \mathbf{1}_{h_v=h_c}$, for internal labels v . Let $B_v = A_{av}A_{vb}$ and $E_v = A_{cv}A_{vd}$, and retain $B_\Sigma = \sum_v B_v$, $E_\Sigma = \sum_v E_v$, $J = \sum_v B_v E_v$. If the two terminal buckets are separated, $x_v y_v = 0$ and $\mathbb{E}x_v y_w = p^2$ for $v \neq w$. If they merge, $x_v = y_v$ and $\mathbb{E}x_v y_v = p$. Consequently the constant coefficient channels are exactly

$$p^2(B_\Sigma E_\Sigma - J), \quad p^2 B_\Sigma E_\Sigma + p(1-p)J. \quad (17)$$

The separated and merged sectors themselves have masses $p^2(1-p)$ and p^3 . These are conditional coefficient channels, with the sector masses still to be included in a full Gram energy. The join is not eliminated by the Parseval identity for either coherent sum.

3.4 A frame contraction that really survives repeated visits

For a specified internal label set I , put $H_v = u_v u_v^*$, $G_I = \sum_{v \in I} H_v \preceq I_d$, and $\mu_I = \max_{v \in I} \ell_v$. Empty sums and maxima are zero, and the convention is $\mu_I^0 = 1$ even when $\mu_I = 0$. The positive

tensors

$$\Gamma_t = \sum_{v \in I} H_v^{\otimes t}, \quad \Gamma_{t+1} \preceq \mu_I \Gamma_t \otimes I_d, \quad \|\Gamma_t\| \leq \mu_I^{t-1}, \quad \text{tr } \Gamma_t \leq d \mu_I^{t-1} \quad (18)$$

follow term by term from $0 \preceq H_v \preceq \mu_I I$. They determine all even homogeneous sums in one row. Their tensor entries must not be reshuffled and treated as a positive map without checking the relevant slot ordering.

For an external label $a \notin I$, define

$$(V_a x)_v = A_{av} x, \quad D_a = \bigoplus_{v \in I} A_{av}, \quad g_a = \left\| \sum_{v \in I} A_{av}^2 \right\|, \quad \theta_a = \max_{v \in I} \|A_{av}\|.$$

Then $\|V_a\| = \sqrt{g_a}$ and $\|D_a\| = \theta_a$.

Theorem 3.2 (Live-label contraction). *For every boundary sequence $a_1, \dots, a_r$ outside I , $r \geq 2$,*

$$\sum_{v \in I} A_{a_1 v} \cdots A_{a_r v} = V_{a_1}^* D_{a_2} \cdots D_{a_{r-1}} V_{a_r}, \quad (19)$$

$$\left\| \sum_{v \in I} A_{a_1 v} \cdots A_{a_r v} \right\| \leq \sqrt{g_{a_1} g_{a_r}} \prod_{j=2}^{r-1} \theta_{a_j}. \quad (20)$$

Moreover $g_a \leq (d+3)\ell_a$ and $\theta_a \leq \min\{1, 2\sqrt{\ell_a \mu_I}\}$. At even length $r = 2t$ this gives

$$\left\| \sum_{v \in I} A_{a_1 v} \cdots A_{a_{2t} v} \right\| \leq (d+3) 4^{t-1} \mu_I^{t-1} \prod_{j=1}^{2t} \|u_{a_j}\|.$$

Proof. Block-diagonal multiplication keeps the same row label at every visit and proves (19); taking norms proves (20). Direct expansion gives

$$\sum_{v \in I} A_{av}^2 = (\text{tr } G_I) H_a + H_a G_I + G_I H_a + \ell_a G_I.$$

As a subsum of positive squares it is dominated by the sum over the full frame, including the algebraic loop $A_{aa} = 2H_a$ for that calculation. The latter is $(d+2)H_a + \ell_a I$ and has norm $(d+3)\ell_a$. The two potentially nonzero eigenvalues of A_{av} are $\langle u_a, u_v \rangle \pm \sqrt{\ell_a \ell_v}$, giving the leverage bound. For distinct labels it is also a compression of the norm-one matrix $e_a e_v^* + e_v e_a^*$, giving the bound by one. Substitution proves the final inequality. $\square$

For arbitrary compatible matrices C_v , the related positive update

$$\sum_v C_v A_{av}^2 C_v^* \preceq \theta_a^2 \sum_v C_v C_v^*$$

is preserved under every later congruence, including noncommuting coefficient congruences; no stability under arbitrary one-sided multiplication in Loewner order is asserted. The live-register factorization is a genuine all-length upper rule for one live label. It is not a recurrence on the norm of a previously summed, possibly cancelling join. The operator register or equivalent higher tensor remains part of its state.

3.5 The exact acceptance test for a proposed compression

Theorem 3.3 (Descent through an encoding). *Let $S : E \rightarrow \mathcal{H}$ encode coefficient arrays, $G = S^* S$, and let $S' : E' \rightarrow \mathcal{H}'$ be the encoding after a proposed update $T : E \rightarrow E'$. There is a well-defined linear map $\bar{T}$ on $\text{range } S$ with $S' T = \bar{T} S$ if and only if*

$$\ker G \subseteq \ker(S' T). \quad (21)$$

An induced map of norm at most L exists if and only if

$$T^* G' T \preceq L^2 G, \quad G' = S'^* S'. \quad (22)$$

Proof. The proposed value $S'Tx$ depends only on Sx precisely when it annihilates every vector in $\ker S = \ker G$. The norm assertion is the inequality $\|S'Tx\|^2 \leq L^2\|Sx\|^2$ for every representative x . $\square$

Common multiplication of a genuinely zero random matrix preserves zero. Label-dependent multiplication of its separately indexed summands need not. This distinction is the reason a current random-word Gram can be exact yet insufficient for the next operation on a reduced coefficient state.

An operational sufficient condition is *last incidence*. If X_0 is proposed for retirement, Y_0 is retained, and every remaining factor depends only on Y_0 and fresh primitives independent of (X_0, Y_0) , then

$$\mathbb{E}[L(Y_0)F(X_0, Y_0)R(Y_0) \mid Y_0] = L(Y_0)\mathbb{E}[F \mid Y_0]R(Y_0).$$

Thus elimination at last incidence preserves every later ordered expectation. For a Frobenius energy this rule must be applied to the complete paired expression F^*F , not separately to a prematurely compressed half.

For example, sweep a fixed single-block r -edge word in multiplication order, with each formal vertex assigned a row from a specified eligible set of size at most n . If these eligible sets are pairwise disjoint and at most w vertices are live simultaneously, retain the actual live row choices, their actual hash partition, a sign-parity bit per live label, and a full matrix message. Introducing a fresh hash joins each of c existing classes with probability $1/b$ or creates a new class with probability $(b - c)/b$. An edge tests actual equality, toggles parity and multiplies its matrix in order. At last incidence, odd parity is discarded and the partition is restricted, preserving induced equalities on remaining labels. These updates are exact by conditional expectation and colour symmetry. The state count is at most

$$K = n^w 2^w \text{Bell}_b(w), \quad \text{Bell}_b(w) = \sum_{c=0}^{\min(b,w)} \left\{ \begin{matrix} w \\ c \end{matrix} \right\},$$

with $O(d^2 K)$ matrix storage, $O(wK)$ state bookkeeping, and conservative arithmetic cost $O(rn(w + 1)K(d^3 + w))$. This is a tractable width restriction, not an ambient-uniform result. Shared eligible sets or paired copies need additional alias and exclusion information. A label returning with the same row index uses the same primitive even after a long absence; it is not fresh.

The following chapters turn these representation and closure principles into exact-law and analytic certificates. Their different structural costs are part of the theorem statements, not implementation details to be suppressed.

4 A signed Gram calculus with exact and approximate law interfaces

The reusable object in this part of the report is a small matrix algebra, equipped with a joint probability law and, when necessary, an error radius. These three components have different responsibilities. The algebra preserves ordered coefficient contractions. The joint law preserves reused random constraints. The radius transfers a statement about a tractable nominal matrix to the same original input. Keeping the components separate makes it possible to reuse the algebra without incorrectly transferring a probability calculation from one model to another.

4.1 Model and deterministic algebra

Let $U \in \mathbb{R}^{n \times d}$ have rows u_i^T and satisfy $U^T U = I_d$. In each block $g \in \{1, \dots, s\}$, let h_{gi} be uniform on $\{1, \dots, b\}$ and let σ_{gi} be a uniform sign. All these primitives are independent across

their actual indices (g, i) . Set

$$A_{ij} = u_i u_j^T + u_j u_i^T, \quad E_{g,ij} = s^{-1} \sigma_{gi} \sigma_{gj} \mathbf{1}_{\{h_{gi}=h_{gj}\}} A_{ij}, \quad X = \sum_g \sum_{i < j} E_{g,ij}. \quad (23)$$

Thus X is the centered SparseStack Gram error. The primitive identity includes the block index: repeated use of $E_{g,ij}$ reuses randomness, whereas (g, i) and (g', i) have independent signs when $g \neq g'$.

A matrix circuit below has scalar multiples of I_d and specified random matrix inputs, and uses addition, real scalar multiplication, transpose, and ordered multiplication. Its size J counts inputs and gates. Its formal degree is zero on scalar identities, one on designated generators, the maximum at addition, and the sum at multiplication. Deterministic matrix insertions other than scalar identities require their own representation; they are not silently included in this definition.

Theorem 4.1 (The signed Gram algebra). *Fix $V \in \mathbb{R}^{q \times d}$ and $G = VV^T$, with no rank assumption. If every designated input has the form $V^T C V$, every circuit value has the form*

$$\mathcal{P}(\alpha, C) = \alpha I_d + V^T C V. \quad (24)$$

The exact operations on its coefficient state are

$$\begin{aligned} (\alpha, C) + (\beta, D) &= (\alpha + \beta, C + D), \\ a(\alpha, C) &= (a\alpha, aC), \quad (\alpha, C)^T = (\alpha, C^T), \\ (\alpha, C)(\beta, D) &= (\alpha\beta, \alpha D + \beta C + CGD). \end{aligned} \quad (25)$$

These rules are associative and valid under arbitrary further circuit extension. A state uses $q^2 + 1$ real coefficients. Dense evaluation costs $O(J(q+1)^3)$ arithmetic operations and, without recomputation, at most $O(J(q+1)^2)$ intermediate storage.

For a possibly nonsymmetric state put

$$F = \alpha(C + C^T) + C^T G C, \quad r = \text{rank } G. \quad (26)$$

For $t \geq 0$ its exact norm test is

$$\|\mathcal{P}(\alpha, C)\| \leq t \iff (t^2 - \alpha^2)G - GFG \succeq 0, \quad \text{and, if } r < d, \quad t^2 \geq \alpha^2. \quad (27)$$

In particular, if $\alpha = 0$ and $C = C^T$, the test is simply $tG - GCG \succeq 0$ and $tG + GCG \succeq 0$.

Proof. Multiplication in physical space gives

$$(\alpha I + V^T C V)(\beta I + V^T D V) = \alpha\beta I + V^T(\alpha D + \beta C + C(VV^T)D)V.$$

The other operations are immediate, and induction proves the circuit claim. The coefficient product $C \star D = CGD$ is associative. Although the representation may be nonunique, the kernel of the representation map is an algebra ideal, so two equivalent states stay equivalent after any later operation. The operation and storage bounds follow from dense matrix multiplication; no expansion into words is required.

For the norm test, direct multiplication gives $\mathcal{P}^T \mathcal{P} = \alpha^2 I + V^T F V$. The matrix $K = (t^2 - \alpha^2)I - V^T F V$ preserves the orthogonal decomposition $\mathbb{R}^d = \text{range}(V^T) \oplus \ker V$. On the first summand, $K \succeq 0$ is equivalent to $V K V^T \succeq 0$, because V^T maps onto that summand. This is the matrix inequality in (27). On the second summand K is the scalar $t^2 - \alpha^2$, and this summand is present exactly when $r < d$. Finally $K \succeq 0$ is equivalent to the asserted norm bound. For a symmetric output apply the same range argument to $tI \pm V^T C V$. $\square$

The scalar condition in (27) is necessary only when $r < d$. For example, $V = I_d$, $\alpha = 1$, $C = -I_d$ represents zero, so an unconditional requirement $t \geq |\alpha|$ would be false. Conversely, an identity minus a projector onto a proper subspace has norm one. The oracle uses neither G^{-1} nor a square root of G . Rational inputs and rational thresholds therefore permit exact rational positive semidefiniteness tests, including equality at a strict-tail boundary.

There are also exact rules for enlarging or identifying the geometric list. If $V' = [V; W]$, embed C as $\text{diag}(C, 0)$ and retain the bordered Gram

$$G' = \begin{pmatrix} G & VW^T \\ WV^T & WW^T \end{pmatrix}. \quad (28)$$

If an identification gives $V_{\text{old}} = HV_{\text{new}}$, transport a coefficient by $C_{\text{new}} = H^T C_{\text{old}} H$. Both assertions follow by substitution in (24). The cross inner products in (28) are indispensable. Identifying two equal geometric vectors does not identify their originally distinct random signs.

4.2 The probability interface: retain a joint feature law

Theorem 4.2 (Exact law transfer and its two finite implementations). *Suppose a finite random descriptor Θ has a known joint law and determines all the coefficient inputs of Theorem 4.1, including their shared occurrences. The row list V and its Gram may depend on the descriptor atom and are held fixed during that atom's circuit evaluation. It then determines every circuit's exact norm law by deterministic evaluation. In particular,*

$$\mathbb{P}\{\|P\| > t\} = \sum_{\theta} \mathbb{P}\{\Theta = \theta\} \mathbf{1}_{\{\|P(\alpha_{\theta}, C_{\theta})\| > t\}}. \quad (29)$$

The following are two implementations for the actual SparseStack law.

1. Arbitrary geometry, bounded collision rank. Let π_g be the complete equality partition of the n hashes in block g , and let

$$r_g = n - |\pi_g|, \quad R = \sum_g r_g.$$

Retain the full partitions, the identities and Gram matrix of all row copies belonging to nonsingleton buckets, and one relative sign per nonanchor row in each such bucket. There are at most $2R$ active copies and exactly R independent uniform relative signs. This descriptor determines every circuit in the named inputs $E_{g,ij}$.

2. Repeated coefficient types, an additive feature list. Let type $a \in \{1, \dots, K\}$ have M_a labeled rows all equal to $v_a \in \mathbb{R}^d$, and put $h = \sum_a M_a$. A bucket has type-count vector n and signed-total vector z . Specify in advance integer features $\phi_j(n, z)$, $1 \leq j \leq q_f$, satisfying

$$\phi_j(0, 0) = 0, \quad |\phi_j(n, z)| \leq C_j \left(\sum_a n_a \right)^{p_j}, \quad p_j \in \mathbb{N}, \quad p_j \geq 1. \quad (30)$$

Retain their joint aggregate $T_j = \sum_{c=1}^b \phi_j(n_c, z_c)$. Its exact law is given by the labeled convolution below. Every circuit whose designated matrix inputs are proved functions of this joint feature vector then satisfies (29), with no omitted collision outcomes.

Proof. The first statement is the pushforward of the finite primitive law through an exact deterministic map. It preserves products of dependent inputs because their coefficients are evaluated on the same atom.

For the first implementation, a bucket of size $m \geq 2$ contributes m active copies but only $m - 1$ to collision rank, and $m \leq 2(m - 1)$. A common sign flip of a bucket fixes all its edge

inputs. Fixing one anchor sign leaves $m - 1$ independent uniform relative signs. With V the active row list, every nonzero input is $V^T C V$, where for its two active indices a, c ,

$$C = s^{-1} \sigma_{gi} \sigma_{gj} (e_a e_c^T + e_c e_a^T).$$

Inactive inputs are zero. This proves sufficiency, including triangles, repeated labels, and cross-block geometric coincidences.

For the second implementation define $F_b(m, t)$ to count assignments of a fixed labeled population with type counts m to b labeled buckets and independent signs, having aggregate feature t . For one bucket,

$$F_1(m, t) = \sum_{0 \leq r \leq m} \left(\prod_a \binom{m_a}{r_a} \right) \mathbf{1}_{\{\phi(m, m-2r)=t\}}. \quad (31)$$

The product of binomial coefficients chooses the negative signs. For two disjoint families of buckets the exact rule is

$$F_{a+b}(m, t) = \sum_{r, u} \left(\prod_j \binom{m_j}{r_j} \right) F_a(r, u) F_b(m - r, t - u). \quad (32)$$

The binomial factor chooses which original labels enter the first family, and each assignment on the two disjoint label sets is counted once. This proves the rule bijectively, and also its associativity. The zero-bucket identity is $F_0(0, 0) = 1$ and zero elsewhere. Consequently

$$\mathbb{P}\{T = t\} = F_b(M, t) / (2b)^h. \quad (33)$$

All features remain joint throughout the calculation. This proves the claimed implementation and completes the proof. $\square$

The two implementations control different costs. The first restricts the number of active random interactions while allowing arbitrary coefficient geometry. The second aggregates arbitrarily many collisions using an explicit symmetry of the coefficient population. Neither follows from polynomial degree alone.

For a circuit of formal edge degree D in (23), the unreduced sign degree is at most $2D$. Writing a hash equality as $\sum_{c=1}^b \mathbf{1}_{\{h_{gi}=c\}} \mathbf{1}_{\{h_{gj}=c\}}$ gives unreduced total degree at most $4D$ in signs and one-hot hash variables. Their actual relations remain enforced by the law. The coefficient-state size is independent of D , while circuit length, probability-state complexity, and output bit lengths are separate costs.

4.2.1 Collision-rank truncation as a two-sided certificate

For a specified complete partition tuple and its relative signs, the exact mass is

$$w(\pi, \epsilon) = 2^{-R} \prod_{g=1}^s \frac{(b)_{n-r_g}}{b^n}, \quad (34)$$

where $(b)_j = b(b-1) \cdots (b-j+1)$, $(b)_0 = 1$, and $(b)_j = 0$ for $j > b$. Indeed distinct classes of a complete partition must occupy distinct buckets. An equality word alone has a different probability: three edges of one triangle impose two independent equalities, whereas (34) also specifies all inequalities between distinct classes.

For a circuit P and budget $k \geq 0$, evaluate only the atoms $R \leq k$ and define

$$T_{k,P}(t) = \sum_{R \leq k, \epsilon} w(\pi, \epsilon) \mathbf{1}_{\{\|P(\pi, \epsilon)\| > t\}}.$$

Splitting the original event into retained and omitted atoms gives

$$T_{k,P}(t) \leq \mathbb{P}\{\|P\| > t\} \leq T_{k,P}(t) + \mathbb{P}\{R > k\}. \quad (35)$$

In particular the lower certificate is for the same input; it is not worst-case sharpness demonstrated by a different example.

Write $N_e = s\binom{n}{2}$ and $\lambda = N_e/b$. If $R \geq j$, spanning trees within equality classes contain j block-labeled forest edges. Each such forest imposes exactly j independent equalities and has probability b^{-j} . A union bound over at most $\binom{N_e}{j}$ forests proves

$$\mathbb{P}\{R \geq j\} \leq \min \left\{ 1, \binom{N_e}{j} b^{-j} \right\} \leq \min\{1, \lambda^j/j!\}. \quad (36)$$

Cyclic edge sets are not assigned the forest probability. The exact remainder is also available: with $\left\{ \begin{smallmatrix} n \\ q \end{smallmatrix} \right\}$ a Stirling number,

$$A(z) = b^{-n} \sum_{r=0}^{n-1} \left\{ \begin{smallmatrix} n \\ n-r \end{smallmatrix} \right\} (b)_{n-r} z^r, \quad \mathbb{P}\{R = r\} = [z^r] A(z)^s. \quad (37)$$

This follows by first counting partitions with $n - r$ classes and then using block independence.

For the particular output X there is a sharper deterministic envelope,

$$\|X\| \leq R/s \leq n - 1. \quad (38)$$

A coordinate collision block on a bucket of size $q \geq 2$ is $\epsilon\epsilon^T - I_q$ and has norm $q - 1$. Thus its block norm is at most r_g , compression by the isometry U is contractive, and the triangle inequality proves (38). The omitted term in (35) may therefore be replaced, for this output, by

$$\mathbb{P}\{R \geq \max(k + 1, \lfloor st \rfloor + 1)\}.$$

The integer $\lfloor st \rfloor + 1$ is forced by the strict threshold. For every real $\rho > 0$ the same law supplies the moment certificate

$$0 \leq \mathbb{E}\|X\|^\rho - \mathbb{E}[\|X\|^\rho \mathbf{1}_{\{R \leq k\}}] \leq s^{-\rho} \sum_{r > k} r^\rho [z^r] A(z)^s. \quad (39)$$

General circuits admit analogous remainders by propagating a scalar norm envelope through their gates; (38) is not such an envelope for every polynomial output.

The state count is explicit. Encode each nonsingleton class by the star joining its least label to all other labels. This injects rank- r partition tuples into r -element subsets of the N_e possible edges. The encoding is a star, but all clique edges of its bucket are used in the input matrices. There are at most

$$\sum_{r=0}^{\min(k, N_e)} 2^r \binom{N_e}{r} \quad (40)$$

signed states. After input access and precomputation of the partition weights, constructing a Gram per partition and evaluating its sign choices gives the conservative arithmetic bound

$$O \left(\sum_{r \leq k} \binom{N_e}{r} [\text{poly}(s, k) + k^2 d + 2^r (J + 1)(k + 1)^3] \right). \quad (41)$$

It is polynomial for fixed k , with an exponent depending on k ; it is not a fixed-parameter bound with a universal polynomial exponent. The remainder alone is much cheaper: the near-diagonal recurrence

$$f_r(n) = \left\{ \begin{smallmatrix} n \\ n-r \end{smallmatrix} \right\} = f_r(n-1) + (n-r)f_{r-1}(n-1)$$

computes ranks through k in $O(n(k+1))$ operations, followed by degree- k truncated polynomial powering. Falling factorials and b^n , needed for the partition weights, can be prepared in $O(n)$ additional arithmetic operations; their integer bit lengths are charged separately.

As a concrete accuracy regime, if $n \geq 2$, $\beta, A_0, B_0 > 0$ are fixed, $b \geq (s/2)n^{2+\beta}$, and $\beta(k+1) \geq A_0 + B_0$, then $\mathbb{P}(R > k) \leq n^{-A_0 - B_0}$. At every computed threshold with $T_{k,P}(t) \geq n^{-A_0}$, the two bounds in (35) differ by a factor at most $1 + n^{-B_0}$. This is an explicit two-sided regime for arbitrary, potentially noncommuting frame geometry. It requires many buckets and does not assert an optimal embedding dimension.

One can first marginalize primitives absent from the circuit. If block g uses only m_g row labels after deleting identically zero generators, replace n by m_g in that block, N_e by $\sum_g \binom{m_g}{2}$, and $A(z)^s$ by the product of its block-specific rank polynomials. A block with $m_g = 0$ has the single empty partition and rank polynomial $A_g(z) = 1$. The unused independent primitives integrate to one. This exact pruning removes zero-row padding; discarding small nonzero rows instead requires the perturbation interface below.

4.2.2 Typed populations and fixed features

For repeated types let V now have the K rows v_a^T , and let $D_M = \text{diag}(M_1, \dots, M_K)$. The quadratic feature is the entire symmetric signed score

$$S_g = \sum_{c=1}^b z_{gc} z_{gc}^T, \quad H = V^T D_M V, \quad X = V^T \left(s^{-1} \sum_g S_g - D_M \right) V. \quad (42)$$

The identity follows by expanding each bucket's signed vector sum and removing each physical row's diagonal once. In particular $z_a^2 - n_a = 2 \sum_{i < j \text{ of type } a} \sigma_i \sigma_j$. The off-diagonal entries of S_g can be negative and must be retained.

Partition the blocks into R_b specified groups of sizes s_r and keep $T_r = \sum_{g \text{ in group } r} S_g$. The complete type-pair inputs are

$$\begin{aligned} Y_{r,aa} &= (T_{r,aa} - s_r M_a) v_a v_a^T, \\ Y_{r,ab} &= T_{r,ab} (v_a v_b^T + v_b v_a^T), \quad a < b. \end{aligned} \quad (43)$$

They are the unnormalized original complete edge sums with those types and blocks: each $Y_{r,ab}$ is s times the corresponding complete sum of the normalized inputs $E_{g,ij}$ in (23). Their ordered circuits use Theorem 4.1 with a $K^2 + 1$ coefficient state. No per-edge independence approximation is involved. Different groups have independent score laws, each obtained by ordinary convolution of the one-block law. This block convolution has no binomial label-allocation factor: every block uses the whole physical row list.

For completeness the complexity of the general feature construction is controlled by

$$B_\phi = \prod_a (M_a + 1) \prod_{j=1}^{q_f} (2 \lfloor C_j h^{p_j} \rfloor + 1). \quad (44)$$

Indeed $|T_j| \leq C_j \sum_c (\sum_a n_{ca})^{p_j} \leq C_j h^{p_j}$, including partial populations. Pairwise convolution therefore costs $O((K + q_f) B_\phi^2 \log(b+1))$ arithmetic, plus local feature evaluation. The local table considers at most $(h+1)^{2K}$ count/sign-total pairs. Binary powering can keep every intermediate bucket count at most b , so all count integers have $O(h \log(2b) + 1)$ bits.

For the quadratic score alone, a sharper count/score bound is

$$B_K = \prod_a (M_a + 1) \prod_a (M_a^2 + 1) \prod_{a < c} (2M_a M_c + 1) \leq 2^{K(K-1)/2} (h+1)^{K^2 + 2K}. \quad (45)$$

If $p = K(K+1)/2$, an s -block score has support at most $D_s = (2sh^2 + 1)^p$. Generating R_b group laws costs at most $O(R_b p D_s^2 \log(s+1))$ operations after the one-block law, and evaluating a size- J circuit over their joint support costs at most

$$O(D_s^{R_b}(J+1)(K+1)^3). \quad (46)$$

These are polynomial bounds in explicit population and stack size for fixed K, R_b and a fixed feature list. They are not polynomial in binary multiplicity lengths or uniformly in the number of types or profiles. If all rows of a Parseval frame have these K types then $d \leq K$; a typed retained subframe may instead have $K < d$.

The feature requirement is a precise boundary on closure. Four same-type rows with signs $+, +, -, -$ in one bucket have signed quadratic score zero and four within-type triangles. Putting two opposite-sign pairs in separate buckets preserves the score and leaves no triangle. Thus even a type-permutation-invariant primitive polynomial need not be determined by the quadratic score. Appending

$$\tau_a = \sum_c \binom{n_{ca}}{3}$$

repairs this particular generator. With unnormalized single-block edges $\tilde{E}_{ij} = \sigma_i \sigma_j \mathbf{1}_{\{h_i=h_j\}} A_{ij}$,

$$\sum_{i < j < k \text{ of type } a} \tilde{E}_{ij} \tilde{E}_{jk} \tilde{E}_{ik} = \tau_a (2v_a v_a^T)^3. \quad (47)$$

All three signs cancel, and all three equalities require one bucket. The new feature has range at most $\binom{M_a}{3}$, so its state cost is explicit. Further products of the complete triangle sum are then exact, including products of triangles that share physical labels. Keeping its marginal law separately would not suffice: for four scalar rows $v_i = 1/2$ and two buckets, direct conditioning on bucket sizes gives

$$\mathbb{E}\tau = 1, \quad \mathbb{E}X^2 = 3/4, \quad \mathbb{E}[\tau X^2] = 9/8 \neq (\mathbb{E}\tau)(\mathbb{E}X^2).$$

The bucket-size classes $(4, 0), (3, 1), (2, 2)$ have probabilities $1/8, 1/2, 3/8$, triangle counts $4, 1, 0$, and conditional second moments $3/2, 3/4, 1/2$, respectively, proving these identities.

4.2.3 A noncommuting population beyond every fixed rank budget

The rational vectors

$$w_0 = (6, -2)^T/7, \quad w_1 = (-2, 3)^T/7, \quad w_2 = (-3, -6)^T/7 \quad (48)$$

satisfy $\sum_a w_a w_a^T = I_2$. Put $A_{ac} = w_a w_c^T + w_c w_a^T$. Direct multiplication yields

$$[A_{01}, A_{02}] = \frac{240}{343} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \text{tr}(A_{01} A_{02} A_{01} A_{02}) = -\frac{137952}{7^8}. \quad (49)$$

In the corresponding single-block random trace word all sign multiplicities are even and its two distinct equality edges form a tree. Its expectation is therefore the negative displayed coefficient times b^{-2} . Positive fourth moments are obtained only after the complete ordered sum; individual surviving words need not be positive.

Use $M_a = q^2$ copies of $v_a = w_a/q$ for each of the three types and two buckets. This is Parseval with $h = 3q^2$ rows. Every hash outcome has rank at least $h - 2$, so every fixed collision budget eventually retains no outcome. Nevertheless its full quadratic score law has at most $(q^2 + 1)^6$ simple parameter tuples. Encode the two buckets by an independent sign β_i and put $\xi_i = \sigma_i$,

$\zeta_i = \sigma_i \beta_i$. The pair (ξ_i, ζ_i) consists of two independent signs. If a, c are their type totals, the original bucket totals are $(a + c)/2, (a - c)/2$, and hence

$$S = \frac{1}{2}(aa^T + cc^T). \quad (50)$$

All coordinates of a, c are independent binomial sign totals with the actual multiplicities. This proves the tuple count and the exact weights. Each allowed matrix circuit is thus computed in polynomial arithmetic time in this growing, noncommuting population, even in its dense-collision regime. The transform determines the quadratic score; it does not recover unsigned counts needed for arbitrary lifted features.

4.2.4 A second primitive law: independent edge gates

The coefficient algebra also applies to a distinct random model, provided its probability interface is proved separately. Give each labeled row an independent sign and each unordered pair an independent gate $\xi_{ij} \sim \text{Bernoulli}(p_{ac})$, where a, c are its types. Consider

$$Y = \sum_{i < j} \xi_{ij} \sigma_i \sigma_j (v_{t(i)} v_{t(j)}^T + v_{t(j)} v_{t(i)}^T).$$

Let a_t count positive signs in type t . Conditional on these counts, the signed score T_{tu} for a type pair is a difference of independent binomials,

$$T_{tu} \stackrel{d}{=} \text{Bin}(P_{tu}, p_{tu}) - \text{Bin}(Q_{tu}, p_{tu}), \quad (51)$$

independently across type pairs, where for $t < u$

$$P_{tu} = a_t a_u + (M_t - a_t)(M_u - a_u), \quad Q_{tu} = a_t(M_u - a_u) + (M_t - a_t)a_u,$$

and $P_{tt} = \binom{a_t}{2} + \binom{M_t - a_t}{2}$, $Q_{tt} = a_t(M_t - a_t)$. To prove this, first fix the entire sign assignment. Positive and negative edge-sign classes use disjoint gates, so give independent binomials. Their counts depend only on a ; averaging sign positions at fixed a leaves the same product law. Finally mix over the independent probabilities $2^{-M_t} \binom{M_t}{a_t}$. Thus, writing $f_{P,Q,p}$ for the law of the difference in (51),

$$\mathbb{P}\{T = x\} = 2^{-h} \sum_{0 \leq a \leq M} \left(\prod_t \binom{M_t}{a_t} \right) \prod_{t < u} f_{P_{tu}, Q_{tu}, p_{tu}}(x_{tu}). \quad (52)$$

The coefficient matrix has diagonal $2T_{tt}$ and off-diagonal T_{tu} , so $Y = V^T C(T) V$ and every complete type-pair circuit again uses Theorem 4.1. Its unreduced primitive degree is at most $3D$ for formal edge degree D , since each edge uses one Bernoulli gate and two vertex signs. If $p = K(K + 1)/2$, a direct mixture algorithm costs at most

$$O\left(\prod_t (M_t + 1)[p(h^2 + 1)^2 + p(h^2 + 1)^p]\right)$$

arithmetic operations before circuit evaluation. Rational gate probabilities have a common law denominator $2^h \prod_{t \leq u} b_{tu}^{E_{tu}}$ when $p_{tu} = a_{tu}/b_{tu}$ and E_{tu} is the number of possible edges of that type pair.

The mixing preserves shared-sign cycle constraints. With one vertex in each of three types,

$$\mathbb{E}[T_{12}T_{13}T_{23}] = p_{12}p_{13}p_{23}.$$

The three vertex signs cancel and three independent gates remain. A hash triangle instead has probability b^{-2} , not b^{-3} . The common theorem is the coefficient algebra and exact-law transfer; SparseStack positive-semidefinite covariance properties are not imported into this independent-gate model.

4.3 One same-input error and confidence transfer principle

Exact aggregation is useful beyond exact types or small collision sectors if its error is coupled to the original primitives. The following result isolates the deterministic and probabilistic parts of that transfer.

Theorem 4.3 (Radius closure and same-input tail certificates). *Let $Z_1, \dots, Z_m$ be actual matrix inputs and let $A_1(\Theta), \dots, A_m(\Theta)$ be nominal inputs with an exact Gram representation. On a retained descriptor atom θ , suppose*

$$\mathbb{P}\{\|Z_j - A_j\| > \epsilon_{j,\theta} \mid \Theta = \theta\} \leq \eta_{j,\theta}.$$

Put $\eta_\theta = \min(1, \sum_j \eta_{j,\theta})$. The nominal circuit is evaluated by (25). A radius e_P for its actual value propagates by

$$\begin{aligned} e_{P+Q} &= e_P + e_Q, & e_{cP} &= |c| e_P, & e_{PT} &= e_P, \\ e_{PQ} &= a_P e_Q + a_Q e_P + e_P e_Q, \end{aligned} \quad (53)$$

where $a_P \geq \|P_0\|$ and $a_Q \geq \|Q_0\|$ bound the exact nominal norms. Scalar identities have radius zero. Every node is certified on the one common input-success event, whose conditional failure is at most η_θ .

Let the retained atom masses be w_θ , let $r_ = 1 - \sum_\theta w_\theta$, and write $a_\theta = \|P_0(\theta)\|$ and e_θ for the final norm and radius. Define*

$$\beta = \sum_\theta w_\theta \eta_\theta, \quad T_-(t) = \sum_\theta w_\theta \mathbf{1}_{\{a_\theta - e_\theta > t\}}, \quad T_+(t) = \sum_\theta w_\theta \mathbf{1}_{\{a_\theta + e_\theta > t\}}.$$

Then for $t \geq 0$,

$$[T_-(t) - \beta]_+ \leq \mathbb{P}\{\|P\| > t\} \leq \min\{1, T_+(t) + r_* + \beta\}. \quad (54)$$

Let $Y_\pm$ have values $(a_\theta - e_\theta)_+$ and $a_\theta + e_\theta$, respectively, on retained atoms, and zero on omitted mass. For the strict quantile $Q_Y(\delta) = \inf\{t \geq 0 : \mathbb{P}(Y > t) \leq \delta\}$,

$$Q_{Y_-}(\delta + \beta) \leq Q_{\|P\|}(\delta) \leq Q_{Y_+}(\delta - r_* - \beta), \quad r_* + \beta < \delta < 1 - \beta. \quad (55)$$

Proof. The union bound produces a common event where all input errors are bounded, without independence assumptions. At a product gate write $P = P_0 + \Delta_P$, $Q = Q_0 + \Delta_Q$. The identity $PQ - P_0Q_0 = P_0\Delta_Q + \Delta_PQ_0 + \Delta_P\Delta_Q$ and submultiplicativity prove (53). The other gates follow from elementary norm inequalities. Induction proves the whole circuit on the same event; repeated inputs introduce no additional failures. On a successful retained atom, the triangle and reverse triangle inequalities sandwich the actual norm between $a_\theta \pm e_\theta$. Summing these implications and charging unsuccessful retained mass and all omitted mass proves (54). For the lower quantile, every threshold below $Q_{Y_-}(\delta + \beta)$ has strict tail greater than $\delta + \beta$ and therefore actual tail greater than δ . The upper quantile follows by evaluating the upper bound at the finite-law upper quantile. This proves (55), including its atom conventions. $\square$

For a length- D word with nominal factor norms at most M and input radii at most ϵ , induction in (53) gives

$$e_{\text{out}} \leq (M + \epsilon)^D - M^D \leq D\epsilon(M + \epsilon)^{D-1}. \quad (56)$$

A circuit can propagate its own scalar majorant without expanding into words. Thus degree affects sensitivity rather than introducing a new probability union penalty. A small input error need not be a small relative output error when the nominal polynomial cancels. At degrees of order $\log(d/\delta)$, this distinction is quantitatively relevant.

4.3.1 A conditional residual calculation that preserves all mixed edges

Split the deterministic row identities into retained and light sets and write

$$H = \sum_{i \text{ retained}} u_i u_i^T, \quad L = \sum_{i \text{ light}} u_i u_i^T = I - H, \quad \ell = \text{tr } L, \quad M_L = \sum_{i \text{ light}} \|u_i\|^2 u_i u_i^T, \quad q_L = \text{tr } M_L.$$

Let A be the retained-retained part of X , and let $W = X - A$ contain every light-light and retained-light edge. Conditional on all retained primitives, set

$$h_{gc} = \sum_{i \text{ retained}: h_{gi}=c} \sigma_{gi} u_i, \quad Q = s^{-1} \sum_{g,c} h_{gc} h_{gc}^T = H + A.$$

Define $D_L = \ell^2 + \text{tr}(L^2) - 2q_L$. The exact conditional identities are

$$\mathbb{E}[W^2 \mid \text{retained primitives}] = \frac{1}{sb} \{ \ell L + L^2 - 2M_L + (\text{tr } Q)L + \ell Q + QL + LQ \}, \quad (57)$$

$$\mathbb{E}[\|W\|_F^2 \mid \text{retained primitives}] = \frac{D_L + 2\ell \text{tr } Q + 2 \text{tr}(LQ)}{sb} =: e(Q). \quad (58)$$

Here and below $\|W\|_F$ denotes the Frobenius norm.

To prove these formulas, distinct unordered light edges have orthogonal sign products, even when their equality constraints overlap. This gives the light-light covariance $(\ell L + L^2 - 2M_L)/(sb)$. The retained-light part is exactly

$$s^{-1} \sum_g \sum_{i \text{ light}} \sigma_{gi} (u_i h_{g,h_{gi}}^T + h_{g,h_{gi}} u_i^T).$$

Distinct light primitives are orthogonal by their signs. For each fixed vector v , direct expansion gives

$$\sum_{i \text{ light}} (u_i v^T + v u_i^T)^2 = \|v\|^2 L + \ell v v^T + v v^T L + L v v^T.$$

Averaging the light hash contributes $1/b$, and summing (g, c) gives the remaining terms of (57). Mixed products of the light-quadratic and light-linear parts have zero expectation by odd sign parity. The trace proves (58). In particular QL and LQ have not been replaced by commuting products.

The diagonal correction $-2q_L$ is exact. With one unit light row and no retained row it makes $D_L = 0$, as required because there is no edge. Averaging the retained primitives gives $\mathbb{E}Q = H$ and therefore

$$\mathbb{E}\|W\|_F^2 = \frac{2(d+1)\ell - \ell^2 - \text{tr}(L^2) - 2q_L}{sb} \leq \frac{2(d+1)\ell}{sb}. \quad (59)$$

The bound depends on discarded leverage and covariance, rather than the number of light rows. Splitting fixed leverage among ever more rows does not by itself force the residual to vanish.

Define the unnormalized block covariance $Q_g = \sum_c h_{gc} h_{gc}^T$, so $Q = s^{-1} \sum_g Q_g$. Let W_g be the unnormalized single-block sum of all light-light and retained-light error edges, so $W = s^{-1} \sum_g W_g$. For deterministic block weights ω_g , the corresponding full residual $W_\omega = \sum_g \omega_g W_g$ satisfies

$$\mathbb{E}[\|W_\omega\|_F^2 \mid \text{retained primitives}] = \frac{1}{b} \sum_g \omega_g^2 [D_L + 2\ell \text{tr } Q_g + 2 \text{tr}(LQ_g)]. \quad (60)$$

Conditional light blocks are independent and centered, proving the squares of the weights; this does not make different weighted inputs independent. Markov's inequality supplies an input radius ϵ_θ with failure $\min(1, e_\theta/\epsilon_\theta^2)$, and zero energy permits zero radius and zero failure.

If $A = V^T C V$ on a retained atom, (58) uses only $G = V V^T$ and the additional weighted Gram $G_L = V L V^T$, since $\text{tr } A = \text{tr}(CG)$ and $\text{tr}(LA) = \text{tr}(CG_L)$. The identity survives a coarsening of

the retained descriptor whenever its right side is measurable for that coarsening, by the tower property. In particular for typed retained rows and total score T ,

$$G_L = G - GD_M G, \quad e(T) = \frac{D_L}{sb} + \frac{2}{s^2 b} [\ell \operatorname{tr}(GT) + \operatorname{tr}(G_L T)]. \quad (61)$$

Both the exact retained law and its residual energy are then small-state functions. Theorem 4.3 applies with no omitted mass for the full typed law, or with the explicitly computed omitted mass for a retained collision budget. This proves a second-moment interface, not an exact conditional residual law at all orders.

4.3.2 Approximate coefficient types on the same primitives

There is a complementary interface when every row is retained but is replaced by a nearby typed nominal vector. Let $U, V \in \mathbb{R}^{n \times d}$ be any two deterministic arrays on the same row identities, use the same SparseStack primitives, and define their centered errors by $X_U = U^T(S^T S - I_n)U$ and $X_V = V^T(S^T S - I_n)V$. No Parseval assumption is needed for this definition. Put

$$D_{ij} = u_i u_j^T + u_j u_i^T - v_i v_j^T - v_j v_i^T, \quad \mathcal{D} = \sum_{i < j} \|D_{ij}\|_F^2.$$

The same sign-orthogonality argument proves

$$\mathbb{E}(X_U - X_V)^2 = (sb)^{-1} \sum_{i < j} D_{ij}^2, \quad \mathbb{E}\|X_U - X_V\|_F^2 = \mathcal{D}/(sb). \quad (62)$$

This keeps both mixed terms in the pathwise expansion, with $E = U - V$,

$$X_U - X_V = V^T(S^T S - I)E + E^T(S^T S - I)V + E^T(S^T S - I)E.$$

It also retains signed coefficient cancellation before taking norms.

The energy is computable without enumerating all pairs. Define

$$F(W) = (\operatorname{tr} W^T W)^2 + \operatorname{tr}((W^T W)^2) - 2 \sum_i \|w_i\|^4, \\ C(U, V) = (\operatorname{tr} U^T V)^2 + \operatorname{tr}((U^T V)^2) - 2 \sum_i (u_i^T v_i)^2.$$

Then

$$\mathcal{D} = F(U) + F(V) - 2C(U, V). \quad (63)$$

Indeed $\langle ab^T + ba^T, cd^T + dc^T \rangle_F = 2(a^T c)(b^T d) + 2(a^T d)(b^T c)$, and summing this identity over unordered pairs gives the three terms. The cross Gram $U^T V$ need not be symmetric; $\operatorname{tr}((U^T V)^2)$ cannot be replaced by its squared Frobenius norm. For instance a ninety-degree rotation of I_2 has cross-Gram square trace -2 , not 2 . Conversely $U = -V$ gives zero replacement error on every outcome, which the signed formula preserves.

A convenient distance estimate follows by putting $W = (U + V)/2$ and $\tau = \|U - V\|_F$. The pair difference is $A(w_i, e_j) + A(e_i, w_j)$. The inequality $\|P + Q\|_F^2 \leq 2\|P\|_F^2 + 2\|Q\|_F^2$, followed by summation and deletion of nonnegative diagonal terms, yields

$$\mathcal{D} \leq 4\tau^2(\|W\|_F^2 + \|W\|^2). \quad (64)$$

When both arrays are Parseval, an improved bound is

$$\mathcal{D} \leq 2(d + 2)\tau^2. \quad (65)$$

To verify it, let $K = U^T V$, $J_0 = (K - K^T)/2$, and $d_0 = \sum_i \|u_i u_i^T - v_i v_i^T\|_F^2$. The relation $K + K^T = 2I - E^T E$ and expansion of (63) give exactly

$$\mathcal{D} = 2(d+1)\tau^2 - \frac{1}{2}[\tau^4 + \text{tr}((E^T E)^2)] + 2\|J_0\|_F^2 - 2d_0.$$

The skew projection is a Frobenius contraction and $\|U^T E\|_F \leq \tau$, so $\|J_0\|_F \leq \tau$; dropping the nonpositive terms proves (65).

For any $\epsilon > 0$, (62) gives a same-input failure bound $\eta = \min(1, \mathcal{D}/(s\epsilon^2))$. If $T_V(t) = \mathbb{P}(\|X_V\| > t)$, with $T_V(t) = 1$ for $t < 0$, then

$$[T_V(t + \epsilon) - \eta]_+ \leq \mathbb{P}(\|X_U\| > t) \leq \min\{1, T_V(t - \epsilon) + \eta\}. \quad (66)$$

This follows from the same coupled norm inequality as Theorem 4.3 and requires no independence between nominal matrix and error. Its quantile consequence, for $\eta < \delta < 1 - \eta$, is

$$[Q_{\|X_V\|}(\delta + \eta) - \epsilon]_+ \leq Q_{\|X_U\|}(\delta) \leq Q_{\|X_V\|}(\delta - \eta) + \epsilon.$$

For a nominal atom at $v > 0$, the explicit margin conditions $T_V(v) \leq \delta - \eta$ and $\mathbb{P}(\|X_V\| \geq v) > \delta + \eta$ certify a quantile within ϵ of v . An atom endpoint with no probability margin has no such automatic stability.

When the nominal rows have fixed type count, their exact law is already available from (33). Formula (63) can then be prepared in one pass over the original rows, using type sums and matched row inner products; for known Parseval U this costs $O(nd + K^2 d)$ arithmetic. A hash-only conditional version of (62) also holds, but it must not be reused as a conditional-on-signed-score energy: the latter conditioning reveals signs as well. The unconditional coupled certificate (66) avoids that mistake.

4.4 What the calculus establishes and what remains open

The same algebra now has three proved probability uses: exact signed partition laws on arbitrary frames, exact joint feature laws for typed hash or independent-gate models, and coupled approximation to those nominal laws. Lower bounds arise by measuring original events or by charging the probability of cancellation by the residual. They do not follow merely from positive energy in an auxiliary sector.

The size of a coefficient state is independent of word length, but the cost of its law need not be. The retained collision algorithm costs $(sn^2)^{O(k)}$ rather than a bound uniform in collision rank. The typed algorithm is polynomial only at fixed type, feature, and profile counts. Feature lists must be sufficient for the specified input generators; neither quadratic scores nor type-permutation symmetry alone imply universal sufficiency. Full laws require all correlations between retained features. Circuit arithmetic complexity is also distinct from bit complexity: repeated squaring can produce exponentially large degree and output coefficients from a short circuit. Arbitrary real moments are finite expressions from the law, not automatically rational numbers with a cheap exact representation.

The remaining methodological problem is consequently precise: contract or sum many signed geometries without an exponent growing with the collision budget or type count, while retaining the cross geometry needed by future products and the probability margins needed by strict confidence bounds. The results above provide exact closure and checkable two-sided regimes. They do not yet provide a uniformly sharp norm theorem for arbitrary coefficient geometry in the dense-collision regime.

5 From exact bucket geometry to two-sided bounds

The preceding algebra retains the random constraints and the order of the coefficient products. For SparseStack, a second representation makes a substantial part of that information tractable:

condition on the actual hashes, and retain the independent signed vector sum in each bucket. This section develops that representation into bounds for the *full* error. Its three operations are conditional concentration, projections of the original sign law, and integration over the hash geometry. They give both a diffuse two-sided theorem and a separate rare-collision theorem. The different hypotheses of these conclusions are essential.

Let $u_1, \dots, u_n \in \mathbb{R}^d$ satisfy $\sum_i u_i u_i^T = I_d$, put $\ell_i = \|u_i\|^2$ and $\mu = \max_i \ell_i$, and let $m = sb$. Every pair $(g, i) \in [s] \times [n]$ has its own uniform hash $h_g(i) \in [b]$ and independent uniform sign $\sigma_{gi} \in \{-1, 1\}$. The hashes and signs are mutually independent. Throughout this section,

$$X = \frac{1}{s} \sum_{g=1}^s \sum_{i < j} \mathbf{1}_{h_g(i)=h_g(j)} \sigma_{gi} \sigma_{gj} (u_i u_j^T + u_j u_i^T) \quad (67)$$

is the original Gram error. In particular, equal coefficient vectors with different row labels still carry different random primitives. Write $\|X\|_{L_p} = (\mathbb{E}\|X\|^p)^{1/p}$.

5.1 The conditional descriptor and its upper contraction

Fix every hash. For a block–bucket pair $j = (g, c)$ define

$$y_j = \frac{1}{\sqrt{s}} \sum_{i: h_g(i)=c} \sigma_{gi} u_i, \quad G_j = \sum_{i: h_g(i)=c} u_i u_i^T, \quad C_j = G_j/s. \quad (68)$$

The y_j are independent conditional on these hashes, and

$$\sum_j C_j = I_d, \quad X = \sum_j (y_j y_j^T - C_j). \quad (69)$$

No favorable sign event has been imposed. The equalities include all interactions within every bucket, including noncommuting products in powers of X .

It is useful to state the analytic result for arbitrary signed buckets $y_j = \sum_i \sigma_{ji} a_{ji}$, where the vectors $a_{ji} \in \mathbb{R}^d$ are fixed and all signs are independent. Let $C_j = \sum_i a_{ji} a_{ji}^T$ and suppose $\sum_j C_j = I_d$. Empty buckets are permitted. The following cap is a tractable part of the conditional descriptor; it is not asserted to determine the whole norm law.

Lemma 5.1 (Conditional full-matrix upper bound). *If $\max_j \|C_j\| \leq a$, then, with $D_d = d \log 9 + \log 2$,*

$$\mathbb{P}\{\|X\| > 4\sqrt{a(D_d + t)} + 4a(D_d + t)\} \leq e^{-t} \quad (t \geq 0), \quad (70)$$

$$\|X\|_{L_p} \leq 12\{\sqrt{a(d+p)} + a(d+p)\} \quad (p \geq 1). \quad (71)$$

All real moment orders are allowed.

Proof. For a unit vector x , put $z_j = x^T y_j$ and $v_j = x^T C_j x$. Then $\sum_j v_j = 1$, $v_j \leq a$, and $\sum_j v_j^2 \leq a$. The scalar sign identities

$$\mathbb{E} e^{tz_j} \leq e^{t^2 v_j/2}, \quad \mathbb{E} z_j^4 = 3v_j^2 - 2 \sum_i (x^T a_{ji})^4$$

follow respectively from $\cosh t \leq e^{t^2/2}$ and expansion. Using an independent standard Gaussian g only as an integration device,

$$\mathbb{E} e^{\theta z_j^2} = \mathbb{E}_g \mathbb{E} e^{\sqrt{2\theta} g z_j} \leq (1 - 2\theta v_j)^{-1/2}.$$

Thus for $0 \leq \theta < (2v_j)^{-1}$,

$$\log \mathbb{E} e^{\theta(z_j^2 - v_j)} \leq \frac{\theta^2 v_j^2}{1 - 2\theta v_j}.$$

We also need the negative tail. If $Y \leq v$, $\mathbb{E}Y = 0$, then

$$e^{\theta Y} - 1 - \theta Y \leq \frac{Y^2}{v^2} (e^{\theta v} - 1 - \theta v).$$

For $0 \leq Y \leq v$ this follows from the power series; for $Y < 0$ the left side is at most $\theta^2 Y^2/2$. Apply it to $Y = v_j - z_j^2$, whose variance is at most $2v_j^2$, and use $e^r - 1 - r \leq r^2/[2(1 - r/3)]$. The case $v_j = 0$ is identically zero. Independence now gives the common two-sided bound

$$\log \mathbb{E} e^{\pm \theta x^T X x} \leq \frac{a\theta^2}{1 - 2a\theta}, \quad 0 \leq \theta < (2a)^{-1}. \quad (72)$$

Substitution of $\theta = \sqrt{u/a}/(1 + 2\sqrt{au})$ in Chernoff's inequality yields $\mathbb{P}\{|x^T X x| > 2\sqrt{au} + 2au\} \leq 2e^{-u}$. A $1/4$ -net of the unit sphere has at most 9^d points, by the usual disjoint-ball volume comparison, and $\|X\| \leq 2 \max_{x \text{ in the net}} |x^T X x|$. Taking $u = D_d + t$ proves (70).

For the moment statement, (70) stochastically dominates $\|X\|$ by $4\sqrt{a(D_d + E)} + 4a(D_d + E)$, where E has the mean-one exponential distribution. For $r \geq 1$, $\|E\|_{L_r} \leq 2r$: $x^r \leq (2r/e)^r e^{x/2}$ and $\mathbb{E}e^{E/2} = 2$ suffice. For $p \geq 2$, Minkowski gives $\|\sqrt{D_d + E}\|_{L_p} \leq \sqrt{D_d + p}$; for $1 \leq p < 2$, Jensen gives $\sqrt{D_d + 1}$. In either case this is at most $\sqrt{3(d + p)}$. Also $\|D_d + E\|_{L_p} \leq 3(d + p)$. Another application of Minkowski proves (71). $\square$

The net contributes d inside the deviation parameter. There is no additional multiplier $\log d$ in this particular conditional bound. More refined geometry may improve it in special inputs; its value here is that its coefficient sums were performed before taking a norm.

5.2 Lower certificates for the same matrix

A bucket covariance cap cannot imply a lower bound. For example, in dimension one, put one coefficient $1/\sqrt{m}$ in each of m buckets. Then $C_j = 1/m$, but $X = 0$ for every sign outcome. The missing information is already visible in the fourth cumulant of the primitive signed sums.

Lemma 5.2 (Full-error covariance and small-ball certificate). *For arbitrary signed buckets as above, define*

$$M_j = \sum_i \|a_{ji}\|^2 a_{ji} a_{ji}^T, \quad R_4 = \sum_{j,i} \|a_{ji}\|^4.$$

Then

$$V := \mathbb{E}X^2 = \sum_j \{(\text{tr } C_j)C_j + C_j^2 - 2M_j\} \succeq 0. \quad (73)$$

If $v = \|V\| > 0$, then

$$\mathbb{E}\|X\| \geq \sqrt{v/15}, \quad \mathbb{P}\{\|X\| > \sqrt{v/2}\} \geq 1/60. \quad (74)$$

If $v = 0$, $X = 0$ for every sign realization. If there are at most m buckets and $R_4 \leq d^2/(4m)$, then, writing $\alpha = d/m$,

$$\mathbb{E}\|X\| \geq \max\{\sqrt{\alpha/30}, \alpha - 1\} \geq (\sqrt{\alpha} + \alpha)/14, \quad (75)$$

$$\mathbb{P}\{\|X\| > (\sqrt{\alpha} + \alpha)/6\} \geq 1/60. \quad (76)$$

Proof. The diagonal sign terms in $y_j y_j^T$ sum to C_j . Different unordered pairs of primitive signs are orthogonal. Expanding the square of $ab^T + ba^T$ and summing the pairs in one bucket gives (73). Every pair contributes a positive semidefinite square, which also proves the zero case. In particular,

$$\text{tr } V = \sum_j \{(\text{tr } C_j)^2 + \text{tr}(C_j^2)\} - 2R_4. \quad (77)$$

We record the elementary chaos inequality used for the lower witness. For $f = \sum_{i < k} b_{ik} \sigma_i \sigma_k$,

$$\mathbb{E} f^4 \leq 15(\mathbb{E} f^2)^2. \quad (78)$$

To prove it, expand the fourth power, replace b_{ik} by $|b_{ik}|$, and then replace signs by independent standard Gaussians. Each surviving coefficient is now nonnegative, and every even Gaussian moment is at least the corresponding sign moment. Set $B_{ii} = 0$ and $B_{ik} = |b_{ik}|/2$. Orthogonal diagonalization and Gaussian invariance give

$$\mathbb{E}(g^T B g)^4 = 12(\text{tr } B^2)^2 + 48 \text{tr } B^4 \leq 60(\text{tr } B^2)^2 = 15 \left(\sum_{i < k} b_{ik}^2 \right)^2.$$

Here $\text{tr } B = 0$; the identity follows from $\mathbb{E}(g^2 - 1)^2 = 2$ and $\mathbb{E}(g^2 - 1)^4 = 60$. Applying Cauchy–Schwarz to pairs of coordinates extends (78) to any finite-dimensional vector chaos Q : $\mathbb{E} \|Q\|^4 \leq 15(\mathbb{E} \|Q\|^2)^2$.

Choose a unit top eigenvector x of V and let $T = \|Xx\|^2$. Then $\mathbb{E} T = v$ and $\mathbb{E} T^2 \leq 15v^2$. The strict form of Paley–Zygmund follows directly from

$$(1 - \theta)\mathbb{E} T \leq \mathbb{E}[T \mathbf{1}_{T > \theta \mathbb{E} T}] \leq \sqrt{\mathbb{E} T^2 \mathbb{P}(T > \theta \mathbb{E} T)}.$$

Take $\theta = 1/2$. Hölder also gives $\mathbb{E} \sqrt{T} \geq (\mathbb{E} T)^{3/2} / (\mathbb{E} T^2)^{1/2}$. This proves (74).

Since $\sum_j \text{tr } C_j = d$, (77) gives $\text{tr } V \geq d^2/m - 2R_4 \geq d^2/(2m)$, hence $v \geq \alpha/2$. This supplies the square-root terms in (75)–(76). For the linear term put $S = \sum_j y_j y_j^T$ and $T_0 = \text{tr } S$. Pointwise, $\|X\| \geq T_0/m - 1$ because $S \succeq 0$ has rank at most m . Furthermore

$$\mathbb{E} T_0 = d, \quad \mathbb{E} T_0^2 = d^2 + 2 \sum_j \text{tr}(C_j^2) - 2R_4 \leq d^2 + 2d \leq 3d^2.$$

Here $\sum_j \text{tr}(C_j^2) \leq \text{tr}(\sum_j C_j)^2 = d$. Consequently $\mathbb{E} \|X\| \geq \alpha - 1$, and $\mathbb{P}\{T_0 > d/2\} \geq 1/12$ gives $\mathbb{P}\{\|X\| > \alpha/4\} \geq 1/12$ when $\alpha \geq 4$. For $\alpha \leq 4$, $\sqrt{\alpha} + \alpha \leq 3\sqrt{\alpha}$; for $\alpha \geq 4$, it is at most $3\alpha/2$. These prove (76). For (75), split instead at $\alpha = 2$: below it use $(1 + \sqrt{2})\sqrt{30} < 14$, and above it use $\alpha - 1 \geq \alpha/2$. $\square$

The correction $-2M_j$ is indispensable. It exactly cancels the apparent variance in the singleton example. At the other extreme, positive V certifies a probability statement about the full matrix; it does not prevent $X = 0$ on a positive-probability set.

Lemma 5.3 (Dispersed sign sums). *Let $S = \sum_i c_i \sigma_i$, $v = \sum_i c_i^2 > 0$, and let $q \geq 1$ be an integer. If $\max_i c_i^2 \leq v/(2q)$, then $\mathbb{E} S^{2q} \geq (q/6)^q v^q$.*

Proof. Retain in the moment expansion the words having q distinct labels, each appearing twice. Their ordered squared-coefficient mass is at least $v^q \prod_{j=0}^{q-1} (1 - j/(2q)) \geq 2^{-q} v^q$: after j choices at most $j \max_i c_i^2$ mass has been removed. All omitted surviving terms are nonnegative. Thus

$$\mathbb{E} S^{2q} \geq \frac{(2q)!}{4^q q!} v^q \geq \frac{q!}{2^q} v^q \geq (q/6)^q v^q,$$

where $q! \geq (q/3)^q$ follows by integrating $\log x$ and using $e < 3$. $\square$

Theorem 5.4 (A reusable conditional two-sided theorem). *Suppose the signed buckets satisfy*

$$\sum_{j=1}^m C_j = I_d, \quad \frac{I_d}{2m} \preceq C_j \preceq \frac{2I_d}{m}, \quad \max_{j,i} \|a_{ji}\|^2 \leq \frac{1}{6mp},$$

where $p \geq 1$ is any requested real order. Then, for $F_p = \sqrt{(d+p)/m} + (d+p)/m$,

$$\frac{1}{230} F_p \leq |||X|||_{L_p} \leq 24 F_p. \quad (79)$$

Proof. The upper bound is Lemma 5.1. The atom bound implies $R_4 \leq d/(6mp) \leq d^2/(4m)$, so Lemma 5.2 supplies $(\sqrt{d/m} + d/m)/14$ as a lower bound at every order.

For the additional order dependence, first fix an integer $q \geq 1$ with $\lambda := \max \|a_{ji}\|^2 \leq 1/(12mq)$, and fix a unit vector x . Set $c_{ji} = x^T a_{ji}$ and $v_j = \sum_i c_{ji}^2 \geq 1/(2m)$. Greedily split each bucket into L_j, R_j so that each part has squared coefficient mass at least $v_j/3$. Indeed the first part to reach $v_j/3$ has mass at most $v_j/3 + \lambda \leq v_j/2$. Let S_{L_j} and S_{R_j} be its two signed sums, and put $Z = 2 \sum_j S_{L_j} S_{R_j}$. Flipping all signs in all L_j gives an equidistributed quadratic form Q' from $Q = x^T X x$, and $Q - Q' = 2Z$. Therefore

$$|||X|||_{L_r} \geq \|Q\|_{L_r} \geq \|Z\|_{L_r} \quad (r \geq 1). \quad (80)$$

This is a projection of the actual input law, rather than a comparison with a different matrix model.

The variables $Z_j = 2S_{L_j}S_{R_j}$ are independent and symmetric, and their two factors are independent. Both factor variances are at least $1/(6m)$, so Lemma 5.3 gives $\|Z_j\|_{L_{2q}} \geq q/(18m)$. Conditional Jensen thus gives $\|Z\|_{L_{2q}} \geq q/(18m)$. If $q \leq m/2$, distinct-bucket pairings, using $\mathbb{E}Z_j^2 \geq 1/(9m^2)$, also give

$$\mathbb{E}Z^{2q} \geq \frac{(2q)!}{2^q} \binom{m}{q} (9m^2)^{-q} \geq (q/(54m))^q.$$

For $q > m/2$, the preceding linear bound controls the square-root term as well. In both cases, $|||X|||_{L_{2q}} \geq [\sqrt{q/m} + q/m]/54$. For $p \geq 2$, choose $q = \lfloor p/2 \rfloor$, so $p/4 \leq q \leq p/2$, and obtain $|||X|||_{L_p} \geq [\sqrt{p/m} + p/m]/216$. Combining this with the dimension lower bound costs $14 + 216 = 230$. For $1 \leq p < 2$, $F_p \leq 3(\sqrt{d/m} + d/m)$, so the dimension bound alone suffices. $\square$

The explicit atom condition is an order horizon. A fixed finite sign polynomial is bounded; a lower expression growing with p cannot hold at every order for such an input. The theorem allows arbitrary vector directions and genuine noncommutation.

5.3 Integrating the random geometry: the diffuse theorem

We use one standard concentration input. For independent centered self-adjoint matrices of dimension d , each of norm at most R , and $\|\sum_i \mathbb{E}Z_i^2\| \leq v$, the two-sided matrix Bernstein bound is

$$\mathbb{P}\left\{\left\|\sum_i Z_i\right\| > t\right\} \leq 2d \exp\{-t^2/[2(v + Rt/3)]\}. \quad (81)$$

This is the bounded-summand theorem of Tropp [1, Theorem 6.1(ii)]; its concentration proof is the external input. All coefficient, normalization, and conditioning calculations needed here follow below.

Lemma 5.5 (Hash geometry and all-outcome integration). *For (68), set $\Lambda = \max_{g,c} \|G_{gc}\|$ and $\Delta = \max_{g,c} \|G_{gc} - I_d/b\|$. For $t > 0$,*

$$\mathbb{P}\{\Delta > t\} \leq 2dm \exp \left\{ -\frac{t^2}{2(\mu/b + \mu t/3)} \right\}. \quad (82)$$

For every real $q \geq 1$, define

$$L_q = \log(2dm) + q + 1, \quad K_q = \min \left\{ 1, \frac{1}{b} + \sqrt{\frac{2\mu L_q}{b}} + \frac{2\mu L_q}{3} \right\}.$$

Then $\|\Lambda\|_{L_q} \leq K_q$, and without any diffuseness restriction,

$$\| \|X\| \|_{L_q} \leq 12 \left\{ \sqrt{\frac{(d+q)K_q}{s}} + \frac{(d+q)K_q}{s} \right\}. \quad (83)$$

Proof. For one bucket, the centered independent matrices are $Z_i = (\mathbf{1}_{h_g(i)=c} - 1/b)u_i u_i^T$. They have norm at most μ , and

$$\sum_i \mathbb{E} Z_i^2 = \frac{1}{b} (1 - 1/b) \sum_i \ell_i u_i u_i^T \preceq \frac{\mu}{b} I_d.$$

Apply (81) and union-bound the m buckets. Their mutual independence is not required. Bernstein's exponent is at least u at $t = \sqrt{2\mu u/b} + 2\mu u/3$. Thus Δ is stochastically dominated by this expression at $u = \log(2dm) + E$, with E exponential. The bound $\|E\|_{L_q} \leq q + 1$ follows by comparison with the integer $j = \lceil q \rceil$, since $\mathbb{E} E^j = j! \leq j^j$; use Jensen for square-root moments of order below one. Minkowski gives the asserted bound for $\Lambda \leq 1/b + \Delta$. The deterministic bound $\Lambda \leq 1$ supplies the minimum in K_q . Finally apply Lemma 5.1 conditionally on every hash outcome, with cap $a = \Lambda/s$, and integrate. The square-root term uses $\|\Lambda\|_{L_{q/2}} \leq \|\Lambda\|_{L_q}$, also valid for positive moments with $q/2 < 1$. This proves (83); no exceptional hash event was discarded. $\square$

Theorem 5.6 (Matching full SparseStack moments and quantiles). *Let $d, s \geq 1$, $b \geq 2$, $m = sb$, and $L = \log(4dm)$. For every requested real $p \geq 1$, if*

$$\mu \leq \frac{1}{16b(L+p)}, \quad (84)$$

then the complete error (67) satisfies

$$\frac{1}{460} \left\{ \sqrt{\frac{d+p}{m}} + \frac{d+p}{m} \right\} \leq \| \|X\| \|_{L_p} \leq 24 \left\{ \sqrt{\frac{d+p}{m}} + \frac{d+p}{m} \right\}. \quad (85)$$

Define the strict-tail quantile by $Q_\delta = \inf\{u \geq 0 : \mathbb{P}(\|X\| > u) \leq \delta\}$. For $0 < \delta \leq 1/120$, put $t = \log(1/\delta)$. If $\mu \leq [32b(L+t)]^{-1}$, then

$$\frac{F_t}{50000} \leq Q_\delta \leq 50F_t, \quad F_t = \sqrt{(d+t)/m} + (d+t)/m. \quad (86)$$

In fact the lower displayed threshold has strict exceedance probability at least 2δ .

Proof. The hypothesis implies $\mu \leq [4b\{\log(2dm) + p + 1\}]^{-1}$. Consequently $K_p \leq 2/b$, since $\sqrt{1/2} + 1/6 < 1$. Lemma 5.5 proves the upper moment bound. Substitution of $t = 1/(2b)$ in (82) shows that

$$\mathbb{P} \left\{ \frac{I_d}{2b} \preceq G_{gc} \preceq \frac{3I_d}{2b} \text{ for all } g, c \right\} \geq \frac{1}{2}$$

whenever $\mu \leq 3/[28b \log(4dm)]$. Condition (84) implies this and also $\mu/s \leq 1/(6mp)$. On the displayed event, Theorem 5.4 applies to the same conditional input. Integration of its lower moment costs $(1/2)^{1/p} \geq 1/2$, proving (85).

For the quantile upper bound, (82) with $t = 1/b$ gives $G_{gc} \preceq 2I_d/b$ simultaneously with failure at most $\delta/2$ under the stated leverage condition. Conditional sign concentration, at parameter $\log(2/\delta)$, costs another $\delta/2$ and gives threshold $24F_{t+\log 2} \leq 48F_t$.

For the lower bound, use the moment theorem at orders q and $2q$. Strict Paley–Zygmund applied to $W = \|X\|^q$, with $\theta = 2^{-q}$, gives

$$\mathbb{P}\{\|X\| > F_q/920\} \geq \frac{1}{4}(22080)^{-2q} \geq e^{-22q} \quad (q \geq 1), \quad (87)$$

because $F_{2q} \leq 2F_q$ and $4(22080)^2 < e^{22}$. If $t \geq 22 + \log 2$, choose $q = (t - \log 2)/22$. Then $2q \leq t$, both leverage hypotheses hold, and $q \geq t/44$ gives $F_q \geq F_t/44$. The lower probability is 2δ at threshold $F_t/40480$. If instead $\log 120 \leq t < 22 + \log 2$, the scaled fourth-atom sum is $R_4 = s^{-1} \sum_i \ell_i^2 \leq \mu d/s \leq d^2/(4m)$ for every hash outcome. Lemma 5.2 gives $\mathbb{P}\{\|X\| > F_0/6\} \geq 1/60 \geq 2\delta$. Since $F_t < 24F_0$ in this range, a threshold $F_t/144$ suffices. The common smaller threshold in (86) therefore works. The probability margin deals with possible atoms at the quantile. $\square$

The hypothesis is nonvacuous for noncommuting Parseval frames. For example, stack four orthogonal bases and divide all rows by 2, with at least one basis a rational $3/5, 4/5$ rotation. Replace every row by J^2 distinct rows equal to that row divided by J . The frame remains Parseval, $\mu = 1/(4J^2)$, and the noncommuting geometry remains present. Every new row receives independent hashes and signs; this describes a new input family, not a replacement of the law of a fixed input.

The moment descriptor requires only (d, s, b, μ, p) and is computable from row norms in $O(nd)$ arithmetic. Realized bucket matrices give a finer certificate at cost $O(snd^2 + md^3)$ with dense linear algebra. For $0 < \epsilon \leq 1$, the quantile theorem identifies the scale $(d + \log(1/\delta))/\epsilon^2$ while its leverage hypothesis holds at the chosen $b = m/s$. It is not an unrestricted SparseStack sampling formula.

5.4 A second route to lower bounds: invariant sign projections

The preceding diffuse result uses dispersion to make a full-error projection quantitatively large. Orthogonal groups yield another extension-stable mechanism, without a moment horizon. This also gives a direct noncommuting class on which a small overlap descriptor is sharp.

Theorem 5.7 (A finite union of orthogonal bases). *Consider one block and the vertical stack*

$$U = \begin{pmatrix} w_1 Q_1 \\ \vdots \\ w_J Q_J \end{pmatrix}, \text{ where each } Q_\alpha \text{ is a real } d \times d \text{ orthogonal matrix, } w_\alpha \geq 0, \text{ and } \sum_\alpha w_\alpha^2 = 1. \text{ Each}$$

basis row has its own independent hash and sign. Conditional on the hashes let $n_{\alpha c}$ count the rows of basis α in bucket c . Define

$$W = \max \left\{ w_\alpha^2 (\max_c n_{\alpha c} - 1), \quad w_\alpha w_\beta \max_c \sqrt{n_{\alpha c} n_{\beta c}} : \alpha, \alpha < \beta \right\}.$$

Then, for every real $p \geq 1$,

$$W \leq \| \|X\| \|_{L_p(\sigma)} \leq J^2 W, \quad \|X\| \leq J^2 W \quad \text{pointwise.} \quad (88)$$

After averaging the original hashes, for every $t \geq 0$,

$$\mathbb{P}\{\|X\| > J^2 t\} \leq \mathbb{P}\{W > t\} \leq 96 \mathbb{P}\{\|X\| > t/\sqrt{2}\}. \quad (89)$$

There is no restriction on b or on the moment order.

Proof. Let $s_{\alpha c}$ be the signed sum of the unweighted orthonormal basis rows in bucket c . For fixed α , these vectors are orthogonal and $\|s_{\alpha c}\|^2 = n_{\alpha c}$. The full error is the sum of

$$A_\alpha = w_\alpha^2 \left(\sum_c s_{\alpha c} s_{\alpha c}^T - I_d \right), \quad C_{\alpha\beta} = w_\alpha w_\beta \sum_c (s_{\alpha c} s_{\beta c}^T + s_{\beta c} s_{\alpha c}^T).$$

Signed all-ones blocks give $\|A_\alpha\| = w_\alpha^2 (\max_c n_{\alpha c} - 1)$. Furthermore $\|\sum_c s_{\alpha c} s_{\beta c}^T\| = \max_c \sqrt{n_{\alpha c} n_{\beta c}}$. Triangle inequality yields $\|X\| \leq [J + 2\binom{J}{2}]W = J^2 W$.

Conditional expectation over all bases except α projects X onto A_α , so Jensen supplies its lower moment. To retain a pair in bucket c , flip all signs of basis β by an involution F_β and only the bucket- c signs of basis α by $F_{\alpha c}$. These flips commute and preserve the actual conditional law. The projection

$$\frac{1}{4}(X - X \circ F_\beta - X \circ F_{\alpha c} + X \circ F_\beta \circ F_{\alpha c}) = w_\alpha w_\beta (s_{\alpha c} s_{\beta c}^T + s_{\beta c} s_{\alpha c}^T)$$

is contractive in every L_p . The exact rank-two identity $\|uv^T + vu^T\| = \|u\|\|v\| + |u^T v|$ gives the remaining lower entries in W .

For the strict tail, choose a maximizing entry using hashes alone. If it is a pair entry, at least one of the four flip images has norm greater than $W/\sqrt{2}$, giving conditional probability at least $1/4$. If it is a within-basis entry, fix that basis's signs and a unit eigenvector x witnessing $\|A_\alpha\| = W$. As a function of the remaining signs, $f = x^T X x$ has mean a_0 with $|a_0| = W$ and has sign degree at most two. Its centered part has fourth moment at most 15 times squared variance: homogenize the linear terms with one extra sign and use (78). With variance v , Minkowski and Cauchy–Schwarz give

$$\mathbb{E}f^4 \leq (|a_0| + 15^{1/4}\sqrt{v})^4 \leq (1 + \sqrt{15})^2(a_0^2 + v)^2 < 24(\mathbb{E}f^2)^2.$$

Strict Paley–Zygmund gives probability at least $1/96$ of $|f| > \sqrt{\mathbb{E}f^2/2} \geq W/\sqrt{2}$. Thus $\mathbb{P}_\sigma\{\|X\| > W/\sqrt{2}\} \geq 1/96$ whenever $W > 0$. Integrating on $\{W > t\}$ proves (89). $\square$

The hash descriptor retains *aligned* occupancies across bases; separate maxima cannot recover its cross entries. For example, take two bases with a 90-degree relative rotation in dimension two, and hash their corresponding row indices to the same two distinct buckets. Then $W = w_1 w_2$, while X is zero with probability $1/2$ and has norm $2W$ otherwise. Thus even this theorem permits exact cancellation and compares tails with explicit rescaling of their probabilities. It does not give matching quantiles at an unchanged confidence level from occupancy alone. The coefficient directions may change the exact law although the estimates hold for every orientation.

5.5 An anisotropic interface for a retained heavy part

The same scalar contraction controls extensions of an exactly retained heavy realization. This is useful when uniform diffuseness fails; it does not assume that the norm of a sum equals the sum of two scales.

Fix a deterministic partition of the frame rows into heavy and light sets $\mathcal{H}, \mathcal{L}$, and put $H_0 = \sum_{i \in \mathcal{H}} u_i u_i^T$, $L_0 = I_d - H_0$, $\ell = \|L_0\|$, and $\mu_L = \max_{i \in \mathcal{L}} \|u_i\|^2$. Condition on all heavy hashes and signs. Write h_{gc} and l_{gc} for the unnormalized heavy and light signed bucket sums, respectively, and set

$$\hat{H} = s^{-1} \sum_{g,c} h_{gc} h_{gc}^T, \quad A = \hat{H} - H_0, \quad \nu = \|\hat{H}\|.$$

The complete input decomposes exactly as $X = A + B + C$, with

$$B = s^{-1} \sum_{g,c} (l_{gc} l_{gc}^T - G_{gc}^L), \quad C = s^{-1} \sum_{g,c} (h_{gc} l_{gc}^T + l_{gc} h_{gc}^T).$$

Proposition 5.8 (Conditional diffuse residual). *Suppose the light hashes obey $\mathbb{P}\{\max_{g,c}\|G_{gc}^L\| \leq \kappa\} \geq 1 - \beta$. For $0 < \eta < 1$, put $a = \kappa/s$ and $u = d \log 9 + \log(4/\eta)$. For every conditioned heavy realization,*

$$\mathbb{P}\{\|B + C\| > 4\sqrt{a\ell u} + 4au + 4\sqrt{2a\ell u} \mid \text{heavy data}\} \leq \beta + \eta. \quad (90)$$

For $\ell > 0$, a sufficient condition for $\kappa = 2\ell/b$ is

$$\mu_L \leq \frac{3\ell}{8b \log(2dm/\beta)}. \quad (91)$$

The light covariance may be singular. If $L_0 = 0$, the residual vanishes identically.

Proof. On a good light-hash outcome, normalize the light vectors by $s^{-1/2}$ and set $f_j = h_j/\sqrt{s}$. Their covariance matrices satisfy $\sum_j C_j = L_0$, $C_j \preceq aI_d$, while $\sum_j f_j f_j^T = \hat{H}$. For a unit x , the proof of (72) now uses $\sum_j (x^T C_j x)^2 \leq a\ell$, giving a self-error threshold $4\sqrt{a\ell(D_d + t)} + 4a(D_d + t)$ with failure e^{-t} . The cross quadratic form is a linear light-sign sum with exact variance

$$4 \sum_j (x^T f_j)^2 x^T C_j x \leq 4a\ell.$$

The scalar sign mgf and the same sphere net give its threshold $4\sqrt{2a\ell(D_d + t)}$, also with failure e^{-t} . Use $t = \log(2/\eta)$, add the sign failures, and then add β . No independence between B and C was used.

For the light-hash cap, centered one-bucket variance is at most $b^{-1}(1 - b^{-1})\mu_L L_0$, of norm at most $\mu_L \ell/b$. Equation (81), union over the m buckets, and $t = \ell/b$ give (91). Independence of the light hashes from the conditioned heavy data makes the same cap valid for every heavy realization. $\square$

The mixed scale depends on $\sqrt{\ell\nu}$, while the light self scale is proportional to ℓ under (91). A small light self-error therefore does not remove the cross interaction. Combining this proposition with an exact heavy law requires both triangle inequalities and an explicit residual failure budget. It does not by itself give a lower certificate for $A + B + C$.

6 Rare collisions and the sharp confidence exponent

The diffuse theorem and the rare-collision theorem answer different questions. To formulate the latter, keep d, s and $0 < \epsilon \leq 1$ fixed and allow $b \rightarrow \infty$. Define

$$\Phi_b(d, s, \epsilon) = \sup_{n \geq d} \sup_{U^T U = I_d} \mathbb{P}\{\|X\| > \epsilon\}.$$

The frame is fixed independently of the sketch, but the supremum may choose a different frame for each b .

Theorem 6.1 (Sharp fixed-parameter rare-event exponent). *Put $r = \lceil s\epsilon \rceil$. At fixed (d, s, ϵ) ,*

$$\Phi_b(d, s, \epsilon) = \Theta_{d,s,\epsilon}(b^{-r}). \quad (92)$$

More explicitly, for any integer $k \geq 1$ with $\gamma = \epsilon - (k-1)/s > 0$, put $L_* = 1 + \epsilon$ and

$$A_{s,\epsilon,k} = \frac{4^{k+1} k^{2k+2} k! L_*^{k+1}}{s \gamma^{2k+2}} + 2 \left(\frac{4ekL_*}{\gamma^2} \right)^k.$$

Uniformly over all finite frames and every $b \geq 1$,

$$\mathbb{P}\{\|X\| > \epsilon\} \leq \min\{1, A_{s,\epsilon,k} d^{k+1} b^{-k}\}. \quad (93)$$

In particular, define the least sufficient number of rows by $m_*^{\text{SS}} = s \min\{b \in \mathbb{N} : \Phi_b(d, s, \epsilon) \leq \delta\}$. It has order $\delta^{-1/r}$ as $\delta \downarrow 0$ at fixed (d, s, ϵ) , and is at most $s \lceil A_{s,\epsilon,r}^{1/r} d^{1+1/r} \delta^{-1/r} \rceil$.

The proof uses a second standard concentration input: the matrix martingale Bennett inequality [2, Theorem 3.1]. For self-adjoint martingale differences of norm at most R , with predictable quadratic variation bounded by vI_d , it implies

$$\mathbb{P}\{\|M\| > t\} \leq 2d \exp\left\{-\frac{v}{R^2}h\left(\frac{Rt}{v}\right)\right\}, \quad h(x) = (1+x)\log(1+x) - x. \quad (94)$$

We state the imported result explicitly; the proof below supplies every model-specific martingale and variance estimate.

Proof of the upper bound. Set $a_0 = (k-1)/s$ and $\tau = s\gamma^2/(4k^2L_*)$. Call a row heavy if $\ell_i > \tau$; there are at most $H = \lfloor d/\tau \rfloor$ such rows. Reveal all their hashes and signs. Connect two heavy labels when they share a bucket in some block, and for a connected component C define

$$R(C) = \sum_g (|C| - |\{h_g(i) : i \in C\}|).$$

The uncompressed heavy error is block diagonal over these components. A bucket of size c has signed hollow Gram norm $c-1$ for $c \geq 2$, and zero for $c = 1$. Summing its block norms and compressing by the heavy rows of U , an operator contraction, gives $\|X_H\| \leq s^{-1} \max_C R(C)$.

If some $R(C) \geq k$, choose a spanning tree inside each bucket in each block. Their colored union remains connected and has $R(C)$ edges. It contains a connected k -edge subset which is a forest in each block. Such a specified witness imposes exactly k independent equalities and has probability b^{-k} . There are at most $s^k k! H^{k+1}$ witnesses: choose a root, and at step j choose an existing endpoint in at most j ways, another label in at most H ways, and its block in at most s ways. Invalid sequences only increase this count. Except on a set of probability at most $s^k k! H^{k+1} b^{-k}$, we therefore have $\|X_H\| \leq a_0$.

Condition on any such good heavy realization and reveal the light block-row pairs in a deterministic order. Maintain the normalized partial bucket vectors z_{gc} and $Q = \sum_{g,c} z_{gc} z_{gc}^T$. Initially $Q = P_H + X_H \succeq 0$ and $\|Q\| \leq 1 + a_0 < L_*$. Stop just after the first update with $\|Q\| > L_*$, freezing all later increments. Before an active exposure of light row i in block g , the increment is

$$Q_{\text{new}} - Q = s^{-1} u_i u_i^T + D_t, \quad D_t = \frac{\sigma}{\sqrt{s}} (u_i z_{g,h}^T + z_{g,h} u_i^T).$$

The stopping indicator is predictable and the fresh sign is centered, so the stopped D_t are martingale differences. Since $z_{gc} z_{gc}^T \preceq Q \preceq L_* I_d$ before an active step,

$$\|D_t\| \leq 2\sqrt{\tau L_*/s} = \gamma/k =: R.$$

Write $Q_g = \sum_c z_{gc} z_{gc}^T \preceq Q$. From $(B + B^T)^2 \preceq 2(BB^T + B^T B)$ and averaging the fresh hash,

$$\mathbb{E}_{t-1} D_t^2 \preceq \frac{2}{sb} \{\text{tr}(Q_g) u_i u_i^T + \ell_i Q_g\} \preceq \frac{2L_*}{sb} (du_i u_i^T + \ell_i I_d).$$

Summing all possible light exposures gives predictable variance at most $4L_* d I_d / b =: v I_d$, uniformly in the stopping history.

Let M be the terminal stopped martingale. The accumulated diagonal matrix always lies between 0 and I_d . If stopping occurs, $Q_{\text{stop}} = P_{\text{partial}} + X_H + M$ implies $\lambda_{\max}(M) > L_* - (1 + a_0) = \gamma$. If stopping does not occur, $X = X_H + M$, so failure also implies $\|M\| > \gamma$. Apply (94). For positive t ,

$$\exp\{-vR^{-2}h(Rt/v)\} \leq (ev/(Rt))^{t/R},$$

which follows by writing $x = Rt/v$ and $(1+x)\log(1+x) - x \geq x \log x - x$. With $t = \gamma$ and $R = \gamma/k$ this costs $2d(4ekL_*d/\gamma^2)^k b^{-k}$. Add the heavy-witness probability and use $H \leq 4k^2 L_* d / (s\gamma^2)$ to obtain (93). $\square$

Lower bounds and strict endpoints. If $s\epsilon$ is not an integer, use the scalar Parseval vector $(1, 1)^T/\sqrt{2}$. A block contributes $+1$ or -1 with probability $1/(2b)$ each, and zero otherwise. Exactly r colliding blocks with agreeing signs give the lower bound

$$2^{1-r} \binom{s}{r} b^{-r} (1 - 1/b)^{s-r}. \quad (95)$$

Adjoin disjoint coordinate columns to obtain the same Rayleigh certificate in any dimension d .

For an integer endpoint $\epsilon = k/s$, use instead a scalar unit vector with two amplitudes $a_b = \sqrt{(1 - 1/b)/2}$ and b additional amplitudes $1/b$. For $b \geq \max\{k, 3\}$, require the two large rows to collide with positive sign product in the first k blocks, and to have distinct hashes elsewhere. This costs $H_b = (2b)^{-k} (1 - 1/b)^{s-k}$. Let $c_g = 1$ for $g \leq k$ and $c_g = 2$ otherwise. A small row is special if it hits the heavy bucket in block 1 and avoids all heavy buckets in the other blocks; it is background if it avoids all heavy buckets. Their disjoint probabilities are

$$A_b = b^{-1} \prod_{g=2}^s (1 - c_g/b), \quad Z_b = \prod_{g=1}^s (1 - c_g/b).$$

Require exactly k special rows, the other $b - k$ rows background, and align the special signs in block 1 with the common heavy sign. This costs $\binom{b}{k} A_b^k Z_b^{b-k} 2^{-k}$ conditionally. On this event the scalar error obeys

$$sX = k + (4a_b k - k)/b + L_b,$$

where L_b is the light-light hollow error before division by s . Conditional on the special labels and heavy data,

$$\begin{aligned} \mathbb{E}L_b &= k(k-1)/b^2, \\ \text{Var}(L_b) &= \frac{2(b-k)(b-k-1)}{b^4(b-1)} + \sum_{g=2}^s \frac{2b(b-1)}{b^4(b-c_g)} \leq \frac{2s}{b^2(b-2)}. \end{aligned}$$

Indeed block 1 has only background random pair terms; in the other blocks all small rows are independently uniform on the allowed $b - c_g$ buckets. Remaining distinct sign pairs are orthogonal. The deterministic surplus is at least k/b because $a_b \geq 1/2$. Chebyshev therefore supplies the strict lower bound

$$\mathbb{P}\{X > k/s\} \geq H_b \binom{b}{k} A_b^k Z_b^{b-k} 2^{-k} \left(1 - \frac{2s}{k^2(b-2)}\right)_+. \quad (96)$$

Here $x_+ = \max\{x, 0\}$. At fixed s, k , its ratio to b^{-k} tends to $2^{-2k} e^{-(2s-k)}/k! > 0$. Again adjoin disjoint coordinates for dimension d . Together with (93) at $k = r$, these prove (92).

For the row-count inversion, no fixed finite b has zero worst-case failure: three equal scalar coordinates can all collide with aligned signs in every block, producing error 2. Thus the least feasible b tends to infinity as $\delta \downarrow 0$, and the two asymptotic powers can be inverted without assuming monotonicity in b . $\square$

Remark 6.2 (Which endpoint is being quantified?). For fixed finite n , let R be the sum of the collision ranks $n - \#\{h_g(i) : i \in [n]\}$ over blocks. Compression of the signed bucket adjacencies gives $\|X\| \leq R/s$. Hence strict failure requires $R \geq \lfloor s\epsilon \rfloor + 1$, and finite forest counting gives $O_{n,s,\epsilon}(b^{-\lfloor s\epsilon \rfloor - 1})$. At integer $s\epsilon$, this differs by one power from the supremum over all finite frames. The b -dependent diffuse cloud in (96) explains the distinction; replacing the strict event by a weak inequality or exchanging these quantifiers loses it.

Proposition 6.3 (Dimension power in an iterated rare-event limit). *Fix s, ϵ , put $r = \lceil s\epsilon \rceil$ and $K = r + 1$. For every fixed $d \geq K$,*

$$\liminf_{b \rightarrow \infty} b^r \Phi_b(d, s, \epsilon) \geq c_{s, \epsilon} s \binom{d}{K},$$

where $c_{s, \epsilon} = 1$ off integer $s\epsilon$, and at an integer endpoint it can be taken as $(K/2) \exp\{-K[1 + K(s-1)]\}$. Thus the upper dimension power $d^{1+1/r}$ for the required row count is optimal in the iterated limit: first $\delta \downarrow 0$ at fixed d , then $d \rightarrow \infty$.

Proof. For d distinguished rows, require exactly one clique of K rows in one block, all other distinguished hashes distinct from each other and the clique, and all distinguished hashes distinct in other blocks. The events for different cliques and blocks are disjoint and have total probability

$$p_b = s \binom{d}{K} \frac{(b)_{d-r}}{b^d} \left(\frac{(b)_d}{b^d} \right)^{s-1}, \quad b^r p_b \rightarrow s \binom{d}{K}.$$

Off an endpoint, $U = I_d$ has an eigenvalue $r/s > \epsilon$ on this event, independently of signs.

At $\epsilon = k/s$, set $K = k + 1$ and use distinguished rows $a_b e_j$, with $a_b = \sqrt{1 - 1/(k^2 b)}$, supplemented for each coordinate by b rows $e_j/(kb)$. The frame is Parseval. On the distinguished clique event use its signed unit eigenvector, supported on the K clique coordinates. Only their $N = Kb$ small rows contribute to this Rayleigh quotient. A relevant small row is special if it hits the clique in its block and avoids the K relevant distinguished buckets in every other block. A background row also avoids the clique bucket. Their probabilities are $A_b = b^{-1}(1 - K/b)^{s-1}$ and $Z_b = (1 - 1/b)(1 - K/b)^{s-1}$. Exactly one special row, all others background, and its sign aligned with its distinguished coordinate have probability

$$N A_b Z_b^{N-1} / 2 \rightarrow (K/2) e^{-K[1+K(s-1)]}.$$

The signed Rayleigh error, before division by s , is $k + (2a_b - 1)/(kb) + L_b$. Remaining sign-pair orthogonality gives $\mathbb{E}L_b = 0$ and $\text{Var}(L_b) \leq 2s/[k^4 b^2(b - K)]$. Chebyshev shows that the positive surplus survives with conditional probability tending to one. Multiply the two event probabilities. Finally $\binom{d}{K} \geq (d/K)^K$ and Theorem 6.1 give the stated dimension power after inversion. $\square$

The threshold in b in the preceding lower asymptotic may depend on d . No uniform joint asymptotic follows. A concrete guard is $d = s = 1$, $u_i = 1/\sqrt{K}$ for $1 \leq i \leq K$, $b = K^4$, and $\epsilon = 1/K$. Exactly one colliding pair and all other labels distinct occur with probability

$$\binom{K}{2} \frac{(K^4)_{K-1}}{(K^4)^K} \geq \frac{1}{8K^2} \quad (K \geq 2).$$

For this inequality, $\binom{K}{2} \geq K^2/4$, and the product $\prod_{j=0}^{K-2} (1 - j/K^4)$ is at least $1 - \sum_{j=0}^{K-2} j/K^4 \geq 1/2$. On every sign realization in this event, $|X| = 2/K > \epsilon$. Here $b\mu = K^3$, so the diffuse theorem's hypothesis fails. This example simultaneously rules out extrapolating a diffuse formula and silently replacing fixed-parameter constants by uniform ones.

The common methodology is now explicit. Preserve the exact conditional coefficient sums; use a cap for upper contraction and a separate same-input projection, covariance, or Rayleigh witness for lower bounds; then integrate the original constraint law. When diffuse geometry fails, count independent heavy equalities and control the remaining ordered interaction by a stopped process. These are proved bridges for the stated classes. A single tractable descriptor giving matching joint-parameter quantiles for every Parseval frame remains open.

7 From an exact law to exact constants

The finite feature law and the analytic diffuse estimates answer different questions. A finite law preserves lattice thresholds and cancellation. An analytic estimate gives uniform inequalities over a regime. A third operation is to pass the exact descriptor through a limit and compute its limiting law. This section supplies a complete benchmark for that operation; it also marks the point at which fixed-confidence convergence ceases to justify rare-tail asymptotics. Gaussian limits for sparse sketches have prior literature, including Ahfock, Astle, and Richardson [6]; no priority claim is made for the limit below.

Fix a real Parseval frame $u_1, \dots, u_K \in \mathbb{R}^d$. For each integer $n \geq 1$, replace u_t by n distinct physical copies of $u_t/\sqrt{n}$, all with independent signs and hashes. Take two buckets and one block, and denote the Gram error by X_n . Encode the bucket by $\beta_{ti} \in \{-1, 1\}$. The map $(\sigma, \beta) \mapsto (\xi, \zeta) = (\sigma, \sigma\beta)$ is a bijection of the four sign pairs. Hence all the ξ_{ti}, ζ_{ti} are independent. Put

$$r_n = n^{-1/2} \sum_{t,i} \xi_{ti} u_t, \quad q_n = n^{-1/2} \sum_{t,i} \zeta_{ti} u_t.$$

An orthogonal transform of the two original sketch rows gives the pointwise identity

$$X_n = \frac{1}{2}(r_n r_n^* + q_n q_n^*) - I_d. \quad (97)$$

These independent transformed rows are derived from the original primitives; independence of the untransformed hash buckets has not been assumed.

Theorem 7.1 (A fixed-dimensional exact-law limit). *If G is a $2 \times d$ matrix of independent standard Gaussian entries, then*

$$X_n \implies X_\infty = G^*G/2 - I_d, \quad \mathbb{E}\|X_n\|^p \longrightarrow \mathbb{E}\|X_\infty\|^p \quad \text{for every fixed } p > 0.$$

For $d = 2$ and every fixed $0 < \delta < 1$, the strict quantiles also converge: $Q_{\|X_n\|}(\delta) \rightarrow Q_{\|X_\infty\|}(\delta)$.

Proof. For each fixed $\theta \in \mathbb{R}^d$,

$$\mathbb{E}e^{i\langle \theta, r_n \rangle} = \prod_t \cos(\langle \theta, u_t \rangle / \sqrt{n})^n \longrightarrow e^{-\|\theta\|^2/2},$$

using $\log \cos z = -z^2/2 + O(z^4)$ and Parseval. Since r_n, q_n are independent for every n , their joint limit is a pair of independent standard Gaussian vectors. Continuity of the Gram map proves weak convergence. The bound $\cosh z \leq e^{z^2/2}$ also gives $\mathbb{E}e^{\langle \theta, r_n \rangle} \leq e^{\|\theta\|^2/2}$. For an independent standard Gaussian g and $0 < c < 1/2$, Gaussian integration and Tonelli yield

$$\mathbb{E}e^{c\|r_n\|^2} = \mathbb{E}_g \mathbb{E}e^{\sqrt{2c}\langle g, r_n \rangle} \leq (1 - 2c)^{-d/2}.$$

The same bound holds for q_n , independently. Equation (97) then gives uniform integrability of every fixed power of $\|X_n\|$. For $d = 2$, the density calculation below proves a continuous strictly increasing limiting norm distribution on $(0, \infty)$. Weak convergence at points on either side of its unique $(1 - \delta)$ quantile proves the final assertion, with the strict-tail convention (1). $\square$

7.1 An explicit two-dimensional calculation

For $d = 2$, let $\lambda_1 > \lambda_2 > 0$ be the eigenvalues of G^*G . Their joint density on this ordered region is

$$f(\lambda_1, \lambda_2) = \frac{\lambda_1 - \lambda_2}{4\sqrt{\lambda_1\lambda_2}} e^{-(\lambda_1 + \lambda_2)/2}. \quad (98)$$

Here is a direct derivation of the normalizing constant. Write the independent Gaussian columns as $x = re_\theta$ and $y = ue_\theta + ve_\theta^\perp$. After integrating the angle, the density is $r(2\pi)^{-1}e^{-(r^2+u^2+v^2)/2}$. The Gram coordinates $(a, c, d_0) = (r^2, ru, u^2+v^2)$ have Jacobian magnitude $4r^2|v|$. Summing over the signs of v gives $(4\pi\sqrt{ad_0 - c^2})^{-1}e^{-(a+d_0)/2}$. Diagonalizing with the angle of an eigenvector in $[0, \pi)$ contributes Jacobian $\lambda_1 - \lambda_2$, which yields (98).

Set $\lambda_1 = R(1 + T)$, $\lambda_2 = R(1 - T)$. The joint density becomes

$$Re^{-R} \frac{T}{\sqrt{1-T^2}}, \quad R > 0, \quad 0 < T < 1.$$

Thus R and T are independent and

$$\|X_\infty\| = |R/2 - 1| + RT/2. \quad (99)$$

Every fixed norm level has zero probability by the continuous eigenvalue density. Each nonempty norm interval in $(0, \infty)$ contains an open eigenvalue rectangle of positive density. This proves the continuity and strict increase used in Theorem 7.1.

For $\varepsilon \geq 1$, the lower-eigenvalue condition is automatic and success means $\lambda_1 \leq L = 2(1 + \varepsilon)$. On setting $\lambda_1 = x^2$, $\lambda_2 = y^2$, the density becomes $(x^2 - y^2)e^{-(x^2+y^2)/2}$ on $0 < y < x$. Integrate in y and use $I_0(x) = \int_0^x e^{-y^2/2} dy$ to obtain $e^{-x^2/2}[(x^2 - 1)I_0(x) + xe^{-x^2/2}]$. One integration by parts gives

$$\mathbb{P}\{\|X_\infty\| > \varepsilon\} = e^{-2(1+\varepsilon)} + \sqrt{\pi(1+\varepsilon)}e^{-(1+\varepsilon)}\operatorname{erf}\sqrt{1+\varepsilon}. \quad (100)$$

For $0 < \varepsilon < 1$, integrating the R density over its allowed interval gives the exact success probability

$$\int_0^\varepsilon \frac{t}{\sqrt{1-t^2}} [(1+r_-)e^{-r_-} - (1+r_+)e^{-r_+}] dt, \quad r_- = \frac{2(1-\varepsilon)}{1-t}, \quad r_+ = \frac{2(1+\varepsilon)}{1+t}. \quad (101)$$

The finite strict tails converge to these formulas at every fixed positive threshold. In particular, explicit integration in (99) gives

$$\begin{aligned} \lim_n \mathbb{E}\|X_n\| &= \frac{\pi}{4} + 4e^{-2}, \\ \lim_n \mathbb{E}\|X_n\|^2 &= \frac{3}{2} - \frac{\pi}{4} + \frac{9\pi}{2}e^{-2}. \end{aligned} \quad (102)$$

Indeed $\mathbb{E}T = \pi/4$, $\mathbb{E}T^2 = 2/3$, $\mathbb{E}R = 2$, $\mathbb{E}R^2 = 6$, $\mathbb{E}|R/2 - 1| = 4e^{-2}$, and $\mathbb{E}[R|R/2 - 1|] = -1 + 18e^{-2}$; these values give (102) by expansion.

This calculation shows what the finite-law route can supply beyond constant-factor bounds. It does not prove a saddle-point approximation uniform in a moving threshold or in $\delta_n \rightarrow 0$. Such a result needs a local limit theorem with lattice span, nondegeneracy, and boundary errors controlled in the chosen joint limit. These hypotheses must be checked for the actual signed score, not for an independently resampled substitute.

8 Exact contraction with a controlled live boundary

The descent criterion in Theorem 3.3 has a constructive positive counterpart for specified matrix-product programs. The structural parameter is the number of original primitive variables still needed by future factors; neither total character rank nor polynomial degree replaces this parameter.

8.1 Ordered parity messages and norm moments

Let $z_1, \dots, z_D$ be independent uniform signs and write $\chi_A(z) = \prod_{i \in A} z_i$. The supplied program has positive integer length L and is

$$F_j(z) = \sum_{A \in \mathcal{T}_j} C_{j,A} \chi_A(z), \quad P = F_1 \cdots F_L, \quad R = U P V,$$

where the $C_{j,A}$ are arbitrary complex $n \times n$ matrices, and U, V are deterministic readouts giving a $d \times d$ output. Equal local character labels are merged by matrix addition. Put $V_j = \bigcup_{A \in \mathcal{T}_j} A$,

$$B_j = \left(\bigcup_{h \leq j} V_h \right) \cap \left(\bigcup_{h > j} V_h \right), \quad w = \max_j |B_j|, \quad t = \max_j |\mathcal{T}_j|.$$

Zero factors are handled separately, giving $R = 0$; otherwise $t \geq 1$. The boundary schedule belongs to this factor order. Finding a better schedule or a smaller program is a separate problem; matrix factors cannot generally be reordered. The reduced primitive degree is at most $\min(D, \sum_j \max_{A \in \mathcal{T}_j} |A|)$.

Theorem 8.1 (Exact boundary transfer). *Initialize $M_0(\emptyset) = I_n$. For $b \subseteq B_j$, set*

$$M_j(b) = \sum_{\substack{s \subseteq B_{j-1}, A \in \mathcal{T}_j \\ s \Delta A = b}} M_{j-1}(s) C_{j,A}. \quad (103)$$

Terms whose XOR is not a subset of B_j are discarded. Then

$$\mathbb{E}[F_1 \cdots F_j \mid z_{B_j}] = \sum_{b \subseteq B_j} M_j(b) \chi_b(z), \quad M_L(\emptyset) = \mathbb{E}P.$$

For $n \geq 2$, $d \leq n$, and integer $p \geq 1$, the same primitive boundary computes $S_p = \mathbb{E} \operatorname{tr}((R^ R)^p)$ exactly in*

$$O(L 2^w t^{2p} n^{6p}) \text{ arithmetic operations,} \quad O(2^w n^{4p}) \text{ streaming working storage.} \quad (104)$$

Consequently

$$(S_p/d)^{1/(2p)} \leq (\mathbb{E} \|R\|^{2p})^{1/(2p)} \leq S_p^{1/(2p)}, \quad \mathbb{P}\{\|R\| > \eta^{-1/(2p)} S_p^{1/(2p)}\} \leq \eta. \quad (105)$$

Proof. A variable removed before stage j occurs in no later factor. Its expectation can therefore be taken before right multiplication by F_j . Character multiplication is XOR, and integrating an eliminated nonempty parity gives zero. This proves (103) and the conditional identity inductively. In particular, equal surviving channels combine their entire ordered matrix sums; a zero innovation weight is not permission to delete that coefficient.

For the moment, keep the original order in every tensor leg and use

$$\mathcal{F}_j = F_j^{\otimes p} \otimes \overline{F}_j^{\otimes p}, \quad \prod_j \mathcal{F}_j = P^{\otimes p} \otimes \overline{P}^{\otimes p}.$$

The bar is entrywise conjugation, not adjoint. These factors use only V_j , have dimension n^{2p} , and expand into at most t^{2p} local terms. After the lifted readouts, apply the fixed linear functional

$$\Lambda_p(T) = \sum_{a_1, \dots, a_p; b_1, \dots, b_p=1}^d T_{(a_1, \dots, a_p, a_1, \dots, a_p), (b_1, \dots, b_p, b_2, \dots, b_p, b_1)}.$$

On $Q^{\otimes p} \otimes \overline{Q}^{\otimes p}$ this is $\operatorname{tr}((Q^* Q)^p)$: summing each a_i forms one entry of $Q^* Q$, and the remaining indices form its cyclic trace. This functional need not be positive on intermediate messages; the final identification proves $S_p \geq 0$. Dense matrix multiplication and two consecutive message arrays give (104). Regenerating local tensor terms avoids storing all local expansions. The singular-value inequalities $\|R\|^{2p} \leq \operatorname{tr}((R^* R)^p) \leq d \|R\|^{2p}$ and Markov's inequality give (105). $\square$

At $p = \max(1, \lceil \log(d/\eta) \rceil)$, $0 < \eta < 1$, the confidence threshold in (105) is at most $\sqrt{e}$ times the actual L^{2p} norm. The arithmetic bound becomes

$$O\left(L2^w n^6 t^2 (d/\eta)^{6 \log n + 2 \log t}\right).$$

It is polynomial for fixed n, t, w ; increasing internal dimension can make it expensive at logarithmic order. Scalar programs can be padded to $n = 2$. Full D -bit masks add $O(\lceil D/b_{\text{word}} \rceil)$ machine-word work per transition. Rational coefficient bit growth and input parsing are extra costs, not covered by unit-cost arithmetic.

Proposition 8.2 (A separate same-input lower certificate). *For a deterministic unit vector v , let $Q = \|Rv\|^2$, $\mu = \mathbb{E}Q > 0$ and $\nu = \mathbb{E}Q^2$. Both moments are computable by the same transfer, and for $0 < \theta < 1$,*

$$\mathbb{P}\{\|R\| > \sqrt{\theta\mu}\} \geq (1 - \theta)^2 \mu^2 / \nu.$$

One can also find an original sign assignment with $Q \geq \mu$ using at most $2D$ additional transfers for μ .

Proof. The event $Q > \theta\mu$ implies the stated norm event. Splitting $\mathbb{E}Q$ over that event and its complement and using Cauchy–Schwarz proves the probability bound. For the assignment, fix each bit to a value whose conditional mean is at least the current mean; one of its two values always has this property. Substitution removes variables and never increases the declared boundary. At the end the conditional mean is the actual value of Q . This is one witness, not a typical-input claim. $\square$

Chains of factors using $\{j, j+1\}$ have $w \leq 1$; a closed cycle has $w \leq 2$. Consecutive overlapping triangles have $w \leq 2$ when each triangle is one factor, or $w \leq 3$ when its three edges are separate factors. Arbitrary noncommuting coefficients and repeated shared signs are allowed. A sum of ordered triples is also a program: use four block states, retain identity blocks $(0, 0), (3, 3)$, and in three consecutive factors add respectively blocks $(0, 1) = X_1, (1, 2) = X_2, (2, 3) = X_3$. Concatenation accumulates the triple products in block $(0, 3)$. Its actual enlarged dimension is charged. Appending an interaction that reuses an eliminated bit requires a new boundary schedule; no completed expectation recovers that bit’s correlations.

8.2 Even common full rank does not replace named constraints

Here is a sharper test than comparing a graph with a tree. It keeps the entire unlabelled graph and its character rank fixed.

Theorem 8.3 (Equal-rank extension obstruction). *For $q = 2^s$, $s \geq 1$, there are two assignments of $k = q^3$ named degree-two sign features to the same graph on $D = 2q^2$ primitive signs, and identical balanced noncommuting Parseval-frame coefficients, such that the full named word Gram on reduced subsets of names of size at most two and the full character rank $D - 1$ agree. The same ordered cubic circuit has outputs of primitive degree two with L^2 norm ratio $\sqrt{q-1}$. Their expected-norm ratio is at least this, and their strict 1/10-tail quantile ratio is at least $\sqrt{(q-1)/10}$.*

Proof. Use the incidence graph with vertices $P(x, y), L(a, b)$ over $\mathbb{F}_q$ and edges $y = ax + b$. It is connected and bipartite, has q^3 edges, and has no four-cycle, since two points lie on at most one nonvertical line. Assign to an edge vw the character $\zeta_v \zeta_w$. Its full character rank is $D - 1$: every edge has even coordinate parity, and the edges of a spanning tree are independent by successive leaf removal.

For $a \neq 0$ compare the three-edge paths

$$\begin{aligned} C_a &: P(0, 0) - L(a, 0) - P(a^{-1}, 1) - L(0, 1), \\ I_a &: P(0, 0) - L(a, 0) - P(1, a) - L(0, a). \end{aligned}$$

In each family these $q - 1$ paths are edge disjoint. Give consecutive edges the same three abstract names in the two families and extend each assignment to a bijection of all k names onto graph edges. The first family telescopes to one common endpoint character; the second telescopes to $q - 1$ independent endpoint characters. Every nonempty set of at most five graph edges has a nonzero boundary, so both word Grams indexed by reduced subsets of at most two names are identity. Ordered words with the same reduced subset are one Gram channel, not distinct orthogonal channels. The selected triple words are orthogonal to that old space. Their actual next Grams are $\mathbf{11}^T$ and I_{q-1} .

Put $w_1 = e_1, w_2 = e_2, w_3 = (e_1 + e_2)/\sqrt{2}, w_4 = (e_1 - e_2)/\sqrt{2}$ and $P_j = w_j w_j^T$. Give successive path edges types 1, 3, 2 and fill unused names so each type occurs $k/4$ times; this is possible because $q - 1 \leq k/4$. Let $u_e = \sqrt{2/k} w_j$ and $A_e = \sqrt{2/k} P_j$ at a type- j name. Then $\sum_e u_e u_e^T = I_2$, $\|u_e\|^2 = 2/k$ and $\sum_e A_e^2 = I_2$. The matrices do not commute: $P_1 P_3 P_2 = e_1 e_2^T/2$ while $P_1 P_2 P_3 = 0$. For $X_e = \varepsilon_e A_e$ use the same circuit summing every forward path product and its reverse. With $n_0 = q - 1$, $\gamma = \sqrt{2}/k^{3/2}$ and $J_2 = e_1 e_2^T + e_2 e_1^T$, its outputs are

$$H_C = n_0 \gamma \zeta_{P(0,0)} \zeta_{L(0,1)} J_2, \quad H_I = \gamma \zeta_{P(0,0)} \sum_{a \neq 0} \zeta_{L(0,a)} J_2.$$

The vector e_2 attains their norms pointwise. The first norm is $n_0 \gamma$, the second has squared mean $n_0 \gamma^2$, giving the L^2 and expectation claims. Since n_0 is odd, the second norm is always positive; Markov bounds its strict 1/10 quantile by $\gamma \sqrt{10 n_0}$. The first quantile is exactly $n_0 \gamma$. $\square$

The missing data are the association of named constraints with later ordered coefficients, not merely the unlabelled incidence graph. The named incidence map and the new Gram distinguish these models and are not refuted. The four-block program above has internal dimension eight and boundary at most four on these path sums, so the positive transfer detects the obstruction at bounded width. This is a polynomial-matrix separation, not a Walsh RIP sample-complexity lower bound.

8.3 Coherent cell sums: a small closure with an explicit boundary

A second positive calculus closes a deliberately smaller class of operations. For distinct Walsh frequencies s_i , a fixed real unit coefficient vector c , and an original row z , set

$$M = \|c\|_1^2, \quad p_i = |c_i|/\sqrt{M}, \quad \varepsilon_i = \text{sign}(c_i) \chi_{s_i}(z), \quad D = \sum_i p_i \varepsilon_i.$$

Zero coefficients are omitted. The directional energy is $Z = M D^2$. Average the indicator that an auxiliary iid list of q labels drawn with weights p_i has all its ε signs equal. Conditional on the original row, this coherent positive sum is exactly

$$U_q = \left(\frac{1+D}{2} \right)^q + \left(\frac{1-D}{2} \right)^q, \quad q \geq 1.$$

It includes repeated labels and every actual overlap. With $A_0 = 2, L_0 = 0$, the exact extension is

$$\begin{pmatrix} A_{q+1} \\ L_{q+1} \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ D^2/2 & 1/2 \end{pmatrix} \begin{pmatrix} A_q \\ L_q \end{pmatrix}, \quad A_q = U_q. \quad (106)$$

Indeed $L_q = D[(1+D)/2]^q - ((1-D)/2)^q$ gives the recursion directly. The seed $A_0 = 2$ is algebraic, not an empty-list probability. Products of these unmarked sums remain functions of D^2 , so its actual one-row law suffices for their joint transforms. This is list-extension closure for one fixed coefficient direction, not arbitrary matrix-word closure.

For signed-flat coefficients and uniform distinct q -subsets, $1 \leq q \leq k$, put $T = \sum_i \varepsilon_i$, $n_{\pm} = (k \pm T)/2$. Then

$$U_q^{\text{dist}} = \frac{\binom{n_+}{q} + \binom{n_-}{q}}{\binom{k}{q}}.$$

The numerators $A_q = (n_+)_q + (n_-)_q$ and $L_q = T[(n_+)_q - (n_-)_q]$ evolve by the same two-state rule with diagonal entries $k/2 - q$ and lower-left entry $T^2/2$. On a basis support, $(k - T)/2 \sim \text{Bin}(k, 1/2)$, so all transforms through list length q cost $O(kq)$ arithmetic with $O(k)$ stored values. The cubic law of Proposition 19.3 instead gives three energy atoms and $O(q)$ work for the positive-flat direction and its actual row modulations; arbitrary signed-flat coefficients need not have that three-atom law. The update is common to these two models; the law is model-specific. For arbitrary supports, building that law need not be inexpensive, and unequal coefficients need not have $k + 1$ possible energies.

Proposition 8.4 (A clipping repair on independent-character supports). *Let $s_i = e_i$ be basis characters and let c be a fixed real unit vector with $M = \|c\|_1^2 \geq 65536$. Put $q = \lceil 4 \log_2 M \rceil$ and*

$$F_q(D) = \frac{((1+D)/2)^q + ((1-D)/2)^q}{(3/4)^q + (1/4)^q}, \quad Y = M \min\{1, F_q(D)\}.$$

Pointwise $Z1_{\{Z \geq M/4\}} \leq Y$, and for independent original rows,

$$\mathbb{P} \left\{ \frac{1}{m} \sum_{j=1}^m Y(z_j) > 1/4 \right\} \leq e^{-m/4096} \quad (m \geq 1). \quad (107)$$

Proof. The numerator is even and nondecreasing in $|D|$, and equals its denominator at $|D| = 1/2$. Thus $Y = M$ on the specified layer and $Z \leq M$ everywhere. Under the basis law, $\mathbb{E}e^{t(x,c)} \leq e^{t^2/2}$ implies $\mathbb{P}\{|D| > 1/8\} \leq 2e^{-M/128}$. On $|D| \leq 1/8$,

$$F_q(D) \leq 2(3/4)^q \leq 2M^{-3/2} \leq 1/(8M).$$

Here $(3/4)^8 < 1/8$, $q \geq 4 \log_2 M$, and $M \geq 256$ suffice. With $\lambda = 1/256$, we have $Y \leq 1/8$ on that event and always $Y \leq M$, so

$$\mathbb{E}e^{\lambda Y} \leq e^{\lambda/8} + 2e^{-M/256} \leq e^{3\lambda/16}.$$

The last step follows from $e^{3\lambda/16} - e^{\lambda/8} \geq \lambda/16 = 1/4096$ and $2e^{-M/256} \leq 1/4096$. Independence and exponential Markov give (107). $\square$

A predetermined family of J such directions has failure at most $Je^{-m/4096}$. In particular, for $k \geq 65536$, all signed-flat directions on the fixed basis support satisfy this layer bound once $m \geq 4096[k \log 2 + \log(1/\eta)]$. This does not bound smaller energy layers or prove full isometry for all coefficients. Clipping is substantive: without it, the normalization at an all-equal row is $F_q(1) = k^{4 \log_2(4/3) + o(1)}$ for flat coefficients. At $m = \lfloor k F_q(1) \rfloor$ a single such row already forces the unclipped majorant average above $1/4$, with probability of order at least $m2^{-k}$. The exact coherent sum alone does not remove that high-confidence overshoot.

Named successors need more state. For $k = 5$, the two original basis rows $(1, 1, -1, -1, 1)$ and $(1, -1, 1, -1, 1)$ have the same $T^2 = 1$ and every unmarked coherent sum, but the named equality $\varepsilon_1 = \varepsilon_2$ holds only in the first. Their inner product is one, so their rank-one matrices do not commute. Retaining only T^2 cannot support multiplication by that further interaction. A precise basis-model repair is to retain the signs on a predetermined set of b named labels and the number of negative signs among the others: there are $2^b(k - b + 1)$ states with masses $2^{-k} \binom{k-b}{w}$. This state represents expressions invariant under permutations of unnamed coordinates, rather than arbitrary unequal weights on those coordinates. Products, named edges within the boundary, and uniformly averaged unnamed extensions preserve this state. Newly named labels enlarge it. For general supports the masses are an actual conditional sign-count law; basis binomial probabilities cannot be reused.

9 Computable calculi: symmetric lifts, tensor factors, and interacting chains

The boundary transfer of Theorem 8.1 keeps the correct ordered information, but its full tensor lift is expensive at high moment order. Three subsequent results give different answers to that cost problem. Symmetric powers make bounded internal matrix programs polynomial in the moment order. Tensor factorization permits an exponentially large irreducible coefficient algebra without expanding the ambient matrix. Matrix-product operators add full sums and interactions between physical sites, at an explicitly charged replica cost. These are distinct structural classes; the most economical state in one class need not survive the operations allowed in another.

The tensor identities and matrix-product arithmetic used here are standard; see [19, 20]. The project-specific conclusions concern the specified original random law, the complete update rules, quantitative costs, and certificates for the same matrix. No claim of priority for the underlying linear-algebra identities is intended.

9.1 A common schedule and the distinction between moments and quantiles

Let $z_1, \dots, z_D$ be independent uniform signs. A supplied ordered description declares local variable sets V_j , $1 \leq j \leq L$, and retains the original names in every character $\chi_A(z)$. Define

$$B_j = \left(\bigcup_{i \leq j} V_i \right) \cap \left(\bigcup_{i > j} V_i \right), \quad E_j = B_{j-1} \cup V_j, \quad w = \max_j |B_j|. \quad (108)$$

Here $B_0 = B_L = \emptyset$. For a quantity $f(u)$ on E_j , write

$$(\mathcal{A}_j f)(b) = 2^{-|E_j \setminus B_j|} \sum_{u \in \{\pm 1\}^{E_j} : u|_{B_j} = b} f(u). \quad (109)$$

Every eliminated sign has no further occurrence, so this is its actual conditional average. Its normalization is determined by the original signs, including all reused names. The schedule is supplied; optimizing it is not a free operation. If $|V_j| \leq a$, then $|E_j| \leq w + a$. Reduced random degree is at most $\min(D, La)$, whereas length, coefficient dimensions, and bit sizes remain separate costs.

We will repeatedly use the following elementary certificate. Suppose a nonnegative statistic of the same matrix satisfies, pointwise,

$$\|Z\|^{2p} \leq T_p \leq K_p \|Z\|^{2p}, \quad p \in \mathbb{N}, \quad p \geq 1. \quad (110)$$

If $M_p = \mathbb{E}T_p$, $N_p = \mathbb{E}T_p^2$, and $\mathcal{F}_{2p} = M_p^{1/(2p)}$, then

$$K_p^{-1/(2p)} \mathcal{F}_{2p} \leq (\mathbb{E}\|Z\|^{2p})^{1/(2p)} \leq \mathcal{F}_{2p}, \quad (111)$$

$$\mathbb{P}\{\|Z\| > \eta^{-1/(2p)} \mathcal{F}_{2p}\} \leq \eta, \quad 0 < \eta < 1. \quad (112)$$

If $M_p > 0$ and $0 < \theta < 1$, there is the separate strict lower certificate

$$\mathbb{P}\{\|Z\| > (\theta M_p / K_p)^{1/(2p)}\} \geq (1 - \theta)^2 M_p^2 / N_p. \quad (113)$$

The first assertions follow by averaging and Markov. For the last, split $\mathbb{E}T_p$ over $\{T_p > \theta M_p\}$ and its complement and apply Cauchy–Schwarz on the former. This proves strict Paley–Zygmund, and (110) gives the asserted event inclusion. The lower confidence range can be small; matching moments do not imply matching quantiles. The case $M_p = 0$ gives $Z = 0$ on every original sign assignment.

9.2 Symmetric powers remove the exponential moment cost at bounded size

For $1 \leq \ell \leq J$, supply $n_\ell \times n_\ell$ factors $F_{\ell j} = \sum_{A \subseteq V_j} C_{\ell j, A} \chi_A$, with at most t nonzero terms and $n_\ell \leq b$. Arbitrary signed complex coefficients are allowed. Supply readouts U_ℓ of size $d_\ell \times n_\ell$ and W_ℓ of size $n_\ell \times d_\ell$, where $1 \leq d_\ell \leq n_\ell$, and put

$$R_\ell = U_\ell F_{\ell 1} \cdots F_{\ell L} W_\ell, \quad Z = \text{diag}(R_1, \dots, R_J), \quad d = \sum_\ell d_\ell. \quad (114)$$

The blocks may share all their signs. Fixed unitary matrices on either side preserve the norm statement; fixed common conjugation also preserves the ordered algebra. A hidden block decomposition is not being inferred from a dense input.

Let $h_p(s) = \binom{s+p-1}{p}$, $h = h_p(b)$, and $H = \sum_\ell h_p(d_\ell)$. The map $\text{Sym}^p(A)$ is the restriction of $A^{\otimes p}$ between symmetric tensor spaces, in orthonormal bases. It respects products and adjoints, including rectangular maps. Set $S_{\ell j} = \text{Sym}^p(F_{\ell j})$ and

$$C_{\ell 0} = \text{Sym}^p(U_\ell)^* \text{Sym}^p(U_\ell), \quad T_\ell = \text{Sym}^p(W_\ell) \text{Sym}^p(W_\ell)^*.$$

Theorem 9.1 (Positive symmetric-power transfer). *The positive semidefinite matrix messages*

$$C_{\ell j}(b) = \mathcal{A}_j \left[S_{\ell j}(u)^* C_{\ell, j-1}(u|_{B_{j-1}}) S_{\ell j}(u) \right] (b) \quad (115)$$

give

$$M_p = \sum_\ell \text{tr}(C_{\ell L} T_\ell) = \mathbb{E} \sum_\ell \|\text{Sym}^p(R_\ell)\|_F^2.$$

For this statistic, (110)–(113) hold with $K_p = H$. In particular the comparison factor is $H^{1/(2p)}$.

Proof. Conditioning on B_j , induction identifies $C_{\ell j}$ with the conditional mean of

$$(S_{\ell 1} \cdots S_{\ell j})^* C_{\ell 0} (S_{\ell 1} \cdots S_{\ell j}).$$

Only signs with no future use are integrated in (115). Conjugation preserves positivity. The final trace and multiplicativity of the symmetric lift give the displayed expectation.

If R_ℓ has singular values $\sigma_1, \dots, \sigma_{d_\ell}$, its lift has singular values $\prod_i \sigma_i^{\alpha_i}$ for $\alpha_i \geq 0$, $\sum_i \alpha_i = p$. This follows by lifting a singular value decomposition: lifted unitaries are unitary, and the diagonal map is diagonal in occupation vectors. Its squared Frobenius norm is therefore $\sum_{|\alpha|=p} \prod_i \sigma_i^{2\alpha_i}$. One term is $\|R_\ell\|^{2p}$ and all $h_p(d_\ell)$ terms are at most that number. Summing over blocks proves (110) for the largest block of the same Z , with constant H . $\square$

The second statistic N_p is computed by all ordered pairs of blocks: use $S_{\ell j} \otimes S_{\nu j}$, $C_{\ell 0} \otimes C_{\nu 0}$, and $T_\ell \otimes T_\nu$, on the *same* original assignment at each stage, and sum the final traces. Multiplying separately averaged blocks would be incorrect. If $p \geq b + \log(J/\eta)$, then $H \leq J2^{p+b-1}$ implies

$$(H/\eta)^{1/(2p)} \leq \sqrt{2e}.$$

Thus the upper-tail threshold is at most $\sqrt{2e}$ times the actual L^{2p} norm at an order comparable to $\log(d/\eta)$ for fixed b .

Arithmetic and exact bit work. A lift can be formed in $O(bph^2)$ operations: for each occupation index, multiply its p linear forms by a degree-by-degree monomial recurrence, with at most h monomials per degree and b choices per multiplication. Using dense matrix products, the first and second statistics respectively cost

$$O\left(J(L+1)2^{w+a}[tb^2 + bph^2 + h^3]\right), \quad O(2^w h^2) \text{ working entries}, \quad (116)$$

$$O\left(J^2(L+1)2^{w+a}[tb^2 + bph^2 + h^6]\right), \quad O(2^w h^4) \text{ working entries}. \quad (117)$$

Blocks are processed serially; supplied input storage is additional. Original-name masks, parity evaluation and the schedule add at most $O(J(L+1)2^{w+a}a(t+1) + JL(D+1))$ elementary bit operations with a naive implementation.

The orthonormal lift does not require irrational arithmetic. In the unnormalized occupation basis its metric is $G_s(\alpha, \alpha) = \alpha!/p!$. The raw lift $A_{[p]}$ has column α equal to the coefficients of $\prod_j (\sum_i A_{ij}x_i)^{\alpha_j}$, and respects products. Use $C_0 = U_{[p]}^* G_d U_{[p]}$, conjugate by raw factor lifts, and terminate with $\text{tr}(C W_{[p]} G_d^{-1} W_{[p]}^*)$. For Gaussian rational input with common denominator Q and numerator magnitudes and Q bounded by 2^B , all intermediate lengths are

$$O(p(L+2)[B + \log_2(t+b+2)] + D + p \log_2(p+b+2) + \log_2(J+2)) = O(B_*). \quad (118)$$

Indeed a matrix path has $L+2$ coefficient factors, each evaluated entry has numerator at most $t2^B$ per real or imaginary part, and each matrix sum contributes at most a factor $2b$. Lifting raises these bounds to order p ; the squared or paired statistic changes only an absolute constant. Averaging contributes denominator at most 2^D , the metrics contribute factorials of order p , and the block sum contributes at most $2 \log_2 J$ bits. Common denominators avoid spurious fraction growth. Schoolbook bit work is (116) or (117) times $O(B_*^2)$, plus bookkeeping. At fixed b, a, t , since $h \leq (p+b)^{b-1}$, these are polynomial in $p, J, L, D, B, 2^w$.

Closure with the new internal size charged. For matching output blocks, ordered multiplication concatenates factor programs with bridge $W_\ell U'_\ell$; padding permits internal size $\max(b, b')$. Addition uses direct sums of internal programs and adding readouts, with size at most $b+b'$, identity padding, and the union of declared local variables. Adjoint reverses the factor order and exchanges the adjoint readouts. Primitive identifications substitute names, XOR repeated sign occurrences, and add equal characters as matrices. They define the newly aliased model, not conditioning of the old model. All operations require the updated schedule: reusing c formerly eliminated names increases a conservative boundary bound by c . Concatenating two programs with c shared names has boundary at most $w+w'+c$. Different output block partitions require a supplied common decomposition or a larger coupled internal block. Exact updates have zero mathematical approximation error; repeated additions need not retain small b . Ordered collision factors

$$I + \sigma_v \sigma_w (\mathbf{1}_{h_v=h_w} - 1/q)(u_v u_w^* + u_w u_v^*)$$

with $q = 2^c$ buckets give a noncommuting Parseval-frame example. They have at most q characters and $2(c+1)$ local bits. A sliding strip of triangles, processed consecutively with no nonlocal vertex identifications, has boundary at most $3(c+1)$. The actual frame dimension is the charged internal size; this is a collision product, not a claim about the complete SparseStack error.

9.3 Tensor factors and growing irreducible coefficient algebras

Supply local square matrices $F_j = \sum_{A \subseteq V_j} C_{j,A} \chi_A$ of sizes $b_j \leq b$, at most t terms each, and put

$$Z(z) = \bigotimes_{j=1}^L F_j(z), \quad d = \prod_j b_j. \quad (119)$$

No independence between factors is assumed. Their signed noncommuting coefficient matrices, physical positions, and common primitive names are retained. Define the nonnegative local table $f_{j,p} = \text{tr}((F_j^* F_j)^p)$ and update the scalar message

$$m_0 = 1, \quad m_j(b) = \mathcal{A}_j \left[m_{j-1}(u|_{B_{j-1}}) f_{j,p}(u|_{V_j}) \right] (b). \quad (120)$$

Theorem 9.2 (Tensor norm-moment calculus). *The final value is $M_p = \mathbb{E} \operatorname{tr}((Z^* Z)^p)$. Consequently (111)–(113) hold with $K_p = d$, and N_p is computed by squaring each local weight in (120). Arithmetic cost for M_p is*

$$O\left(2^a \sum_j [tb_j^2 + b_j^3 \log(p+1)] + L2^{w+a}\right), \quad (121)$$

The working storage is $O(2^w + 2^a + b^2)$ entries, and the same type of cost computes N_p . At the order

$$p \geq \max(1, \lceil \log(d/\eta) \rceil),$$

the upper confidence threshold is at most $\sqrt{e}$ times the same input's L^{2p} norm.

Proof. Entrywise multiplication of tensor products gives $(\otimes_j A_j)(\otimes_j B_j) = \otimes_j (A_j B_j)$, and traces multiply. Hence $T_p = \operatorname{tr}((Z^* Z)^p) = \prod_j f_{j,p}$. Conditional elimination proves (120), with original shared signs still present at their future uses. The singular values give $\|Z\|^{2p} \leq T_p \leq d\|Z\|^{2p}$ pointwise. Squaring the local weights computes $\mathbb{E} T_p^2$. Repeated squaring forms local matrix powers in $O(b_j^3 \log(p+1))$ operations, and the boundary update visits at most 2^{w+a} assignments per site. Finally $(d/\eta)^{1/(2p)} \leq \sqrt{e}$ at the stated order. $\square$

With the common-denominator convention above, exact intermediate bit lengths are bounded by a constant times

$$\mathcal{B} = pL[B + \log_2(t + b + 2)] + D + \log_2(d + 1) + 1. \quad (122)$$

To see this, a local Gram power uses $2p$ evaluated entries and denominator Q^{2p} ; its sums cost at most a fixed power of $2b(t2^B)$ of order $2p$. Multiply over L factors, include the trace dimensions and at most D averaging bits. Squaring the statistic doubles these exponents. Thus (121) times $O(\mathcal{B}^2)$ bounds schoolbook bit work, plus $O(L2^{w+a}a(t+1) + L(D+1))$ for naive label bookkeeping. The positive update also permits a certified approximation rule: if $0 \leq f_j^- \leq f_j \leq f_j^+$ and $f_j^+ \leq e^{\epsilon_j} f_j^-$, including exact zeros, then

$$M_p^- \leq M_p \leq M_p^+ \leq e^{\sum_j \epsilon_j} M_p^-.$$

The resulting root interval has ratio at most $\exp(\sum_j \epsilon_j / (2p))$. This requires rigorous local intervals.

At a common tensor partition, ordered multiplication is exactly $\otimes_j (F_j G_j)$; adjoint is factor-wise. Local variable sets are unions and expanded character budgets multiply before merging. The old scalar weights cannot themselves be multiplied to implement this matrix operation. Tensor append, local additions, and primitive aliases have exact updates, with the new boundary charged. Arbitrary full sums are not covered by this factorization. Different partitions may be coarsened by merging overlapping factor blocks, but a merged local dimension is the product of its constituents. Two overlapping interaction layers can force a single large block. A small random boundary does not control this coefficient cost.

Example 9.3 (Irreducibility without expanding the ambient matrix). Let $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Z_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and $F_j = I_2 + s_j X + t_j Z_0$ with independent local signs. The expanded coefficient set of $\otimes_j F_j$ contains every single-site X_j and $(Z_0)_j$. Their products include all tensor products of I, X, Z_0, XZ_0 , a basis of the full algebra $M_{2L}(\mathbb{C})$. Thus there is no proper common invariant subspace. Nonetheless $b = 2$, $w = 0$, and the moment calculation is polynomial in L, p . Large irreducible coefficient dimension is therefore compatible with compact computation on this class.

A separate exact strict-quantile subclass. Suppose certified local tables give either $F_j = 0$ or $\|F_j\| = \gamma^{s_j}$, where $\gamma > 1$ is common and $s_j \in \{-R, \dots, R\}$. Tensoring singular-value decompositions proves $\|Z\| = \gamma^{\sum_j s_j}$ whenever no factor is zero. Replace $f_{j,p}$ in (120) by the monomial x^{s_j+R} , and by zero at a zero factor. The final polynomial's coefficients are the exact probabilities of nonzero norm levels, with exponent shift LR ; missing mass is the zero atom. Monomial multiplication only shifts arrays, so the cost is $O(L^2(R+1)2^{w+a})$ rational operations and $O(L(R+1)2^w)$ working coefficients. Probabilities have $O(D + \log(LR+2))$ bits, apart from construction of the certified tables. Scanning atoms, including zero, gives exactly

$$\inf\{u \geq 0 : \mathbb{P}(\|Z\| > u) \leq \eta\}.$$

A positive transition path recovers an original assignment for a supported atom; tensoring maximizing local right singular vectors realizes its norm in the same input. The lattice hypothesis survives tensor append and aliases, but need not survive arbitrary local matrix multiplication.

For actual bucket overlaps take

$$F_{vw} = I_2 + \mathbf{1}_{h_v=h_w} P_{\sigma_v}, \quad P_+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_- = \frac{1}{25} \begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix},$$

where $h_v \in [q]$, $q = 2^c$, and σ_v are original independent vertex variables. These two rank-one projectors do not commute. Here the norm is nevertheless orientation-independent: $\|\otimes_e F_e\| = 2^{N_{\text{col}}}$. One triangle has

$$\mathbb{P}(N_{\text{col}} = 0, 1, 3) = \frac{1}{q^2}((q-1)(q-2), 3(q-1), 1),$$

with no count-two atom, because two equalities imply the third. At $q = 4$ the norm atoms are 1, 2, 8, of masses 6/16, 9/16, 1/16. The strict $\eta = 1/16$ quantile is 2, whereas it is 8 for $0 < \eta < 1/16$. Replacing edges by independent Bernoulli constraints changes this law.

9.4 Matrix-product operators retain coherent sums and physical interactions

To allow sums of tensor products without merging the whole physical space, supply local arrays $A_j^{a,b}(z)$ of $s_j \times s_j$ matrices, $s_j \leq s$, with bond sizes $b_0 = b_L = 1$, $b_j \leq \beta$. Each site array uses at most t characters on $|V_j| \leq a_0$ original signs. Define the matrix-product operator by the actual finite sum

$$Z = \sum_{\alpha_1, \dots, \alpha_{L-1}} A_1^{1, \alpha_1} \otimes A_2^{\alpha_1, \alpha_2} \otimes \dots \otimes A_L^{\alpha_{L-1}, 1}, \quad d = \prod_j s_j. \quad (123)$$

Virtual paths may interfere; neither their signed sums nor the original shared signs are replaced by independent copies. No small-bond representation is promised for an arbitrary dense matrix.

At bond j , let $\omega_j = (\alpha_j^1, \dots, \alpha_j^p; \beta_j^1, \dots, \beta_j^p) \in [b_j]^{2p}$. Define the local replica transfer, on the same local sign assignment, by

$$Q_{j,p}[\omega_{j-1}, \omega_j] = \text{tr}(M_p \cdots M_1), \quad M_h = (A_j^{\beta_{j-1}^h, \beta_j^h})^* A_j^{\alpha_{j-1}^h, \alpha_j^h}. \quad (124)$$

Here M_h denotes only a local matrix, not the final moment statistic. With endpoint bonds fixed, initialize $m_0 = 1$ and set

$$m_j(b, \omega_j) = \mathcal{A}_j \left[\sum_{\omega_{j-1}} m_{j-1}(u|_{B_{j-1}}, \omega_{j-1}) Q_{j,p}(u)[\omega_{j-1}, \omega_j] \right] (b). \quad (125)$$

The intermediate message can be signed or complex.

Theorem 9.4 (Exact cyclic-replica calculus). *The terminal scalar is real and nonnegative and equals $M_p = \mathbb{E} \operatorname{tr}((Z^* Z)^p)$. Hence (111)–(113) hold with $K_p = d$. An exact direct implementation has arithmetic cost*

$$O\left(L2^{w+a_0}[t\beta^2 s^2 + ps^3 \beta^{4p}]\right) \quad (126)$$

and working storage $O(2^w \beta^{2p} + \beta^2 s^2 + ps^2)$ entries. For N_p , use two trace cycles on the same signs; replacing exponents $2p, 4p$ by $4p, 8p$ gives corresponding bounds.

Proof. Expand the physical trace as

$$\operatorname{tr}((Z^* Z)^p) = \sum_{x_1, \dots, x_p, y_1, \dots, y_p} \prod_{h=1}^p Z[x_h, y_h] \overline{Z[x_h, y_{h+1}]}, \quad y_{p+1} = y_1.$$

Substituting (123), the sum over each site's physical indices is

$$\sum_{x_1, \dots, x_p, y_1, \dots, y_p} \prod_h A_j^{\alpha_j^h - 1, \alpha_j^h} [x_h, y_h] \overline{A_j^{\beta_j^h - 1, \beta_j^h} [x_h, y_{h+1}]}.$$

Summing x_h forms $M_h[y_{h+1}, y_h]$, giving exactly the reverse product in (124). Summing virtual labels contracts these transfers; conditional elimination gives (125). The pointwise singular-value comparison proves the certificate. Two copies of the trace have local transfer $Q_{j,p} \otimes Q_{j,p}$ under one shared assignment; this generally differs from the single cycle $Q_{j,2p}$. There are at most β^{4p} entries per local transfer, each requiring $O(ps^3)$ operations. Evaluate one assignment and one entry at a time and accumulate the next boundary message to obtain the stated cost and storage. $\square$

With input common denominator and current coefficient length B , message bit lengths are

$$O(pL[B + \log_2(t + s + \beta + 2)] + D + 1) = O(\mathcal{B}). \quad (127)$$

Every transfer contains $2p$ evaluated coefficient factors, a physical trace sum bounded by s^{2p} , and at most $\beta^{2p(L-1)}$ full virtual paths. A stage- j common denominator is Q^{2pj} times the averaging powers of two. Bounding before cancellation proves (127); the two-cycle statistic only doubles exponents. Thus (126) times $O(\mathcal{B}^2)$ bounds schoolbook integer work. Label and parity bookkeeping adds a polynomial in the written input and $L2^{w+a_0}(t+1)(D+p\log_2(\beta+1))$.

All exact functionals in this section can be returned as a rational radicand with a specified positive root. Relative numerical root precision requires additional polynomial bit work by rational bisection. Moreover, B is the *actual current* length after updates: repeatedly squaring the constant two has degree zero and doubles its bit length. A polynomial number of operations and bounded reduced random degree are insufficient to ensure polynomial bit cost. A polynomial bound on the unreduced expression's input-leaf count, with polynomial initial coefficient lengths, is sufficient; otherwise the current lengths must be charged directly. Signed replica messages have no positive-message relative-error guarantee. The theorem has zero exact approximation error; any numerical truncation needs its own certified global error estimate.

At $p = \max(1, \lceil \log(d/\eta) \rceil)$ the comparison threshold remains $\sqrt{e}$, and

$$\beta^{4p} \leq \beta^4 (d/\eta)^{4 \log \beta}, \quad \beta^{8p} \leq \beta^8 (d/\eta)^{8 \log \beta}.$$

For fixed β, s, a_0 , logarithmic boundary and polynomial coefficient length, this is polynomial in $L + D + d/\eta$, with the displayed exponents. It is not generally polynomial in $\log(1/\eta)$, or in a compact chain description when $d = 2^L$. In contrast, the tensor case $\beta = 1$ has the compact-input efficiency of Theorem 9.2. Both regimes avoid enumeration of all 2^D original assignments at controlled boundary.

Reusable sums, products, and overlapping interactions. For a common physical partition, the ordered product of arrays A and B is represented locally by

$$C_j^{(a,c),(b,e)} = A_j^{a,b} B_j^{c,e}. \quad (128)$$

Indeed tensor products multiply sitewise before the two path sums are performed. The new bond is at most $\beta_A \beta_B$, arities are unions, and expanded term budgets are at most $t_A t_B$; dense local products cost the corresponding number of s^3 operations. For a full sum and $L \geq 2$, concatenate first-site outgoing arrays, put interior arrays in block diagonal form, and stack final-site incoming arrays. Its paths are exactly those of one old operator or the other, giving bond at most $\beta_A + \beta_B$. At $L = 1$ add the local matrices directly. Adjoint takes local adjoints with unchanged virtual labels. Physical tensor append joins scalar endpoint bonds; neighboring sites may be regrouped by contracting their intermediate bonds and multiplying their dimensions.

These rules prove exact word extension and closure under arbitrary finite sums and ordered products, with bond growth charged. Original aliases use XOR and signed coefficient addition, followed by schedule recomputation; they are distinct from affine conditioning. Reuse of c forgotten primitive names can be accommodated by a conservative increase of c in historical boundary width. Different physical partitions require common coarsening and its larger local dimensions. No automatic small-bond compression is presumed. A terminal expectation is not a descriptor for a later interaction with forgotten names.

As a genuinely interacting example, take the three-state site array

$$A_j = \begin{pmatrix} I & z_j Z_0 & z_j z_{j+1} X \\ 0 & 0 & z_j X \\ 0 & 0 & I \end{pmatrix},$$

with initial state one and terminal state three. Its paths give

$$H(z) = \sum_{j=1}^L z_j z_{j+1} X_j + \sum_{j=1}^{L-1} z_j z_{j+1} (Z_0)_j X_{j+1}.$$

The bond is three and the random boundary is at most one. Adjacent interactions overlap, X_j and $(Z_0)_j X_{j+1}$ do not commute, and the same signs couple the fields and interactions. Coherent sums of tensor chains and their ordered products are further examples governed by the same theorem, rather than separately estimated path norms.

9.5 An actual continuation disproves full replica symmetrization

The successful symmetric-power reduction for sequential matrix programs does not extend by simply symmetrizing the virtual replicas above.

Proposition 9.5 (A reached-state obstruction). *For a deterministic two-site matrix-product operator with local size two and bond four, replacing a reached $p = 3$ replica message by its ordinary simultaneous symmetric projection changes the final trace moment from 4096 to 2560 after a legal continuation.*

Proof. Let the first site's blocks be $(I, X, Z_0, X Z_0)$, with left bond one and right bond four, and let q denote its reached replica vector. Fixing all conjugate labels to I , the unconjugated triples $(X, Z_0, X Z_0)$ and $(Z_0, X, X Z_0)$ give respectively

$$\text{tr}((X Z_0) Z_0 X) = 2, \quad \text{tr}((X Z_0) X Z_0) = -2.$$

These labels have the same paired-replica occupation counts. Append the same four blocks as a second site with outgoing bond one. The actual matrix is

$$Z = I \otimes I + X \otimes X + Z_0 \otimes Z_0 + (XZ_0) \otimes (XZ_0) = \begin{pmatrix} 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 \end{pmatrix}.$$

It has rank one and norm four, so $\text{tr}((Z^*Z)^3) = 4^6 = 4096$. Its second-site replica vector is again q .

For the exact projected value, label blocks by $v = (a, b) \in \mathbb{F}_2^2$ as $X^a Z_0^b$ and use their commutation form $[v, w] = ab' + ba'$. Put $\delta_h = \alpha_h + \beta_h$ for each replica pair. The entry q vanishes unless $\delta_1 + \delta_2 + \delta_3 = 0$; otherwise it is ± 2 . There are $4^5 = 1024$ nonzero entries, so $\|q\|^2 = 4096$. Cyclic permutations leave q invariant. A transposition, after a cyclic rotation if necessary, changes its nonzero sign by $(-1)^{[\delta_1, \delta_2]}$. For each prescribed delta triple there are 4^3 original paired-label choices. Its inner product with any transposed vector is therefore

$$4 \cdot 4^3 \sum_{\delta_1, \delta_2 \in \mathbb{F}_2^2} (-1)^{[\delta_1, \delta_2]} = 4 \cdot 4^3 \cdot 4 = 1024.$$

Averaging the three cyclic and three transposed inner products gives $\langle P_{\text{sym}} q, q \rangle = (3 \cdot 4096 + 3 \cdot 1024)/6 = 2560$. Thus the information lost by symmetrization affects the same matrix after a permitted extension. $\square$

Cyclic invariance is valid, but length- p words over β^2 paired symbols have at least β^{2p}/p cyclic orbits. This counts that specific invariant representation, not the dimension of every reachable family and not a lower bound for every possible algorithm. The obstruction rules out one exact occupation-count reduction. A more informative compression with an explicit approximation loss remains a separate problem.

The three calculi therefore establish complementary parts of the general target. At controlled internal size, symmetric powers give economical high moments; tensor factors allow growing irreducible geometry and a separate exact lattice confidence law; finite-bond operators close under full sums and genuine interactions at a stated, sometimes larger cost. None supplies a useful bounded-size or bounded-bond representation for arbitrary Parseval-compressed SparseStack or for the all-support Walsh relation sum. That missing representation or analytic contraction is a mathematical limitation, not a suppressed computational cost.

10 A complete norm and strict-quantile calculus at controlled block sparsity

The preceding exact transfers retain large ordered coefficient tensors when the matrix has a compact product representation. A different class permits a dimension-free comparison before its probability law is integrated. The result below realizes all three requirements in Section 28: reusable operations, useful costs at logarithmic confidence order, and a separate distributional argument for the same matrix's strict quantiles. Its restrictions are block sparsity and local random-variable width. It is not a reduction of arbitrary dense matrices to those parameters.

10.1 The retained class and a constructive pointwise certificate

Partition the coordinates of a square matrix Z of dimension d into n supplied blocks of size at most b . Let $\mathcal{E}$ be its supplied possible block support, with $M = |\mathcal{E}|$, maximum block-row degree

Δ_r and maximum block-column degree Δ_c . Each block is

$$Z_{ij}(z) = \sum_{A \subseteq V_{ij}} C_{ij,A} \chi_A(z), \quad |V_{ij}| \leq a, \quad (129)$$

with at most t supplied character terms in independent uniform signs $z_1, \dots, z_D$. Identical names always mean the same original sign. Coefficients may be complex, signed, nonsymmetric and noncommuting. Duplicate characters are added as matrices. The actual reduced degree is at most a ; local evaluation programs may instead be supplied, with their degree, evaluation work and actual bit lengths charged separately.

The descriptor retains every coefficient, block position and primitive name, together with an order $e_1, \dots, e_M$ of the blocks. Write

$$B_j = \left(\bigcup_{i \leq j} V_{e_i} \right) \cap \left(\bigcup_{i > j} V_{e_i} \right), \quad w = \max_j |B_j|. \quad (130)$$

The partition and schedule are part of the input; discovering a better one is not assumed easy. Unused signs can be removed, giving $D \leq Ma$. The case $\mathcal{E} = \emptyset$ is the zero matrix and is immediate.

Define

$$S_e(z) = \|Z_e(z)\|_{\mathbb{F}}^2, \quad G(z) = \max_{e \in \mathcal{E}} \sqrt{S_e(z)}, \quad \Lambda = \sqrt{\Delta_r \Delta_c}. \quad (131)$$

Lemma 10.1 (A same-input sparse-block certificate). *Every original assignment satisfies*

$$b^{-1/2} G(z) \leq \|Z(z)\| \leq \Lambda G(z). \quad (132)$$

The lower side has a coordinate-vector witness in the original matrix.

Proof. A block attaining G has a column with Euclidean norm at least $G/\sqrt{b}$. The same coordinate column of the full matrix is at least as long. This proves the lower inequality constructively.

Let C be the nonnegative scalar matrix with entries $C_{ij} = \|Z_{ij}\|$. If x has coordinate blocks x_j , then $\|(Zx)_i\|_2 \leq \sum_j C_{ij} \|x_j\|_2$, so $\|Z\| \leq \|C\|$. Weighted Cauchy–Schwarz gives

$$\sum_i \left| \sum_j C_{ij} u_j \right|^2 \leq \left(\max_i \sum_j C_{ij} \right) \left(\max_j \sum_i C_{ij} \right) \|u\|_2^2.$$

Every $C_{ij} \leq G$ and the support degrees are supplied, proving the upper inequality. Neither independence nor commutation of the coefficients was used in this deterministic step. $\square$

The factor $\sqrt{b \Delta_r \Delta_c}$ is a controlled structural loss, not an exact-constant assertion. Scalar all-ones blocks on a complete $\Delta \times \Delta$ support have $\|Z\| = \Delta G$. A single identity block of size b has $G = \sqrt{b} \|Z\|$. These examples show why those parameters cannot simply be omitted from this readout. They do not give lower complexity bounds for every possible descriptor.

10.2 An exact envelope law and matching strict quantiles

Assume first that real and imaginary coefficient parts are rational. Evaluate the actual matrix sum in each block on its at most 2^a local assignments. The squared scores S_e are nonnegative rationals. Their ordered union, including zero, is

$$0 = \tau_0 < \tau_1 < \dots < \tau_K, \quad K + 1 \leq M 2^a + 1.$$

There can be fewer globally attainable maxima than local score levels. For each threshold τ use the true local indicator $f_{j,\tau} = \mathbf{1}_{\{S_{e_j} \leq \tau\}}$ and set $E_j = B_{j-1} \cup V_{e_j}$. The transfer is

$$m_j(v) = 2^{-|E_j \setminus B_j|} \sum_{u: u|_{B_j} = v} m_{j-1}(u|_{B_{j-1}}) f_{j,\tau}(u|_{V_{e_j}}), \quad m_0 = 1. \quad (133)$$

Its output is $F(\tau) = \mathbb{P}(G^2 \leq \tau)$. Indeed the message is the conditional expectation of the product of the processed indicators, given the live original signs. Integrate a sign only after its last declared use. Induction gives the entire joint event. Reused signs, overlapping equalities and triangles therefore survive; the calculation does not multiply marginal block probabilities.

Omitting the normalizing factors in (133) gives integer counts of completions of forgotten variables. The final count is divided by 2^D . Every update then uses addition and multiplication by zero or one. Define

$$\pi_0 = F(0), \quad \pi_l = F(\tau_l) - F(\tau_{l-1}) \quad (l \geq 1).$$

These are exactly the masses of G^2 . Local levels of mass zero can be retained without changing any formula.

Theorem 10.2 (All moments and all strict confidence levels). *For every real $p > 0$, let*

$$\mathcal{F}_p(\mathcal{D}(Z)) = \left(\sum_{l=0}^K \pi_l \tau_l^{p/2} \right)^{1/p}. \quad (134)$$

Then the same input matrix obeys

$$\boxed{b^{-1/2} \mathcal{F}_p \leq (\mathbb{E} \|Z\|^p)^{1/p} \leq \sqrt{\Delta_r \Delta_c} \mathcal{F}_p.} \quad (135)$$

For $0 < \eta < 1$, use the strict convention $q_Y(\eta) = \inf\{u \geq 0 : \mathbb{P}(Y > u) \leq \eta\}$. If $l(\eta)$ is the first index with $1 - F(\tau_l) \leq \eta$, then

$$q_G(\eta) = \sqrt{\tau_{l(\eta)}}, \quad \boxed{b^{-1/2} q_G(\eta) \leq q_{\|Z\|}(\eta) \leq \sqrt{\Delta_r \Delta_c} q_G(\eta).} \quad (136)$$

Zero atoms and confidence exactly equal to an upper-tail mass are included.

Proof. Raise (132) to p and integrate the same original law. This proves (135). The finite CDF calculation identifies the first admissible atom in the strict quantile definition. Pointwise order implies quantile order, and positive scaling commutes with that definition. Applying these facts twice to (132) proves (136). This is a distributional proof of the same norm's quantiles, separate from its moment comparison. $\square$

To recover an original assignment realizing a specified positive-probability maximum τ , append a Boolean flag recording that some block score equals τ to the $\leq \tau$ transfer. A positive path ending with the flag true realizes that atom. This at most doubles the state. An ordinary CDF path could instead realize a smaller level. On the recovered assignment, Lemma 10.1 supplies its coordinate-vector lower witness.

Markov also gives $\mathbb{P}\{\|Z\| > \Lambda \eta^{-1/p} \mathcal{F}_p\} \leq \eta$. At $p \geq \log(1/\eta)$ its extra factor is at most e . The direct quantile calculation is stronger when its local tables and boundary are manageable; no extreme lower quantile is inferred from moments alone.

10.3 Useful arithmetic, bit costs and error control

Put all rational coefficient parts over a common positive denominator Q . Charge its actual length: suppose Q and numerator magnitudes are at most 2^B . Local squared scores have denominator Q^2 and numerator length

$$B_s = O(B + \log(bt + 2)).$$

Local score formation costs $O(M2^a tb^2)$ arithmetic. Sorting at most $M2^a + 1$ integer numerators costs $O((M2^a + 1) \log(M2^a + 2))$ comparisons. The complete CDF sweep costs

$$O((M2^a + 1)M2^{w+a}) \quad (137)$$

integer additions, each on counts of at most $D + 1$ bits, plus score comparisons. The bit cost of this part is bounded by (137) times $O(D + B_s + 1)$. Local arithmetic can be charged at $O(B_s^2)$ schoolbook work per operation. Local tables use $O(M2^a B_s)$ bits, a current message uses $O(2^w(D + 1))$, and the finite law uses $O((M2^a + 1)(D + B_s + 1))$. Thresholds can be processed sequentially. Explicit mask bookkeeping contributes conservatively $O(M(D + 1) + M2^a a(t + 1))$ bit operations; input storage is separate.

After the law is computed, an even moment of order $2h$, $h \geq 1$ integer, uses the rational radicand $\sum_l \pi_l \tau_l^h$. Repeated squaring costs $O((K + 1) \log(h + 1))$ arithmetic, with common-denominator bit length

$$O(hB_s + D + \log(K + 2)).$$

The exact functional can be returned as this rational and a positive $2h$ th-root instruction. Numerical root precision is separately charged; bisection with rational-power comparisons is sufficient. Formula (134) for other real orders is a finite expression, not a claim of rational unit-cost arbitrary real exponentiation.

Strict quantiles need only rational comparisons with the written η , followed by square-root expressions. They do not iterate $1/\eta$ times. Thus fixed $b, \Delta_r, \Delta_c, a, t$, logarithmic w and polynomial actual coefficient lengths give polynomial computation in the written input and $\log(1/\eta)$, including h comparable to $\log(d/\eta)$. Both comparison factors remain dimension-free. Exponential dependence on local arity or live width is explicit.

The exact algorithm has zero approximation error. If the final local scores instead have certified intervals $s^- \leq s \leq s^+ \leq e^\epsilon s^-$ with exact zeros, then $G^- \leq G \leq G^+ \leq e^{\epsilon/2} G^-$. Both moment and quantile brackets acquire at most this extra factor. Relative errors on individual matrix summands cannot be carried through cancellation without a new proof; the certified intervals must concern the actual final score.

10.4 Full operations and final structural losses

Keep a common supplied block partition. Full addition and subtraction union supports, add row/column degrees at worst, union primitive sets, and add actual signed coefficients. For multiplication retain

$$(AB)_{ij} = \sum_{k: (i,k) \in \mathcal{E}_A, (k,j) \in \mathcal{E}_B} A_{ik} B_{kj}. \quad (138)$$

The new support is this relational composition. Its row degree is at most $\Delta_{r,A} \Delta_{r,B}$ and column degree at most $\Delta_{c,A} \Delta_{c,B}$. If at most ν intermediate blocks contribute to an output, then

$$a_{AB} \leq \nu(a_A + a_B), \quad t_{AB} \leq \nu t_A t_B, \quad \nu \leq \min(\Delta_{r,A}, \Delta_{c,B}).$$

Degrees add, characters multiply by XOR, and coefficient matrices multiply in the displayed order. All intermediate-block contributions and duplicate characters are summed before a norm

is taken. The expanded update costs its actual number of term pairs times $O(b^3)$ arithmetic. A retained evaluation program is permitted with its work charged instead.

Adjoint transposes the support and conjugates/transposes coefficients. Word extension is the corresponding product. Different partitions may be replaced by a supplied common coarsening; block sizes become sums of the merged coordinate sizes, and support degrees and local arities are recounted. No inexpensive optimal coarsening is assumed.

Primitive identification substitutes common new independent names, reduces characters by XOR, adds duplicates, and recomputes (130). This defines the new aliased model; it is not conditioning the former independent model. Local arity does not increase. If c old names belong to nonsingleton identification classes, the new boundary in the same order is at most $w + c$. Whole bucket identifications use complete bit strings. An affine conditional law may be represented by a supplied independent parametrization only with its resulting arity and width charged.

The comparison factor is always computed from the *final* structural parameters. It is not multiplied once per earlier operation. Exact closure does not promise that all parameters stay small: products can make the support dense, and long-range reuse can enlarge the live boundary. Coefficient bit growth is another independent cost. A product adds input lengths and $O(\log(\nu t_A t_B b + 2))$ summation bits. Polynomially many updates are polynomial in bit cost only when actual coefficient lengths remain polynomial; polynomial unreduced leaf count is a sufficient condition. Repeated squaring of the constant two has degree zero and exponentially increasing length, so degree alone cannot justify a cost claim.

Scores alone are not the update state. Take complementary rank-one projections P, Q and a rank-one projection R not commuting with P . The factors P, Q have equal Frobenius norm, but the common continuation $P(\text{factor})R$ is nonzero for P and zero for Q . Full coefficient matrices, retained in the descriptor, resolve this ambiguity.

10.5 Banded polynomials and collision thresholds

Let $A_1, \dots, A_g$ have initial block size b_0 and half-bandwidth $h_0 \geq 1$, each initial block using at most a_0 signs and t_0 terms. Supply a sum of T noncommutative words, each of length at most $R \geq 1$. Every entry is a sum over weighted walks of at most R steps, so the output has half-bandwidth $H = Rh_0$. This elementary matrix-walk locality is the classical sparse-matrix mechanism discussed in [18]. Grouping consecutive sites into chunks of H gives block size at most Hb_0 and at most three neighboring blocks per row and column. Hence

$$(Hb_0)^{-1/2} \mathcal{F}_p \leq (\mathbb{E} \|P(A_1, \dots, A_g)\|^p)^{1/p} \leq 3\mathcal{F}_p, \quad (139)$$

with the same factors for all strict quantiles.

For any output block, restrict the inputs to its endpoint chunks and a margin of H sites. At most $4H + 1$ sites occur, and every contributing walk stays there. Thus the local evaluation preserves complete matrix sums and costs, per original local assignment, at most

$$O(g(4H + 1)(2h_0 + 1)t_0b_0^2 + TR((4H + 1)b_0)^3).$$

Its relevant primitive arity is at most $a_0g(4H + 1)(2h_0 + 1)$. With bounded-range primitive placement, the row order has bounded boundary for fixed R and initial locality. Arbitrary reuse pays its actual boundary. Growing R can make local tables expensive; that dependence is not hidden. These inputs include noncommuting block Jacobi matrices, commutators, and fixed-degree polynomials in several banded matrices, as well as nearby bucket-collision edges forming overlapping triangles. They need not be tensor products or block diagonal.

Exact checks compare full original laws and ordered matrix updates on finite noncommuting examples, verify same-input upper PSD inequalities and lower coordinate columns, and retain

strict endpoint changes. A 257-sign sparse path is evaluated without its 2^{257} assignments: its envelope is zero with probability 2^{-256} and two otherwise, giving $\mathbb{E}G^{256} = 2^{256} - 1$. The full matrix's moments and quantiles are bracketed by the theorem, not asserted equal to those envelope values.

The diffuse integer-endpoint diagnostic also survives. With k equal scalar frame entries, bucket count k^4 , and distortion $1/k$, exactly one colliding pair gives absolute error $2/k$ for either pair sign. Its probability is

$$\binom{k}{2} \frac{(k^4)_{k-1}}{(k^4)^k} \geq \frac{1}{8k^2}, \quad k \geq 2.$$

Indeed $\binom{k}{2}/k^4 \geq 1/(4k^2)$ and the remaining product is at least $1 - (k-2)(k-1)/(2k^4) \geq 7/8$. The count holds for all integer k . For powers of two k , its exact binary encoding has local arity of order $k \log k$, so this scalar example is not a cheap constant-arity input to the theorem. Uniform non-power-of-two buckets are not silently encoded by finitely many fair bits. Full dense SparseStack and arbitrary Walsh support selection remain outside the affordable regime of this sparse-block calculus.

11 A dense calculus from complete permutation sectors

Bounded coordinate-block sparsity is not necessary for a useful two-sided calculus. A different sufficient structure is a supplied partition of physical sites into permutation-invariant groups, together with a controlled set of marked sites permitting arbitrary local interactions. Complete spin sectors preserve the entire noncommuting coefficient algebra on this class. An exact joint law for additive integer scores preserves its overlapping random inputs. Their combination gives both norm moments and strict quantiles for the same dense matrix, with comparison arbitrarily close to one at a charged readout order.

Spin and Schur decompositions are established methods; see, for example, [23]. The elementary decomposition, multiplicities, centralizer identification and classical arithmetic needed here are proved below. No quantum algorithm, efficient recognition of an unspecified dense input, or general compression of the replica states in Section 9 is assumed.

11.1 The supplied matrix class and its original random inputs

The physical space is $(\mathbb{C}^2)^{\otimes L}$, of dimension $d = 2^L$. Mark t supplied sites and partition the others into g nonempty groups of sizes $M_1, \dots, M_g$, so $t + \sum_j M_j = L$. A supplied site permutation places the marked factor first. On group j use

$$H_j = \sum_{i \in j} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_i, \quad E_j = \sum_{i \in j} |0\rangle\langle 1|_i, \quad F_j = E_j^*. \quad (140)$$

The subscript inserts the matrix at that site with identities elsewhere. Within a group, $[H_j, E_j] = 2E_j$, $[H_j, F_j] = -2F_j$, and $[E_j, F_j] = H_j$. Different groups commute, but the resulting ordered polynomial need not factor across them. Arbitrary matrices on the marked 2^t -dimensional factor are allowed and need not commute with one another.

Let $z_1, \dots, z_D$ be the actual independent uniform signs. Supply local integer maps and their evaluation order:

$$\phi_v : \{\pm 1\}^{V_v} \longrightarrow \{-K, \dots, K\}^c, \quad |V_v| \leq a, \quad S(z) = s_0 + \sum_{v=1}^m \phi_v(z_{V_v}), \quad (141)$$

where $c, K \geq 1$ and the integer offset s_0 is retained. Exact tables or local programs may be supplied; table formation is charged. Remove unused primitive names. A repeated name is the

same sign, including across different score coordinates. With the supplied order define

$$B_v = \left(\bigcup_{u \leq v} V_u \right) \cap \left(\bigcup_{u > v} V_u \right), \quad w = \max_v |B_v|, \quad U = (2mK + 1)^c. \quad (142)$$

The input $Z(z) = P(\{H_j, E_j, F_j\}_j; S(z))$ is a supplied ordered matrix polynomial, with arbitrary marked coefficient matrices, or an arithmetic circuit evaluating such an expression. Score arguments are scalars. For explicit term-list costs, write

$$Z(z) = \sum_{\nu=1}^T S(z)^{\alpha_\nu} C_\nu \otimes W_\nu(\{H_j, E_j, F_j\}_j), \quad |\alpha_\nu| \leq q, \quad |W_\nu| \leq R. \quad (143)$$

The C_ν are Gaussian rational marked matrices. Several marked factors in a term may first be multiplied in their original order, with this construction cost and the resulting entry lengths charged. Each local score function has multilinear sign degree at most a , so the reduced random degree is at most qa . Neither this degree nor the number of groups replaces the actual expression size, word length, coefficient lengths, or live random boundary. A constant matrix can already be expensive to compute from a long repeated-squaring circuit.

11.2 All spin sectors, their metric, and their positive multiplicities

Consider one group of length M . For $\ell = M, M-2, \dots, 0$ or 1 , define a space with rational basis $e_0, \dots, e_\ell$, metric and matrices

$$G_\ell = \text{diag} \left(\binom{\ell}{k}^{-1} : 0 \leq k \leq \ell \right), \quad A^\sharp = G_\ell^{-1} A^* G_\ell, \quad (144)$$

$$H_\ell e_k = (\ell - 2k)e_k, \quad E_\ell e_k = k e_{k-1}, \quad F_\ell e_k = (\ell - k)e_{k+1}.$$

Out-of-range vectors are zero. The physical adjoints satisfy $H_\ell^\sharp = H_\ell$ and $E_\ell^\sharp = F_\ell$. Using an ordinary Euclidean adjoint in these rational coordinates would change the represented operator.

Lemma 11.1 (Complete one-group decomposition). *The physical generators simultaneously decompose into the sectors (144), orthonormalized by $G_\ell^{1/2}$, with multiplicity*

$$\mu_{M-2u} = \binom{M}{u} - \binom{M}{u-1} > 0, \quad 0 \leq u \leq \lfloor M/2 \rfloor, \quad \binom{M}{-1} = 0. \quad (145)$$

For any ordered polynomial, this is an intertwining of the actual physical matrix, not merely a matching spectrum for its separate generators.

Proof. The physical weight space of H at $M-2u$ has dimension $\binom{M}{u}$. Choose a unit highest-weight vector v , with $E v = 0$ and $H v = \ell v$. Such a vector exists by taking a largest weight in a nonzero invariant subspace. The commutators imply inductively

$$E F^k v = k(\ell - k + 1) F^{k-1} v, \quad H F^k v = (\ell - 2k) F^k v.$$

Since $E^* = F$, $\|F v\|^2 = \ell \|v\|^2$ implies $\ell \geq 0$; ℓ is an integer weight of the physical H . The same recursion gives

$$\|F^k v\|^2 = \frac{k! \ell!}{(\ell - k)!} \quad (0 \leq k \leq \ell), \quad F^{\ell+1} v = 0.$$

Distinct weights are orthogonal. Put $e_k = (\ell - k)! F^k v / \ell!$; the actions and squared norms become (144). This chain is invariant under all three generators. Their adjoint relations make

its orthogonal complement invariant as well, so the construction repeats until the finite physical space is exhausted.

At weight $M - 2u \geq 0$, every chain with highest weight at least that weight contributes exactly one dimension. Subtracting consecutive physical weight dimensions gives (145); positivity follows from the increasing binomial coefficients up to the middle. Thus every listed sector occurs. Simultaneous intertwining of the generators intertwines every ordered expression in them. $\square$

Tensor these actual decompositions across the groups, leaving the marked space unchanged. For a sector tuple ℓ , put

$$\begin{aligned} n_\ell &= 2^t \prod_j (\ell_j + 1), & G_\ell &= I_{2^t} \otimes \bigotimes_j G_{\ell_j}, & \mu_\ell &= \prod_j \mu_{\ell_j}, \\ D_{\text{eff}} &= \sum_\ell n_\ell = 2^t \prod_{j=1}^g \left\lfloor \frac{(M_j + 2)^2}{4} \right\rfloor. \end{aligned} \quad (146)$$

Indeed $\sum_{\ell=M, M-2, \dots} (\ell + 1) = \lfloor (M + 2)^2 / 4 \rfloor$ by summing an arithmetic progression. If $A_\ell(s)$ is the full evaluation of P in the marked and group sector matrices, then a fixed physical unitary gives

$$Z(z) \simeq \bigoplus_\ell \left(G_\ell^{1/2} A_\ell(S(z)) G_\ell^{-1/2} \right) \otimes I_{\mu_\ell}. \quad (147)$$

The retained blocks contain arbitrary signed couplings between their group factors. Only the proved identity multiplicities are discarded. They can be exponentially large, and physical Schatten traces must still include them. At $t = L$, the group list is empty and the product convention gives $D_{\text{eff}} = 2^L$, recovering the entire physical matrix.

11.3 The coefficient algebra is the full group-permutation centralizer

Proposition 11.2 (Exact algebra characterization). *For one group, polynomials in H, E, F are exactly all physical matrices commuting with every site permutation. In the decomposition above their algebra is*

$$\bigoplus_{\ell=M, M-2, \dots} M_{\ell+1}(\mathbb{C}) \otimes I_{\mu_\ell}. \quad (148)$$

For marked sites and several groups, tensor this algebra across the groups and with $M_{2^t}(\mathbb{C})$. Thus sector coefficients need not be diagonal or mutually commuting.

Proof. The Casimir expression $\mathcal{C} = H^2 + 2(EF + FE)$ acts on sector ℓ as $\lambda_\ell I$, where $\lambda_\ell = \ell(\ell + 2)$, by (144). The numbers λ_ℓ are distinct. Hence

$$\Pi_\ell = \prod_{\ell' \neq \ell} \frac{\mathcal{C} - \lambda_{\ell'} I}{\lambda_\ell - \lambda_{\ell'}}$$

selects the full sector, including its identity multiplicity. Inside it, interpolation in the distinct weights $\ell - 2k$ of H gives each coordinate projector. Left multiplication by suitable powers of E or F , divided by their nonzero ladder coefficients, gives every matrix unit $e_a e_b^T$ in rational sector coordinates. This proves that the generated algebra contains (148); simultaneous decomposition gives the reverse containment.

Its dimension is $\sum_\ell (\ell + 1)^2 = \binom{M+3}{3}$. For a direct verification, subtract the identity at $M - 2$: both sides change by $(M + 1)^2$, and $M = 0, 1$ give the bases. On the other hand, a physical matrix commutes with site permutations exactly when its entries are constant on simultaneous permutation orbits of pairs of binary strings. An orbit is specified by the four counts of pairs 00, 01, 10, 11, summing to M . There are $\binom{M+3}{3}$ orbits. The collective generators commute with

permutations, so inclusion and equal dimensions identify the entire centralizer. Separate group orbit counts multiply, and marked row and column indices are unrestricted, proving the tensor assertion. $\square$

This identifies an algebra, not a free input-compression algorithm. The calculus requires a supplied expression, circuit, or explicit sector coefficient data with its construction charged. For the latter, factored Casimir interpolation, weight interpolation and ladder operations represent each one-group matrix unit with $O(M)$ generator operations and rational constants of $O(M \log(M + 2))$ bits. Tensor these constructions for several groups and add the supplied coefficients. Every coefficient entry and sector size is still paid for; explicitly distributing all products can increase the expression size. Reading an arbitrary 2^L by 2^L array, testing its symmetry, or extracting a compact representation is not included in the evaluation theorem.

11.4 A same-input moment and strict-quantile theorem

For an integer readout order $h \geq 1$, use the physical metric adjoint in each retained block and define

$$\tau_h(s) = \sum_{\ell} \text{tr} \left((A_{\ell}(s)^{\sharp} A_{\ell}(s))^h \right), \quad G_h(s) = \tau_h(s)^{1/(2h)}. \quad (149)$$

These are traces with each sector counted once, not the physical trace weighted by multiplicities. Similarity by the square root of the metric identifies every summand with a positive Gram power, so $\tau_h \geq 0$. For the supplied rational data it is rational, computable without taking any square roots of metric entries.

Theorem 11.3 (Dense grouped-sector calculus). *Retain the marked set, partition, full signed ordered coefficient expression, score maps, original primitive names and schedule. For every matrix of this class, every integer $h \geq 1$, and every original assignment,*

$$D_{\text{eff}}^{-1/(2h)} G_h(S(z)) \leq \|Z(z)\| \leq G_h(S(z)). \quad (150)$$

If $\pi(s) = \mathbb{P}(S = s)$ and $p > 0$ is real, the functional

$$\mathcal{F}_{p,h}(\mathcal{D}(Z)) = \left(\sum_s \pi(s) \tau_h(s)^{p/(2h)} \right)^{1/p} \quad (151)$$

satisfies the same-input comparison

$$D_{\text{eff}}^{-1/(2h)} \mathcal{F}_{p,h} \leq (\mathbb{E} \|Z\|^p)^{1/p} \leq \mathcal{F}_{p,h}. \quad (152)$$

For $q_Y(\eta) = \inf\{u \geq 0 : \mathbb{P}(Y > u) \leq \eta\}$ and every $0 < \eta < 1$, one separately has

$$D_{\text{eff}}^{-1/(2h)} q_{G_h(S)}(\eta) \leq q_{\|Z\|}(\eta) \leq q_{G_h(S)}(\eta). \quad (153)$$

The exact joint score law and these readouts are computable at the costs below. All permitted updates preserve the theorem with their final structural parameters.

Proof. By (147), the norm of the original Z is the largest norm of an orthonormalized retained sector. Its largest singular value contributes one term to (149). Each of the D_{eff} retained singular values is at most that same maximum, so

$$\|Z\|^{2h} \leq \tau_h(S) \leq D_{\text{eff}} \|Z\|^{2h}.$$

This proves (150). Raising it to p and integrating gives (152). Pointwise order implies order of strict quantiles, and multiplication by a positive constant commutes with their definition. Apply these facts twice to obtain (153). No moment-to-tail conversion is used for that conclusion; zero atoms and confidence equal to an atom's upper-tail mass are included. $\square$

The lower witness can be placed in the same physical matrix. In one group, an explicit copy of sector $\ell = M - 2u$ is the tensor product of u singlets $(|01\rangle - |10\rangle)/\sqrt{2}$ and the symmetric subspace on the other ℓ sites. Collective generators annihilate each singlet, and their action on the symmetric part is (144). Tensor these embeddings across groups and leave the marked factor intact. Inserting a maximizing vector of a retained full block, including entanglement between marked and group coordinates, realizes its norm in the original matrix. It may be returned in this compact representation; writing all 2^L coordinates has its actual output cost.

For any $\epsilon > 0$, choose

$$h \geq \max\left(1, \left\lceil \frac{\log D_{\text{eff}}}{2 \log(1 + \epsilon)} \right\rceil\right). \quad (154)$$

The ratio in both the moment and strict-quantile comparisons is then at most $1 + \epsilon$. The precision cost is paid through h ; it is not a hidden universal constant. If an upper-tail estimate is desired solely from a moment, Markov gives $\mathbb{P}\{\|Z\| > \eta^{-1/p} \mathcal{F}_{p,h}\} \leq \eta$, but (153) uses the stronger direct finite-law calculation. At $p = 2h$, the moment functional has the rational radicand $\sum_s \pi(s) \tau_h(s)$. General real p gives the finite expression (151), not unit-cost rational evaluation of arbitrary real powers.

11.5 Exact transfer of the overlapping score law

Let $E_v = B_{v-1} \cup V_v$. An unnormalized count message stores a live assignment b and exact partial score s . Initialize $N_0(\emptyset, s) = \mathbf{1}_{s=s_0}$ and update

$$N_v(b, s) = \sum_{u \in \{\pm 1\}^{E_v} : u|_{B_v} = b} N_{v-1}(u|_{B_{v-1}}, s - \phi_v(u|_{V_v})). \quad (155)$$

The final distribution is $\pi(s) = 2^{-D} N_m(\emptyset, s)$. To prove this, interpret N_v as the number of assignments to already forgotten primitive names producing its partial score at the given live assignment. A newly occurring sign is introduced with both of its actual values. A sign is summed out only after its last occurrence. The recurrence counts every compatible original assignment once, so induction gives the claim. A reused equality or overlapping triangle is processed through its shared original names, with no independent replacement or guessed penalty.

At stage v the score lies in the translate of $[-vK, vK]^c$; hence at most U score keys and $2^w U$ count fields are needed. The direct transfer uses

$$O(mc 2^{w+a} U) \quad (156)$$

integer count/score operations, apart from forming local tables. This enumerates neither the full original sign law nor the ambient matrix. The exponential dependence on score count c and boundary w remains explicit.

Evaluate $\tau_h(s)$ for every supported score, sort the rational values, and combine ties by adding their exact probabilities. The first atom with strict upper-tail mass at most η gives $q_{G_h(s)}(\eta)$ as the positive $2h$ -th root of a rational number. A zero atom is treated in the same scan. Counts have at most $D + 1$ bits, so strict mass endpoints need no floating-point decision. A positive predecessor path in (155) recovers an original sign assignment for a specified score. Together with the sector embedding, it provides the promised same-input witness; marginal score distributions alone would not suffice.

11.6 Arithmetic, bit costs, and useful confidence orders

Write

$$N = 2^t \prod_j (M_j + 1), \quad C_{\text{sec}} = 2^{3t} \prod_j (M_j + 1)^4. \quad (157)$$

The largest retained matrix has at most N rows, and the sum of its dense cubic multiplication costs obeys

$$\sum_{\ell} n_{\ell}^3 = 2^{3t} \prod_j \sum_{\ell_j=M_j, M_j-2, \dots} (\ell_j + 1)^3 \leq C_{\text{sec}}.$$

This charges sector dimensions as well as their number. For the explicit term list (143), evaluate the actual polynomial and its physical adjoint at each score; adjoint reverses words, swaps E_j, F_j , keeps H_j , and takes marked coefficient adjoints. It has the same word length and avoids irrational metric changes. Repeated squaring computes the h -th Gram power. The total arithmetic bound is

$$O\left(mc 2^{w+a}U + U[(T(R+1) + \log(h+1))C_{\text{sec}} + Tc \log(q+2)]\right), \quad (158)$$

plus local-table and input-leaf construction and $O(U \log(U+1))$ comparisons for sorting. At most $T2^{2t}$ marked coefficient entries already occur in a dense input list. Initial sector generator matrices and multiplicities are elementary explicit data from (144)–(145); constructing them and any supplied coefficient expressions is included in leaf formation, not a free symmetry-recognition oracle. One sector at a time uses $O(2^wU + N^2 + U)$ working count/coefficient fields, besides input and score tables. For a circuit with G charged evaluation gates, replace the term-evaluation cost by $O(UGC_{\text{sec}})$, with corresponding physical adjoint gates and actual leaf costs.

For an exact term-list bit bound, place all coefficient parts over a common positive denominator Q , with Q and numerator magnitudes at most 2^{B_0} . If fractions are supplied separately, first charge the actual length of that common denominator. Set

$$\begin{aligned} B_{\text{sc}} &= \lceil \log_2(2 + \|s_0\|_{\infty} + mK) \rceil, \\ B_A &= O(B_0 + qB_{\text{sc}} + R \log_2(N+2) + \log_2(T+2)), \\ B_{\tau} &= O(h[B_A + \log_2(N+2)] + \log_2(D_{\text{eff}}+2)). \end{aligned} \quad (159)$$

Sector generator entries are integers bounded by N , and an entry of a word of length R has at most N^R summands before cancellation. The score monomial has magnitude at most $2^{qB_{\text{sc}}}$. These bounds prove the formula for both A and the independently evaluated physical adjoint, with common denominator Q . A Gram power uses $2h$ coefficient factors, its matrix sums add $O(h \log(N+2))$ bits, and the sector traces add $O(\log(D_{\text{eff}}+2))$ bits. Thus all readouts can be placed over Q^{2h} with lengths $O(B_{\tau})$. For arbitrary sector leaves, their metric adjoints must also be constructed and their actual resulting lengths charged; they are not automatically Euclidean adjoints of equal-cost physical data.

Multiply the matrix-arithmetic part of (158) by $O(B_{\tau}^2)$ for a schoolbook bit bound. Count-message work has the additional factor $O(D + cB_{\text{sc}} + 1)$. Common denominators permit sorting in $O(U \log(U+1)B_{\tau})$ bit operations. Probability and moment aggregation require polynomial work in U, D, B_{τ} ; rational confidence comparisons also charge the written bit length of η . Roots may remain symbolic. Numerical root precision requires additional polynomial bit work in the radicand length, h and requested precision. Naive primitive-label masks and schedule construction add polynomial bookkeeping in their written sizes.

For arbitrary circuits, the actual current intermediate coefficient lengths replace the term-list estimate. Polynomially many gates and bounded reduced random degree alone do not give polynomial bit growth: repeated squaring of two has degree zero and doubles its bit length at each step. A polynomial unreduced input-leaf count with polynomial initial lengths is a sufficient alternative; otherwise current lengths are required explicitly.

At fixed g, c, a , with logarithmic t, w , polynomial score range K , and polynomial actual expression and coefficient lengths, the algorithm is polynomial in the compact input parameters and the chosen order. In particular $h = O(\log(d/\eta)) = O(L + \log(1/\eta))$ is affordable. One may combine that choice with (154); the additional accuracy cost is polynomial in $1/\epsilon$ through h . The bounds are polynomial in the numerical site count L and range K , not merely their binary

encoding lengths. Their explicit exponents in group count, marked sites, score count and primitive width do not support uncontrolled growth of those parameters. Exact rational computation has zero accumulated approximation error; no unspecified numerical sector truncation is used.

11.7 Extension, symmetry breaking, and common refinement

The state consists of the full expressions and score maps, not their norm readouts. On fixed marked sites and groups, sums, differences, ordered products, adjoints and commutators act on those expressions. Sector evaluation is an algebra homomorphism, so each operation remains valid under every later operation. Term counts add under sums and multiply under distributed products; word lengths and score degrees add under products. Circuits may preserve factored expressions, with their gate and bit costs charged.

For different score collections, concatenate score coordinates and maps with zero padding while retaining every original shared name. The resulting c , ranges, and joint boundary determine the new cost. Independently multiplying marginal score laws is generally wrong. Additional score interactions append actual local maps and require a new schedule. Named primitive identification substitutes the same independent quotient name throughout all maps; it defines a new model rather than conditioning the old one. Local arity cannot increase merely by aliases, but boundary can. Affine conditioning instead requires a supplied independent parametrization of the true conditional sign law, retaining its affine phases and charging the induced local maps, arity and boundary. Reusing a forgotten primitive requires revisiting the retained description.

To append a matrix interaction on formerly unmarked sites, first mark its sites. If J is the newly marked subset of one old group, the exact identities on the same physical matrix are

$$H_{\text{old}} = H_{\text{remaining}} + \sum_{i \in J} H_i, \quad E_{\text{old}} = E_{\text{remaining}} + \sum_{i \in J} E_i, \quad F_{\text{old}} = F_{\text{remaining}} + \sum_{i \in J} F_i. \quad (160)$$

Lift old marked matrices by identities and use the required site permutation. The new single-site matrices are marked coefficients. Perform these substitutions before the new addition or multiplication, so every earlier cancellation and matrix order survives. Rebuild all sectors on the remaining groups. This is an exact continuation of the same old operator, followed by the new interaction, not a replacement model having only the same norm.

For two different marked sets or group partitions, take the union of marked sites. On the remaining sites take every nonempty intersection of an old group from each partition; there are at most $g_A g_B$ such groups. Rewrite each old collective generator as the sum of its generators on these intersection cells, together with its newly marked contributions. Then both expressions use a common representation and can be added or multiplied in their true order. Lift marked coefficients by identities on new marked coordinates. Empty unmarked groups are removed. Substitution can be kept as a circuit; an explicit distributive expansion is charged.

All these operations have zero algebraic error. The comparison factor is computed from the *final* D_{eff} once, rather than multiplied over earlier readouts. The same is true of the final score law and bit costs. Arbitrary symmetry breaking can eventually mark every site and require 2^{3L} dense arithmetic per multiplication. Arbitrary partition refinement can increase g . Exact closure therefore comes with a precise boundary on useful computation.

11.8 Dense full-rank geometry and exact verification examples

For one unmarked group put $X_c = E + F$, choose a bounded positive integer score s , and integers $A > sL$, $B > L$ for every possible score. Then

$$Z = (AI + sX_c)^L (BI + H) \quad (161)$$

is dense and invertible in the product basis. For any two binary strings, their Hamming distance is at most L ; the corresponding power of the hypercube adjacency X_c supplies a positive walk in the binomial expansion. All summands are nonnegative, so every entry of the first factor is positive. Its eigenvalues are positive under $A > sL$, while $BI + H$ is positive diagonal, proving density and invertibility. These factors do not commute: $(A + sx)^L$ is injective on the spectrum of X_c , so commuting with its power would imply commuting with X_c by interpolation. But $[H, X_c] = 2(E - F) \neq 0$. Every row has 2^L nonzero entries. Thus bounded coordinate-block sparsity is genuinely absent, while the sector description remains polynomial-sized.

The same theorem covers coupled groups, arbitrary marked matrices, and noncommuting local extensions after marking. Exact checks in the companion source compare the original physical operators before and after marking a new site in a five-site example and then append a noncommuting gate. A six-site, two-group calculation checks coupled sector embeddings and physical trace multiplicities; twelve named refinement identities are verified. Full physical trace checks include the multiplicities in (145), even though the useful norm readout omits them.

The earlier Bell continuation from Proposition 9.5 is also retained. At $L = 2$,

$$Z = E^2 + F^2 + \frac{1}{2}H^2 = I \otimes I + X \otimes X + Z_0 \otimes Z_0 + (XZ_0) \otimes (XZ_0),$$

where $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. The $\ell = 0$ sector vanishes; the $\ell = 2$ sector has rank one and norm four. Thus $\tau_3 = 4096$, equal to the true sixth physical trace moment in this example. The invalid symmetric replica projection gave 2560. Complete spin sectors discard only proved identity multiplicities and do not make that invalid reduction.

A larger exact benchmark uses 65 original signs, 63 overlapping binary-collision triangles and 64 score atoms. At $L = 24$, $h = 34$, the entire readout law gives an order-68 moment certificate and a strict quantile certificate at $\eta = 2^{-24}$ on dimension $2^{24} = 16,777,216$, with

$$D_{\text{eff}}^{1/(2h)} = 169^{1/68} < 1.079.$$

Neither the 2^{65} original law nor the physical matrix is enumerated. This large collective example is distinct from the dense full-rank family (161), which is checked separately at smaller sizes. The saved suite reports PASS for exact embeddings, rational metrics, physical adjoints, all multiplicities, centralizer projectors and matrix units, original joint score laws, strict endpoints, and marking/refinement updates. Those finite checks support the implementation; the all-size claims follow from the proofs above.

This is a dense norm-and-quantile calculus for a specified full coefficient algebra and score representation. General matrix-product operators, arbitrary Parseval-compressed SparseStack, and the full Walsh support family are not inferred to possess that symmetry or score structure. Their useful representation remains an additional mathematical problem.

12 Two complementary dense calculi: rank and Clifford geometry

Dense coefficient geometry can remain tractable for reasons other than a small fixed block decomposition. This section gives two such mechanisms. A matrix-product operator with a certified rank bound admits an exact scalar Frobenius transfer and matching norm moments. A homogeneous Clifford expression admits a small signed normal form after evaluation, even when its original coefficient algebra is irreducible in exponentially growing dimension. Both retain the original random names and full ordered coefficient information; their closure conditions and confidence conclusions differ.

12.1 Dense low-rank matrix-product programs

Use $L \geq 1$ tensor sites of physical sizes $s_j \leq s$, with $d = \prod_j s_j$. Original independent signs are $z_1, \dots, z_D$. An open-boundary matrix-product operator is supplied as

$$A(z) = \sum_{\alpha_1, \dots, \alpha_{L-1}} A_1^{1, \alpha_1}(z) \otimes A_2^{\alpha_1, \alpha_2}(z) \otimes \dots \otimes A_L^{\alpha_{L-1}, 1}(z). \quad (162)$$

The bonds satisfy $\beta_0 = \beta_L = 1$, $\beta_j \leq \beta$. Every local $s_j \times s_j$ array is a signed sum of at most t characters in a declared set V_j of at most a original signs. One set contains all primitives used at that site. Repeated names are the same variable. The supplied site order has live random boundary at most w ; neither order nor bond optimization is free. Physical adjoints and traces in this subsection are those of the supplied orthonormal tensor bases.

In addition supply a pointwise rank certificate

$$\text{rank } A(z) \leq R \quad \text{for every original assignment.} \quad (163)$$

This is a verifiable annotation, not an assumption about typical numerical rank. One certificate grammar starts with vector outer products, adds caps under sums, takes their minimum under products, and preserves them under adjoint and scalar multiplication. Outer products of matrix-product vectors of bonds χ_v, χ_u have operator bond at most $\chi_v \chi_u$. Alternatively an explicit factorization $A = VU^*$ with R columns certifies the cap, with the representation and verification of that factorization charged. We consider both the pure class $Z = A$, $R \geq 1$, and the class

$$Z(z) = c(z)I_d + A(z), \quad 0 \leq R < d, \quad (164)$$

where c has a scalar matrix-product program of bond at most γ in the same or a padded common schedule. The shifted output can have full rank, be dense and nonnormal, and have noncommuting coefficients.

Define the paired virtual transfer

$$D_j[(u, v), (u', v')] = \text{tr}((A_j^{v, v'})^* A_j^{u, u'}). \quad (165)$$

Its dimensions are $\beta_{j-1}^2 \times \beta_j^2$. Expanding the original operator and using multiplicativity of tensor traces gives

$$\|A(z)\|_F^2 = D_1(z) \cdots D_L(z). \quad (166)$$

Every signed matrix sum is evaluated before (165); the local transfers may be complex or signed. Only the terminal scalar is necessarily nonnegative. In the shifted case use instead $T = |c|^2 + \|A\|_F^2$. Its scalar program is the direct sum of the paired programs for $|c|^2$ and $\|A\|_F^2$: concatenate first-site arrays, block-diagonalize internal arrays, and stack final arrays. For $L = 1$, add the two scalar outputs directly. The maximal scalar bond is therefore

$$q_F = \beta^2 \quad \text{in the pure case,} \quad q_F = \beta^2 + \gamma^2 \quad \text{in the shifted case.} \quad (167)$$

Schedule unions and the actual local evaluation work are charged after combining the programs.

Theorem 12.1 (Symmetric Frobenius transfer and norm comparison). *For integer $p \geq 1$, the preceding descriptor computes exactly $M_p = \mathbb{E}T^p$ by messages of lifted bond at most $h = \binom{q_F + p - 1}{p}$. Put $F_{2p} = M_p^{1/(2p)}$. Then*

$$R^{-1/2} F_{2p} \leq (\mathbb{E}\|A\|^{2p})^{1/(2p)} \leq F_{2p}, \quad Z = A, \quad R \geq 1, \quad (168)$$

$$(1 + 4R)^{-1/2} F_{2p} \leq (\mathbb{E}\|cI + A\|^{2p})^{1/(2p)} \leq \sqrt{2} F_{2p}, \quad Z = cI + A, \quad R < d. \quad (169)$$

All comparisons hold pointwise before integration and concern the same original input matrix.

Proof. For each scalar-program bond q_j , take the ordinary degree- p monomial basis, of size $h_j = \binom{q_j+p-1}{p}$. Substituting linear forms gives $\text{Sym}^p(DE) = \text{Sym}^p(D) \text{Sym}^p(E)$. The endpoint bonds are one, so the scalar endpoint is raised to power p :

$$T(z)^p = (\text{Sym}^p D_1(z)) \cdots (\text{Sym}^p D_L(z)).$$

This monomial basis need not be orthonormal, because no intermediate norm or Euclidean adjoint is used. It is an exact scalar polynomial identity. If B_j is the original live boundary after site j and $E_j = B_{j-1} \cup V_j$, initialize the row message $m_0 = 1$ and set

$$m_j(b) = 2^{-|E_j \setminus B_j|} \sum_{u: u|_{B_j} = b} m_{j-1}(u|_{B_{j-1}}) \text{Sym}^p D_j(u). \quad (170)$$

Induction integrates exactly the independent original variables with no future occurrence and retains every reused one. The terminal scalar is $M_p = \mathbb{E}T^p$.

For a matrix of rank at most R , its actual singular values give $\|A\|^2 \leq \|A\|_F^2 \leq R\|A\|^2$. This proves (168). Under $R < d$, a unit vector in $\ker A$ gives $\|cI + A\| \geq |c|$. Thus $\|A\| \leq 2\|cI + A\|$, and

$$T \leq (1 + 4R)\|cI + A\|^2, \quad \|cI + A\| \leq |c| + \|A\|_F \leq \sqrt{2T}.$$

Integrating these inequalities proves (169). $\square$

The low-rank lower witness is a maximizing singular vector of the same A . For a shift, a kernel vector witnesses $|c|$; when $\|A\| > 2|c|$, a top singular vector also gives $\|(cI + A)v\| > \|A\|/2$. Constructing an expanded d -dimensional witness is not included in the compact transfer cost. The condition $R < d$ cannot be dropped: $A = -cI_d$ gives zero output but positive T . Likewise small operator bond alone is not a rank certificate: the identity has bond one and loses $\sqrt{d}$ in the Frobenius-to-operator comparison.

The legitimate symmetry here applies to $\mathbb{E}(\text{tr } A^* A)^p$, which has separate physical trace cycles. It does not project the cyclic transfer for $\mathbb{E} \text{tr}(A^* A)^p$ from Section 9. For the two-site Bell continuation of rank one and norm four, both sixth-order traces equal 4096, and (170) returns 4096. The rejected cyclic-message symmetrization returned 2560. A scalar functor and a projection of a cyclic replica state are different operations.

12.2 Rank updates, computational cost, and confidence scope

MPO sums have bond at most $\beta_A + \beta_B$; ordered products pair virtual indices and multiply physical matrices in their actual order, with bond at most $\beta_A \beta_B$. Adjoint adjoints each physical matrix without reversing tensor sites. Scalar multiplication pairs its scalar and operator bonds. Character sets are unioned and equal characters aggregated before the next contraction. Ranks update by

$$R_{A+B} \leq R_A + R_B, \quad R_{AB} \leq \min(R_A, R_B), \quad R_{A^*} = R_A.$$

For two shifted operands the ordered update is

$$(cI + A)(eI + B) = ceI + [A(eI + B) + cB]. \quad (171)$$

The residual range lies in $\text{ran } A + \text{ran } B$, so its cap is at most $R_A + R_B$, rather than a separate charge for each expanded term. Its MPO bond still follows the actual sum/product construction. If the final cap reaches d , the shifted theorem needs a smaller valid certificate. Comparison factors use the final rank once, not a product of the factors assigned to earlier updates.

Primitive identification substitutes the same new independent sign at every occurrence and reduces characters by XOR. Rank identities survive pointwise, but the live boundary is

recomputed. If c_{alias} old names belong to nonsingleton alias classes, the same schedule has width at most $w + c_{\text{alias}}$. Actual affine conditioning $Ay = b$ is handled by binary elimination, an origin y_0 , and an injective parametrization $y = y_0 + Nx$ with independent uniform new bits. Every substituted character retains its affine sign. This is the normalized original conditional law; it can have larger local arity and width. Inconsistency gives a zero event, and an unnormalized expectation pays the exact original mass $2^{-\text{rank } A}$.

For a local assignment, forming the pure paired transfer costs at most $O(t\beta^2s^2 + \beta^4s^2)$ arithmetic. Add $O(t\gamma^2 + \gamma^4)$ for the scalar program when present, or charge its actual evaluation cost. Denote the resulting bound by E_{loc} . Multiplying the p linear forms of an occupation monomial by coefficient convolution forms the full lift in $O(q_Fph^2)$ arithmetic. Hence

$$\text{work} = O(L2^{w+a}[E_{\text{loc}} + q_Fph^2]), \quad \text{storage} = O(2^wh + h^2 + \beta^2s^2 + \gamma^2) \quad (172)$$

working entries, apart from inputs, named schedule data and optional caches. Omit γ in the pure case. With a supplied common coefficient denominator Q , numerator and denominator lengths at most B , a conservative intermediate bit bound is

$$B_* = O(pL[B + \log(t + s + q_F + 2)] + D + p\log(p + q_F + 2) + 1). \quad (173)$$

Indeed after j sites the denominator divides Q^{2pj} times the powers of two already integrated. Numerators sum products of at most $2pj$ input coefficients, with physical, virtual, character and occupation choices bounded by the displayed logarithmic terms. This also bounds partial lift construction. Schoolbook bit work is (172) times $O(B_*^2)$, plus polynomial work in the written primitive names. Converting separately encoded rationals to the common denominator is charged in B . The exact output is a rational radicand and a positive-root instruction; numerical root precision is separate.

At fixed $\beta, \gamma, a, s, t, R$, the bound $h \leq (p + q_F)^{q_F-1}$ is polynomial in p . Thus logarithmic w and polynomial actual coefficient lengths permit polynomial work in the compact input and in p , including $p \asymp \log(d/\eta)$ when $d = s^L$. The exponent depends on the bond square q_F and is not suppressed. Signed intermediate messages have no automatic relative-error guarantee through cancellation. Exact arithmetic has zero algebraic error; repeated circuit updates charge their actual coefficient lengths. Degree-zero repeated squaring can still produce exponentially long constants.

Write the structural constants in Theorem 12.1 as $c_0F_{2p} \leq \|Z\|_{L^{2p}} \leq C_0F_{2p}$. Markov gives an upper threshold $C_0\eta^{-1/(2p)}F_{2p}$. For $p \geq \max(1, \lceil \log(1/\eta) \rceil)$, its ratio to the same norm moment is at most $\sqrt{e}C_0/c_0$. A separate strict lower event uses the same transfer at order $2p$: if $M_p > 0$ and $0 < \theta < 1$, then

$$\mathbb{P}\{\|Z\| > c_0(\theta M_p)^{1/(2p)}\} \geq (1 - \theta)^2 M_p^2 / M_{2p}. \quad (174)$$

This follows from Paley–Zygmund on T^p and the pointwise lower bound. It is not a matching extreme-quantile theorem for an arbitrary program.

The class includes dense entangled outer products, their noncommuting sums and products, and full-rank scalar identity shifts. It need not have a small common coefficient algebra. At a three-dimensional site, the character coefficients of $(1, s, t)(1, s, t)^T$ are I_3 and the three symmetric off-diagonal units. Squaring those units and taking linear combinations produces every diagonal unit, and subsequent multiplication produces all oriented matrix units. Tensoring yields the full algebra M_{3L} , although every realization has rank one. Bounded-bond entangling unitary conjugation preserves rank and irreducibility with its product-bond cost charged. The saved suite also evaluates a 32-site, 33-sign entangled rank-one input of dimension 2^{32} and bond two without constructing the ambient matrix or enumerating the full law. These are diagnostics of the proved mechanism; they do not imply that a centered sum of rank-one Walsh samples has low rank.

12.3 An even-copy homogeneous Clifford class

Let $\xi = U^T \varepsilon \in \mathbb{R}^n$, $U \in \mathbb{R}^{k \times n}$, $U^T U = I_n$, $n \geq 2$, under the original affine law and even-innovation hypotheses of Theorem 17.3. Retain the full signed frame, original names, affine phases and junction state. Put $\rho = \|\xi\|_2$. The needed analytic inputs are precisely isotropy and

$$\mathbb{E} \xi \xi^T = I_n, \quad \mathbb{E} |v^T \xi|^{2h} \leq (2h-1)!! (v^T K v)^h, \quad I_n \preceq K \preceq \kappa I_n, \quad (175)$$

at every required integer order. Any original vector law with these proved inputs is allowed. The affine construction obtains isotropy from the absence of weight-one and weight-two codewords and obtains K from its eliminated even private kernels.

Fix an *even* integer $q \geq 2$, put $N = qn$ and $D_0 = 2^{N/2}$. The Pauli matrices X, Y, Z_0 give Hermitian generators

$$\begin{aligned} \Gamma_{2j-1} &= Z_0^{\otimes(j-1)} \otimes X \otimes I^{\otimes(N/2-j)}, \\ \Gamma_{2j} &= Z_0^{\otimes(j-1)} \otimes Y \otimes I^{\otimes(N/2-j)}. \end{aligned} \quad (176)$$

Each squares to identity; the first differing nonidentity site supplies the unique sign change for distinct generators. Thus $\Gamma_a \Gamma_b + \Gamma_b \Gamma_a = 2\delta_{ab} I$ in the physical Euclidean metric. The symbolic tensor representation avoids a dense array. Use the same original vector in the consistently named pencils

$$A_j = \sum_{i=1}^n \xi_i \Gamma_{(j-1)n+i}, \quad A_j A_l + A_l A_j = 2\rho^2 \delta_{jl} I, \quad A_j^* = A_j. \quad (177)$$

The last identities follow by expanding the anticommutators. A fixed common unitary conjugation preserves them and may be retained symbolically.

On an external coefficient space of dimension b , allow any homogeneous degree- R ordered expression, $R \geq 1$,

$$Z = \sum_{\nu=1}^T C_\nu \otimes A_{j_{\nu,1}} \cdots A_{j_{\nu,R}}, \quad C_\nu \in \mathbb{C}^{b \times b}. \quad (178)$$

The C_ν are unrestricted and may not commute. Inserted external coefficient matrices are multiplied in their actual order; they commute with the Clifford letters only because they act on a different physical factor. Arbitrary matrices on the ambient Clifford factor are not thereby allowed. A homogeneous arithmetic circuit may replace the list, with its written size and actual intermediate coefficient lengths charged. The full dimension is $d = b2^{qn/2}$.

Let $\gamma_1, \dots, \gamma_q$ be (176) with $N = q$, acting on $s_C = 2^{q/2}$ dimensions. Keep the full deterministic signed normal form

$$C_P = \sum_{\nu} C_\nu \otimes \gamma_{j_{\nu,1}} \cdots \gamma_{j_{\nu,R}}, \quad c_P = \|C_P\|. \quad (179)$$

Its physical matrix dimension is bs_C , independently of n .

Theorem 12.2 (Same-input Clifford factorization). *For every original assignment,*

$$\|Z\| = c_P \rho^R. \quad (180)$$

When $\rho > 0$, the full matrix is unitarily equivalent to $\rho^R (C_P \otimes I_{D_0/s_C})$, up to a tensor-factor permutation. The unitary may depend on the actual ξ .

Proof. For $\rho > 0$, the matrices $B_j = A_j/\rho$ are Hermitian anticommuting involutions. If $S \subseteq [q]$ is increasing, write $B_S = \prod_{j \in S} B_j$. Every nonempty B_S has trace zero: for even $|S|$, conjugate

by an element of S ; for odd $|S|$, conjugate by an element outside S . The latter exists because q is even. Thus the 2^q monomials are orthogonal under normalized trace, since $B_S^* B_T$ is a signed nonempty monomial for $S \neq T$. The same argument applies to γ_S , and $s_C^2 = 2^q$, so those monomials form a basis of $M_{s_C}(\mathbb{C})$.

The map $\gamma_S \mapsto B_S$ preserves multiplication and adjoint by the same anticommutation rules, and is an injective unital star-homomorphism. Images of diagonal matrix units are orthogonal projections summing to identity; images of off-diagonal units identify their ranges isometrically. Those ranges consequently all have dimension D_0/s_C . Choosing a basis in one range and transporting it with the matrix units constructs a unitary realizing the map as $C \mapsto C \otimes I_{D_0/s_C}$. Matrix amplification preserves this description. Homogeneity factors out ρ^R , proving the theorem. At $\rho = 0$ all positive-degree letters and Z vanish. $\square$

A maximizing right singular vector of the small C_P , transported by the assignment's representation unitary, witnesses the norm in this same Z . Writing that dense ambient vector is not included in compact computation. This reduction is not a fixed common small-block decomposition of the original coefficients. The coefficient matrices $\sum_i U_{a,i} \Gamma_{(j-1)n+i}$ span every original Γ because U has rank n . Their increasing monomials form a basis of $M_{D_0}(\mathbb{C})$ by the same trace argument. Hence the original coefficient algebra is irreducible in exponentially growing dimension.

Theorem 12.3 (Homogeneous norm moments). *For integer $p \geq 1$, put $\mathcal{F}_{2p} = c_P n^{R/2}$. Under (175),*

$$\mathcal{F}_{2p} \leq (\mathbb{E}\|Z\|^{2p})^{1/(2p)} \leq \mathcal{F}_{2p} \kappa^{R/2} \left[\prod_{j=0}^{pR-1} \left(1 + \frac{2j}{n} \right) \right]^{1/(2p)}. \quad (181)$$

The upper comparison factor is at most $[\kappa(1 + 2(pR - 1)/n)]^{R/2}$. It includes all degree and moment-order dependence. Both norm bounds vanish with the exact cancellation $C_P = 0$.

Proof. For an independent standard Gaussian vector g , exchange expectations in (175), using $\mathbb{E}_g |g^T \xi|^{2h} = (2h - 1)!! \rho^{2h}$. This gives

$$n^h \leq \mathbb{E} \rho^{2h} \leq \mathbb{E} (g^T K g)^h \leq \kappa^h \prod_{j=0}^{h-1} (n + 2j).$$

The lower bound is Jensen and $\mathbb{E} \rho^2 = n$; the last product is the Gaussian radial moment, obtained from $(1 - 2t)^{-n/2}$ or integration by parts. Set $h = pR$ in (180) and take a $2p$ -th root. Bounding each factor by its largest one proves the stated simpler bound. $\square$

The retained anisotropic K also yields

$$\mathbb{E} (g^T K g)^h \leq \prod_{j=0}^{h-1} (\text{tr } K + 2j \|K\|). \quad (182)$$

Indeed, for $S = \sum_i \lambda_i g_i^2$, integration by parts gives $\mathbb{E} S^h = \text{tr } K \mathbb{E} S^{h-1} + 2(h-1) \mathbb{E}[(\sum_i \lambda_i^2 g_i^2) S^{h-2}]$. The latter quadratic form is at most $\|K\| S$, and induction proves (182).

For fixed q, R , take $p = \lceil \log(d/\eta) \rceil$ and assume $\log(b/\eta) \leq An$ for a fixed $A \geq 0$. Since $n \geq 2$, $p/n \leq q \log 2/2 + A + 1/2$, so the comparison is at most

$$[\kappa(1 + R(q \log 2 + 2A + 1))]^{R/2}. \quad (183)$$

It stays bounded as n, d grow when the stated parameters and κ are bounded. More extreme confidence retains the explicit p/n loss.

For $c_P > 0$, the same radial variable gives a separate lower event. Using $\mathbb{E}\rho^4 \leq \kappa^2 n(n+2) \leq 2\kappa^2 n^2$ and strict Paley–Zygmund,

$$\mathbb{P}\{\|Z\| > c_P(n/2)^{R/2}\} \geq \frac{1}{8\kappa^2}, \quad \mathbb{E}\|Z\| \geq \frac{2^{-R/2}}{8\kappa^2} \mathcal{F}_{2p}. \quad (184)$$

Markov gives failure at most η above $\eta^{-1/(2p)}$ times the upper moment bound; this adds at most $\sqrt{\epsilon}$ when $p \geq \log(1/\eta)$. Matching extreme quantiles require the radial law of the same vector, rather than following from these two estimates alone.

12.4 Graded closure and consistent pencil names

Store a $b \times b$ coefficient for each Clifford subset and its original homogeneous degree. With increasing words,

$$\gamma_S \gamma_T = (-1)^{\#\{(s,t) \in S \times T : s > t\}} \gamma_{S \Delta T}, \quad \gamma_S^* = (-1)^{|S|(|S|-1)/2} \gamma_S. \quad (185)$$

Moving letters past one another and cancelling adjacent squares proves these formulas. An ordered product channel is the full matrix sum

$$(CD)_V = \sum_{S \Delta T = V} (-1)^{\#\{(s,t) : s > t\}} C_S D_T.$$

Equal-degree sums and differences, homogeneous ordered products and adjoints therefore close exactly; products add degrees even when their Clifford words shorten. The final c_P detects arbitrary signed cancellation before any norm is estimated.

Mixed-degree sums retain the exact small matrix polynomial $C(\rho) = \sum_j \rho^j C_j$ and still have $\|Z\| = \|C(\rho)\|$ pointwise. This is algebraic closure, but its norm moments require the actual radial law or a new valid certificate. The homogeneous functional cannot be reused across different degrees.

Both operands must refer to the same consistently defined named pencils. A common new real Parseval vector or a common orthogonal transformation can be supplied and recertified. Separate orthogonal transformations in the different Clifford copies also preserve (177), since the coordinate blocks stay orthogonal and every radius remains ρ , provided each transformed pencil is replaced consistently throughout the expression. Equal radii do not identify two differently transformed definitions in the *same* copy when those operands are multiplied. Unequal radii, unrelated compressions, or arbitrary entangling matrices outside the specified external-factor/Clifford representation need a new state. Primitive extensions, aliases and affine restrictions retain phases and rebuild the original vector law. The algebraic identity survives pointwise; the moment theorem is invoked only after isotropy and its scalar proxy are recertified.

A circuit of written size L_{in} needs at most $O(L_{\text{in}} 4^q b^3)$ coefficient arithmetic; each product also costs $O(4^q q)$ binary sign work. Addition costs $O(2^q b^2)$. A word list with T degree- R terms can be sign-reduced in $O(TRq + Tb^2)$ work after the coefficients are supplied, with any inserted matrix products additionally charged. The small matrix has size $b2^{q/2}$. Its norm is represented by the square root of the largest eigenvalue of $C_P^* C_P$, or enclosed by verified rational PSD certificates at requested precision. This spectral work is polynomial in the small dimension, actual bit lengths and requested precision; it is not a unit-cost irrational operation. Test exact zero before asking for a relative enclosure.

Junction construction, evenness tests, local distance certificates and $O(kn^2)$ covariance assembly retain the costs of Theorem 17.3. If a retained-subcode comparator replaces the full original Walsh law, (181) requires agreement of scalar moments through degree $2pR$, supplied by the strict horizon $g(\mathcal{C}) > 2pR$ for these even powers. The separate lower event also requires

$g(\mathcal{C}) > 4$. Raw-feature circuits use their full relation code and affine phases. An arbitrary-support horizon search is not free.

For common-denominator coefficient lengths B_1, B_2 , a product has length $B_1 + B_2 + O(q + \log(b + 1))$; aggregation of at most 2^q equal channels adds $O(q)$ numerator bits. Word lists with polynomial unreduced leaf count have polynomial output lengths; general circuits charge their actual intermediate lengths. The scalar product in (181) takes $O(pR)$ integer products with bit lengths $O(pR[\log(n + pR + 1) + B_K])$, where B_K is the length of a rational cost certificate. Schoolbook arithmetic multiplies the operation bounds by the square of current lengths. Root precision is separate. At fixed q, R, w and polynomial coefficient lengths the work is polynomial in $L_{\text{in}}, k, n, b, \log(1/\eta)$ at logarithmic confidence orders, even though d is exponential in n .

If original features have primitive degree at most δ , the output degree is at most $R\delta$; the width-four chain has $\delta \leq 3$. An arbitrary affine parametrization may increase both primitive degree and width. Exact arithmetic incurs zero algebraic error. A final certified relative enclosure of c_P of ratio $1 + \epsilon$ gives that same ratio in every norm and moment scale, after complete signed aggregation. Relative errors assigned separately to cancelling summands have no such guarantee.

12.5 Examples and the mixed-degree obstruction

The commutator $A_1A_2 - A_2A_1 = 2A_1A_2$ has norm $2\rho^2$, whereas the anticommutator is identically zero. At degree three, the normal form of $C_1 \otimes A_1A_2A_1 + C_2 \otimes A_2A_1A_2$ is $-C_1 \otimes \gamma_2 - C_2 \otimes \gamma_1$, preserving noncommuting external coefficient orientation. For a large compact instance, the original 40-primitive-sign innovation chain has 78 named features. Compression to $n = 16$, two copies and $b = 2$ gives ambient dimension 131072, but a four-dimensional normal form. Its confidence certificate uses the proved junction proxy without enumerating 2^{40} rows or the ambient matrix; its norm law is bracketed analytically, not asserted exactly distributed.

The radial information needed for mixed degrees is indispensable. Let U be the first four columns of $I_8 - \mathbf{1}\mathbf{1}^T/4$, a fixed real Parseval frame. Its first two row outer products have commutator entry $-1/8$. For independent signs $a_1, \dots, a_4, \eta$, compare the two original affine laws

$$\begin{aligned} x^{(0)} &= (a_1, a_2, a_3, a_4, \eta a_1, \eta a_2, \eta a_3, \eta a_4), \\ x^{(1)} &= (a_1, a_2, a_3, a_4, \eta a_1, \eta a_2, -\eta a_3, -\eta a_4). \end{aligned} \tag{186}$$

Their check rowspace has rank three and is identical; only the affine phase changes. Both have identity covariance. Retaining the first pair of labels $\{1, 5\}$ as separator and the others as private pairs gives the same even-innovation certificate $K = 2I - U_{\{1,5\}}^T U_{\{1,5\}}$, with $\|K\| = 2$.

If $A = a_1 + \dots + a_4$ and B is the sum of the final four signs, direct expansion of this Householder compression gives $\rho^2 = 4 + (B^2 - A^2)/4$. Thus the first law has $\rho^2 = 4$ pointwise. For the second law, write $L = a_1 + a_2$, $R' = a_3 + a_4$. Then

$$\rho^2 = 4 - LR', \quad \mathbb{P}\{\rho^2 = 0, 4, 8\} = \{1/8, 3/4, 1/8\}. \tag{187}$$

For any permitted even number of copies, apply the same written mixed-degree continuation $Q = A_1^2 - 4I = (\rho^2 - 4)I$. It vanishes in the first law, while in the second its norm is four with probability $1/4$:

$$(\mathbb{E}\|Q^{(0)}\|^{2p})^{1/(2p)} = 0, \quad (\mathbb{E}\|Q^{(1)}\|^{2p})^{1/(2p)} = 4 \cdot 4^{-1/(2p)}.$$

Its second strict quantile is four for $0 < \eta_{\text{conf}} < 1/4$ and zero for $\eta_{\text{conf}} \geq 1/4$; the first is always zero. These endpoints come directly from the original radial atoms, not from norm moments.

Consequently the complete signed deterministic normal form, n , K , the full signed U , and the unphased check rowspace do not support any finite multiplicative mixed-degree norm comparison.

The missing radial law depends on the affine phase. The full descriptor retains that phase already; the counterexample prohibits treating its homogeneous scalar summary as a sufficient mixed-degree state. It does not invalidate either homogeneous theorem or an algorithm that computes the actual radial law.

The saved exact suites check signed/complex physical transfers, rank and identity-shift certificates, original alias and affine laws, the Bell continuation, Pauli anticommutators, signed homogeneous products and adjoints, complete finite radial moments and the preceding strict atoms. Their independent agent reviews are mathematical checks, not formal verification or external peer review. The remaining boundaries are dense full-rank residuals without useful rank certificates, growing bond or primitive width, mixed-degree radial cancellation without an affordable law, and interactions outside the named Clifford geometry. Rank-one individual samples and small coefficient programs alone do not establish any of these missing hypotheses for unrestricted SparseStack or FR-10.

13 Walsh sampling: a complete contraction chain and its uniformity boundary

The Walsh model separates three issues that are easily conflated: contracting the coefficients of one polynomial, controlling a supremum over many supports, and producing a lower witness for the same sampled matrix. The results below form one proof chain addressing all three. The chain gives a matching theorem on a geometric support class, and an explicit uniform upper bound on arbitrary supports. Its remaining loss can be located in one counting step. It does not solve unrestricted FR-10 at intermediate sparsity.

13.1 Model, target, and the retained state

Let $V = \mathbb{F}_2^d$, $N = 2^d$, and identify its dual V^* with the row indices. For $v \in V$ and $x \in V^*$ write $\chi_v(x) = (-1)^{x(v)}$. Throughout this section $x_1, \dots, x_m$ are independent uniform points of V^* , sampled *with replacement*, and $m \geq 1$. The sampled matrix is

$$(A_m)_{iv} = m^{-1/2} \chi_v(x_i).$$

For a frequency set S , let $G_S = A_{m,S}^* A_{m,S}$ and $H_S = G_S - I$. The squared-norm restricted isometry error is

$$\Delta_k = \max_{|S| \leq k} \|H_S\| = \sup_{\substack{\|c\|_2=1 \\ |\text{supp } c| \leq k}} \|\|A_m c\|_2^2 - 1\|. \quad (188)$$

Norms of matrices without a subscript are spectral norms. All logarithms are natural. Real coefficients suffice: every $G_S - I$ is real symmetric, so the same bounds hold for complex coefficients by separating real and imaginary parts.

Let $m_*(N, k, \delta; 1 - \eta)$ be the smallest positive integer m for which $\mathbb{P}\{\Delta_k \leq \delta\} \geq 1 - \eta$. It exists because the empirical frequencies of the finitely many rows converge to their uniform probabilities. No monotonicity of the success probability in m is assumed. The FR-10 target, at $\delta = 1/2$ and $\eta = 1/10$, is comparison with

$$\mathcal{D}(N, k) = k \log(2k) \log(2N/k). \quad (189)$$

For $k = 1$ one has $\Delta_1 = 0$ exactly and $m_* = 1$.

The actual Gram retains additive identities among named frequencies:

$$(G_S)_{uv} = \frac{1}{m} \sum_{i=1}^m \chi_{u+v}(x_i). \quad (190)$$

The same empirical character can appear in many entries. Translating S by one frequency multiplies each sampled row by a common sign, leaving its Gram unchanged. We may therefore anchor a support at zero and represent its remaining frequencies in their actual binary span. Quotienting that span can make different named frequencies parallel; those labels must remain separate coefficient coordinates.

13.2 An exact extension and a closed upper potential

First consider one support, before maximizing over supports. A fresh binary coordinate has independent signs on the original sample indices conditional on every old row coordinate. This produces an exact extension rule.

Proposition 13.1 (Mixed-support extension). *Fix old rows $z_1, \dots, z_m$ and two nonempty old frequency sets S_0, S_1 of sizes a, b . They may overlap. Let A, B be their sampled matrices normalized by $m^{-1/2}$, and let $G_j = I + H_j$ be their Grams. With independent signs ε_i , the new sampled matrix is $[A, D_\varepsilon B]$. For every real θ , set*

$$L = \log \cosh(\theta \sqrt{ab}/m), \quad \gamma_0 = \frac{m}{a}L, \quad \gamma_1 = \frac{m}{b}L.$$

Then, conditionally on the complete old sample,

$$\mathbb{E}_\varepsilon \operatorname{tr} e^{\theta H_S} \leq e^{\gamma_0} \operatorname{tr} e^{(\theta+\gamma_0)H_0} + e^{\gamma_1} \operatorname{tr} e^{(\theta+\gamma_1)H_1}, \quad \gamma_0 \leq \frac{b\theta^2}{2m}, \quad \gamma_1 \leq \frac{a\theta^2}{2m}. \quad (191)$$

Moreover,

$$\mathbb{E}_\varepsilon \operatorname{tr} H_S^2 = \operatorname{tr} H_0^2 + \operatorname{tr} H_1^2 + \frac{2ab}{m}. \quad (192)$$

Proof. Write $u_i = A_{i,:}^*$ and $v_i = B_{i,:}^*$. Then

$$H_S = D + \sum_i \varepsilon_i Y_i, \quad D = \begin{pmatrix} H_0 & 0 \\ 0 & H_1 \end{pmatrix}, \quad Y_i = \begin{pmatrix} 0 & u_i v_i^* \\ v_i u_i^* & 0 \end{pmatrix},$$

and

$$\|u_i\|^2 = a/m, \quad \|v_i\|^2 = b/m, \quad Y_i^2 = \operatorname{diag}((b/m)u_i u_i^*, (a/m)v_i v_i^*), \quad Y_i^3 = (ab/m^2)Y_i.$$

Thus $Y_i^2/(ab/m^2)$ is an orthogonal projection, and functional calculus gives

$$\log \mathbb{E}_\varepsilon e^{\theta \varepsilon_i Y_i} = \frac{m^2 L}{ab} Y_i^2, \quad \sum_i \log \mathbb{E}_\varepsilon e^{\theta \varepsilon_i Y_i} = \operatorname{diag}(\gamma_0 G_0, \gamma_1 G_1).$$

We import the matrix cumulant-generating-function inequality [1, Lemma 3.4]: for independent Hermitian X_i and fixed Hermitian D_0 ,

$$\mathbb{E} \operatorname{tr} \exp(D_0 + \sum_i X_i) \leq \operatorname{tr} \exp(D_0 + \sum_i \log \mathbb{E} e^{X_i}).$$

Taking $D_0 = \theta D$ proves (191); $\log \cosh t \leq t^2/2$ gives its scalar bounds. Expanding the square, using centered signs and $\operatorname{tr} G_0 = a$, $\operatorname{tr} G_1 = b$, proves (192). These arguments require neither commuting row matrices nor independent child Grams. $\square$

Corollary 13.2 (All orders for a fixed support). *For every fixed S of size $k \geq 2$, every $u > 0$, and every real $p \geq 1$,*

$$\mathbb{P} \left\{ \|H_S\| > \sqrt{\frac{2(k-1)u}{m}} + \frac{(k-1)u}{2m} \right\} \leq 2ke^{-u}, \quad (193)$$

$$(\mathbb{E} \|H_S\|^p)^{1/p} \leq \sqrt{\frac{2(k-1)(\log(2k) + 2p)}{m}} + \frac{(k-1)(\log(2k) + 2p)}{2m}. \quad (194)$$

Proof. Split S recursively by binary coordinates, retaining common child labels in separate children, and suppress unary nodes. A singleton has zero Gram error. On a root-to-leaf path let a_j be the chosen child size and b_j its sibling size. Then $\sum_j b_j = k - 1$. Apply (191) recursively with parameter updates

$$\theta_{j+1} = \theta_j + \frac{m}{a_j} \log \cosh(\theta_j \sqrt{a_j b_j / m}).$$

Linearity of expectation, rather than independence between children, gives $\mathbb{E} \operatorname{tr} e^{\theta H_S} \leq \sum_{v \in S} e^{\theta v - \theta}$. Put $t = |\theta| < 2m/(k-1)$. If s is the sum of preceding sibling sizes divided by $2m$, the accumulated nonnegative shift is at most $t/(1-ts) - t$. Indeed a further $h = b_j/(2m)$ contributes at most

$$h \left(\frac{t}{1-ts} \right)^2 \leq \frac{t}{1-t(s+h)} - \frac{t}{1-ts}.$$

The argument works for either sign of θ . Consequently

$$\mathbb{E} \operatorname{tr} e^{\theta H_S} \leq k \exp \left\{ \frac{(k-1)t^2}{2m[1 - (k-1)t/(2m)]} \right\}. \quad (195)$$

Chernoff optimization at the two signs gives (193). The tail is dominated by $\sqrt{2v(\log(2k) + E)} + c(\log(2k) + E)$, with E exponential of mean one, $v = (k-1)/m$, and $c = (k-1)/(2m)$. Since $\|E\|_{L^p} \leq 2p$ for real $p \geq 1$, Minkowski and monotonicity of moments give (194). $\square$

The parameter recursion costs at most $2k - 2$ scalar updates after constructing a tree from $O(kd)$ input bits. This is a tractable upper certificate for one named support. Maximizing it by a union bound over every support replaces u by $\log(2k \binom{N}{k} / \eta)$, leading to order $k^2 \log(eN/k)$ samples. Closure of a fixed-support potential therefore does not settle the uniform problem.

One can also identify the first geometry discarded by this potential. For fixed Hermitian D, Y_i , the difference between its matrix-cgf upper bound and the actual trace exponential satisfies

$$\begin{aligned} & \operatorname{tr} \exp \left(\theta D + \sum_i \log \cosh(\theta Y_i) \right) - \mathbb{E} \operatorname{tr} \exp \left(\theta [D + \sum_i \varepsilon_i Y_i] \right) \\ &= \frac{\theta^4}{24} \left(\sum_i \|[D, Y_i]\|_F^2 + \sum_{i < j} \|[Y_i, Y_j]\|_F^2 \right) + O(\theta^5). \end{aligned} \quad (196)$$

To verify this, expand $\log \cosh(\theta Y) = \theta^2 Y^2/2 - \theta^4 Y^4/12 + O(\theta^6)$ in the ordered trace series. The fourth-degree difference for a pair P, Q is $[\operatorname{tr}(P^2 Q^2) - \operatorname{tr}(P Q P Q)]/12 = \|[P, Q]\|_F^2/24$; all terms through degree three agree. This local expansion is not a globally valid subtraction rule. Retaining its deficit would require controlling its subsequent interactions.

13.3 A density filtration that keeps quotient multiplicities

A different contraction organizes the named dependence structure before counting supports. A *flat* of a represented matroid is the set of all named input columns in the span of a given subset. Ranks below are binary linear ranks, and contraction means passage to a quotient with the named elements retained.

Lemma 13.3 (Dyadic density filtration). *Let S consist of $n \geq 1$ distinct nonzero binary vectors. There is a flag of flats $\emptyset = P_0 \subsetneq \cdots \subsetneq P_s = S$, with $B_i = P_i \setminus P_{i-1}$, prefix ranks $r_i = \operatorname{rank} P_i$, and powers of two $A_i \geq 1$, such that*

$$A_{i+1} \leq A_i/2, \quad |B_i| \geq A_i(r_i - r_{i-1}),$$

and B_i partitions into $q_i = 2A_i$ independent sets modulo $\operatorname{span} P_{i-1}$. Writing $Q_i = q_i \log(2q_i)$,

$$\sum_i Q_i r_i \leq 4n \log(4n), \quad s \leq 1 + \lfloor \log_2 n \rfloor. \quad (197)$$

Proof. For a nonempty loopless matroid M , maximize $|T|/r_M(T)$ over nonempty T , and call the maximum μ . A maximizer is a flat, since adding an element in its closure would increase its density. If F, G maximize the density, rank submodularity gives

$$\mu(r(F) + r(G)) = |F \cup G| + |F \cap G| \leq \mu(r(F \cup G) + r(F \cap G)) \leq \mu(r(F) + r(G)).$$

Thus their union is a maximizer. There is a unique inclusion-maximal maximizing flat. Contract it and repeat. There are no remaining loops. Every remaining T satisfies

$$|F| + |T| \leq \mu(r(F) + r_{M/F}(T)),$$

so its quotient density is at most μ , and is strictly smaller for the maximal choice of F . The resulting fine flag therefore has decreasing densities μ_j and layers of sizes μ_j times their rank increments.

Group consecutive densities in the same interval $[A, 2A)$, where A is a power of two. This gives the claimed decreasing A_i and lower bounds on $|B_i|$. At the beginning of each group the maximum density of the entire remaining quotient is less than $2A_i$; hence each subset of B_i has cardinality at most q_i times its quotient rank. We import Edmonds's matroid partition theorem [3, Theorem 1]: those subset rank inequalities are equivalent to partitionability into q_i independent sets. It applies to the named quotient matroid, including parallel elements.

Put $R_j = r_j - r_{j-1}$. Swapping finite sums gives

$$\sum_i Q_i r_i = \sum_j R_j \sum_{i \geq j} 2A_i \log(4A_i) \leq 4 \sum_j R_j A_j \log(4A_j) \leq 4n \log(4n).$$

Here geometric decrease bounds $\sum_{i \geq j} A_i$ by $2A_j$, and $A_j R_j \leq |B_j|$. The same decrease gives the number of layers. $\square$

The filtration can be constructed in polynomial time in the explicit binary input size using constructive matroid partition. For detail, maximum density is one of the $O(n^2)$ ratios a/b with $1 \leq a, b \leq n$. To test a candidate p/q , replace each input element by q parallel labeled copies and test partitionability into p independent sets. Edmonds's criterion is precisely $q|T| \leq p \text{rank } T$. When the maximum density is one the remaining matroid is independent and can be taken at once. Otherwise a failing partition supplies a violating set; at the candidate immediately below the maximum its original elements maximize density. Successive maximizing contractions, merging equal-density steps if needed, produce the flag. Each replicated instance has at most n^2 elements. This constructs the deterministic filtration, not a polynomial algorithm for searching over all ambient supports or all input laws.

Choose sample coordinates adapted to the flag. For a layer polynomial $F_i = \sum_{v \in B_i} c_v \chi_v$ and $w_i = \sum_{v \in B_i} c_v^2$, quotient independence and Hölder's inequality give

$$\mathbb{E}_U[F_i \mid \text{old coordinates}] = 0, \quad \mathbb{E}_U[e^{sF_i} \mid \text{old coordinates}] \leq e^{q_i w_i s^2 / 2}, \quad \mathbb{E}_U F_i^2 = w_i. \quad (198)$$

Indeed within each independent piece the signs are independent after fixing the old coordinates. Hölder with exponent q_i across pieces gives the displayed proxy. The square identity is global orthogonality, and is generally false after conditioning. For example $F = \chi_{e_2}(1 + \chi_{e_1})$ has global square mean 2, but conditional square mean 0 or 4 after revealing the first coordinate. The distinction between the conditional mean estimate and the global square estimate is essential.

13.4 The scalar input and additive Schur assembly

Lemma 13.4 (A variance-normalized square estimate). *Suppose under a reference law that $\mathbb{E}X^2 = 1$ and $\mathbb{E}e^{sX} \leq e^{q s^2 / 2}$ for all real s , with $q \geq 1$. Write $Q = q \log(2q)$. Then*

$$\log \mathbb{E} e^{\theta(X^2 - 1)} \leq \frac{20Q\theta^2}{1 - 4Q|\theta|}, \quad |\theta| < (4Q)^{-1}. \quad (199)$$

For any alternative law ν of relative entropy D with respect to the reference law,

$$|\mathbb{E}_\nu X^2 - 1| \leq \sqrt{80QD} + 4QD. \quad (200)$$

Proof. Set $Y = X^2$ and $T = 4q \log(2q)$. Markov using $\mathbb{E}Y = 1$, and Chernoff using the subgaussian bound, give $\mathbb{P}(Y > u) \leq \min(1/u, 2e^{-u/(2q)})$. For $u \geq T$ the second expression is at most $(4/T)e^{-u/T}$. The ratio is largest at $u = T$; the required endpoint inequality reduces to $2e \log(2q)/q \leq 4$, whose maximum occurs at $q = e/2$. Splitting the tail integral at T gives, for integers $p \geq 2$,

$$\mathbb{E}Y^p \leq \frac{p}{p-1} T^{p-1} + 4p! T^{p-1} \leq 5p! T^{p-1}.$$

The pointwise inequality $|y - 1|^p \leq y^p - py + p - 1$ for $y \geq 0$ follows by differentiating separately on $[0, 1]$ and $[1, \infty)$. Since $\mathbb{E}Y = 1$, $\mathbb{E}|Y - 1|^p \leq \mathbb{E}Y^p$. Expanding the centered exponential and summing the geometric series yields

$$\log \mathbb{E}e^{\theta(Y-1)} \leq \frac{5T\theta^2}{1 - T|\theta|},$$

which is (199). The entropy inequality $\mathbb{E}_\nu F \leq D + \log \mathbb{E}e^F$ follows from Jensen applied to the density ratio. Applying it to either sign of $\theta(Y - 1)$ and optimizing gives $2\sqrt{5TD} + TD$, proving (200). The case $D = 0$ is immediate. $\square$

The logarithm in Q is the price justified by these scalar hypotheses. The normalization $\mathbb{E}X^2 = 1$ is responsible for improving a rough square proxy q^2 to $q \log(2q)$; conditional normalization cannot be substituted for it.

Lemma 13.5 (Additive squared cost). *For $x, b \geq 0$ and $0 \leq c \leq L$,*

$$\lambda_{\max} \begin{pmatrix} L\sqrt{x} & c\sqrt{b} \\ c\sqrt{b} & L\sqrt{b} \end{pmatrix} \leq L\sqrt{x + 4b}.$$

Proof. Put $T = L\sqrt{x + 4b}$. For $L, b > 0$ the second diagonal gap is at least $T/2$, while

$$T - L\sqrt{x} = \frac{4Lb}{\sqrt{x + 4b} + \sqrt{x}} \geq \frac{2L^2b}{T}.$$

Their product is at least $L^2b \geq c^2b$. The two diagonal gaps and the determinant are nonnegative, so the difference from TI is positive semidefinite. The zero cases are direct. $\square$

This elementary Schur step lets a new diagonal estimate and an old/new cross estimate add to a squared error budget, without multiplying the preceding budget at every extension.

Theorem 13.6 (A deterministic entropy certificate for every support). *Let $S \subset \mathbb{F}_2^r$ contain zero, span the space, and have $h \geq 2$ distinct frequencies. For a law ν on the row space, put*

$$\Gamma_S(\nu) = \mathbb{E}_\nu \phi_S \phi_S^* - I, \quad \kappa_r(\nu) = \max_{1 \leq j \leq r} \max_{P: \mathbb{F}_2^r \rightarrow \mathbb{F}_2^j} \frac{D(P\nu \| U_j)}{j \log 2},$$

where $\phi_S = (\chi_v)_{v \in S}$ and U_j is uniform. For the density filtration of $S \setminus \{0\}$ define $C(S) = 4 \sum_i Q_i r_i \leq 16(h-1) \log(4(h-1))$. If $z = \kappa_r(\nu) \log 2$ satisfies $zC(S) \leq 1/256$, then

$$\|\Gamma_S(\nu)\| \leq 16\sqrt{zC(S)}. \quad (201)$$

In particular, for $0 < \delta < 1$,

$$\|\Gamma_S(\nu)\| \geq \delta \implies \kappa_r(\nu) \geq \frac{\delta^2}{4096(\log 2)(h-1) \log(4(h-1))}. \quad (202)$$

Proof. For prefix marginals set $D_i = D(\nu_{\leq i} \| U_{r_i})$ and $H_i = \mathbb{E}_{\nu_{< i}} D(\nu_{i|< i} \| U_{r_i - r_{i-1}})$. The entropy chain rule gives $0 \leq H_i \leq D_i \leq z r_i$. Use (200) on $F_i / \sqrt{w_i}$ under the entire uniform prefix law, not its conditional law. With $b_i = z Q_i r_i$,

$$|\mathbb{E}_\nu F_i^2 - w_i| \leq w_i(9\sqrt{b_i} + 4b_i).$$

Let G be a polynomial on the old prefix with uniform square norm w , and suppose its Gram error is at most $\epsilon \leq 1$. The conditional mgf in (198), conditional entropy inequality, and Cauchy–Schwarz under the actual old marginal give

$$|\mathbb{E}_\nu G F_i| \leq \sqrt{2q_i w_i H_i \mathbb{E}_\nu G^2} \leq \sqrt{2q_i w_i w(1 + \epsilon) D_i} \leq 3\sqrt{b_i w_i w}.$$

For $b_i \leq 1$, the absolute error for $G + F_i$ is therefore bounded by the quadratic form at $(\sqrt{w}, \sqrt{w_i})$ of

$$\begin{pmatrix} \epsilon & 3\sqrt{b_i} \\ 3\sqrt{b_i} & 13\sqrt{b_i} \end{pmatrix}.$$

Starting with the constant character, apply Lemma 13.5 with $L = 16$, $c = 3$, and $x = z C_{i-1}$, where $C_i = 4 \sum_{j \leq i} Q_j r_j$. The new bound is $16\sqrt{z C_i}$. The assumption $z C(S) \leq 1/256$ ensures that every old error is at most one and every b_i is at most one, completing the induction. If the right side of (202) exceeded $\kappa_r(\nu)$ strictly, the upper bound on $C(S)$ and (201) would give a norm strictly smaller than δ . $\square$

The theorem retains the complete support and its prefix laws; $C(S)$ is a cost certificate, not a claim that ranks alone reconstruct the Gram. It proves coefficient contraction for arbitrary overlapping binary dependencies. It does not make projected entropy small for an empirical law: a high-rank projection may have a huge alphabet while the empirical law has at most m atoms. The next argument controls the necessary empirical interfaces directly.

13.5 From the same contraction to a uniform finite sample bound

Theorem 13.7 (Uniform empirical filtration bound). *For $2 \leq k \leq N$, $0 < \delta \leq 1$, and $0 < \eta < 1$, put $n = k - 1$. Every integer sample size satisfying*

$$m \geq 16384 \delta^{-2} n \log(4n) \{6(d + k) + \log(1/\eta)\} \quad (203)$$

obeys $\mathbb{P}\{\Delta_k > \delta\} \leq \eta$.

Proof. We establish simultaneous diagonal estimates and conditional cross estimates for every possible current prefix and its immediately preceding span. This avoids counting an entire history.

Fix such a flagged prefix. Its old frequencies include zero, its new frequencies lie outside the old span, and the new named quotient frequencies partition into $q \geq 2$ independent pieces. Put $Q = q \log(2q)$. For a unit new coefficient vector, independence of the original rows and (199) imply

$$\mathbb{P} \left\{ \left| \frac{1}{m} \sum_{\ell} F(x_{\ell})^2 - 1 \right| > \sqrt{80Qv/m} + 4Qv/m \right\} \leq 2e^{-v}.$$

A fixed $1/4$ -net of the unit sphere in dimension h has at most 9^h points, by the disjoint-ball volume argument. Approximating a maximizing quadratic form or each argument of a bilinear form loses at most one half of its operator norm. Thus, with $a = Qv/m$, the new-layer Gram error obeys

$$\|D_{\text{new}}\| \leq 2\sqrt{80a} + 8a \leq 26\sqrt{a} \quad (a \leq 1) \quad (204)$$

except with probability $2 \cdot 9^k e^{-v}$.

For the cross block C , condition on all old sample coordinates. Fix unit old and new coefficient vectors with evaluations g_ℓ, f_ℓ . Equation (198) gives

$$\mathbb{E} \left[\exp \left(\frac{\theta}{m} \sum_{\ell} g_{\ell} f_{\ell} \right) \middle| \text{old sample} \right] \leq \exp \left(\frac{q\theta^2}{2m^2} \sum_{\ell} g_{\ell}^2 \right).$$

On the old-good event $\|D_{\text{old}}\| \leq 1$, the last sum is at most $2m$. Chernoff and a product of two fixed nets therefore give

$$\|C\| \leq 4\sqrt{qv/m} \leq 4\sqrt{a}, \quad (205)$$

with exceptional probability at most $2 \cdot 9^k e^{-v}$ intersected with the old-good event. That event is measurable in the old sample, so the conditional calculation is valid. It assumes no unconditional independence of the two blocks.

A rank- r flagged prefix can be described by an injective map $\mathbb{F}_2^r \rightarrow \mathbb{F}_2^d$, an old dimension $0 \leq j < r$, and an ordered list of at most n distinct nonzero frequencies in $\mathbb{F}_2^r$. The old subspace uses the first j coordinates. The old set consists exactly of the listed frequencies in that subspace, because the preceding prefix is a flat. Dropping restrictions only increases the count, giving

$$M_r \leq n^2 2^{r(d+n)}, \quad 1 \leq r \leq \min(d, n). \quad (206)$$

For each description choose its smallest dyadic quotient-cover number at least two. This deterministic number is at most $2n$. Bounds imposed for it also hold for any larger dyadic number from Lemma 13.3; no union over partitions or coefficient values is required.

Set $u = \log(1/\eta)$ and $v_r = 6r(d+k) + u$. The union of failures in (204) and (205) has probability at most

$$4 \sum_{r=1}^{\min(d,n)} M_r 9^k e^{-v_r} \leq e^{-u}. \quad (207)$$

Indeed $\log 4 + 3 \log n \leq 2k$, and hence

$$\log 4 + 3 \log n + r(d+n) \log 2 + k \log 9 < 6r(d+k).$$

Each summand is at most e^{-u}/n , and there are at most n ranks. This constructs a simultaneous event; we do not condition on that event to infer independence.

On this event, take any anchored support and its density filtration. Put $a_i = Q_i v_{r_i}/m$ and $A = \sum_i a_i$. If $A \leq \delta^2/4096$, induction gives

$$\|D_{\leq i}\| \leq 64 \sqrt{\sum_{j \leq i} a_j}. \quad (208)$$

The old bound is at most $\delta \leq 1$, so (205) applies. The new diagonal bound is at most $26\sqrt{a_i}$. Apply Lemma 13.5 with $L = 32$, $c = 4$, and $x = 4 \sum_{j < i} a_j$ to obtain (208). Finally, since $r_i \geq 1$, Lemma 13.3 yields

$$A \leq \frac{4n \log(4n)}{m} \{6(d+k) + u\}.$$

Condition (203) makes this at most $\delta^2/4096$. Translation invariance of each actual Gram covers supports not anchored at zero, proving the theorem. $\square$

The cost $d+k$ has a specific source: the factor 2^{rn} in (206). The proof contracts arbitrary coefficients, keeps parallel quotient labels, and avoids an exponential empirical-type alphabet. Its current support selection step still pays for too many distinct named patterns when $k \gg d$. A sharper estimate for one fixed pattern does not remove this cost.

13.6 Confidence is a separate, cheaper operation

Proposition 13.8 (Concentration of the full RIP supremum). *Let $k \geq 2$, $q = k - 1$, and $\mu = \mathbb{E}\Delta_k$. For $u > 0$,*

$$\mathbb{P}\left\{\Delta_k > \mu + \sqrt{\frac{2q(1+2\mu)u}{m}} + \frac{qu}{3m}\right\} \leq e^{-u}. \quad (209)$$

For every real $p \geq 1$,

$$\|\Delta_k\|_{L^p} \leq \mu + \sqrt{\frac{2q(1+2\mu)}{m}} \Gamma(1+p/2)^{1/p} + \frac{q}{3m} \Gamma(1+p)^{1/p}. \quad (210)$$

Proof. For a real unit vector on at most k characters put $f_c(x) = (\sum_v c_v \chi_v(x))^2 - 1$. Orthogonality and Cauchy–Schwarz give

$$\mathbb{E}f_c = 0, \quad |f_c| \leq q, \quad \mathbb{E}f_c^2 = \mathbb{E}(\sum_v c_v \chi_v)^4 - 1 \leq k - 1 = q.$$

For the last inequality, the square of the polynomial is pointwise at most k and its expectation is one. Take countably many dense coefficient directions on each of the finitely many supports; continuity preserves the exact supremum. Include both signs of f_c/q .

We import Bousquet’s bounded empirical-process inequality [4, Theorem 2.3]: for a countable centered class with absolute envelope one, pointwise variances at most σ^2 , and $Z = \sup_f \sum_{i=1}^m f(x_i)$,

$$\mathbb{P}\{Z > \mathbb{E}Z + \sqrt{2(m\sigma^2 + 2\mathbb{E}Z)u} + u/3\} \leq e^{-u}.$$

Here $Z = m\Delta_k/q$ and $\sigma^2 = 1/q$, giving (209). Its upper deviation is stochastically dominated by $\sqrt{2q(1+2\mu)E/m} + qE/(3m)$ for an exponential E of mean one. Integrating tails and using $\mathbb{E}E^s = \Gamma(1+s)$ proves (210). $\square$

Corollary 13.9 (Separated confidence cost). *For $2 \leq k \leq 2^d$ and $0 < \eta < 1$, it suffices that*

$$m \geq \max\left\{2^{25}(k-1)\log(4(k-1))(d+k), \quad 256(k-1)\log(1/\eta)\right\} \quad (211)$$

to have $\mathbb{P}\{\Delta_k > 1/4\} \leq \eta$.

Proof. Theorem 13.7 at $\delta = 1/16$ and $\eta_0 = e^{-2(d+k)}$ applies above the first sample threshold in (211). Every k -column Gram is positive semidefinite with trace k , so $0 \leq \Delta_k \leq k - 1$. Consequently

$$\mu \leq 1/16 + (k-1)e^{-2(d+k)} < 1/8.$$

The second threshold and Proposition 13.8 bound its tail threshold by $1/8 + \sqrt{5/512} + 1/768 < 1/4$. For an exact strict comparison, $(95/768)^2 - 5/512 = 3265/589824 > 0$. $\square$

The baseline still contains $k \log k(d+k)$, but confidence beyond that baseline costs only $k \log(1/\eta)$. This is consequential when combining many geometric reductions: the logarithm of the number of reductions enters a confidence term, without an extra factor $\log k$.

13.7 An overlap certificate on the original rows

The lower side uses a different object: a collection of explicit sparse vectors that vanish on the same realized sample. Exact overlap counting allows their dependence to be retained until after centering.

Theorem 13.10 (Uniform empty-cell lower certificate). *For $2 \leq k \leq N$ and every $0 < \delta < 1$,*

$$1 \leq m \leq \frac{\mathcal{D}(N, k)}{50000} \implies \mathbb{P}\{\Delta_k \leq \delta\} < 1/10. \quad (212)$$

The certificate family consists of flat signed coefficient vectors on binary frequency subspaces of cardinality at most k .

Proof. First let $h = 2^r$, $1 \leq r \leq d$, put $n = d - r$ and $p = 1/h$. For an r -dimensional frequency space W and $z \in W^*$, define the row cell $C(W, z) = \{x : x|_W = z\}$. If this cell is empty in the actual sample, then the unit coefficient vector

$$c_v = h^{-1/2}(-1)^{z(v)} \mathbf{1}_{\{v \in W\}} \quad \text{satisfies} \quad (A_m c)_i = \sqrt{h/m} \mathbf{1}_{\{x_i|_W = z\}} = 0. \quad (213)$$

Let Z count all empty such cells. There are $L = h \binom{d}{r}_2$ cells, each empty with probability $\alpha = (1 - p)^m$. Here $\binom{b}{a}_2$ denotes the number of a -dimensional binary subspaces of a b -dimensional space; an invalid binomial is zero.

If two frequency spaces intersect in dimension j , a fraction 2^{-j} of their ordered label pairs agree on the intersection. Compatible cells have intersection mass $2^j p^2$, so their joint empty probability is $\beta_j = (1 - 2p + 2^j p^2)^m$; incompatible cells are disjoint and have joint empty probability $\beta_- = (1 - 2p)^m$. For a fixed W , the number of other spaces meeting it in dimension j is

$$n_j = 2^{(r-j)^2} \binom{r}{j}_2 \binom{n}{r-j}_2.$$

To see this, choose the intersection, choose the image of the other space in the quotient by W , and then choose a linear graph into W modulo the intersection. The last two vector spaces each have dimension $r - j$. Write $w_j = n_j / \binom{d}{r}_2$ and $c_j = 2^{-j} w_j$. The exact centered variance is

$$\frac{\text{Var } Z}{(\mathbb{E} Z)^2} = \sum_{j=0}^r w_j \left[2^{-j} \left(\frac{\beta_j}{\alpha^2} - 1 \right) + (1 - 2^{-j}) \left(\frac{\beta_-}{\alpha^2} - 1 \right) \right]. \quad (214)$$

In particular the $j = 0$ compatible channel has zero covariance. Since $\beta_- \leq \alpha^2$, dropping only incompatible nonpositive covariances and separating the diagonal gives

$$\frac{\text{Var } Z}{(\mathbb{E} Z)^2} \leq \frac{1}{L\alpha} + \sum_{j=1}^{r-1} c_j (e^{4\lambda 2^{j-r}} - 1), \quad \lambda = mp. \quad (215)$$

Indeed the proper-channel ratio is $[1 + (2^j - 1)p^2 / (1 - p)^2]^m$, and $p \leq 1/2$. Keeping the minus one in each channel is necessary for the summable estimate.

The product formula for Gaussian binomials gives

$$2^{a(b-a)} \leq \binom{b}{a}_2 \leq 4 2^{a(b-a)}. \quad (216)$$

For the upper bound, the denominator product $\prod_{i \geq 1} (1 - 2^{-i})$ exceeds $1/4$: its first three factors multiply to $21/64$ and the remaining product is at least $1 - \sum_{i \geq 4} 2^{-i} = 7/8$. The lower bound follows by comparing corresponding factors in the finite product. Thus, with $H = r(n + 1) \log 2$,

$$e^H \leq L \leq 4e^H, \quad c_j \leq 16 2^{-j(n-r+j+1)}.$$

For a feasible proper channel set $\ell = r - j$ and $E_\ell = (r - \ell)(n - \ell + 1)$. Here $1 \leq \ell \leq \min(r - 1, n)$, the positive integers E_ℓ are distinct, and

$$\frac{r(n + 1)2^{-\ell}}{E_\ell} \leq \frac{(\ell + 1)^2}{2^\ell} \leq \frac{9}{4}, \quad \sum_{\ell} E_\ell 2^{-E_\ell/2} \leq 4 + 3\sqrt{2}. \quad (217)$$

The first ratio decreases in r and n at their allowed lower endpoints, and $(\ell + 1)^2/2^\ell$ is maximal at $\ell = 2$. The second inequality sums over all positive integers instead of only these distinct exponents.

Put $c = 1/10000$ and assume $1 \leq m \leq chH$. Since $\log(1 - p) \geq -2p$, $1/(L\alpha) \leq e^{-(1-2c)H}$. Use $e^x - 1 \leq xe^x$, (217), and $c \leq 1/18$ to bound the proper channels in (215) by

$$144c \log 2 \sum_{\ell} E_{\ell} 2^{-(1-9c)E_{\ell}} \leq 144c \log 2 (4 + 3\sqrt{2}) < 840c = 0.084.$$

Every nonvacuous instance has $d \geq 11$: for $d \leq 10$, $chH \leq 2^d d \log 2/10000 < 1$. Also $r(n+1) \geq d$, so $e^{-(1-2c)H} < 2^{-10}$. Chebyshev gives

$$\mathbb{P}\{\Delta_h \leq \delta\} \leq \mathbb{P}\{Z = 0\} \leq \frac{\text{Var } Z}{(\mathbb{E}Z)^2} < 2^{-10} + 0.084 < 1/10.$$

This includes $r = d$, when the proper-channel sum is empty.

For an arbitrary $k \geq 2$, choose $h = 2^r \leq k < 2h$ and $n = d - r$. Then

$$\mathcal{D}(N, k) \leq 2h(r+2)(n+1)(\log 2)^2 \leq 6hr(n+1)(\log 2)^2.$$

Since $6 \log 2 < 5$, the threshold in (212) is smaller than $hH/10000$. Every witness in (213) is k -sparse, giving the claim. When the interval contains no positive integer the statement is vacuous, and the corresponding strict lower bound for m_* follows from $m_* \geq 1$. $\square$

The empty-subspace mechanism and its target scale have a substantial prior literature; see Błasiok, Lopatto, Luh, Marcinek and Rao [5]. The calculation here is stated directly for the fixed- m with-replacement model and all affine row cells. No Bernoulli-row theorem is transferred by identifying an expected row count with m , and no novelty claim is made.

13.8 A matching theorem from exact core-quotient geometry

The preceding upper and lower modules become matching estimates when the support admits a complete core factor and a quotient containing relatively few frequencies. The core itself is allowed to vary, so its selection cost must be paid.

Definition 13.11. Choose $0 \leq t \leq d$, and write $a = 2^t$, $D = d - t$, $M = 2^D$, $1 \leq K \leq M$, and $k = aK \geq 2$. Let $\mathcal{C}_{t,K}$ consist of all coefficient vectors whose support is contained in at most K cosets of some t -dimensional frequency subspace $L \leq V$. Both L and the cosets vary. Coefficients within each coset are arbitrary and may vanish at any positions. Let $m_*^{\text{core}}(d, t, K; \eta)$ be the minimum positive sample size for

$$\frac{1}{2} \|c\|_2^2 \leq \|A_m c\|_2^2 \leq \frac{3}{2} \|c\|_2^2 \quad \text{simultaneously for every } c \in \mathcal{C}_{t,K}$$

with probability at least $1 - \eta$.

Theorem 13.12 (Matching geometric and confidence scales). *Let $0 < \eta \leq 1/10$ and $\mathcal{R} = \mathcal{D}(N, k) + k \log(1/\eta)$. If*

$$K = 1 \quad \text{or} \quad 2 \leq K \leq \min(2D, 2^D), \quad (218)$$

then

$$\frac{\mathcal{R}}{400000} < m_*^{\text{core}}(d, t, K; \eta) \leq \lceil 2^{32} \mathcal{R} \rceil. \quad (219)$$

More informative sufficient bounds, valid for every K , are

$$m \geq 12a \log(2aH/\eta) \quad (K = 1), \quad (220)$$

$$m \geq a \left[2^{26} (K-1) \log(4(K-1))(D+K) + 1024(K-1) \log(4aH/\eta) \right] \quad (K \geq 2), \quad (221)$$

where $H = \begin{bmatrix} d \\ t \end{bmatrix}_2 \leq 4 \cdot 2^{tD}$. The lower bound in (219) is valid for every K with $aK \geq 2$.

Proof. Fix a core L and a linear complement Q , so $V = L \oplus Q$. For a quotient support $T \subset Q$, the complete containing support is $L + T$. Write an actual sampled row in dual coordinates as (z_i, y_i) . Let U be the normalized Walsh transform on L . Direct character orthogonality sends that row, under $U \otimes I_T$, to $\sqrt{a} e_{z_i}^\top \otimes \phi_T(y_i)^\top$. Therefore

$$(U \otimes I_T)^* G_{L+T} (U \otimes I_T) = \bigoplus_{z \in L^*} \frac{an_z}{m} G_{T,z}, \quad n_z = |\{i : z_i = z\}|. \quad (222)$$

For $n_z > 0$, $G_{T,z} = n_z^{-1} \sum_{i: z_i = z} \phi_T(y_i) \phi_T(y_i)^*$. An empty bucket contributes the zero block, without defining an empty normalized Gram. Given the entire assignment $(z_1, \dots, z_m)$, the y_i are independent uniform quotient rows. Character features within a row retain all their dependencies.

For each fixed core, binomial Chernoff and a union over buckets give

$$\mathbb{P} \left\{ \max_z |an_z/m - 1| > 1/8 \right\} \leq 2a \exp[-m/(192a)]. \quad (223)$$

Suppose $K \geq 2$, and set $v = \log(4aH/\eta)$ and $\rho = e^{-v}$. Condition (221) implies

$$\frac{m}{2a} \geq 2^{25}(K-1) \log(4(K-1))(D+K), \quad \frac{m}{2a} \geq 512(K-1)v,$$

and also $m/(192a) \geq v$. A union over the H cores bounds any occupancy failure by $\eta/2$. For a fixed core and a fixed assignment satisfying its occupancy condition, each $n_z \geq 7m/(8a) \geq m/(2a)$. Corollary 13.9, in quotient dimension D , bounds the conditional probability of quotient K -RIP error exceeding $1/4$ in one bucket by ρ . It controls all quotient supports T at once. Integrate over the good assignments for this core and sum over cores and buckets, costing at most $H\rho = \eta/4$. This calculation conditions separately for each fixed core, never on the joint good event for all cores.

Outside these exceptions the eigenvalues of every block in (222) lie in

$$[(1 - 1/8)(1 - 1/4), (1 + 1/8)(1 + 1/4)] = [21/32, 45/32] \subset [1/2, 3/2].$$

Compression controls arbitrary coefficients on subsets of the complete containing support. This proves (221). If $K = 1$, every nonempty quotient Gram is the scalar one, and Chernoff at relative error $1/2$ gives total failure at most $2aHe^{-m/(12a)}$, proving (220).

For comparison with the target, put $A = \log(2k)$, $B = \log(2N/k)$ and $u = \log(1/\eta)$. When $2 \leq K \leq 2D$, the integer inequality $D^3 \leq 4^D$ implies

$$B = (D+1) \log 2 - \log K \geq D \log 2/3.$$

The integer inequality holds at $D = 1, 2$ and then follows by induction since $((D+1)/D)^3 \leq 27/8 < 4$ for $D \geq 2$. Also $D+K \leq 3D$, $\log(4(K-1)) \leq 2A$, and $a(K-1) \leq k$. Thus the first term of (221) is at most $(18 \cdot 2^{26}/\log 2)kAB < 2^{31}kAB$. The core count gives

$$\log(4aH) \leq \log 16 + t(D+1) \log 2 \leq (8/\log 2)AB.$$

For the last inequality use $t \log 2 \leq A$, $D+1 \leq 2D \leq 6B/\log 2$, and $AB \geq 2(\log 2)^2$. The second term of (221) is consequently at most $2^{14}kAB + 1024ku$. Their sum is less than $2^{32}(kAB + ku)$. When $K = 1$, $k = a \geq 2$ and $\log(2aH) \leq 2AB/\log 2$, so (220) also fits that bound. This proves the upper side of (219).

For the lower side set $h = a2^{\lfloor \log_2 K \rfloor}$, so $k/2 \leq h \leq k$ and $h \geq 2$. Every frequency subspace W of size h contains a t -dimensional core and is a union of at most K of its cosets. Thus all size- h witnesses (213) belong to $\mathcal{C}_{t,K}$. The proof of Theorem 13.10 transfers through this actual witness family, not through failure of a larger vector class. It rules out every

$m \leq h \log(2h) \log(2N/h)/50000$, and hence every $m \leq kAB/200000$. The latter comparison uses $h \geq k/2$, $\log(2h) \geq A/2$, and $\log(2N/h) \geq B$.

For confidence, retain just one cell. Its probability of being empty is $(1 - 1/h)^m > e^{-2m/h}$. Every $m \leq ku/4$ therefore has failure probability strictly greater than η . Thus $m_*^{\text{core}} > kAB/200000$ and $m_*^{\text{core}} > ku/4$, which imply the lower bound in (219). Thresholds below one are covered by the minimum positive-integer convention. $\square$

For a fixed core chosen in advance, the same upper proof replaces H by one. The geometric lower bound just proved varies the core and is not asserted for that fixed-core subclass.

The theorem includes nontrivial compressed, noncommuting geometry. Define $\kappa_t(S) = \min_{\dim L=t} |\pi_L(S)|$. Anchoring at a representative of the first occupied coset shows

$$\text{rank}_{\text{aff}}(S) \leq t + \kappa_t(S) - 1. \quad (224)$$

A support of full affine rank thus needs at least $D + 1$ quotient cosets. The range $K \leq 2D$ includes such supports. In particular, $T = \{0, 1, 2, 3, 4\} \subset \mathbb{F}_2^3$ has full affine rank and is admissible at $K = 5$. Its quotient row errors $R_y = \phi_T(y)\phi_T(y)^* - I_5$ retain

$$\text{tr}((R_0 R_1)^3) = -4. \quad (225)$$

Indeed $\langle \phi_T(0), \phi_T(1) \rangle = 1$. On their two-dimensional span the product has trace -7 and determinant 16 ; on its three-dimensional complement it is the identity. Its cubed trace is $(-7)^3 - 3 \cdot 16(-7) + 3 = -4$. In a core bucket with at least two sample indices, the word $1, 2, 1, 2, 1, 2$ reuses those two observations; the row-pattern event has conditional probability M^{-2} . The negative coefficient is one ordered trace word, not the full nonnegative sixth moment. The proof has not replaced the quotient matrices by commuting scalar errors.

13.9 Why completing every support to core cosets is insufficient

Proposition 13.13 (Sidon obstruction to efficient completion). *Call $S \subset \mathbb{F}_2^d$ Sidon if every nonzero difference occurs for at most one unordered pair of distinct points. If $|S| = k$ and a core of size a meets S in K occupied cosets, then*

$$k^2 \leq K(k + 2a - 2). \quad (226)$$

If $k \geq 4d$ and $K \leq 2d$, the complete coset cover has size $aK \geq k^2/4$. For every $r \geq 3$, the explicit support

$$S_r = \{(x, x^3) : x \in \mathbb{F}_{2^r}\} \subset \mathbb{F}_2^{2r}$$

is Sidon and has full affine rank.

Proof. Let $s_1, \dots, s_K$ be the occupied coset sizes. Each pair in the same coset has a nonzero difference in the core, and no such difference can be used twice. Hence $\sum_j \binom{s_j}{2} \leq a - 1$. Cauchy–Schwarz gives $k^2 \leq K \sum_j s_j^2 \leq K(k + 2a - 2)$. If $K \leq 2d \leq k/2$, rearrangement yields $2aK \geq k^2 - Kk + 2K \geq k^2/2$.

For the graph S_r , fix a nonzero first-coordinate difference h . The second-coordinate difference is $(x + h)^3 + x^3 = hx^2 + h^2x + h^3$. Equal values at x, y imply $h(x + y)(x + y + h) = 0$, so $y = x$ or $y = x + h$. Thus each difference belongs to at most one unordered pair. To prove full affine rank, note that the graph contains zero. Any annihilating binary linear functional has the form $\text{Tr}(\alpha x + \beta x^3)$, where Tr is the field trace. Polarization gives $\text{Tr}(\beta(x^2z + xz^2)) = 0$ for every x, z . The trace pairing is nondegenerate, and $\text{Tr}(ax^2) = \text{Tr}(a^{2^{r-1}}x)$; therefore $(\beta z)^{2^{r-1}} + \beta z^2 = 0$ for all z . Squaring gives $\beta z + \beta^2 z^4 = 0$ throughout a field with more than four elements. The root bound for a nonzero polynomial forces $\beta = 0$, and then $\alpha = 0$. Nondegeneracy of the trace pairing follows because the trace polynomial is a nonzero polynomial of degree below 2^r ; for $\alpha \neq 0$, multiplication by α is a bijection. $\square$

For $r \geq 6$, this family has $d = 2r$, $k = 2^r = \sqrt{N}$, and $k \geq 4d$. Any completion admissible in (218) has size at least $N/4$. Applying a target-rate theorem to that completed space may cost order $N \log N$, whereas (189) for the original support is order $\sqrt{N}(\log N)^2$. The diverging ratio is an obstruction to this completion strategy; it is not a sample-complexity lower bound for S_r .

The original Sidon support has benign fourth moments:

$$\mathbb{E}_U \left(\sum_{v \in S} c_v \chi_v \right)^4 = 3 \left(\sum_v c_v^2 \right)^2 - 2 \sum_v c_v^4 \leq 3 \quad (\|c\|_2 = 1). \quad (227)$$

In expanding the fourth moment, four distinct frequencies summing to zero would violate the Sidon property; only the pair multiplicities remain. This low-order property does not itself control the supremum of sampled quadratic forms.

There is a complementary finite-family description. If one pattern of size k is fixed in binary dimension s , then it has at most 2^{sd} injective linear embeddings in $\mathbb{F}_2^d$. Frequency translations leave each sampled Gram unchanged, so no translation multiplicity is charged. Corollary 13.2 and a union bound show that all these embedded supports have distortion at most $\delta \leq 1$ with probability $1 - \eta$ whenever

$$m \geq 16\delta^{-2}(k-1)\{sd \log 2 + \log(2k/\eta)\}. \quad (228)$$

For J prescribed patterns, add $\log J$ in braces. For the single family S_r at $s = d = 2r$, this is $O_\delta(kr^2 + k \log(1/\eta))$, the candidate original scale at fixed distortion. Efficient core covers and a small collection of affine patterns are therefore complementary descriptors. Neither presently controls the number and shared events of all possible supports.

13.10 What is proved, and the remaining global question

The uniform lower bound (212) holds at every sparsity. The upper theorem (203) matches it up to universal constants for $2 \leq k \leq d$, since $d + k \leq 2d$ and $\log(2N/k)$ is comparable with d in that range. At the other endpoint, conjugating the complete Walsh Gram gives

$$G_V \simeq \text{diag}(NC_x/m : x \in V^*),$$

where C_x are the original row occupancies. Chernoff and a union bound show that $m \geq 12N \log(20N)$ gives half distortion with probability at least 0.9. This matches (189) for $N/2 \leq k \leq N$ by compression and Theorem 13.10. Theorem 13.12 additionally matches the target, including confidence, on its whole specified geometric class even when $k \gg d$.

For unrestricted supports in the remaining intermediate regime, a sharp upper estimate has not been proved here. Equations (195), (201), and (208) are three valid ways to contract coefficients while exposing different losses. The first closes an upper spectral potential but erases the commutator information in (196); the second retains projected laws but can be costly to evaluate; the third turns that contraction into a finite simultaneous event but pays the named-pattern count in (206). The exact quotient factorization removes this cost for a substantial class, and the Sidon proposition proves why that factorization alone cannot cover all supports at comparable size.

The next global problem is consequently precise: replace the current enumeration of flagged patterns by a controlled description of their shared bad events, or develop an alternative extension state that performs this selection while retaining useful signed coefficient contractions. Such an argument must apply to the same original samples, retain quotient aliases and multiplication order, and allow moment orders and confidence costs to grow explicitly. A local exact identity, an average over supports, or a lower witness from another matrix does not supply that missing step.

14 Larger geometric classes and one-sided uniform control

14.1 Unrelated affine-flat directions

The common-core theorem extends to a bounded number of covering flats whose directions can differ. The proof controls each entire union Gram, including all cross interactions, rather than assembling marginal isometries.

Theorem 14.1 (A bounded number of unrelated flats). *Let $d \geq 1$, $0 \leq t \leq d$ and $J \geq 1$ be integers, $N = 2^d$, $h = 2^t$, and $k = \min(N, Jh) \geq 2$. Consider all coefficient vectors supported in a union of at most J affine frequency flats, each of size at most h , allowing intersections and arbitrary real or complex coefficients. For original independent uniform Walsh rows sampled with replacement, let m_* be the smallest positive integer giving simultaneous half squared-norm distortion with probability at least $1 - \eta$, $0 < \eta \leq 1/10$. Put $A = \log(2k)$, $B = \log(2N/k)$ and $u = \log(1/\eta)$. Then*

$$\frac{kAB + ku}{400000} < m_* \leq \left\lceil \max\{2^{14}JkAB, 32(k-1)u\} \right\rceil. \quad (229)$$

Every integer above the upper threshold succeeds; no sample monotonicity is assumed. In particular the scale matches for each fixed J .

Proof. Enlarge each covering flat to dimension t and repeat flats if necessary. Translating every frequency by the same vector multiplies an original row by a common sign and preserves the Gram. Normalize the first affine offset to zero. With $D = d - t$, $M = N/h$ and the binary Gaussian coefficient $H_0 = \begin{bmatrix} d \\ t \end{bmatrix}_2$, the number of normalized ordered covers is at most $P = H_0^J M^{J-1}$. Each cover is replaced by its full distinct-frequency union S , of size $s \leq k$; duplicate labels are used only once.

For $s \geq 2$, the original centered row errors divided by m have norm $(s-1)/m$ and total matrix variance $(s-1)I_s/m$. Matrix Bernstein [1], applied to the full union and then to all covers, gives for the supremum error Δ and every $\delta > 0$,

$$\mathbb{P}\{\Delta > \delta\} \leq 2kP \exp\left[-\frac{m\delta^2}{2(k-1)(1+\delta/3)}\right].$$

Size-one unions have zero error. At $\delta = 1/16$, a sufficient threshold for probability at most $1/(16k)$ is $1024(k-1)\log(32k^2P)$. Alternatively the full Walsh Gram diagonalizes to the original occupancy counts; binomial Chernoff gives the threshold $768N\log(32Nk)$. Thus at

$$\mathcal{B} = \min\{1024(k-1)\log(32k^2P), 768N\log(32Nk)\}$$

we have $\mathbb{E}\Delta < 1/8$, since $0 \leq \Delta \leq k-1$. The Bousquet confidence argument already used for Walsh sampling applies to this restricted coefficient class, on exactly the same rows. For $m \geq 32(k-1)u$ its threshold is at most $1/8 + \sqrt{5/64} + 1/96 < 1/2$.

For completeness, $\mathcal{B} \leq 2^{14}JkAB$ follows in two regimes. Write $r = N/k$. If $r \geq J$, then $k = Jh$, $D\log 2 = \log(Jr) \leq 2B$ and $(t+1)\log 2 \leq A$. Using $H_0 \leq 4 \cdot 2^{tD}$ gives $\log(32k^2P) \leq 13JAB/(2\log 2) < 10JAB$. If $r < J$, then $\log r \leq A$, and $768N\log(32Nk) \leq 3456rkA < 5184JkAB$. These inequalities prove the upper rate.

For the lower rate let $g = h2^{\min(\lfloor \log_2 J \rfloor, d-t)}$, so $k/2 \leq g \leq k$ and $g \geq 2$. Every size- g frequency subspace is a union of at most J size- h affine flats. All empty-cell vectors from the original-row overlap lower certificate therefore belong to this class. That proof rules out success probability 0.9 when $m \leq g\log(2g)\log(2N/g)/50000$, giving $m_* > kAB/200000$. One such cell is empty with probability $(1 - 1/g)^m > e^{-2m/g}$; hence $m_* > ku/4$. Combining the two inequalities gives (229). These are witnesses in the same class and matrix. $\square$

This includes $S = U \cup W$ for complementary t -dimensional subspaces in dimension $2t$, with $J = 2$ and $|S| = 2h - 1$. An admissible common-core cover from the earlier matching theorem needs completion size at least $h^2/(4t)$: for core dimension q , the intersections with U, W have dimensions summing to at most q , so at least $h/\sqrt{2^q}$ cosets are occupied; the earlier admissible coset budget is at most $4t$. This proves a cost limitation of that reduction, not a sampling lower bound on S . Sampling all diagonal rows (z, z) makes both marginal flat Grams identity while their union has a two-column kernel. Marginal control alone therefore fails. Taking $h = 1, J = k$ in (229) admits arbitrary sparse supports but loses a factor k ; it does not give full FR-10.

14.2 All-Sidon lower isometry and the fourth-moment input

For a Sidon support, meaning no four distinct frequencies sum to zero, the actual Walsh character average gives for real coefficients

$$\mathbb{E} \left| \sum_{v \in S} c_v \chi_v(Y) \right|^4 = 3\|c\|_2^4 - 2 \sum_v c_v^4 \leq 3\|c\|_2^4. \quad (230)$$

Only pairings and a fourfold repeated index survive, with the displayed correction for overcounted repetitions. This identity alone is not an upper isometry theorem. It does supply a uniform lower theorem through an explicit classical input.

Theorem 14.2 (Imported lower covariance theorem). *Let B_i be independent identically distributed positive semidefinite $r \times r$ matrices with entrywise finite second moments and mean Σ . If $b > 1$ satisfies $\sqrt{\mathbb{E}(v^T B_1 v)^2} \leq b v^T \Sigma v$ for every real v , then with probability at least $1 - \rho$,*

$$\frac{1}{m} \sum_i B_i \succeq \left[1 - 7b \sqrt{\frac{r + 2 \log(2/\rho)}{m}} \right] \Sigma.$$

This is Oliveira's Theorem 3.1 [16]; it is imported.

Corollary 14.3 (Simultaneous lower isometry). *Let $1 \leq k \leq N$, $0 < \delta < 1$ and $0 < \eta < 1$. If*

$$m \geq 147\delta^{-2}[k + 2k \log(eN/k) + 2 \log(2/\eta)],$$

then, with probability at least $1 - \eta$, every unit vector on every Sidon support of size at most k has sampled energy at least $1 - \delta$.

Proof. For one support take $B_i = \phi_S(Y_i) \phi_S(Y_i)^T$, $\Sigma = I$ and $b = \sqrt{3}$ using (230). Entries have finite second moments by Cauchy–Schwarz. For a family of J_0 allowed supports a union bound with $\rho = \eta/J_0$ gives the sufficient count $147\delta^{-2}[k + 2 \log J_0 + 2 \log(2/\eta)]$. All nonempty subsets of size at most k number at most $(eN/k)^k$: use $x = k/N \leq 1$ in $x^k \sum_{j=0}^k \binom{N}{j} \leq (1+x)^N \leq e^k$. This counts every smaller Sidon support and requires no extension to size k . Real symmetric Grams give the same lower eigenvalue for complex vectors. $\square$

More generally the same argument works for an isotropic finite dictionary with fourth-moment constant Λ , replacing 147 by 49Λ ; at $\Lambda = 1$ use the limiting closed events as $b \downarrow 1$. No deterministic row-length cap is required by this imported theorem. The result is one-sided: rare upper-energy events and simultaneous upper isometry remain separate tasks.

15 Ordered moment corrections and the cost of summing them

This section isolates what is now closed in the actual Walsh Gram and what still requires an all-order estimate. General ordered-integration lemmas are proved completely. Longer finite classification arguments are presented as explicit proof outlines; the corresponding proof notes and saved exact counts are listed in the report's source map.

15.1 The actual sample-count polynomial

Fix $k \geq 2$ distinct frequencies $S = (s_1, \dots, s_k) \subset \mathbb{F}_2^d$ and let $Y(z) = \phi_S(z)\phi_S(z)^T - I_k$, $H_m = m^{-1} \sum_{i=1}^m Y(z_i)$, on the original iid uniform sampled rows. A diagonal entry of Y is zero; for $a \neq b$, its entry is $\chi_{s_a+s_b}(z)$. Expanding a cyclic trace before expectation gives the exact identity

$$\mathbb{E} \operatorname{tr} H_m^{2p} = m^{-2p} \sum_{h=1}^{\min(m,p)} \binom{m}{h} a_h(S, p). \quad (231)$$

Here a_h counts ordered cyclic column lists of length $2p$, with adjacent columns distinct, and onto assignments of their edges to h named sample indices, each used at least twice. For every name, the XOR of its edge labels must vanish. Each accepted original entry term has expectation one; otherwise it has expectation zero. This proves $a_h \geq 0$ without asserting positivity of individual matrix trace words. Distinct sample names can take equal row values; they are not identified in this count.

There is also a compact exact program in the sample count. Form the shear

$$\mathcal{S}(Y) = \begin{pmatrix} I_k & Y \\ 0 & I_k \end{pmatrix}, \quad T_p = \mathbb{E} \mathcal{S}(Y)^{\otimes 2p} = I + N_p.$$

The product of shears has upper-right block $\sum_i Y_i$. Extracting this block in every tensor leg and identifying the column indices cyclically is a fixed linear readout Λ_p for $\operatorname{tr}(\sum_i Y_i)^{2p}$. Every nonidentity term in N_p changes at least two tensor legs in the upper-triangular direction, because $\mathbb{E}Y = 0$. A leg can change only once. Consequently $N_p^{p+1} = 0$, and independence gives

$$\mathbb{E} \operatorname{tr} H_m^{2p} = m^{-2p} \Lambda_p((I + N_p)^m), \quad a_h = \Lambda_p(N_p^h).$$

This proves the finite polynomial without enumerating the md sampled bits. A dense construction checks each entry's original XOR constraints in $O(pd(2k)^{4p})$ bit-level label work and evaluates the required matrix powers in $O(p(2k)^{6p})$ arithmetic, with $O((2k)^{4p})$ storage. Integer bit growth and binomial evaluation in m are additional. At logarithmic moment order this dependence on k is generally expensive; nilpotence alone is not a useful uniform norm bound.

15.2 A finite hierarchy stable under further paired names

Assume now that S is Sidon. Write $j = p - h$ for the deficit from the top sample-count term. A relation pattern $\mathcal{B}$ has b ordered active vertices and t ordered parity subsets $\Gamma_1, \dots, \Gamma_t$. Retain the actual injective joint count

$$\Psi_{\mathcal{B}}(S) = \#\{(v_1, \dots, v_b) \in S^b : v_i \text{ distinct, } \bigoplus_{i \in \Gamma_a} v_i = 0 \ (1 \leq a \leq t)\}.$$

The empty pattern has count one.

Theorem 15.1 (Bounded-deficit coefficient descriptor). *For $p \geq 1$ and $0 \leq j \leq p - 1$, there are universal nonnegative integers $C_{p,j,\mathcal{B},v}$ such that*

$$a_{p-j}(S, p) = \sum_{\mathcal{B}} \sum_{v=b}^{2p} C_{p,j,\mathcal{B},v} (k-b)_{v-b} \Psi_{\mathcal{B}}(S). \quad (232)$$

It suffices to retain $b \leq 6j$, $t \leq 2j$, each nonempty parity row of even size at least six, and each active vertex in a positive even number of rows. Too-large injective assignments contribute zero. The same pattern family works at every p after arbitrary further pair-used sample names are added.

Proof. If sample name a has multiplicity $n_a \geq 2$, then $\sum_a(n_a - 2) = 2j$. At most $2j$ names have $n_a \geq 3$, and their total number E of edge positions is at most $6j$. Record the exact equality partition of the cyclic column positions, with v distinct classes. A pair-used name satisfies its XOR constraint precisely when its two undirected edges coincide, by Sidon uniqueness. It imposes no further constraint once this abstract check passes.

For each exceptional name record its odd-incidence classes. A nonempty subset of size two or four is rejected. The full cycle has even incidence at every class, and the ordinary paired edges contribute even incidence. Each active class therefore occurs in at least two exceptional parity rows. There are at most $2E$ exceptional endpoint incidences, giving $b \leq E \leq 6j$. Order active classes by first occurrence and exceptional names by their named order. Their labelings number exactly $\Psi_{\mathcal{B}}$; all remaining classes have $(k - b)_{v-b}$ injective completions. Counting the abstract column partitions and named edge assignments producing each record proves (232). Adding a pair-used name increases p, h equally, so the same bounds on b, t remain valid. $\square$

For deficits through J the number of pattern types is at most $T_J = (6J + 1)(2J + 1)2^{12J^2}$. Direct label work is at most $O(T_J(1 + J)^2 k^{6J} d)$, distinct from the number of types. Universal kernels can be precomputed in $O((J + 1)p^2(2p^2)^{2p})$ arithmetic; evaluation costs $O((J + 1)pT_J)$ once they and the descriptor are available. This is polynomial in k for fixed J , but not uniformly inexpensive in growing J, p . Opening the cyclic trace, exposing an inactive named vertex, or inserting arbitrary frame matrices can invalidate the deletion of inactive labels. The theorem concerns this Walsh-generated coefficient array, not arbitrary arrays with the same covariance.

15.3 The first two corrections retain joint relations

Let W be a real symmetric Gaussian matrix with zero diagonal and independent standard-normal upper-triangular entries. The top coefficient is exactly $a_p(S, p) = p! \mathbb{E} \operatorname{tr} W^{2p}$: the Sidon covariance on each pair-used name is the Gaussian pairing covariance, with all multiplication orders retained. Write $S_{\text{free}} = \{e_1, \dots, e_k\} \subset \mathbb{F}_2^k$, whose original rows have independent sign coordinates. For the first correction, $p \geq 2$, if $A_p^{(1)}(k) = a_{p-1}(S_{\text{free}}, p)$, the exact dependence is

$$a_{p-1}(S, p) = A_p^{(1)}(k) + B_p(k)R_6(S),$$

where R_6 counts unordered six-element zero-XOR subsets and the two polynomials have degrees at most $p + 2$ and $p - 3$ for $p \geq 3$; $B_2 = 0$. For $p \geq 3$ and $k \geq (p + 2)(p + 1)$, the proved quantitative bound is

$$0 \leq \frac{a_{p-1}(S, p) - A_p^{(1)}(k)}{A_p^{(1)}(k)} \leq \frac{80p^6 100^{p-3}}{k}. \quad (233)$$

The combinatorial reason is that the only exceptional multiplicities are (4) or (3, 3): a sole nonempty boundary is impossible, and two triple boundaries must be the same six-relation. The quantitative Gaussian contraction proof is given in the first-correction note.

At the second deficit, $p \geq 3$, the exact descriptor is the eleven-entry list

$$(k, R_6, R_8, Q_0, Q_1, Q_2, Q_3, P_{330}, P_{321}, P_{222}, P_{1,221}). \quad (234)$$

Here Q_ℓ counts unordered pairs of distinct six-relations intersecting in ℓ labels. For four distinct six-relations whose symmetric difference is empty, every label occurs in two or four of them. Let a count the fourfold labels; the six pair-membership cells have equal sizes on opposite pairs, with their three sizes x, y, z read up to permutation. The possible types are $a = 0$, $(x, y, z) = (3, 3, 0), (3, 2, 1), (2, 2, 2)$, or $a = 1$ and $(x, y, z) = (2, 2, 1)$. The four P entries count their unordered quadruples.

Classification proof outline. The exceptional multiplicities are

$$(6), \quad (5, 3), \quad (4, 4), \quad (4, 3, 3), \quad (3, 3, 3, 3).$$

Global parity forces the boundary rows to sum to zero. A triple has either empty boundary or a six-relation; a four-used row permits six or eight. Distinct six-relations intersect in at most three labels, since a larger intersection would make their difference a forbidden two- or four-relation. Applying these restrictions to the five sectors gives exactly (234). Grouping abstract equality partitions as in Theorem 15.1 then expresses the complete a_{p-2} as a linear combination of these statistics with nonnegative falling-factorial kernels. The full membership-cell classification and normalization counts are in the second-correction note. Direct support preprocessing has $O(k^8 \log k)$ label operations and $O(k^8)$ records, with the bit cost of the d -bit labels additional; the universal moment-order precomputation is still exponential in $O(p \log(p+1))$.

Unsigned relation weights alone fail even to determine a complete moment. Consider the following encoded supports in $\mathbb{F}_2^9$:

$$\begin{aligned} S_A &= (299, 1, 2, 460, 4, 8, 496, 16, 32, 64, 128, 256), \\ S_B &= (109, 1, 398, 2, 4, 8, 496, 16, 32, 64, 128, 256). \end{aligned}$$

Both have $k = 12$, affine rank eight, linear rank nine, even-relation weight enumerator $1+4u^6+3u^8$, and the same positive-flat scalar law. But (Q_2, Q_3) are $(3, 3)$ and $(6, 0)$. The exact coefficient counts are

$$a_3(S_A, 5) = 594609275160, \quad a_3(S_B, 5) = 594586413720,$$

and the complete difference for every integer $m \geq 1$ is

$$\mathbb{E} \operatorname{tr} H_m(S_A)^{10} - \mathbb{E} \operatorname{tr} H_m(S_B)^{10} = 22861440 \binom{m}{3} / m^{10}. \quad (235)$$

These are exact finite counts using the original cyclic-column/onto-name rule defining (231); their enumeration and the equality of all other coefficients are documented in the companion proof and check suite. They refute an exact scalar-weight-enumerator reduction, not a constant-factor norm theorem. Shared relation incidence is the missing data.

15.4 Ordered integrations that preserve second-moment gains

Lemma 15.2 (One repeated Hermitian input). *If Z is Hermitian, $\|Z\| \leq L$ almost surely and $\mathbb{E}Z^2 \preceq \nu I$, then for $t \geq 2$ and deterministic compatible square matrices B_j ,*

$$\|\mathbb{E}[ZB_1Z \cdots B_{t-1}Z]\| \leq \nu L^{t-2} \prod_j \|B_j\|. \quad (236)$$

The same constant holds for $Z \otimes I_D$ with arbitrary intervening matrices on the amplified space. Centering is not needed.

Proof. For unit vectors u, w , bound all interior copies of Z by L , preserving the two endpoints. The remaining absolute bilinear form is at most

$$L^{t-2} \prod_j \|B_j\| \mathbb{E}(\|Zu\| \|Zw\|) \leq L^{t-2} \prod_j \|B_j\| \sqrt{u^*(\mathbb{E}Z^2)u} \sqrt{w^*(\mathbb{E}Z^2)w}.$$

Take the supremum over u, w . Amplification preserves the two hypotheses. No coefficient has been commuted past another one. $\square$

Lemma 15.3 (Two triple-used inputs in every interleaving). *Let independent Hermitian A, B satisfy $\|A\| \leq L_A$, $\|B\| \leq L_B$, $\mathbb{E}A^2 \preceq v_A I$ and $\mathbb{E}B^2 \preceq v_B I$. In a six-slot word use each input three times, in any order, with deterministic $D_1, \dots, D_5$ between slots. Then, also under arbitrary tensor amplification,*

$$\|\mathbb{E}[Z_1 D_1 Z_2 \cdots D_5 Z_6]\| \leq v_A v_B L_A L_B \prod_j \|D_j\|. \quad (237)$$

Proof. A three-slot product F using A twice and B once has both $\mathbb{E}F^*F$ and $\mathbb{E}FF^*$ bounded by $v_A v_B L_A^2 \prod \|D_j\|^2 I$. To verify the right second moment, suppress the deterministic norm factors. If $F = APAQB$, then

$$F^*F = BQ^*AP^*A^2PAQB;$$

bound its central A^2 by $L_A^2 I$, then the neighboring deterministic squares by their norms, and integrate the remaining A^2 and B^2 using independence. For $F = APBQA$, the square is $AQ^*BP^*A^2PBQA$; the same central removal leaves B^2 , then A^2 . For $F = BPAQA$, first integrate the central B^2 in F^*F , then bound one remaining A^2 by $L_A^2 I$ and integrate the other. Every replacement is inside a positive sandwich. Reversing the words and taking adjoints proves the left bound. Interchanging A, B gives the other mixed multiplicity.

Split the six slots after the third, into F, D_3, G . If both halves are mixed, one has two A slots and the other two B slots. Matrix Cauchy–Schwarz, directly from scalar Cauchy–Schwarz on F^*u and Gw , gives

$$\|\mathbb{E}[FD_3G]\| \leq \|D_3\| \sqrt{\|\mathbb{E}FF^*\| \|\mathbb{E}G^*G\|}.$$

The two half-product bounds give (237); the halves need not be independent. If a half is homogeneous, both are, and independence separates their expectations. Lemma 15.2 gives the two factors $v_A L_A$, $v_B L_B$. These cases exhaust every order. The same proof works after amplification. $\square$

For an independent sign row $x \in \{-1, 1\}^k$, its error $Z = xx^T - I$ has $v = L = k - 1$. Its automatic empty-boundary tensor is exactly the corresponding empty-boundary Walsh channel, not a replacement for the entire Walsh law. An integrated t -used row costs $(k - 1)^{t-1}$; two triple-used automatic rows cost $(k - 1)^4$. In a cyclic word a retained $A_{ab} = e_a e_b^T + e_b e_a^T$ outside the integrated interval has trace norm two and operator norm one, removing a dimension factor in the trace bound. Other random inputs independent of the integrated rows can be conditioned upon before applying these lemmas.

15.5 The complete second correction at logarithmic order

Theorem 15.4 (Quantitative second-deficit contraction). *Let $A_p^{(2)}(k) = a_{p-2}(S_{\text{free}}, p)$ be the complete independent-sign baseline. For every Sidon support, $p \geq 4$, $k \geq (p + 3)(p + 2)$,*

$$0 \leq \frac{a_{p-2}(S, p) - A_p^{(2)}(k)}{A_p^{(2)}(k)} \leq \frac{200p^{12}100^{p-4}}{k}. \quad (238)$$

At $p = 3$ the correction is zero. In particular the relative geometry correction vanishes uniformly for $p \leq c \log k$ whenever $c < 1/\log 100$ is fixed. This controls one complete sample-count coefficient, not the sum of all deficits or the full norm tail.

Quantitative contraction and finite classification outline. Let

$$\begin{aligned} N_{53} &= \frac{(2p)!}{5!3!(2p-8)!}, & N_{44} &= \frac{(2p)!}{(4!)^2(2p-8)!}, \\ N_{433} &= \frac{(2p)!}{4!(3!)^2(2p-10)!}, & N_{3333} &= \frac{(2p)!}{(3!)^4(2p-12)!}. \end{aligned}$$

Impossible position counts and their summands are omitted. Gaussian pairings absorb all ordinary pair-used names. For each exceptional-position choice, joint parity conservation, the two integration lemmas and one retained trace-norm anchor give the following upper bounds before the sample-name factor $(p - 2)!$:

multiplicities	per-position geometry bound	equal-multiplicity divisor
(6)	0	1
(5, 3)	$(14175/2)k^6\mathbb{E}\ W\ ^{2p-8}$	1
(4, 4)	$7110k^6\mathbb{E}\ W\ ^{2p-8}$	2
(4, 3, 3)	$78750k^7\mathbb{E}\ W\ ^{2p-10}$	2
(3, 3, 3, 3)	$3000000k^8\mathbb{E}\ W\ ^{2p-12}$	24

Here $\mathbb{E}\|W\|^0 = 1$. To specify the finite counting input, a fixed six-boundary admits 90 ordered three-edge matchings, an eight-boundary 2520 ordered four-edge matchings, and a six-boundary $1080(k - 6) + 1440 \leq 1080k$ ordered four-edge tuples. An ordered five-edge tuple with fixed six-boundary has at most $14175k^2$ choices: mark the six odd endpoints in $(10)_6$ ways, pair the remaining four endpoints in three ways and divide by the 2^5 edge orientations. Sidon pair completion gives

$$R_6 \leq k^4/360, \quad R_8 \leq k^6/20160, \quad Q_2 \leq k^6/576, \quad Q_3 \leq k^5/108.$$

The first two nonzero sectors follow immediately from these counts. For $(4, 3, 3)$ the channels are a shared six-relation with one automatic row, or two six-relations overlapping in two or three labels. Integrating the automatic row before norms bounds the three contributions by $45k^7$, $1080k^7$, and $(141750 + 324000)k^6$, times the Gaussian norm moment, at most $78750k^7$ for $k \geq 6$. For four triples, the possibilities are two equal nonempty boundaries; three with empty sum and one automatic row; four equal boundaries; two repeated pairs; or four distinct boundaries with empty sum. The two-row lemma handles the first possibility with two interleaved automatic rows. The last case uses the joint bounds

$$P_{330} \leq k^6/1296, \quad P_{321} \leq k^6/72, \quad P_{222} \leq k^6/192, \quad P_{1;221} \leq k^5/16.$$

Together the five possibilities give the last table entry. The detailed membership-cell pivot counts and normalization proof are in the second-deficit and Gaussian-contraction notes; the table is not inferred from low-order numerical tests.

For unit v , $v^T W v$ has Gaussian variance at most two. A $1/4$ -net of size 9^k gives

$$\mathbb{P}\{\|W\| > 4\sqrt{k \log 9 + \log 2 + u}\} \leq e^{-u}.$$

Integration yields $(\mathbb{E}\|W\|^{2p-2s})^{1/(2p-2s)} \leq 10\sqrt{k}$ for $s = 4, 5, 6$, $p > s$, under the displayed size hypothesis. A six-cycle with further paired leaf excursions gives the same complete free baseline $A_p^{(2)}(k) \geq (p - 2)!(k)_{p+3} \geq (p - 2)!k^{p+3}/2$. Restoring sample names and dividing the table by this baseline gives

$$\frac{1}{k} \left[14175N_{53}100^{p-4} + 7110N_{44}100^{p-4} + 78750N_{433}100^{p-5} + 250000N_{3333}100^{p-6} \right].$$

After division by 100^{p-4} its numerator is at most $8200p^8 + (2800/3)p^{10} + (6400/81)p^{12} \leq 200p^{12}$ for $p \geq 4$. Nonnegativity follows from the original entry count. For $p = 3$, the sole sample name has automatic total cycle boundary. $\square$

Adding further paired names changes the Gaussian order while leaving the exceptional relation family fixed. This is a proved extension rule in that direction. Arbitrarily many interleaved automatic names or larger connected relation components are not covered by the one- and two-row lemmas.

15.6 Why a fixed number of corrections cannot finish the norm theorem

There is a decisive obstruction already on the independent-character support $S = \{e_1, \dots, e_k\}$, at the linear sample count $m = k$. Write $T_p(k) = \mathbb{E} \operatorname{tr} H_k^{2p}$ and retain only deficits $0 \leq j \leq J < p$:

$$M_J(k, p) = k^{-2p} \sum_{j=0}^J \binom{k}{p-j} a_{p-j}(S, p).$$

Define the no-singleton noncrossing partition counts

$$N(n, h) = \frac{1}{n+1} \binom{n+1}{h} \binom{n-h-1}{h-1}, \quad \mu_p = \sum_{h=1}^p N(2p, h), \quad B_p = (2p)^{4p}.$$

For $k \geq 2p$, the complete and truncated counts obey

$$(1 - 2p^2/k)\mu_p \leq T_p(k)/k \leq \mu_p + B_p/k, \quad (239)$$

$$M_J(k, p)/k \leq (J+1)4^p(p+J)^{2J} + B_p/k.$$

For p divisible by three and $k \geq \max\{4p^2, (2p)^{4p}\}$,

$$\frac{M_J(k, p)}{T_p(k)} \leq 8(J+1)(2p+1)^2(4p/3+1)(p+J)^{2J}(4/9)^p. \quad (240)$$

For every fixed J this tends to zero at suitable growing orders of size $\log k / \log \log k$. It is a relative trace-moment obstruction, not an unbounded norm-scale ratio after taking $2p$ -th roots.

Counting proof outline. Original entry expansion gives connected bipartite walks with h sample classes and v column classes, with every edge used evenly. Hence $h + v \leq 2p + 1$. At equality the graph is a tree, each edge is used twice, and its contour gives a unique noncrossing sample partition with no singleton. For a partition permutation σ and cyclic successor τ , column classes are cycles of $\tau\sigma^{-1}$. Their exact contribution is $k^{-2p} \sum_h N(2p, h) (k)_h (k)_{2p+1-h}$. Its normalization lies between $1 - 2p^2/k$ and one. At most $(2p)^{4p}$ lower-degree pairs of partitions remain, each costing at most $1/k$ after division by k . The partition count follows from $F = 1 + w \sum_{q \geq 2} z^q F^q$ and coefficient extraction in $U = zF$. Comparison with Catalan numbers gives $N(2p, p-j) \leq 4^p(p+j)^{2j}$; using the central term $h = 2p/3$ gives (240), with the explicit binomial estimates detailed in the truncation-obstruction note.

Retaining these coefficients together with a valid remainder is legitimate. On any Sidon support and at general sample count m , define $M_J(m, p) = m^{-2p} \sum_{j=0}^J \binom{m}{p-j} a_{p-j}(S, p)$, $0 \leq J < p$, and $R_J(m, p) = \mathbb{E} \operatorname{tr} H_m^{2p} - M_J(m, p) \geq 0$. The bounded-deficit note proves for $m \geq \max(2p, 2k)$ the coarse ratio $R_J(m, p)/M_0(m, p) \leq 2(4p^2)^p(k/m)^{J+1}$. Its order dependence is too expensive for the desired confidence range without further work. The next section's complete-moment transfer and the nonbacktracking baseline offer two ways of retaining the full sum. Neither permits discarding uncontrolled overlapping relations on an arbitrary support.

16 Complete moment transfer with general Parseval geometry

The coefficient-contraction problem admits a restricted complete answer when all short random relations can be kept inside a controlled boundary. This section transfers the entire trace moment, including every sample-count coefficient, to a local parity-code model. It then proves upper moments and a separate lower certificate for the same original compressed matrix. The construction permits noncommuting coefficients and supports whose ordinary even relation girth is only six. The finite horizon and the cost of locating the boundary remain explicit.

Limited independence, moment matching, and code-based covariance ensembles are established methods; see, for example, Soloveychik and Tarokh [15]. No historical novelty claim is made for those principles. The transfer horizons, quantitative compression bounds, and obstructions below are derived directly.

16.1 Hard and relative relation horizons

Fix distinct frequencies $S = (s_1, \dots, s_k) \subseteq \mathbb{F}_2^d$ and independent uniform original rows $z_1, \dots, z_m \in \mathbb{F}_2^d$. Write

$$x(z) = (\chi_{s_i}(z))_{i=1}^k, \quad H_S = \frac{1}{m} \sum_{a=1}^m x(z_a)x(z_a)^T - I_k, \quad H_{S,U} = U^T H_S U, \quad (241)$$

where $U \in \mathbb{R}^{k \times n}$ is fixed before sampling and $U^T U = I_n$. Thus n denotes the compressed matrix dimension and d the binary ambient dimension. The norm of $H_{S,U}$ includes the whole coefficient sphere of this one fixed space; its rows need not have equal leverage or commute.

View subsets of $[k]$ as binary vectors, with symmetric difference as addition, and define the even relation code

$$\mathcal{D} = \left\{ A \subseteq [k] : |A| \text{ is even, } \bigoplus_{i \in A} s_i = 0 \right\}.$$

It is the kernel of the augmented columns $(1, s_i)$. Its minimum nonzero weight is the ordinary even relation girth $g(S)$; an empty minimum is ∞ . Odd relations, including a zero frequency, are allowed. Distinctness excludes weight two, and $g(S) > 4$ is the Sidon condition.

For a predetermined boundary $B \subseteq [k]$, set

$$\begin{aligned} b &= |B|, & \mathcal{D}_B &= \{A \in \mathcal{D} : A \subseteq B\}, & r_B &= \dim_{\mathbb{F}_2} \mathcal{D}_B, \\ g_B &= \min\{|A| : A \in \mathcal{D}, A \not\subseteq B\}, & L_B &= U_B^T U_B, & \beta &= \|L_B\| \leq 1. \end{aligned} \quad (242)$$

The inequality $g_B > h$ means that every even relation of size at most h is contained in B . It does not separate or disregard overlapping relations inside that boundary.

Let ε_B be uniform on the sign vectors satisfying all parity checks in $\mathcal{D}_B$, and let ε_{B^c} have independent Rademacher entries, independently of the boundary. Independently repeat this whole hybrid row across samples, and put

$$\tilde{\xi} = U_B^T \varepsilon_B + U_{B^c}^T \varepsilon_{B^c}, \quad \tilde{H}_U = \frac{1}{m} \sum_{a=1}^m \tilde{\xi}_a \tilde{\xi}_a^T - I_n. \quad (243)$$

The exact comparator retains $\mathcal{D}_B$ and the *entire signed matrix* U , including the bulk rows. The covariance used later is an upper-bound certificate, not a replacement for this exact data.

Theorem 16.1 (Complete finite-horizon transfer). *If $g_B > 2L$, expectations of every polynomial of degree at most L in the entries of H_S agree with the corresponding hybrid law. In particular, for every integer $p \geq 1$ and every $m \geq 1$,*

$$g_B > 4p \implies \mathbb{E} \operatorname{tr} H_{S,U}^{2p} = \mathbb{E} \operatorname{tr} \tilde{H}_U^{2p}. \quad (244)$$

For the uncompressed cyclic trace, $g_B > 2p$ suffices. Taking $B = \emptyset$ gives the independent-sign comparator, with $g_B = g(S)$.

Proof. Multiply an original sign row by an independent common sign. Its outer product is unchanged pointwise. The augmented columns show that the resulting sign vector is uniform on $\mathcal{D}^\perp$. Its monomial expectation on a parity subset A is $1_{\{A \in \mathcal{D}\}}$; the hybrid expectation is $1_{\{A \in \mathcal{D}_B\}}$. Under $g_B > 2L$ these rules agree on every subset of size at most $2L$.

An entry monomial of degree at most L has at most $2L$ frequency occurrences. Within each sample name, its parity subset is even and no larger. Independence of different sample names proves equality of every ordered summand, with its actual deterministic coefficient. Compression is a polynomial substitution, proving (244). For the uncompressed trace, a closed column walk of length $2p$ has at most $2p$ named columns. Every sample boundary lies in those columns, which gives the improved cyclic horizon. General coefficient insertions do not preserve that column identification. $\square$

These are equalities of complete moments, not of a fixed collection of leading coefficients. They permit arbitrary fixed ordered interactions as long as the total entry degree stays inside the stated horizon. They do not give pointwise equality of the original and hybrid matrices.

The strict endpoints carry information. If $g(S) \geq 2p$, $p \geq 2$, and $R_{2p}(S)$ counts unordered zero-XOR subsets of size $2p$, then

$$\mathbb{E} \operatorname{tr} H_S^{2p} - \mathbb{E} \operatorname{tr} H_{\text{ind}}^{2p} = \frac{2(2p)! R_{2p}(S) \binom{m}{2}}{m^{2p}}. \quad (245)$$

Indeed a new surviving boundary must contain all $2p$ columns, so the closed walk visits a relation once in a Hamiltonian cycle. A sample name with odd incidence at every vertex uses one alternating perfect matching. The remaining matching must use one other sample name, since splitting it would produce a smaller nonempty even boundary. There are $(2p)!$ ordered column lists, two alternating assignments, and $\binom{m}{2}$ sample pairs. This accounts for every newly surviving term.

Compression has a different endpoint. For

$$S_8 = \{0, e_1, \dots, e_6, e_1 + \dots + e_6\} \subseteq \mathbb{F}_2^6, \quad U = (I_8 - \tfrac{1}{4} \mathbf{1}\mathbf{1}^T)_{[:,1:2]},$$

we have $g(S_8) = 8$ and $U^T U = I_2$, but at $m = 1$

$$\mathbb{E} \operatorname{tr} H_{S_8, U}^4 = \frac{109}{2}, \quad \mathbb{E} \operatorname{tr} H_{\text{ind}, U}^4 = \frac{1609}{32}. \quad (246)$$

Thus $g > 4p$ cannot be replaced by $g \geq 4p$ for arbitrary compression. An exact finite-sum verification is included below with the signed counterexample.

16.2 A boundary certificate stable through the whole moment

Let h_B be the minimum nonzero weight in $\mathcal{D}_B$, with value ∞ if $r_B = 0$. Define

$$\theta_B = \begin{cases} 1, & r_B = 0, \\ \min\{b, 8(r_B + 1), 2^{2r_B/h_B}\}, & r_B > 0, \end{cases} \quad (247)$$

$$K = I_n + (\theta_B - 1)L_B, \quad \kappa = \|K\| = 1 + (\theta_B - 1)\beta \geq 1.$$

A certified positive lower bound for h_B may be used instead of its exact value; if none is supplied, omit the third alternative. The other two alternatives need only code rank and frame covariance.

Lemma 16.2 (All-order scalar moment domination). *The hybrid row is isotropic and, for every $v \in \mathbb{R}^n$ and $j \geq 1$,*

$$\mathbb{E} \widetilde{\xi} \widetilde{\xi}^T = I_n, \quad \mathbb{E} |\langle v, \widetilde{\xi} \rangle|^{2j} \leq (2j - 1)!! (v^T K v)^j. \quad (248)$$

Proof. Write $Z = \langle v, U_B^T \varepsilon_B \rangle$ and $a^2 = v^T L_B v$. First, $Z^2 \leq ba^2$ pointwise. This gives the first alternative in (247); when $r_B = 0$, the usual even-index pairing expansion for independent signs gives the sharper factor one.

For the rank alternative, the boundary law is the independent-sign law conditioned on r_B independent parity checks, an event of probability 2^{-r_B} . Gaussian integration and $\mathbb{E}e^{tZ} \leq e^{t^2 a^2/2}$ under independent signs give $\mathbb{E}_{\text{ind}} e^{Z^2/(4a^2)} \leq \sqrt{2}$ for $a > 0$. Conditioning followed by concavity of $x^{1/(r_B+1)}$ yields

$$\mathbb{E} \exp\left(\frac{Z^2}{4(r_B+1)a^2}\right) \leq (2^{r_B}\sqrt{2})^{1/(r_B+1)} \leq 2.$$

Consequently

$$\mathbb{E}Z^{2j} \leq 2[4(r_B+1)a^2]^j j! \leq (2j-1)!![8(r_B+1)a^2]^j.$$

The case $a = 0$ is immediate.

For the local-girth alternative, all boundary scalar moments of degree $2j < h_B$ agree exactly with independent signs. At $2j \geq h_B$, positivity of Z^{2j} and the conditioning probability give

$$\mathbb{E}Z^{2j} \leq 2^{r_B} \mathbb{E}_{\text{ind}} Z^{2j} \leq (2j-1)!! a^{2j} 2^{r_B} \leq (2j-1)!! [2^{2r_B/h_B} a^2]^j.$$

The same bound holds at lower orders by their exact equality. Each alternative is valid at every order, so their minimum is legitimate. No factor 2^{mr_B} is paid.

Every codeword has even weight, so the boundary law is centrally symmetric. The independent bulk is symmetric and has Gaussian even moment domination with variance $v^T(I - L_B)v$. Expanding the even moments of their independent sum leaves only nonnegative products of even moments. Comparing term by term with two independent Gaussians proves (248). Finally, distinctness of frequencies excludes weight-two checks. The boundary covariance is therefore the identity, as is the covariance of the whole compressed row. $\square$

For example, $b\beta \leq C$ gives $\kappa \leq 1 + C$ regardless of the local relations. Bounded local code rank gives a boundary-size-independent cost, and a single six-relation gives $\theta_B \leq 2^{1/3}$. These are sufficient envelopes, not assertions that κ is the optimal parameter when it grows.

A richer optional certificate is a positive semidefinite Q_B such that

$$Q_B \succeq (U_B^T \varepsilon)(U_B^T \varepsilon)^T \quad \text{for all allowed boundary signs } \varepsilon.$$

The same deterministic proof works with $K = I - L_B + Q_B$. Averaging shows $Q_B \succeq L_B$, so $K \succeq I$. Checking a supplied certificate involves 2^{b-r_B} rank-one comparisons; the closed-form certificate (247) avoids this atom enumeration.

16.3 Two-sided complete norm moments and a separate confidence bound

Theorem 16.3 (A restricted two-sided compressed theorem). *Let $2 \leq n \leq k$, $p \geq 1$ be an integer, and $g_B > 4p$. Put $a_{n,m} = \sqrt{n/m} + n/m$. For every $m \geq 1$,*

$$\frac{1}{4}a_{n,m} \leq \| \|H_{S,U}\| \|_{L^{2p}} \leq 48\kappa n^{1/(2p)} \left[\sqrt{\frac{n+p}{m}} + \frac{n+p}{m} \right]. \quad (249)$$

If additionally $\max\{2, \lceil \log n \rceil\} \leq p \leq n$, then

$$\frac{1}{4}a_{n,m} \leq \| \|H_{S,U}\| \|_{L^{2p}} \leq 160\kappa a_{n,m}, \quad (250)$$

$$\mathbb{P}\{\|H_{S,U}\| > a_{n,m}/8\} \geq \frac{1}{800\kappa^4}, \quad (251)$$

$$\frac{a_{n,m}}{6400\kappa^4} \leq \mathbb{E}\|H_{S,U}\| \leq 160\kappa a_{n,m}. \quad (252)$$

For $0 < \delta \leq 1$, a separate upper confidence consequence is

$$m \geq 65536\kappa^2\delta^{-2}(n+p) \implies \mathbb{P}\{\|H_{S,U}\| > \delta\} \leq n16^{-p}. \quad (253)$$

Upper moments and confidence. For a unit v , set $Z = \langle v, \tilde{\xi} \rangle^2$. Then $\mathbb{E}Z = 1$ and $\mathbb{E}Z^j \leq (2j-1)!!\kappa^j$. Since $(2j-1)!!/j! \leq 2^j$,

$$\log \mathbb{E}e^{\lambda(Z-1)} \leq \frac{4\kappa^2\lambda^2}{1-2\kappa\lambda} \quad (0 \leq \lambda < 1/(2\kappa)).$$

For the negative sign, $e^{-u} \leq 1 - u + u^2/2$ and $\mathbb{E}Z^2 \leq 3\kappa^2$ give a bound $3\kappa^2\lambda^2/2$. Both signs are thus bounded by $8\kappa^2\lambda^2$ for $0 \leq \lambda \leq 1/(4\kappa)$. Exponential Markov with $\lambda = t/(16\kappa^2)$ when $t \leq 4\kappa$, and $\lambda = 1/(4\kappa)$ otherwise, yields

$$\mathbb{P}\{|v^T \tilde{H}_U v| > t\} \leq 2 \exp \left[-m \min \left\{ \frac{t^2}{32\kappa^2}, \frac{t}{8\kappa} \right\} \right]. \quad (254)$$

A maximal $1/4$ -separated subset is a $1/4$ -net of the unit sphere; disjoint radius- $1/8$ balls give cardinality at most 9^n by volume. For a unit vector and a net point within $1/4$, the two quadratic forms differ by at most $\|A\|/2$. Hence $\|A\| \leq 2 \max_v |v^T A v|$ on that net for symmetric A . Writing $A_n = n \log 9 + \log 2$, the resulting norm tail at

$$8\sqrt{2}\kappa\sqrt{(A_n + u)/m} + 16\kappa(A_n + u)/m$$

is at most e^{-u} . Comparison with a mean-one exponential variable E , for which $\|E\|_{L^q} \leq q$ at positive integer q , gives

$$\|\tilde{H}_U\|_{L^{2p}} \leq 8\sqrt{2}\kappa\sqrt{(A_n + p)/m} + 16\kappa(A_n + 2p)/m. \quad (255)$$

Here the square-root term uses the L^p moment and the linear term the L^{2p} moment. Since $A_n \leq 3n$, this is at most $48\kappa[\sqrt{(n+p)/m} + (n+p)/m]$.

Transfer the complete trace moment by Theorem 16.1 and use $\|H\|^{2p} \leq \text{tr } H^{2p} \leq n\|H\|^{2p}$. This proves the upper part of (249). Under $\log n \leq p \leq n$, $n^{1/(2p)} < 2$, $A_n + p \leq 4n$, and $A_n + 2p \leq 5n$ in (255), proving the upper part of (250). Markov before taking the trace-to-norm root gives

$$\mathbb{P}\{\|H_{S,U}\| > \delta\} \leq n \left[\frac{48\kappa}{\delta} \left(\sqrt{\frac{n+p}{m}} + \frac{n+p}{m} \right) \right]^{2p}.$$

Under (253) its bracket is at most $48/256 + 48/65536 < 1/4$, proving the stated failure bound. $\square$

Lower bounds on the original input. The original compressed row $\xi = U^T x$ is isotropic. For $V = \xi\xi^T - I_n$, Jensen gives

$$\mu_2 := \mathbb{E}\|V\|_F^2 = \mathbb{E}\|\xi\|^4 - n \geq n(n-1), \quad \mathbb{E} \text{tr } H_{S,U}^2 = \mu_2/m.$$

In the hard-girth case $g(S) > 4$, the transferred pairing calculation retains the sharper frame identity

$$\mathbb{E} \text{tr } H_{S,U}^2 = \frac{n^2 + n - 2 \sum_{i=1}^k \ell_i^2}{m}, \quad \ell_i = \|u_i\|^2, \quad \sum_i \ell_i = n.$$

Here u_i^T is the i th row of U . Expanding $\varepsilon^T U U^T \varepsilon$ leaves only paired indices, giving $\mathbb{E}\|\xi\|^4 = n^2 + 2n - 2 \sum_i \ell_i^2$. For the general boundary, the preceding isotropic lower bound suffices. Therefore $\|\|H_{S,U}\|\|_{L^{2p}} \geq \sqrt{(n-1)/m}$. The empirical Gram $G = I_n + H_{S,U}$ has rank at most m and expected trace n , so $\mathbb{E}\|H_{S,U}\| \geq n/m - 1$. For $m \geq n/2$, use the square-root bound and $n/m \leq \sqrt{2}\sqrt{n/m}$. For $m < n/2$, use $n/m - 1 \geq n/(2m)$ and $\sqrt{n/m} \leq n/(\sqrt{2}m)$. Both cases give the lower part of (249).

For a probability certificate, $p \geq 2$ and $g_B > 4p$ transfer all row moments through degree eight. Integrating (248) at $j = 4$ over an independent standard Gaussian direction proves

$$\mathbb{E}\|\xi\|^8 \leq \kappa^4 n(n+2)(n+4)(n+6) \leq 24\kappa^4 n^4.$$

Consequently

$$\mu_4 := \mathbb{E}\|V\|_F^4 \leq 49\kappa^4 n^4, \quad \mu_4/\mu_2^2 \leq 196\kappa^4.$$

For independent centered vectors in a real Hilbert space, direct expansion and Cauchy–Schwarz give

$$\mathbb{E} \left\| \sum_{a=1}^m V_a \right\|_F^4 \leq m\mu_4 + 3m(m-1)\mu_2^2.$$

Indeed the centered cross terms vanish, and each surviving squared pair inner product is at most μ_2^2 . Thus, for $Q = \text{tr } H_{S,U}^2/n$, we have $\mathbb{E}Q^2/(\mathbb{E}Q)^2 \leq 199\kappa^4$. Cauchy–Schwarz on $\{Q > \mathbb{E}Q/2\}$, or the elementary Paley–Zygmund inequality, gives probability at least $1/(796\kappa^4)$ that

$$\|H_{S,U}\| > \sqrt{(n-1)/(2m)} \geq \frac{1}{2}\sqrt{n/m}.$$

If $m \geq n/4$, this exceeds $a_{n,m}/8$.

If $m < n/4$, set $T = \text{tr } G$. Isotropy and the fourth row moment give $\mathbb{E}T = n$ and $\mathbb{E}T^2 \leq n^2 + \mathbb{E}\|\xi\|^4/m \leq 3\kappa^2 n^2$. Thus $\mathbb{P}\{T > n/2\} \geq 1/(12\kappa^2)$. On this event, rank gives $\|H_{S,U}\| > n/(2m) - 1 > n/(4m) > a_{n,m}/8$. This proves (251) in every sample regime. Multiplying its threshold by its probability proves the lower expectation bound; the upper expectation follows from (250). $\square$

Bounded κ therefore gives matching complete norm moments and expectations. The exclusion $n = 1$ is necessary: a one-coordinate compression can have deterministic Gram error zero. The lower certificate in (251) is for the same original matrix; it is not a matching extreme lower quantile. To obtain upper failure at most η from (253), one may take $p \geq \lceil \log(n/\eta)/\log 16 \rceil$, provided $g_B > 4p$. Nothing here extends that confidence assertion beyond the certified horizon or chooses U after seeing the sample.

16.4 Explicit low-girth supports and signed obstructions

The horizon condition has explicit constructions with much smaller ambient dimension than an affine-independent support. In a fixed binary representation of $\mathbb{F}_{2^r}$, set

$$T_{r,t} = \{(x, x^3, \dots, x^{2t-1}) : x \in \mathbb{F}_{2^r}\} \subseteq \mathbb{F}_2^{rt}. \quad (256)$$

Its labels are distinct and $g(T_{r,t}) > 2t$. To prove this, suppose an even number $\ell \leq 2t$ of distinct field elements formed a relation. The odd power sums through $2t-1$ vanish. Frobenius implies that every power sum of degree $1, \dots, 2t-1$ vanishes, and the zeroth sum is zero because ℓ is even. The square Vandermonde matrix with powers $0, \dots, \ell-1$ would annihilate the all-ones vector, contradicting its nonzero determinant. Thus $t = 2p$ supplies the compressed horizon $g > 4p$ in ambient dimension $2pr$, with all signed Parseval geometry still present.

Short relations can also remain in the original support. For fixed distinct $A = (a_1, \dots, a_b) \subseteq \mathbb{F}_2^e$ and $T = T_{r,2p}$, put

$$S = \{(1, a_i, 0) : 1 \leq i \leq b\} \cup \{(0, 0, t) : t \in T\}, \quad B = \{1, \dots, b\}. \quad (257)$$

The tag forces every even relation to use an even number of boundary labels and hence an even number of bulk labels. The two XOR equations then separate. Every relation not contained in B contains a nonempty even relation from T , so $g_B > 4p$.

For the concrete choice $A = \{0, 1, 2, 4, 8, 15\} \subseteq \mathbb{F}_2^4$, its only nonempty even relation consists of all six labels. The tagged support is Sidon and satisfies

$$\begin{aligned} k &= 2^r + 6, & d &= 5 + 2pr, & g(S) &= 6, & r_B &= 1, & h_B &= 6, \\ \kappa &= 1 + (\theta_B - 1)\beta \leq 1 + (2^{1/3} - 1)\beta \leq 2^{1/3}. \end{aligned} \quad (258)$$

Theorem 16.3 therefore holds for every predetermined real Parseval compression in its stated dimension and moment range, although the ordinary hard-girth condition fails at every compressed order $p \geq 2$. The ambient dimension is $O(\log^2 k)$ when $p = O(\log k)$; this dimension claim does not apply to arbitrary growing p . The global relation code may have large dimension, but no global conditioning density is paid.

The need to preserve signed coefficients is visible in the small support

$$S = (0, 1, 2, 4, 8, 15, 16, 32) \subseteq \mathbb{F}_2^6, \quad B = \{1, \dots, 6\}. \quad (259)$$

It has only the boundary six-relation, so $g_B = \infty$. Let U be the first two columns of $I_8 - \mathbf{1}\mathbf{1}^T/4$ and let U' change only the sign of its first row. Both are dense Parseval matrices with identical individual row outer products, L_B , bulk covariance, and boundary code. Nevertheless, at $m = 1$,

$$\mathbb{E} \operatorname{tr} H_{S,U}^4 = \frac{1759}{32}, \quad \mathbb{E} \operatorname{tr} H_{S,U'}^4 = \frac{1459}{32}, \quad \mathbb{E} \operatorname{tr} H_{\text{ind},U}^4 = \frac{1609}{32}. \quad (260)$$

The corrections have opposite signs, $\pm 75/16$. This disproves the exact reduction that discards coefficient-row signs while retaining their outer products and the local code. It does not disprove their common upper envelope in Theorem 16.3.

Here is a reproducible finite arithmetic proof of both this example and (246). Under independent signs, write the first two signs as $a, c \in \{-1, 1\}$, let j of the next four signs be negative and let ℓ of the final two signs be negative. Set

$$w = a + c + 6 - 2j - 2\ell, \quad F(a, c, j, \ell) = \left[(a - w/4)^2 + (c - w/4)^2 - 1 \right]^4 + 1.$$

The independent expectation is

$$\frac{1}{256} \sum_{a,c=\pm 1} \sum_{j=0}^4 \sum_{\ell=0}^2 \binom{4}{j} \binom{2}{\ell} F(a, c, j, \ell) = \frac{1609}{32}.$$

Restricting the same sum to $ac(-1)^j = +1$ or -1 and replacing $1/256$ by $1/128$ gives respectively $1759/32$ and $1459/32$. These are precisely the positive and negative six-parity conditional laws: a sign flip in the first coefficient row interchanges them. Restricting instead to $ac(-1)^{j+\ell} = +1$ gives $109/2$, the eight-parity law in (246). Common-sign augmentation changes none of the row outer products, so these are computations for the original Walsh matrices. Actual compressed rows include $(-1, -1)$ and $(0, -2)$; their rank-one matrices do not commute.

16.5 What is tractable, and what remains model-specific

For a supplied boundary, Gaussian elimination on its augmented frequency columns computes $\mathcal{D}_B$ and r_B in polynomial work in b, d . Computing L_B takes $O(bn^2)$ real arithmetic, followed by an eigenvalue calculation. The first two alternatives in (247) enumerate neither moment words nor boundary atoms. The optional local-girth alternative can be certified by enumerating 2^{r_B} codewords in $O(b2^{r_B})$ elementary bit operations, or by a structural proof such as (257). Numerical precision costs are additional.

The hard-girth condition has a real dimension cost. If $g(S) > 2t$, all sums of at most t augmented columns are distinct, since two equal sums would create a nonempty even relation of size at most $2t$. Therefore

$$\sum_{j=0}^{\min(t,k)} \binom{k}{j} \leq 2^{d+1}.$$

A generic direct certificate can enumerate and compare those subset sums, at exponential cost when t grows with $\log k$. For a relative horizon, direct enumeration of subsets of at most $4p$ not contained in B is likewise expensive. Finding a small or low-leverage boundary for an arbitrary support remains a separate problem. The tagged construction supplies that boundary and its horizon by proof, without hiding a generic efficient search.

The random-polynomial degree is at most d : a Gram entry is a character indexed by a frequency difference and hence a monomial in at most d independent row signs. Compression preserves this bound. The field exponent in (256), the moment order p , and this random degree are different quantities.

Within a certified horizon, extension by fixed ordered interactions preserves the exact local-code comparator. A newly exposed short relation may force the boundary to grow; no bound on that growth has been proved for arbitrary inputs. Bounded κ gives a restricted two-sided theorem, but sharp dependence on unbounded κ , matching extreme lower quantiles, and simultaneous control of all Walsh supports are not consequences of it. In particular, full FR-10 and a general SparseStack contraction valid under unrestricted extension remain open.

17 Conditional registers and a relation calculus without depth loss

The complete compressed theorem of Section 16 localizes short relations in one boundary. Two extensions make that mechanism reusable. First, an actual affine constraint has an exact signed Fourier update, so fresh constraints, reused constraints, and later ordered products can be distinguished before taking norms. Second, a supplied tree of small relation interfaces permits an all-order contraction with no multiplier for its depth. The latter gives two-sided bounds for growing-dimensional Parseval covariance matrices and a class of positive matrix polynomials. Its tractability comes from a checkable conditional symmetry, not from polynomial degree alone.

17.1 All independent names may interleave

The earlier one-row and two-triangle contractions extend to any number of independent names. This resolves the integration of specified automatic channels, while leaving the sum of nonempty relation channels to be controlled.

Theorem 17.1 (Constant-one register contraction). *Let $Z_1, \dots, Z_s$ be mutually independent Hermitian $n \times n$ random matrices, with $\|Z_i\| \leq L_i$ almost surely and $\mathbb{E}Z_i^2 \preceq v_i I_n$. For a written word $(i_1, \dots, i_N)$ in which name i occurs $t_i \geq 2$ times, and deterministic insertions D_j ,*

$$\|\mathbb{E}[D_0 Z_{i_1} D_1 \cdots Z_{i_N} D_N]\| \leq \prod_{j=0}^N \|D_j\| \prod_{i=1}^s v_i L_i^{t_i-2}. \quad (261)$$

The same bound holds under arbitrary common tensor amplification and arbitrary deterministic insertions on the amplified space. Centering is unnecessary. A name occurring exactly twice needs only its second-moment bound.

Proof. If A is the set of currently open names, retain the actual Hilbert space $\mathcal{H}_A = L^2(\prod_{i \in A} \Omega_i; \mathbb{C}^n)$ with the product probability law. Process the factors from right to left,

their order of action on a vector. The rightmost occurrence of i creates its register by $C_i f(\omega_A, \omega_i) = Z_i(\omega_i) f(\omega_A)$, whose norm is at most $\sqrt{v_i}$. Intermediate occurrences multiply by the same retained $Z_i(\omega_i)$, at cost L_i . The leftmost occurrence closes the register by $C_i^* g = \int Z_i(\omega_i) g d\mu_i$, again at cost $\sqrt{v_i}$. All other open registers merely amplify these maps. Deterministic matrices act pointwise in their original positions. After every processed suffix the function equals that suffix with exactly the closed variables integrated; Fubini and mutual independence prove this invariant. The final composition is therefore the original expectation, including crossing occurrence intervals. Multiplying its map norms proves the claim. $\square$

For different, possibly non-Hermitian functions at different occurrences of one name, the corresponding endpoint factors are

$$\alpha_i = \|\mathbb{E} X_{i,L} X_{i,L}^*\|^{1/2}, \quad \beta_i = \|\mathbb{E} X_{i,R}^* X_{i,R}\|^{1/2}. \quad (262)$$

Replace v_i by $\alpha_i \beta_i$ in (261), and charge the essential supremum of each interior occurrence. The proof uses the same creation and integration maps, which need not be adjoints of one another. A singleton is replaced by its exact mean; a centered singleton kills the whole expectation.

Interleaving is free in this certificate, but identifying names is not. Merging two identically distributed independent names of multiplicities $t, u \geq 2$ changes $v^2 L^{t+u-4}$ into $v L^{t+u-2}$, a ratio L^2/v when $v > 0$. Scalar Bernoulli variables attain this loss and the constant one in the theorem. Mutual independence also cannot be weakened to pairwise independence. For independent signs $\varepsilon_1, \varepsilon_2$, the three Bernoulli variables

$$A = (1 + \varepsilon_1)/2, \quad B = (1 + \varepsilon_2)/2, \quad C = (1 + \varepsilon_1 \varepsilon_2)/2$$

have the same one- and two-variable laws as independent Bernoulli variables, but $\mathbb{E} A^2 B^2 C^2 = 1/4 > 1/8$.

For an independent-sign row error $Z = x x^T - I_k$, $\|Z\| = k - 1$ and $\mathbb{E} Z^2 = (k - 1) I_k$. Thus s independent automatic names occupying N positions cost $(k - 1)^{N-s}$, with no additional penalty for their crossings. A cyclic trace can retain an insertion C at cost $\|C\|_1$ by trace duality. This observation contracts specified automatic channels; it does not bound the number, signs, or geometry of the remaining relation terms. The same second-moment identities also hold for the complete uncompressed row error of any distinct-frequency Walsh support, by isotropy and $\|x\|^2 = k$; summing the resulting generic word certificates is still too coarse for unrestricted FR-10.

The schedule and scalar certificate require $O(N + s)$ arithmetic work. With supplied rational scalar bounds of at most B bits, their unreduced product has $O(NB)$ bits and costs $O(N^2 B^2)$ schoolbook bit operations. Obtaining those bounds is separate work. Exact register evaluation can be much larger: if each name has at most q atoms and at most w_{open} names are live, applying the exact composition to one vector costs $O(N n^2 q^{w_{\text{open}}})$ arithmetic operations and stores $O(n q^{w_{\text{open}}})$ entries. Computing the whole expected matrix adds a factor n in time. The cheap scalar contraction and the exact operator representation must therefore be distinguished.

17.2 Innovation in the actual affine constraint Gram

Let y be uniform on $\mathbb{F}_2^q$ and condition on the consistent state $\mathcal{E} = \{y : Cy = b\}$. Retain its rowspace R , rank r , origin y_0 , and a linear section of $\mathbb{F}_2^q/R$. The probability of this state is 2^{-r} . For an ordered matrix Fourier polynomial $F(y) = \sum_t A_t (-1)^{t \cdot y}$, define the signed quotient coefficients

$$B_\gamma = \sum_{t+R=\gamma} (-1)^{t \cdot y_0} A_t. \quad (263)$$

These coefficients, the affine signs and the original primitive names are part of the retained state. Quotient characters on $\ker C$ are orthonormal under the actual conditional law, so

$$\mathbb{E}[F \mid \mathcal{E}] = B_0, \quad M_R := \mathbb{E}[F^* F \mid \mathcal{E}] = \sum_{\gamma} B_{\gamma}^* B_{\gamma}, \quad M_L := \mathbb{E}[F F^* \mid \mathcal{E}] = \sum_{\gamma} B_{\gamma} B_{\gamma}^*. \quad (264)$$

This is an identity of matrix sums, before a norm or absolute value is taken.

Append the equation $c \cdot y = d$, put $h = c + R$ and $\eta = (-1)^{d+c \cdot y_0}$. If $h = 0$, the equation is reused when $\eta = 1$ and contradictory when $\eta = -1$. Its conditional probability is respectively one or zero. If $h \neq 0$, it is fresh, has probability $1/2$, and pairs γ with $\gamma + h$. For one representative per pair write $A_{\gamma} = B_{\gamma}$, $D_{\gamma} = B_{\gamma+h}$.

Lemma 17.2 (Signed constraint update). *The new coefficient on a pair is $A_{\gamma} + \eta D_{\gamma}$, followed by the sign adjustment for the newly chosen origin and quotient section. In particular,*

$$\begin{aligned} M_{R,\eta} &= M_R + \eta \sum_{\gamma} (A_{\gamma}^* D_{\gamma} + D_{\gamma}^* A_{\gamma}), \\ M_{L,\eta} &= M_L + \eta \sum_{\gamma} (A_{\gamma} D_{\gamma}^* + D_{\gamma} A_{\gamma}^*), \\ 0 &\preceq M_{R,\eta} \preceq 2M_R, \quad 0 \preceq M_{L,\eta} \preceq 2M_L. \end{aligned} \quad (265)$$

Proof. On the branch, $\chi_h = \eta$, so paired characters have the stated sum. Expand its left and right squared products. The positive inequality $(A + \eta D)^*(A + \eta D) \preceq 2(A^* A + D^* D)$ proves the bound, and similarly on the left. The average of the two branch matrices is the old matrix. $\square$

The factor two is normalized conditional mass, not a guessed collision penalty. For fresh c , the actual Gram of $(1, \chi_c)$ is I_2 and its Schur innovation is one. For reused c it is $\begin{pmatrix} 1 & \mu \\ \mu & 1 \end{pmatrix}$ with $\mu = \pm 1$, and its innovation is zero. Multiplication by χ_h is a self-adjoint unitary T_h in $L^2(\mathcal{E})$; the branch projections are $P_{\eta} = (I + \eta T_h)/2$. Their normalized squared norm is $2\|P_{\eta} f\|_2^2$, whereas their unnormalized norm is $\|P_{\eta} f\|_2^2$. Keeping only individual equation characters without their multiplicative span would lose later reuse.

For a Parseval frame $\sum_i u_i u_i^* = I_n$, distinct original frequencies s_i , and $\xi = \sum_i u_i \chi_{s_i}$, retain the coherent vectors $a_{\gamma} = \sum_{s_i+R=\gamma} (-1)^{s_i \cdot y_0} u_i$. Then

$$K_{\mathcal{E}} = \mathbb{E}[\xi \xi^* \mid \mathcal{E}] = \sum_{\gamma} a_{\gamma} a_{\gamma}^* \preceq m_R I_n, \quad m_R \leq \min(k, 2^r), \quad (266)$$

where m_R is the largest number of original frequencies in a quotient coset. Cauchy-Schwarz in each coherent sum, followed by the original Parseval identity, proves the bound. For $Y = \xi \xi^* - K_{\mathcal{E}}$, the pointwise bound $\|\xi\|^2 \leq k$ gives

$$\mathbb{E}[Y^2 \mid \mathcal{E}] \preceq k K_{\mathcal{E}} - K_{\mathcal{E}}^2, \quad \|Y\| \leq k.$$

Its exact second moment can instead be computed from the signed Fourier coefficients of $\xi \xi^*$, namely $\sum_{\gamma+\rho=\tau} a_{\gamma} a_{\rho}^*$. Recentring after a new condition is an explicit change of the zero coefficient; the fresh-branch comparison applies directly to a frozen expression.

Equations (264) and (262) now give conditional endpoint norms for Theorem 17.1. Conditions within independent primitive blocks preserve independence. The unnormalized expectation pays the actual joint mass $2^{-\sum r_i}$. One fresh equation inside an existing name increases its endpoint product by at most two; its conditional interior supremum cannot increase. A constraint crossing formerly independent names requires merging the connected primitive blocks and recalculating their joint endpoint state. The separate-name gain is not inherited after that merge. For buckets of size $b = 2^c$, an equality between two disconnected components adds exactly c binary equations, while an equality inside a component adds none. A triangle consequently has weight b^{-2} , including when it is part of a later overlapping equality calculation.

Exact closure and charged size. Addition aggregates equal quotient channels; adjoint adjoints every coefficient; ordered multiplication uses

$$(FG)_\tau = \sum_{\gamma+\rho=\tau} B_\gamma D_\rho. \quad (267)$$

Primitive identification substitutes the declared linear maps in frequencies and equations, retaining affine phases and merging affected independent blocks. These updates have zero mathematical error. If a block has q coordinates and T nonzero $n \times n$ coefficients, whose quotient frequencies span rank w , then $T \leq 2^w$ and its state uses $O(q^2 + T(q + n^2))$ fields. A constraint update costs $O((q + T)q^2)$ binary operations and $O(Tn^2)$ coefficient operations; (267) costs $O(T_1 T_2 n^3)$ arithmetic and $O(T_1 T_2 q)$ binary operations. Endpoint formation costs $O(Tn^3)$ arithmetic. Exact interior optimization may enumerate 2^w values; a supplied upper bound or $\sum_\gamma \|B_\gamma\|$ avoids claiming that optimization is cheap.

For common-denominator coefficient lengths B_1, B_2 , a product needs at most $B_1 + B_2 + \lceil \log_2(T_1 T_2 n + 1) \rceil + 3$ bits per entry. A word of ℓ supplied inputs with at most t terms each and length B therefore has $O(\ell[B + \log(tn)])$ -bit coefficients. Schoolbook arithmetic charges the square of the actual intermediate length per operation. An n -dimensional rational PSD upper certificate can be checked by fraction-free elimination in $O(n^3)$ arithmetic and $O(n^5(B_{\max} + \log(n + 1))^2)$ bit operations, using determinant bounds on intermediate minors. For total written input size N , logarithmic $w = O(\log(n + N))$ and polynomially many updates give polynomial arithmetic work. Polynomial bit work additionally requires polynomial actual $B_{\max}$, for example through a polynomial bound on unreduced input-leaf count. Repeated squaring of the constant 2 has random degree zero and one Fourier channel, but doubles its bit length at every step.

17.3 Covariance and pairwise laws do not survive all continuations

The affine phase discarded by a covariance-only reduction can change a later product on the same original matrix. Let

$$U = \frac{1}{13} \begin{pmatrix} 11 & -4 \\ -4 & 5 \\ -4 & -8 \\ -4 & -8 \end{pmatrix}, \quad x = (1, \varepsilon_1, \varepsilon_2, \varepsilon_3)^T, \quad \xi = U^T x, \quad B = \xi \xi^T, \quad \mathcal{E}_\pm = \{\varepsilon_1 \varepsilon_2 \varepsilon_3 = \pm 1\}.$$

The columns of U are the first two columns of the Householder reflection $I_4 - 2vv^T/13$, $v = (1, 2, 2, 2)^T$, so $U^T U = I_2$. Both branches have rank one, mass $1/2$, the same rowspace, and $\mathbb{E}[B \mid \mathcal{E}_\pm] = I_2$. However, their four equiprobable squared row lengths are

$$\mathcal{E}_+ : (306, 306, 514, 226)/169, \quad \mathcal{E}_- : (578, 674, 50, 50)/169. \quad (268)$$

These follow by substituting the four sign triples of each parity. As B is positive rank one, $\|B^\ell\| = \|\xi\|^{2\ell}$. Thus, for integer $\ell \geq 1$ and real $p \geq 1$,

$$\frac{(\mathbb{E}[\|B^\ell\|^{2p} \mid \mathcal{E}_-])^{1/(2p)}}{(\mathbb{E}[\|B^\ell\|^{2p} \mid \mathcal{E}_+])^{1/(2p)}} \geq 4^{-1/(2p)} (337/257)^\ell. \quad (269)$$

The numerator retains its largest atom and the denominator is bounded by its maximum. For $0 < \eta < 1/4$, the strict quantiles are respectively $(674/169)^\ell$ and $(514/169)^\ell$; at $\eta = 1/4$ they become $(578/169)^\ell$ and $(306/169)^\ell$. These are conditional laws of the same input, not moment-based assertions about its unconditional quantile. The example is noncommuting: its individual triple traces include $-2332800/4826809$ on the plus branch and $-56448/4826809$ on the minus branch. Accordingly no positivity of individual ordered trace coefficients is used. The obstruction

identifies a necessary exponential continuation loss when affine phase is discarded; it does not exclude a theorem explicitly allowing that degree loss.

A stronger zero-versus-nonzero obstruction concerns pairwise gluing. On the six-label support $(0, 1, 2, 4, 8, 15) \subset \mathbb{F}_2^4$, write $\eta_1 = x_1x_2 = \chi_1$, $\eta_2 = x_3x_4 = \chi_6$, $\eta_3 = x_5x_6 = \chi_7$. Every pair is independent but $\eta_1\eta_2\eta_3 = 1$ pointwise. The Parseval rows

$$\frac{e_1}{\sqrt{3}}, \frac{e_2}{\sqrt{3}}, \frac{e_1 + e_2}{\sqrt{6}}, \frac{e_1 - e_2}{\sqrt{6}}, \frac{e_1}{\sqrt{3}}, \frac{e_2}{\sqrt{3}}$$

give actual edge coefficients $A_{12} = X/3$, $A_{34} = Z_0/3$, $A_{56} = X/3$, where $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z_0 = \text{diag}(1, -1)$. Since $XZ_0X = -Z_0$, the ordered output

$$P = Z_0 + 27(\eta_1 A_{12})(\eta_2 A_{34})(\eta_3 A_{56}) = (1 - \eta_1\eta_2\eta_3)Z_0$$

is identically zero. Independently resampling the three edge signs preserves their complete one-piece and two-piece laws and every signed coefficient, but gives $P = 0$ or $2Z_0$ with equal probability. No multiplicative two-sided norm characterization can depend only on those retained data. Valid conditional gluing instead retains a shared live variable V and uses $\mathbb{E}[FG \mid V] = \mathbb{E}[F \mid V]\mathbb{E}[G \mid V]$ only when their private variables are conditionally independent given V , keeping the matrix order.

17.4 Retained subcodes and relative horizons

For the distinct Walsh support and fixed signed real Parseval matrix in (241), let $\mathcal{D}$ be its even relation code. Retain a supplied subcode $\mathcal{C} \subseteq \mathcal{D}$ and define

$$g(\mathcal{C}) = \min\{|A| : A \in \mathcal{D} \setminus \mathcal{C}\}, \quad \min \emptyset = \infty.$$

The comparator uses a uniform sign row on $\mathcal{C}^\perp$, independently for each original sample name. All sums of retained generators remain in $\mathcal{C}$; a union of relations in different components is not discarded because it meets several components. The proof of Theorem 16.1 then gives

$$g(\mathcal{C}) > 2L \implies \text{complete transfer of Gram-entry polynomials of degree at most } L. \quad (270)$$

In particular $g(\mathcal{C}) > 4p$ transfers the full compressed trace $\mathbb{E} \text{tr } H^{2p}$.

For a general ordered circuit P , keep each original sample name and its feature occurrence bound $h_a(P)$. Addition takes the maximum of profiles, ordered multiplication adds them, and adjoint preserves them. The trace word $(P^*P)^p$ has profile at most $2ph_a(P)$. Thus the sufficient condition is $g(\mathcal{C}) > 2ph_a(P)$ for every original sample, provided the circuit is even in each original row. Arbitrary raw-feature circuits require the full, possibly odd, kernel of their frequency map. The exact criterion is agreement of full and retained character expectations on every reachable parity tuple; for arbitrary coefficient arrays this is also necessary, by placing a single nonzero coefficient on any failed tuple. Necessity need not hold for one particular frame circuit with cancellations. All these claims follow by expanding (267) and evaluating the actual characters, with no norms taken. A later primitive identification changes the joint kernel and requires a fresh horizon certificate; it need not preserve the iid observations or distinct-frequency isotropy needed for norm bounds.

17.5 An exact tree of affine innovations

We now give a structural condition that turns retained overlap information into an all-order analytic bound. Consider a uniform nonempty affine law

$$y \in \{y \in \mathbb{F}_2^k : Ay = b\}, \quad \varepsilon_i = (-1)^{y_i}. \quad (271)$$

Assume the rowspace of A has no word of weight one or two. Then the sign row is centered and isotropic, irrespective of its affine phases. Let $U \in \mathbb{R}^{k \times n}$, $U^T U = I_n$, be predetermined, $n \geq 2$. Supply a rooted tree of bags V_t , of size at most w , covering the coordinates and all displayed checks. Bags containing a fixed coordinate must form a connected subtree. Each check is assigned to one containing bag. Put $S_t = V_t \cap V_{\text{parent}(t)}$ and $B_t = V_t \setminus S_t$, with empty root separator. The owned sets B_t partition the coordinates.

Eliminate leaves upwards. Let $\mathcal{L}_t$ be the affine set on V_t defined by the checks assigned there and the support messages received from its children. Write

$$\mathcal{L}_t = \{(y_{B_t}, y_{S_t}) : D_{t,B} y_{B_t} + D_{t,S} y_{S_t} = e_t\}, \quad E_t = \text{rowspan } D_{t,B}, \quad r_t = \dim E_t.$$

The upward support message is $\text{proj}_{S_t} \mathcal{L}_t$; inconsistency is a zero state. If $b_t = |B_t|$, each nonempty private fiber has $2^{b_t - r_t}$ points. Every allowed separator assignment has exactly 2^{c_t} subtree completions, where

$$c_t = b_t - r_t + \sum_{u \text{ child of } t} c_u. \quad (272)$$

To prove this, use induction: child subtrees overlap only in the already specified separator coordinates, so their constant completion counts multiply. Summing over the local private fiber gives (272). The original uniform law can consequently be generated root first, choosing each private block uniformly in this affine fiber. Conditioning on all previously generated coordinates in any parent-before-child traversal gives exactly that same local law.

The required analytic condition is

$$E_t \text{ is an even code for every } t, \quad \text{equivalently } D_{t,B} \mathbf{1}_{B_t} = 0. \quad (273)$$

It concerns the eliminated kernel, including descendant constraints, not only the initially displayed checks. Let h_t be the minimum nonzero weight in E_t , with $h_t = \infty$ at rank zero, and set

$$\theta_t = \begin{cases} 1, & r_t = 0, \\ \min\{b_t, 8(r_t + 1), 2^{2r_t/h_t}\}, & r_t > 0, \end{cases} \quad K = \sum_t \theta_t U_{B_t}^T U_{B_t}, \quad \kappa = \|K\|. \quad (274)$$

Empty blocks contribute zero. A certified lower bound on h_t suffices, or the third option may be omitted. Parseval gives $I_n \preceq K \preceq \max(1, \max_t \theta_t) I_n$.

Theorem 17.3 (No-depth-loss relation contraction). *Under (273), for every real vector v and integer $p \geq 1$,*

$$\mathbb{E}|\langle v, U^T \varepsilon \rangle|^{2p} \leq (2p - 1)!! (v^T K v)^p. \quad (275)$$

There is no multiplier depending on tree depth, number of checks, number of bags, or moment order beyond the displayed Gaussian moment.

Proof. On an affine coset with even check code of rank r , the scalar $Z = \sum_{i=1}^b a_i \varepsilon_i$ is symmetric. Put $a^2 = \sum_i a_i^2$. Cauchy–Schwarz gives $|Z|^2 \leq b a^2$; at rank zero, the independent-sign pairing expansion gives the Gaussian proxy a^2 . For the rank bound, Gaussian integration of the independent Rademacher estimate gives $\mathbb{E}_{\text{ind}} \exp(Z^2/(4a^2)) \leq \sqrt{2}$. Conditioning on an event of mass 2^{-r} and then using concavity to power $1/(r+1)$ gives $\mathbb{E} \exp(Z^2/[4(r+1)a^2]) \leq 2$. Its power series, together with $p! \leq (2p-1)!!$ and $2 \leq 2^p$, yields the proxy $8(r+1)a^2$. Below $2p < h$, the moment equals its independent value because no nonempty reachable character belongs to the code. Above that degree, positivity bounds it by 2^r times the independent moment, and $2^r \leq 2^{2pr/h}$. This proves the third proxy uniformly over the affine right side. The case $a = 0$ is immediate.

Enumerate the owned blocks in parent-before-child order and write $Z_t = \langle v, U_{B_t}^T \varepsilon_{B_t} \rangle$, $a_t^2 = \theta_t \|U_{B_t} v\|^2$. Conditional on the preceding blocks, the exact sampling factorization and (273) give

$$\mathbb{E}[Z_t^{2j+1} \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[Z_t^{2j} \mid \mathcal{F}_{t-1}] \leq (2j - 1)!! a_t^{2j}.$$

Expand $(Z_1 + \dots + Z_t)^{2p}$ conditionally. Only even powers of Z_t survive, and their companion powers of the earlier sum are even and nonnegative. Insert these deterministic conditional bounds and induct at all lower even orders. The resulting binomial sum is exactly the Gaussian moment with variance $\sum_{u \leq t} a_u^2$. At the final block this is (275). Independence of the block contributions was never asserted. $\square$

A single separator is a special case. Partition coordinates into Q , private blocks B_j , and free coordinates F . If the retained even code satisfies

$$\mathcal{C} = \mathcal{C}_Q + \sum_j \mathcal{C}_j, \quad \mathcal{C}_Q = \mathcal{C} \cap \mathbb{F}_2^Q, \quad \mathcal{C}_j = \mathcal{C} \cap \mathbb{F}_2^{Q \cup B_j}, \quad E_j = \text{proj}_{B_j} \mathcal{C}_j \text{ even},$$

the root is Q and the other bags are $Q \cup B_j$. An allowed separator assignment extends because two lifts of a private check differ by a pure separator check. Each private fiber has constant size and its actual affine phase is retained. Hence

$$K = \theta_Q U_Q^T U_Q + \sum_j \theta_j U_{B_j}^T U_{B_j} + U_F^T U_F. \quad (276)$$

Neither the number of components nor their total code rank appears in $\kappa \leq \max(1, \theta_Q, \max_j \theta_j)$.

17.6 Two-sided covariance and positive-polynomial consequences

Take m independent whole rows from (271), put $\xi_a = U^T \varepsilon_a$ and $H = m^{-1} \sum_a \xi_a \xi_a^T - I_n$. Alternatively take the original Walsh matrix (241) with a retained-subcode comparator satisfying Theorem 17.3. The three inputs to Theorem 16.3 are isotropy, all-order scalar domination, and full trace transfer. The proof there applies verbatim with the present κ : a single containing boundary was used only to supply those inputs. Writing $a_{n,m} = \sqrt{n/m} + n/m$, for integer $p \geq 1$,

$$\frac{1}{4} a_{n,m} \leq (\mathbb{E} \|H\|^{2p})^{1/(2p)} \leq 48\kappa n^{1/(2p)} \left[\sqrt{\frac{n+p}{m}} + \frac{n+p}{m} \right]. \quad (277)$$

The Walsh comparison requires $g(\mathcal{C}) > 4p$. For the exact affine law there is no horizon, and the upper factor $n^{1/(2p)}$ can be omitted because no trace transfer is needed. In the useful window $\max(2, \lceil \log n \rceil) \leq p \leq n$, the upper bound is $160\kappa a_{n,m}$, and the same input has the separate certificates

$$\mathbb{P}\{\|H\| > a_{n,m}/8\} \geq (800\kappa^4)^{-1}, \quad \frac{a_{n,m}}{6400\kappa^4} \leq \mathbb{E}\|H\| \leq 160\kappa a_{n,m}. \quad (278)$$

For $0 < \delta \leq 1$, the sufficient sample bound

$$m \geq 65536\kappa^2 \delta^{-2} (n+p) \implies \mathbb{P}\{\|H\| > \delta\} \leq n16^{-p} \quad (279)$$

uses the same horizon in the comparator case. Thus confidence η can use $p \geq \lceil \log(n/\eta) / \log 16 \rceil$ when that horizon is certified. The lower event in (278) is not a matching extreme quantile.

The analytic class also permits noncommuting coefficient processing. For fixed positive integer ℓ , let

$$P(H) = a_0 I_n + \sum_{j=1}^{\ell} a_j \Phi_j(H^{2j}), \quad q(t) = \sum_{j=0}^{\ell} a_j t^j, \quad a_j \geq 0, \quad a_{\ell} > 0, \quad (280)$$

where each predetermined map is positive, unital and trace-preserving. A concrete coefficient description is $\Phi_j(A) = \sum_r C_{j,r}^* A C_{j,r}$ with $\sum_r C_{j,r}^* C_{j,r} = \sum_r C_{j,r} C_{j,r}^* = I_n$. The complex Kraus coefficients and the different positive terms need not commute.

Theorem 17.4 (Positive-polynomial norm bounds). *Let $p, \ell \geq 1$ be integers and $R_p = 2\ell p$. In the Walsh comparison assume $g(\mathcal{C}) > 8\ell p$; in the exact affine model no horizon is needed. Put*

$$B_{R_p} = 48\kappa n^{1/(2R_p)} \left[\sqrt{(n + R_p)/m} + (n + R_p)/m \right].$$

Then

$$q((n-1)/m) \leq \mathbb{E}\|P(H)\| \leq (\mathbb{E}\|P(H)\|^{2p})^{1/(2p)} \leq q(B_{R_p}^2), \quad (281)$$

$$\mathbb{P}\{\|P(H)\| > q(n/(4m))\} \geq (800\kappa^4)^{-1}. \quad (282)$$

If $m \geq n$ and $\max(2, \lceil \log n \rceil) \leq 2\ell p \leq n$, these imply

$$q\left(\frac{n}{2m}\right) \leq \mathbb{E}\|P(H)\| \leq (\mathbb{E}\|P(H)\|^{2p})^{1/(2p)} \leq q\left(102400\kappa^2 \frac{n}{m}\right). \quad (283)$$

The ratio of the displayed upper and lower scales is at most $(204800\kappa^2)^\ell$.

Proof. Let $Q = \text{tr } H^2/n$. Positivity, unitality and trace preservation give the pointwise same-input bounds

$$q(Q) \leq \|P(H)\| \leq q(\|H\|^2). \quad (284)$$

For the upper inequality apply each positive unital map to $0 \preceq H^{2j} \preceq \|H\|^{2j} I_n$. For the lower inequality, a positive matrix has norm at least its normalized trace, and the power-mean inequality gives $\text{tr } H^{2j}/n \geq Q^j$. Trace preservation then proves the claim, regardless of commutation.

Write $V_a = \xi_a \xi_a^T - I_n$. Isotropy gives $\mathbb{E}Q = (\mathbb{E}\|\xi\|^4 - n)/(mn) \geq (n-1)/m$. Jensen and convexity of q prove the expectation lower bound. For the strict lower event, directional domination through degree eight gives $\mathbb{E}\|\xi\|^8 \leq 24\kappa^4 n^4$. With $\mu_2 = \mathbb{E}\|V\|_F^2 \geq n(n-1)$ and $\mu_4 = \mathbb{E}\|V\|_F^4 \leq 49\kappa^4 n^4$, the Hilbert-space fourth-moment expansion yields

$$\mathbb{E} \left\| \sum_a V_a \right\|_F^4 \leq m\mu_4 + 3m(m-1)\mu_2^2, \quad \frac{\mathbb{E}Q^2}{(\mathbb{E}Q)^2} \leq 199\kappa^4.$$

Paley–Zygmund at one half gives probability at least $1/(796\kappa^4)$ for $Q > \mathbb{E}Q/2 \geq n/(4m)$. Since $a_\ell > 0$, q is strictly increasing on positive arguments, proving (282).

The highest Gram-entry degree of $P(H)$ is 2ℓ ; its $2p$ -th trace power therefore transfers under (270) at horizon $8\ell p$. For the norm bound, Minkowski, monotonicity of scalar moments, and (277) at index R_p give

$$(\mathbb{E}\|P(H)\|^{2p})^{1/(2p)} \leq a_0 + \sum_j a_j (\mathbb{E}\|H\|^{4jp})^{1/(2p)} \leq q((\mathbb{E}\|H\|^{4\ell p})^{1/(2\ell p)}) \leq q(B_{R_p}^2).$$

In the stated window the covariance norm is at most $160\kappa(\sqrt{n/m} + n/m) \leq 320\kappa\sqrt{n/m}$. Squaring proves (283). Finally $q(Ct) \leq C^\ell q(t)$ for $C \geq 1$. $\square$

Markov gives failure at most e^{-2p} at threshold $e q(102400\kappa^2 n/m)$ in (283). This is an upper confidence statement; (282) remains a separate constant-mass lower certificate. The class is closed under appending nonnegative filtered even-power terms and under composition or convex mixing of normalized positive maps. Arbitrary products of its outputs need not remain positive, so only their exact signed channel representation, with charged horizon and size, is automatically closed.

The oversampling restriction has content. For the same original one-row independent-sign matrix $H = xx^T - I_n$, the identity map applied to H^2 has norm $(n-1)^2$, while the normalized trace map $A \mapsto (\text{tr } A/n)I_n$ gives norm $n-1$ pointwise. Both maps are unital and trace-preserving. Thus those normalizations alone do not retain enough rank-one spectral geometry for matching bounds when $m \ll n$.

17.7 Updates, useful costs, and growing noncommuting examples

Attaching a leaf bag requires its exact projection message, propagation to affected ancestors, and retesting (273). If each old allowed separator has a nonempty new fiber, the old marginal is unchanged. A new check contained in a bag is handled by affine elimination; a check crossing bags can be handled by coarsening a connected subtree into the union of its bags. Running intersection survives, but the new width and local proxies must be charged. New equality checks can destroy isotropy; the exact state still updates but (277) then requires a new justified normalization. Adding rows also does not preserve Parseval normalization implicitly.

In the single-separator setting, a new even check meeting private blocks B_j , $j \in I$, with even intersection with Q admits the merged private block $\bigcup_{j \in I} B_j$. Its projected rank is at most $1 + \sum_{j \in I} r_j$, its size is the sum of their sizes, and its covariance is exactly the sum of their covariances. If its intersection with Q is odd, a safe update moves all incident private blocks into the separator. Both updates retain all XOR successors, and explicitly recompute the cost.

For T bags, A_{count} displayed checks and width w , building the affine messages costs $O((T + A_{\text{count}})w^3)$ binary operations by incremental elimination. Storage is $O((T + A_{\text{count}})w^2 + Tw \log(k + 1))$ bits, with $O(\log(k + 1))$ bits per completion exponent. Exact local minimum distances cost at most $O(\sum_t b_t 2^{r_t})$ additional binary work, or can be replaced by the size/rank alternatives. Forming all coefficient covariances costs $O(kn^2)$ arithmetic operations; signed U uses $O(kn)$ scalars. A full retained code gives infinite horizon immediately. Otherwise generic horizon search through $\sum_{j \leq H} \binom{k}{j}$ subsets can be exponential and is not included in these polynomial costs.

With rational input length B , covariance entries have $O(k[B + \log(k + 1)])$ -bit common-denominator representations. Multiply arithmetic counts by the square of actual intermediate lengths for schoolbook bit bounds. Rational upper approximations $\theta_t \leq \hat{\theta}_t \leq (1 + \epsilon)\theta_t$ give $K \preceq \hat{K} \preceq (1 + \epsilon)K$: the extra covariance norm factor is $1 + \epsilon$, and the positive-polynomial factor is $(1 + \epsilon)^{2\ell}$. Integer size/rank options introduce no error. For R supplied Kraus matrices, normalization checks cost $O(Rn^3)$ arithmetic and storage is $O(Rn^2)$. If their common-denominator length is B_C , the coefficients a_j have length B_a , and a supplied H has length B_H , evaluating (280) costs

$$O((\ell + R)n^3 L^2) \text{ bit operations, } L = O(\ell(B_H + \log n) + B_C + B_a + \log(R + \ell + 1)).$$

Rational upper certificates for the roots in (277) are checked by powering; the oversampling envelope (283) is rational when κ has a rational upper certificate. At fixed w, ℓ and $p = O(\log(n/\eta))$, supplied structural and horizon certificates thus give polynomial work with all confidence and degree losses visible.

The tree also computes exact moments of a supplied scalar direction without assuming (273). For rational c_i , retain the truncated conditional generating polynomial

$$G_t(z; s) = \mathbb{E} \left[\exp \left(z \sum_{i \text{ owned below } t} c_i \varepsilon_i \right) \mid y_{S_t} = s \right] \pmod{z^{h+1}}.$$

Writing $s_t(y_B) = \sum_{i \in B_t} c_i (-1)^{y_i}$, the exact update is

$$G_t(z; s) = 2^{-(b_t - r_t)} \sum_{y_B: (y_B, s) \in \mathcal{L}_t} \left(\sum_{j=0}^h \frac{s_t(y_B)^j z^j}{j!} \right) \prod_{u \text{ child of } t} G_u(z; y_{S_u}) \pmod{z^{h+1}}. \quad (285)$$

The exact conditional factorization and $e^{z(A+B)} = e^{zA} e^{zB}$ prove the update, including odd signed terms. Root coefficients are $\mathbb{E}(\sum_i c_i \varepsilon_i)^j / j!$. Direct convolution costs $O(T2^w[h^2 + w + h])$ arithmetic, with $O(T2^w(h + 1))$ stored coefficients. If the c_i have a common denominator and length B_c , the output at order j has denominator dividing $2^k Q^j j!$ and length $O(k + h[B_c + \log(k + 1) + \log(h + 1)])$. These are directional moments; enumerating directions to compute a norm is not claimed affordable.

Example 17.5 (An arbitrarily long width-four chain). For $L \geq 3$, use independent signs $a_0, \dots, a_{L-1}$ and the named features

$$x = (a_0, a_1) \cup \bigcup_{i=2}^{L-1} (a_i, b_i), \quad b_i = a_i a_{i-1} a_{i-2}, \quad k = 2L - 2.$$

Their distinct Walsh labels are e_0, e_1 and $e_i, e_i + e_{i-1} + e_{i-2}$. All are odd-weight labels; their full even relation code has the $L - 2$ independent basis checks $a_i b_i a_{i-1} a_{i-2} = 1$. Independence follows from the unique b_i in each check, and dimension follows because all L basis labels occur. Thus no relative-horizon search is required. Use root bag $\{a_0, a_1\}$ and chain bags $\{a_{i-2}, a_{i-1}, a_i, b_i\}$. Each owned pair has a rank-one even conditional check, and every descendant always has two extensions. Hence

$$w = 4, \quad K = 2I_n - U_{\{a_0, a_1\}}^T U_{\{a_0, a_1\}}, \quad \kappa \leq 2 \quad (286)$$

for every predetermined signed real Parseval compression, with any $2 \leq n \leq k$. The depth and code rank grow, while the proxy does not. Gram entries have degree at most six in the original independent signs, and (280) has degree at most 12ℓ , independently of L and the number of samples.

For $L = 5$, take labels $(1, 2, 4, 7, 8, 14, 16, 28)$ and the first two columns of $I_8 - \mathbf{1}\mathbf{1}^T/4$. This dense rational Parseval frame has $K = \begin{pmatrix} 11 & 3 \\ 3 & 11 \end{pmatrix}/8$ and $\kappa = 7/4$. Its first two row outer products have commutator entry $-3/16$, demonstrating noncommuting coefficient geometry.

The star family $S_J = \{0, e_0\} \cup \bigcup_{j=1}^J \{e_j, e_0 + e_j\}$ has the same proxy $K = 2I - U_{\{0, e_0\}}^T U_{\{0, e_0\}}$. Its full even code has basis the J four-checks consisting of the root pair and one private pair. This includes arbitrarily many overlapping relations and all their XOR unions, yet $\kappa \leq 2$ for arbitrary predetermined signed Parseval U . Its original Gram entries have degree at most three, and its positive outputs degree at most 6ℓ . The chain shows that the mechanism is not confined to one retained root separator.

17.8 Why bounded overlap alone is insufficient

There is no general replacement of the supplied even-innovation structure by bounded check weight and bounded generator overlap. For $l \geq 2$, let $S = \{e_i + f_j : 1 \leq i, j \leq l\}$, $k = l^2$, so original features are $x_{ij} = \varepsilon_i \delta_j$. Its full relation code consists of subsets with even degree in every row and column. The adjacent plaquettes

$$\{(i, j), (i + 1, j), (i, j + 1), (i + 1, j + 1)\}, \quad i, j < l,$$

form a basis: inspection successively from the first row proves independence, and their number $(l - 1)^2$ equals $l^2 - (2l - 1)$, the kernel dimension. Every check has weight four, every coordinate lies in at most four checks, and every check meets at most eight other checks.

Nevertheless the flat unit direction gives $T = l^{-1}(\sum_i \varepsilon_i)(\sum_j \delta_j)$, with $\mathbb{E}T^2 = 1$. The event $|T| = l$ has mass 2^{2-2l} . Comparing its contribution to the $2l$ -th moment with $(2l - 1)!!\theta^l \leq (2l)^l \theta^l$ forces

$$\theta \geq l/8 = \sqrt{k}/8. \quad (287)$$

Already at order $2t$, $1 \leq t \leq l/2$, the terms using exactly t distinct signs twice give

$$\mathbb{E}T^{2t} \geq [(2t - 1)!!]^2 [(l)_t / l^t]^2.$$

Since $(l)_t / l^t \geq 2^{-t}$ and $(2t - 1)!! \geq t! \geq (t/e)^t$, a Gaussian proxy at this order must satisfy

$$\theta_t \geq t/12. \quad (288)$$

It therefore grows even at logarithmic moment order while all the displayed local generator parameters stay constant. For even $l \geq 4$, append the orthogonal coefficient column $u'_{ij} = a_j/l$, with half $a_j = 1$ and half $a_j = -1$, to the flat column. At $l = 4$ the original compressed rows include $(4, 0)$ and $(2, 2)$, whose outer products do not commute. The same flat direction proves the obstruction inside this fixed two-dimensional Parseval compression. This is an obstruction to the proposed scalar Gaussian proxy, not a lower bound on the covariance sampling threshold.

Finally, evenness must be verified after elimination. If independent root signs u, v determine a final singleton $z = uv$, then $\mathbb{E}[z \mid u, v] = uv$ and $\mathbb{E}uvz = 1$, despite pairwise independence of u, v, z . Its singleton private code is odd and fails (273). The exact messages and (285) remain valid; replacing that kernel by a centered innovation does not. The remaining model-specific work is thus well localized: construct affordable valid decompositions for more general supports, retain conditional drift when evenness fails, and control the signed sums of connected relation channels. Neither the exact register algebra nor the no-depth-loss theorem alone resolves those dense interactions or the all-support FR-10 problem.

18 Summing the baseline before estimating relations

The obstruction to a fixed number of sample-count corrections motivates a different organization: remove every tree excursion before expectation, then retain the exact additional Walsh constraints. Nonbacktracking covariance polynomials are a classical method of Sodin and Feldheim–Sodin [13, 14]; the finite estimates below are derived directly. No novelty is claimed for the underlying recurrence.

18.1 A deterministic recurrence on the original sign matrix

Let X be any real $m \times k$ sign matrix, $m \geq k \geq 3$, and put

$$M = X^T X, \quad H = M/m - I_k, \quad c = m + k - 2, \quad a = (m - 1)(k - 1).$$

Define matrix polynomials

$$\begin{aligned} P_0(M) &= I, & P_1(M) &= M - mI, \\ P_2(M) &= (M - cI)P_1(M) - m(k - 1)I, \\ P_q(M) &= (M - cI)P_{q-1}(M) - aP_{q-2}(M), \quad q \geq 3. \end{aligned} \tag{289}$$

Lemma 18.1 (Exact path contraction). *The entry (u, v) of $P_q(M)$ is the sum of signed weights of all nonbacktracking paths of length $2q$ from column u to column v in the complete bipartite graph with edge weight X_{ij} on row-column edge (i, j) . The nonbacktracking condition is imposed along the written path, not across the terminal cut of a closed path. In particular a root tail is permitted, but no nonempty closed tree-supported walk contributes to $\text{tr } P_q$.*

Proof. Set $\mathcal{A} = \begin{pmatrix} 0 & X^T \\ X & 0 \end{pmatrix}$ and $\mathcal{D} = \text{diag}(mI_k, kI_m)$. The signed nonbacktracking path matrices have $B_0 = I$, $B_1 = \mathcal{A}$, $B_2 = \mathcal{A}^2 - \mathcal{D}$, and

$$B_\ell = \mathcal{A}B_{\ell-1} - (\mathcal{D} - I)B_{\ell-2}, \quad \ell \geq 3.$$

Indeed prepend an edge. The invalid immediate reversals have unit squared weight, and every neighbor except the first outgoing neighbor of the retained remainder is available. The column block of B_2 is P_1 ; that of B_4 is $M^2 - (2m + k - 2)M + m(m - 1)I = P_2$. Eliminating odd path matrices from the two successive recurrences gives (289), with factors $m - 1$ and $k - 1$ on the two sides. Finally, any closed walk on a tree must immediately reverse at a visited vertex of maximal distance from its root. $\square$

This is an all-order pointwise closure, using two consecutive matrix states. The constant edge magnitudes and two fixed degrees are essential hypotheses. After general Parseval compression they need not hold; replacing k by the compressed dimension in (289) is unjustified.

For a predetermined Walsh support $S = \{s_1, \dots, s_k\} \subset \mathbb{F}_2^d$, let $X_{ij} = \chi_{s_j}(z_i)$ on independent uniform original rows z_i . For each closed path let A_i be the subset of column labels incident an odd number of times to row name i . The path expectation is exactly

$$\prod_i \mathbf{1}_{\{\oplus_{s \in A_i} s = 0\}}.$$

Each A_i has even cardinality. The independent-sign comparator survives precisely when all A_i are empty. Thus

$$\mathbb{E} \operatorname{tr} P_q(M_S) = \mathbb{E} \operatorname{tr} P_q(M_{\text{ind}}) + \Delta_q(S, m), \quad \Delta_q(S, m) \geq 0, \quad (290)$$

where Δ_q counts the same original named paths with all constraints satisfied and at least one nonempty boundary. Positivity is obtained after the actual Walsh expectation; individual coefficient words need not have positive traces.

Write $g(S)$ for the smallest nonempty even zero-XOR subset size, with value ∞ if none exists, and $R_q(S)$ for the number of q -element zero-XOR subsets. A closed length- $2q$ path uses at most q column labels, so $g(S) > q$ implies $\Delta_q = 0$. At the first even relation length $g(S) = q \geq 4$,

$$\Delta_q(S, m) = 2q! R_q(S) \binom{m}{2}. \quad (291)$$

To see this, a new path must use all q labels of one relation exactly once around its column cycle. At each row boundary either no cycle edges or the whole cycle boundary survives; hence exactly two row names occupy the two alternating matchings. There are $q!$ column orders and two row assignments for each chosen pair. No lower-degree polynomial term sees the relation. In particular Sidon supports have $\Delta_4 = 0$ and $\Delta_6 = 1440 R_6 \binom{m}{2}$.

18.2 A complete independent baseline at logarithmic order

Theorem 18.2 (Suppressed-path bound). *For even $q \geq 4$, $m \geq k \geq 3$, and $k \geq 2(2q)^{17}$,*

$$0 \leq a^{-q/2} \mathbb{E} \operatorname{tr} P_q(M_{\text{ind}}) \leq 6(2q)^{13}. \quad (292)$$

This includes every sample-name count, rather than a fixed-deficit truncation.

Proof. Consider a surviving path up to separate row and column renaming. Let e, v be the numbers of edges and vertices of its connected simple bipartite graph. Every edge has even multiplicity, so $e \leq q$. Every vertex except possibly the root has degree at least two, and the graph has a cycle, so $v \leq e$. Put $h = q - v$, $s = e - v$, $r = q - e$, all nonnegative with $s, r \leq h$.

Keep the root and all vertices of degree other than two, suppressing every other degree-two vertex into a chain. In the pure-cycle case keep its root, giving a loop. Parallel edges and loops in this reduced graph are allowed. Since $\sum_x (\deg(x) - 2) = 2s$ and only the root can have degree one,

$$V_0 \leq 2s + 2, \quad E_0 = V_0 + s \leq 3h + 2.$$

Each chain is traversed completely, in an even number $t_j \geq 2$ of passages. With chain lengths $\ell_j \geq 1$, $\sum \ell_j = e$ and $\sum \ell_j t_j = 2q$, the reduced written walk has length

$$L_0 = \sum_j t_j \leq 2E_0 + \sum_j \ell_j (t_j - 2) = 2E_0 + 2(q - e) \leq 8h + 4.$$

Label reduced vertices and edges in order of first appearance. Record the ordered endpoints of every edge, its chain length, and the oriented reduced walk. Expanding each chain with fresh internal vertices recovers the full equality pattern, including subsequent reverse traversals of that chain. The root color fixes bipartite colors. For fixed V_0, E_0, L_0 there are at most $(2q)^{3E_0+L_0}$ records; the three integer parameters each have at most $2q$ choices. Counting inconsistent records as well gives

$$N_{q,h} \leq (2q)^{17h+13}. \quad (293)$$

If v_C, v_R count column and row vertices, their degree sums give $2v_R \leq e$ and $2v_C - 1 \leq e$. Because q is even, both are at most $q/2$. Injective assignments number $(k)_{v_C}(m)_{v_R}$, at most

$$k^{v_C} m^{v_R} \leq (mk)^{q/2} k^{-h}, \quad m \geq k.$$

Each of the h missing vertices costs at least k , whichever color it has. Therefore

$$a^{-q/2} \mathbb{E} \operatorname{tr} P_q(M_{\text{ind}}) \leq \left(\frac{mk}{a}\right)^{q/2} (2q)^{13} \sum_{h \geq 0} \left(\frac{(2q)^{17}}{k}\right)^h.$$

The geometric sum is at most two, and the normalization is at most $(k/(k-1))^q \leq e^{q/(k-1)} < 3$. This proves the bound. $\square$

The leading $h = 0$ term is exactly $(q/2 - 1)(k)_{q/2}(m)_{q/2}$: the graph is a cycle of even length from four through q , with a possibly empty root tail, and each edge is used twice. This identity is consistent with the proof, but no limiting law is used. Crucially, (293) uses even multiplicity of every edge and does *not* count the extra Walsh paths in Δ_q .

18.3 A positive certificate for the full spectral edge

Let U_j denote second-kind Chebyshev polynomials and put $U_{-1} = 0$. Direct substitution in (289) gives, for $q \geq 1$,

$$\frac{P_q(x)}{a^{q/2}} = U_q(t) + b_0 U_{q-1}(t) - d_0 U_{q-2}(t), \quad t = \frac{x-c}{2\sqrt{a}}, \quad b_0 = \frac{k-2}{\sqrt{a}}, \quad d_0 = \frac{1}{m-1}, \quad (294)$$

with the last term zero at $q = 1$. Here $b_0 + d_0 \leq 1$. For even q , the expression is nonnegative outside $[-1, 1]$ and at least $-(2q+1)$ everywhere. Inside the interval use $|U_j(t)| \leq j+1$; outside use parity and the increasing positive sequence $U_j(|t|)$.

For $\epsilon > 0$, set $\theta = \operatorname{arcosh}(1 + \epsilon)$ and $\alpha = \min(1/2, 1 - e^{-\theta})$. On $t \geq \cosh \theta$, (294) is at least $(1 - d_0)U_q(t) \geq e^{q\theta}/2$. On $t \leq -\cosh \theta$, use

$$U_j(\cosh \theta) = \frac{\sinh((j+1)\theta)}{\sinh \theta}, \quad \frac{U_{j-1}(\cosh \theta)}{U_j(\cosh \theta)} \leq e^{-\theta}$$

to get a lower bound $(1 - e^{-\theta})e^{q\theta}$. The same estimates hold farther from the interval. Hence

$$Z_q(X) = a^{-q/2} \operatorname{tr} P_q(M) + k(2q+1) \geq 0$$

pointwise, and an eigenvalue with $|\lambda - c| > 2\sqrt{a}(1 + \epsilon)$ forces $Z_q > \alpha e^{q\theta}$.

Theorem 18.3 (Full-moment spectral certificate). *Under the hypotheses of Theorem 18.2, every predetermined Walsh support satisfies*

$$\mathbb{P}\{\|M_S - cI\| > 2\sqrt{a}(1 + \epsilon)\} \leq \frac{k(2q+1) + 6(2q)^{13} + a^{-q/2}\Delta_q(S, m)}{\alpha e^{q\theta}}. \quad (295)$$

If $g(S) > q$, then in particular

$$\mathbb{P}\left\{\|H\| > \frac{k-2}{m} + \frac{2\sqrt{(m-1)(k-1)}}{m}(1 + \epsilon)\right\} \leq \frac{k(2q+1) + 6(2q)^{13}}{\alpha e^{q\theta}}. \quad (296)$$

Proof. Apply Markov's inequality to Z_q and use (290) and Theorem 18.2. For the second assertion use $\Delta_q = 0$ and $c - m = k - 2 \geq 0$ to translate the centered interval to $H = M/m - I$. $\square$

The threshold is at most $k/m + 2(1 + \epsilon)\sqrt{k/m}$. If $m/k \rightarrow \beta \geq 1$ and $g(S_k)/\log k \rightarrow \infty$, choosing sufficiently large logarithmic q and then arbitrarily small fixed ϵ bounds the limsup in probability by $\beta^{-1} + 2\beta^{-1/2}$. This is an upper edge; a matching lower edge and exact finite quantile have not been proved here.

For an explicit regime take $k = 2^{192}$, $m = 32k$, $q = 512$, and $\epsilon = 1/4$. Then $\theta = \log 2$, $\alpha = 1/2$, and $2(2q)^{17} = 2^{171} \leq k$. The threshold is below one half and failure is less than 2^{-307} . The support $\{(x, x^3, \dots, x^{511}) : x \in \mathbb{F}_{2^{192}}\}$ has $g > 512$ by the Vandermonde argument for odd powers, in binary ambient dimension 49152. These conservative large parameters are an explicit mathematical regime, not practical sampling recommendations.

For observed data the recurrence stores two $k \times k$ matrices and costs $O(qk^3)$ arithmetic operations after forming M . Fixed suppressed complexity h has at most $(2q)^{17h+13}$ abstract records reusable across supports. Exact evaluation of the arbitrary-support remainder still retains named column assignments and overlapping constraints and need not be cheap. Its control at $q \asymp \log(k/\eta)$ is the next concrete bridge, and a simultaneous support theorem requires a further shared-event argument. Each original Gram entry has primitive sign degree at most d , and $P_q(M)$ has degree at most qd before Boolean reduction. None of these costs is justified by polynomial degree alone.

19 Sidon confidence and shared cubic coefficient geometry

This section records two different advances beyond the geometric support classes above. A bounded-row amplification converts a polynomially small fixed-space failure probability into sharp confidence for the original full Gram. A second argument controls growing coefficient spaces simultaneously across a cubic orbit, by contracting their peak operators on common sample events. The distinction between a fixed space and a simultaneous family is essential. Throughout, rows are the original independent uniform Walsh rows, sampled with replacement; $G_m(S) = m^{-1} \sum_i X_S(Y_i) X_S(Y_i)^T$ and $X_S(y) = (\chi_s(y))_{s \in S}$. A Sidon support has no four distinct elements with zero XOR. Every norm failure below is the strict event $\|G - I\| > 1/2$, and $0 < \eta < 1$.

19.1 A probability baseline can be amplified without changing the matrix

Lemma 19.1 (Bounded-row partition amplification). *Let X_i be independent identically distributed real vectors with $\|X_i\|_2^2 \leq K$, and let $G_n = n^{-1} \sum_{i \leq n} X_i X_i^T$. Suppose an integer $n_0 \geq 96K$ and $0 < \rho \leq 1$ satisfy $\mathbb{P}\{\|G_n - I\| > 1/4\} \leq \rho$ for every integer $n \geq n_0$. For $m \geq n_0$ and $L = \lfloor m/n_0 \rfloor$,*

$$\mathbb{P}\{\|G_m - I\| > 1/2\} \leq 2^{L+1} \rho^{\lceil L/2 \rceil} \leq 2(4\rho)^{L/2}. \quad (297)$$

Proof. Fix any failing data set, and a unit vector whose empirical row energies $z_i = |\langle X_i, c \rangle|^2 \in [0, K]$ have mean $g > 3/2$ or $g < 1/2$. For a uniform b -element subset J of the m observations,

$$\mathbb{E}_J \bar{z}_J = g, \quad \text{Var}_J(\bar{z}_J) = \frac{m-b}{b(m-1)} \left(\frac{1}{m} \sum_i z_i^2 - g^2 \right) \leq \frac{Kg}{b}.$$

The identity follows from the first and second moments of subset indicators; the case $m = 1$ is immediate. If $g > 3/2$, Chebyshev bounds $\mathbb{P}_J\{\bar{z}_J \leq 5/4\}$ by $24K/b$; if $g < 1/2$, it bounds $\mathbb{P}_J\{\bar{z}_J \geq 3/4\}$ by $8K/b$. Thus every random subset of size at least n_0 inherits quarter-distortion failure with probability at least $3/4$.

Independently permute all observations and divide them into L balanced blocks, each of size at least n_0 . Conditional on the fixed failing data, the expected number of good blocks is at most $L/4$. Markov therefore implies that at least $\lceil L/2 \rceil$ blocks fail with conditional probability at least $1/2$. No conditional independence is used here. Under the original iid law, the disjoint blocks are independent, and each fails with probability at most ρ . A union over subsets of failing blocks gives probability at most $2^L \rho^{\lceil L/2 \rceil}$. Averaging the conditional inheritance inequality proves (297). The auxiliary partition is only a certificate: the tested matrix still uses all m observations with normalization $1/m$. $\square$

For every real vector c on a Sidon support of size k ,

$$\mathbb{E}|\langle X_S, c \rangle|^4 = 3\|c\|_2^4 - 2 \sum_{s \in S} c_s^4, \quad \mathbb{E}X_S X_S^T = I_k, \quad \|X_S\|_2^2 = k. \quad (298)$$

Indeed the only nonzero four-character averages have all labels even: a surviving four-distinct term would be a forbidden Sidon relation.

Theorem 19.2 (Fixed Sidon full-operator confidence). *There is a universal integer $C \geq 96$ such that every predetermined Sidon support of size $k \geq 8$ satisfies, for every integer $m \geq Ck$,*

$$\mathbb{P}\{\|G_m(S) - I\| > 1/2\} \leq \exp\left[-\frac{m \log k}{24Ck}\right]. \quad (299)$$

For $k \geq 64$, the denominator improves to $12Ck$. The constant C depends on an unspecified imported universal constant; no numerical calibration or novelty claim for that input is made. The conclusion covers the entire real coefficient sphere and its complexification.

Proof. The sole imported concentration input is Tikhomirov's Theorem 1 [17], specialized at moment exponent four using (298). For a universal $C_T \geq 1$ and $s = n/k \geq 2$ it gives

$$\mathbb{P}\left\{\|G_n - I\| > C_T[s^{-1} + s^{-1/2}(\log s)^4 + s^{-1/2}]\right\} \leq k^{-1}.$$

If X_S is not centered, an auxiliary independent common sign centers it while leaving each outer product unchanged. Put $A_T = 65538C_T$ and choose $C = \lceil \max\{96, (4A_T)^4\} \rceil$. The elementary bound $x^4 e^{-x/4} \leq 16^4$ shows that the displayed threshold is at most $A_T s^{-1/4} \leq 1/4$ for every $s \geq C$. Apply Lemma 19.1 with $n_0 = Ck$ and $\rho = 1/k$. The direct baseline also bounds the full-sample failure by $1/k$; hence

$$q_m \leq \min\{k^{-1}, 2^{L+1}k^{-\lceil L/2 \rceil}\}, \quad L = \lfloor m/(Ck) \rfloor.$$

For $k \geq 8$, $\log(k/4) \geq (\log k)/3$ makes this at most $e^{-L \log k/12}$: use the second bound for $L \geq 4$ and the first for $L \leq 3$. Since $L \geq m/(2Ck)$, this proves (299). For $k \geq 64$, use $\log(k/4) \geq (2/3) \log k$ and $\log 2 \leq (\log k)/6$ to replace twelve by six in the intermediate bound. $\square$

For a predetermined family of J such spaces, an ordinary union bound therefore suffices when

$$m \geq \max\left\{Ck, \frac{24Ck}{\log k} \log(J/\eta)\right\}. \quad (300)$$

This keeps the shared sampled rows; it assumes no independence between support events. Counting all supports by $\binom{N}{k}$ still costs order $k^2 \log(eN/k)/\log k$. Thus the fixed-space confidence gap is closed, while the difficult support-selection gap remains.

19.2 An exact cubic law and a same-input lower certificate

Let $r \geq 3$ be odd, $k = 2^r$, and

$$A = \{(x, x^3) : x \in \mathbb{F}_k\} \subset \mathbb{F}_k^2 \simeq \mathbb{F}_2^{2r}.$$

This set is Sidon: for a fixed nonzero first-coordinate difference h , equality of two pair differences reduces to $h(x+y)(x+y+h) = 0$, leaving only the same unordered pair. Using the nondegenerate field-trace pairing, an original row has coordinates $X(\alpha, \beta)_x = (-1)^{\text{Tr}(\alpha x + \beta x^3)}$. This is an invertible relabeling of ordinary Walsh rows.

Proposition 19.3 (Three-value law and strict scalar tail). *For $c = k^{-1/2}\mathbf{1}$, the one-row energy $Z = |\langle X, c \rangle|^2$ has the exact law*

$$\mathbb{P}(Z = 0) = \frac{k^2 + k - 2}{2k^2}, \quad \mathbb{P}(Z = 2) = \frac{k - 1}{2k}, \quad \mathbb{P}(Z = k) = \frac{1}{k^2}. \quad (301)$$

For every integer $m \geq 2k$, its original empirical energy R_m satisfies

$$e^{-3(m/k) \log(2k)} \leq \mathbb{P}\{R_m > 3/2\} \leq e^{-(m/(12k)) \log(2k)}. \quad (302)$$

Proof. For $F(x) = \text{Tr}(\alpha x + \beta x^3)$, its polar form is $B(x, z) = \text{Tr}\{\beta(x^2 z + x z^2)\}$. If $\beta \neq 0$, its radical satisfies $\beta z + \beta^2 z^4 = 0$. Odd r makes cubing a bijection on $\mathbb{F}_k^*$, so this radical is $\{0, z_\beta\}$. Consequently the character sum $W = \sum_x (-1)^{F(x)}$ obeys

$$W^2 = \sum_z (-1)^{F(z)} \sum_x (-1)^{B(x, z)} = k \sum_{z \in \{0, z_\beta\}} (-1)^{F(z)} \in \{0, 2k\}.$$

For $\beta = 0$, character orthogonality gives $W = k$ only at $\alpha = 0$ and zero elsewhere. Isotropy gives $\mathbb{E}Z = 1$, determining the counts in (301). Thus there is exactly one energy- k original row.

Let T count that row and $b = \lfloor 3m/(2k) \rfloor + 1$. Then $T = b$ implies the strict event $R_m > 3/2$, and for $m \geq 2k$, $b \leq 2m/k$. The binomial event therefore gives

$$\mathbb{P}\{R_m > 3/2\} \geq \binom{m}{b} k^{-2b} (1 - k^{-2})^{m-b} \geq (2k)^{-2m/k} e^{-2m/k^2} \geq e^{-3(m/k) \log(2k)}.$$

For the upper bound, a centered $V \leq K$ with $\mathbb{E}V^2 \leq \sigma^2$ has $\mathbb{E}e^{\lambda V} \leq \exp[\sigma^2(e^{\lambda K} - 1 - \lambda K)/K^2]$ for $\lambda \geq 0$. This follows because $(e^u - 1 - u)/u^2 = \int_0^1 (1-t)e^{tu} dt$ is increasing. Optimized exponential Markov with $V = Z - 1$, $K = k - 1$, $\sigma^2 = 2(k-1)/k$ gives exponent $2mh(k/4)/(k(k-1))$, where $h(u) = (1+u) \log(1+u) - u$. Use $h(u) \geq (u/2) \log(1+u)$ and $(1+k/4)^3 \geq 2k$ to obtain (302). $\square$

The exact strict witness tail is also a finite multinomial sum. With $n_2 = k(k-1)/2$ and $n_0 = (k^2 + k - 2)/2$, it equals

$$k^{-2m} \sum_{\substack{a, b \geq 0, \\ a+b \leq m \\ 2ka+4b > 3m}} \binom{m}{a} \binom{m-a}{b} n_2^b n_0^{m-a-b}.$$

The equality endpoint is excluded; the two occupancy counts are jointly multinomial, not independent. The lower bound in (302) is a full-operator lower bound on the *same* Gram, because $\lambda_{\max}(G_m) \geq R_m$.

Corollary 19.4 (Matching fixed cubic sample complexity). *Let $m_{A,*}(\eta)$ be the smallest positive integer giving full half-isometry on one fixed cubic space with failure at most η . Then*

$$\frac{1}{16} \left[k + \frac{k}{\log k} \log \frac{1}{\eta} \right] \leq m_{A,*}(\eta) \leq \left\lceil 24C \left[k + \frac{k}{\log k} \log \frac{1}{\eta} \right] \right\rceil. \quad (303)$$

Proof. For $m < k$ rank deficiency forces failure. For $m \geq k$, the same peak event, now using $b \leq 5m/(2k)$, gives failure at least $e^{-4(m/k) \log(2k)}$. Since $\log(2k) \leq (4/3) \log k$, these two necessary bounds imply the lower side of (303). Theorem 19.2 proves its upper side. This argument applies at each integer m and does not assume sample-count monotonicity. $\square$

In particular, bounded directional fourth moments do not permit a universal upper tail Ae^{-am} for this original Gram at $m \asymp k$: its actual repeated-row failure is only polynomially small in k there.

19.3 Shared original cells remove an ambient-rank loss for modulations

Embed A by every injective linear $T : \mathbb{F}_2^t \rightarrow \mathbb{F}_2^d$, where $t = 2r$ and $d \geq t$, and allow every frequency translation a . For $S = a + TA$, consider the row-modulated flat vectors $c_s^{S,z} = \chi_s(z)/\sqrt{k}$. Their signs obey the actual character relations; they are not arbitrary independent coefficient signs. Write $q = d - t + 1$, and let $F_{\text{flat}}(m)$ be simultaneous strict half-isometry failure over this entire predetermined vector family.

Theorem 19.5 (All-ambient cubic modulation theorem). *For every integer $m \geq 1$,*

$$\mathbb{P}(F_{\text{flat}}(m)) \leq 4e^{tq \log 2 - m \log k / (64k)} + 2e^{t(d+1) \log 2 - m/32}. \quad (304)$$

In particular it suffices that $m \geq 128kq + (64k/\log k) \log(8/\eta)$. The minimum successful sample count obeys

$$\frac{1}{16} \left[kq + \frac{k}{\log k} \log \frac{1}{\eta} \right] \leq m_{\text{flat},*}(\eta) \leq \left\lceil 192 \left[kq + \frac{k}{\log k} \log \frac{1}{\eta} \right] \right\rceil. \quad (305)$$

All constants here are explicit and elementary.

Proof. Proposition 19.3 gives, pointwise on the original row space,

$$| \langle X_S(y), c^{S,z} \rangle |^2 = k 1_{C_{T,z}}(y) + R_{T,z}(y), \quad C_{T,z} = z + \ker T^*, \quad R_{T,z} \in \{0, 2\}, \quad \mathbb{E} R_{T,z} = 1 - 1/k.$$

Every affine codimension- t cell occurs as such a peak, with probability $2^{-t} = k^{-2}$. Their number is at most $4 \cdot 2^{t(d-t+1)}$: the Gaussian-binomial product has denominator $\prod_{j \geq 1} (1 - 2^{-j}) > 1/4$. For each peak count B , exponential Markov at $\log(k/4)$ gives

$$\mathbb{P}\{B > m/(4k)\} \leq \exp \left[-\frac{m}{4k} \{\log(k/4) - 1 + 4/k\} \right] \leq e^{-m \log k / (64k)}.$$

The last scalar inequality holds for all $k \geq 8$ by differentiation and its value at eight. The maps and residual modulations number at most 2^{td+t} . A variable in $[0, 2]$ has tilted variance at most one, so integrating its log-mgf twice and optimizing gives deviation probability $2e^{-m/32}$ at threshold $1/4$ for its sample mean. Thus, except with the probability in (304), all peak contributions are at most $1/4$, and all residual means are within $1/4$ of $1 - 1/k$. Every energy lies in $[3/4 - 1/k, 3/2 - 1/k]$. For the sufficient threshold use $t \log 2 = 2 \log k$ and, for odd $r \geq 3$, $r/2^r \leq 3/8$, $r^2/2^r \leq 9/8$; these imply $32t(d+1) \log 2 < 128kq$ and control the residual term as well.

For a same-sample lower bound, any $\min(m, q)$ observations fit in an affine codimension- t cell. Choose a cubic image in its annihilator and the corresponding modulation. Its empirical energy is at least $k \min\{1, q/m\}$. Hence every sample fails when $m < (2/3)kq$. At an admissible count $m \geq k/2$, fixing one peak cell and using $b = \lfloor 3m/(2k) \rfloor + 1 \leq 7m/(2k)$ gives failure at least $e^{-6(m/k) \log(2k)}$. Thus admissibility also requires $m \geq [k/(8 \log k)] \log(1/\eta)$. Taking the maximum proves the lower bound in (305); the sufficient threshold and $\log 8 \leq \log k$ prove the upper bound. $\square$

For *full* coefficient spheres on the same cubic orbit, the general fixed-space family bound (300) counts at most 2^{2rd} maps. Translation preserves the Gram, and gives the sufficient condition

$$m \geq \max \left\{ Ck, 48Ckd + \frac{24Ck}{\log k} \log \frac{1}{\eta} \right\}.$$

The same selected-cell lower witness and the fixed-cubic confidence witness give, when $d \geq 4r$, minimum sample count between $1/16$ and $48C$ times $kd + (k/\log k) \log(1/\eta)$, with a ceiling for the upper bound. At $d = 2r$ this does not match the linear fixed-confidence modulation rate; the whole coefficient sphere in that regime remains a separate question.

19.4 Arbitrary coefficients: a shared near-peak theorem and its limit

The common-cell mechanism also controls a layer for arbitrary real coefficients, without a coefficient net. For a real unit vector on any Sidon support of size at most k , retain the effective size $M = \|c\|_1^2$ and the actual signed equations of agreement lists.

Theorem 19.6 (Uniform weighted near-peak layer). *Let $k \geq 2048$. For $M \geq 2048$ put $R = \lfloor \log_2(M^2/16) \rfloor$, $L = 8R + 1$, and $\tau(M) = M(1 - 1/(2L))^2$. If*

$$m \geq 512k \left[d + 1 + \frac{\log(4/\eta)}{\log k} \right],$$

then with probability at least $1 - \eta$, simultaneously for all those coefficient vectors,

$$\frac{1}{m} \sum_i |\langle X(Y_i), c \rangle|^2 1_{\{|\langle X(Y_i), c \rangle|^2 \geq \tau(M)\}} < 1/4. \quad (306)$$

Proof. Draw an auxiliary ordered list of frequencies with weights $p_v = |c_v|/\sqrt{M}$, retaining $\sigma_v = \text{sign}(c_v)$. The exact collision mass is $\sum_v p_v^2 = 1/M$. A Sidon set has at most $1 + 2^{(a+1)/2}$ points in any affine a -flat, by counting distinct pair differences. Condition the weights on any set of mass at least $3/4$. While a list has rank below R , its span contains at most $5M/16$ support points; Cauchy-Schwarz bounds its conditional mass by $\sqrt{5}/3 < 3/4$. Each of the next $8R$ draws therefore increases rank with conditional probability at least $1/4$ until rank R is reached. For the number J of successes, declaring success after reaching R , $\mathbb{E}2^{-J} \leq (7/8)^{8R}$; hence rank failure has probability at most $[2(7/8)^8]^R < 1/2$.

For a near-peak row, one of the signs $\sigma_v \chi_v(y)$ has weight at least $1 - 1/(4L)$. All L draws agree on that sign with probability at least $3/4$, and conditional rank succeeds with probability $> 1/2$. For the actual signed affine cell

$$C_v = \{y : \chi_{v_j+v_1}(y) = \sigma_{v_j} \sigma_{v_1} \text{ for all } j\},$$

which may be empty, this proves pointwise $1_{\{Z_c \geq \tau(M)\}} \leq 4\mathbb{E}_v[1_{\{\text{arank } v \geq R\}} 1_{C_v}]$. Repeated equations are reduced by their actual rank, not charged anew.

For every integer $a \geq 18$ put $A_a = \min\{k, 4 \cdot 2^{(a+1)/2}\}$. Use the single common event that every affine codimension- a cell has empirical mass strictly below $1/(16A_a)$. Whenever $a \geq R$, $M \leq A_a$, so the last pointwise majorant and $Z_c \leq M$ prove (306) on this event for every c at once. For a fixed cell, binomial Chernoff bounds violation by

$$\exp \left[-\frac{m}{16A_a} \{a \log 2 - \log(16eA_a)\} \right] \leq e^{-ma \log 2 / (512A_a)}.$$

Here $a \geq 18$ gives the last inequality and $A_a/a \leq k/\log_2 k$. The stated sample threshold makes this exponent at least $a(d+1) \log 2 + \log(4/\eta)$. There are at most $4 \cdot 2^{a(d-a+1)}$ cells at that codimension; summing over $a \geq 18$ leaves at most $\eta \sum_{a \geq 18} 2^{-a^2} \leq \eta$. This proves the theorem. $\square$

This controls only a relative $O(1/\log M)$ neighborhood of the bound $Z_c \leq M$, which need not be attained by any row. It does not control the complementary energy or $M < 2048$. The missing middle cannot be filled by collapsing every positive cell certificate to one occupancy cap. Here is a quantitative obstruction on the same original basis-support Gram.

Proposition 19.7 (Cost of a positive common-cap reduction). *On $\mathbb{F}_2^k$ let $E_h = \{z : |z| \leq h\}$, with integers $0 \leq h < k/2$ and $1 \leq R \leq k$. Among pointwise majorants $1_{E_h} \leq \sum_C \lambda_C 1_C$ with $\lambda_C \geq 0$ and all affine cells of codimension at least R , let Λ_* be the minimum total weight. For every $0 \leq u \leq 1/2$,*

$$\Lambda_* \geq \frac{\mathbb{P}\{\text{Bin}(k, u) \leq h\}}{(1-u)^R}, \quad \Lambda_* \leq \frac{\binom{k}{R}}{\binom{k-h}{R}} \quad (R \leq k-h).$$

If $\alpha_k = h/k \rightarrow \alpha \in (0, 1/2)$ and $R = o(\sqrt{k})$, then $\Lambda_* = (1 + o(1))(1 - \alpha_k)^{-R}$.

Proof. Under a product measure of bit bias u , condition on nonpivot variables in the equations for a cell. Each of its at least R pivot bits must take one specified value, so its probability is at most $(1-u)^R$. Integrate the majorant to prove the lower bound. Uniform weights $1/\binom{k-h}{R}$ on the coordinate cells fixing R zeros prove the upper bound. For the asymptotic choose $u = \alpha_k - k^{-1/4}/\sqrt{R}$. Chebyshev gives probability $1 - o(1)$ in the lower bound, while $R(\alpha_k - u) = o(1)$. The logarithm of the upper binomial ratio differs from $R \log(1/(1 - \alpha_k))$ by $O_\alpha(R^2/k) = o(1)$. $\square$

For the positive flat basis vector, energy on E_h is at least $k(1 - 2h/k)^2$. A common empirical cap for all allowed cells is at least $1/m$, because it includes point cells. The best numerical upper bound returned by this particular substitution is therefore at least order $k\Lambda_*/m$. At $h = \lfloor k/4 \rfloor$ and $R = \lceil 2 \log_2 k \rceil$ this forces a certificate sample requirement $k^{1+2 \log_2(4/3)+o(1)} = k^{1.830074\dots+o(1)}$. The actual same-input basis Gram satisfies

$$\mathbb{P}\{\|G_m - I\| > 1/2\} \leq 2 \cdot 9^k e^{-m/128}.$$

Indeed $\mathbb{E}e^{t(X,c)} \leq e^{t^2/2}$; Gaussian integration gives an upper square-energy mgf, and $e^{-v} \leq 1 - v + v^2/2$ with the fourth moment at most three gives its lower counterpart. Exponential Markov yields two-sided scalar deviation at most $2e^{-mu^2/8}$ for $0 < u \leq 1$. A $1/4$ -net of at most 9^k unit vectors and $\|H\| \leq 2 \max_{c \text{ in net}} |c^T H c|$, with $u = 1/4$, prove the display. Thus the obstruction is a loss of the specified reduction, not a sampling lower bound; coherent cell sums, distinct caps and signed interactions are outside the reduction.

19.5 Even every orbit modulation is not a deterministic norm certificate

Proposition 19.8 (All-orbit modulation obstruction). *Let $r \geq 3$ be odd, $k = 2^r$, $t = 2r$, and $N = k^2$. There is a sign vector σ on A and an allowed original sample with positive integer row multiplicities*

$$n_y = k + W_\sigma(y)^2, \quad W_\sigma(y) = \sum_{a \in A} \sigma_a \chi_a(y), \quad m = 2k^3,$$

for which

$$\|G_m(A) - I\| = 1 - 1/k, \quad \max_{B,z} |(c^{B,z})^T (G_m(B) - I) c^{B,z}| \leq \frac{4}{k} \left[(t(t+1) + 2) \log 2 + \frac{1}{4} \right]. \quad (307)$$

Here the maximum includes every invertible image $B = a + TA$ and every modulation z . For odd $r \geq 13$ the latter bound is below $1/4$.

Proof. Sidon uniqueness and Walsh orthogonality give exactly $G_m(A) = I + (\sigma\sigma^T - I)/k$. Write $D(B)$ for its unordered pair differences. On any other orbit support the modulated Rayleigh error is

$$\frac{2}{k^2} \sum_{\substack{\{a,b\} \subset A \\ a+b \in D(B)}} \sigma_a \sigma_b \chi_{a+b}(z).$$

Thus every test is a quadratic sign sum $2Q/k^2$ with coefficients of absolute value at most one. For independent auxiliary signs and $|\lambda| \leq 1/(2k)$, $\mathbb{E}e^{\lambda Q} \leq e^{\lambda^2 k^2}$. To verify this, randomly divide vertices into two parts and apply Jensen to twice the crossing-edge sum. Condition on one part and use $\cosh u \leq e^{u^2/2}$; a Gaussian integral and the same inequality on the other part bound its mgf by $\prod_j (1 - 4\lambda^2 s_j^2)^{-1/2}$, where s_j are singular values of the cross-coefficient matrix. Its squared Frobenius norm is at most $k^2/4$, and $-\log(1 - u) \leq 2u$ for $u \leq 1/4$ proves the claimed mgf. Hence $\mathbb{P}_\sigma\{|Q| > x\} \leq 2e^{1/4 - x/(2k)}$. There are at most $M = 2^{t(t+1)}$ tests. Choosing $x = 2k[\log(4M) + 1/4]$ makes their union probability at most $1/2$, so some signs obey (307). Its bound decreases below $1/4$ at $r = 13$ and thereafter along odd integers. $\square$

A fully specified finite check uses $\mathbb{F}_8$ defined by $x^3 + x + 1$, the encoded support $(0, 9, 26, 35, 44, 53, 62, 23)$, and only $\sigma_9 = -1$. The counts are 44 at eight rows and 12 at the other 56, giving $m = 1024$. Exhaustively checking all zero-constant Boolean quadratics on six bits with weight 28 (a certified superset of all transformed pair-difference sets) gives maximum modulation error exactly $7/16$, while the full operator error is $7/8$. This finite enumeration is recorded separately from the analytic proof above. The general sample has positive iid probability $m!/(N^m \prod_y n_y!)$, potentially tiny. Thus the proposition refutes a deterministic directional reduction; it does not assert a significant iid failure probability at $m = 2k^3$.

19.6 A direct common-event theorem for growing coefficient spaces

The preceding obstruction motivates controlling the retained coefficient sphere directly. Stay at minimal ambient dimension $t = 2r$, now odd $r \geq 5$. For every cubic image $B = a + TA$, choose any L distinct original roots Z and put

$$v_{B,z} = k^{-1/2} X_B(z), \quad E_{B,Z} = \text{span}\{v_{B,z} : z \in Z\}.$$

Let $F_L(m)$ be half-isometry failure on any unit vector in any such space, all spaces being included simultaneously and predetermined.

Theorem 19.9 (Growing full coefficient spheres). *If the positive integer L satisfies $(L-1)\sqrt{2/k} \leq 1/2$, then for every integer $m \geq 1$,*

$$\mathbb{P}(F_L(m)) \leq k^2 e^{-m \log k / (48k)} + 2e^{t^2 \log 2 + Lt \log 2 + L \log 9 - m / (512L)}. \quad (308)$$

If odd $r \geq 11$ and $1 \leq L \leq \lfloor \sqrt{k/r} \rfloor$, it suffices that

$$m \geq 4096 \left[k + \frac{k}{\log k} \log \frac{4}{\eta} \right]. \quad (309)$$

The minimum successful sample count lies between $1/16$ times $k + (k/\log k) \log(1/\eta)$ and the ceiling of 8192 times this quantity. Every retained space has dimension exactly L , and the conclusion covers arbitrary amplitudes and complexification.

Proof. The cubic law implies root coherence $|\langle v_{B,z}, v_{B,w} \rangle| \leq \sqrt{2/k}$ for $z \neq w$. Let V have the selected roots as columns, $M = V^T V$, and Q be an orthonormal basis for their span.

The hypothesis and the elementary off-diagonal quadratic bound give $I/2 \preceq M \preceq 3I/2$. For $\xi(y) = Q^T X_B(y)$, preserve the peak sum as an operator:

$$\sum_{z \in Z} \xi(z) \xi(z)^T = k Q^T V V^T Q, \quad \left\| \sum_{z \in Z} \xi(z) \xi(z)^T \right\| = k \lambda_{\max}(M) \leq 3k/2. \quad (310)$$

The contraction is independent of L . Writing $\tau = \max_z n_z/m$ for original point occupancies, the empirical peak operator is at most $(3/2)k\tau I$, while its population mean is at most $3I/(2k)$.

Off Z , each coordinate of $V^T X_B(y)$ has absolute value at most $\sqrt{2}$, so

$$\|\xi(y)\|_2^2 = (V^T X_B(y))^T M^{-1} (V^T X_B(y)) \leq 4L.$$

Every residual directional energy lies in $[0, 4L]$ and has second moment at most three by (298). For a centered variable D with $|D| \leq K$ and $\mathbb{E}D^2 \leq 3$, its exponential series gives $\log \mathbb{E}e^{\lambda D} \leq 3\lambda^2/[2(1 - |\lambda|K/3)]$. Thus the sample mean has two-sided deviation at level $1/8$ at most $2e^{-m/(384+64L/3)} \leq 2e^{-m/(512L)}$ when $K = 4L$. A $1/4$ -net of size at most 9^L bounds the residual operator error by $1/4$, outside probability $2 \cdot 9^L e^{-m/(512L)}$. There are at most 2^{t^2} maps and $N^L = 2^{tL}$ root sets; frequency translation changes root vectors by scalar signs and adds no event count. This proves the second term in (308).

A single original point count is $\text{Bin}(m, k^{-2})$. Exponential Markov at $\log(k/6)$ gives

$$\mathbb{P}\{n_z > m/(6k)\} \leq \exp \left[-\frac{m}{6k} \{\log(k/6) - 1 + 6/k\} \right] \leq e^{-m \log k/(48k)} \quad (k \geq 32).$$

Union over the k^2 original points gives the first term in (308). On its complementary event, all empirical peak operators are at most $I/4$ simultaneously. Positivity on the appropriate side, together with the residual bound, gives

$$(1 - 1/4 - 3/(2k))I \preceq Q^T G_m(B)Q \preceq (1 + 1/4 + 1/4)I.$$

This proves strict half-isometry failure is excluded.

It is enough for the two probabilities to be at most $\eta/2$ that

$$\begin{aligned} m &\geq 96k + \frac{48k}{\log k} \log(2/\eta), \\ m &\geq 512L[t^2 \log 2 + L(t \log 2 + \log 9) + \log(4/\eta)]. \end{aligned}$$

For odd $r \geq 11$ one has $r^3 \leq k$. If $L \leq \sqrt{k/r}$, then $Lr^2 \leq k$, $L^2 r \leq k$, $L \leq k/\log k$, and the coherence hypothesis holds. With $\log 2 < 7/10$ and $\log 9 < 3$, the second threshold is at most $2304k + (512k/\log k) \log(4/\eta)$; both are bounded by (309). Every modulation vector belongs to one retained L -dimensional space, so the same-family lower bound in Theorem 19.5, at $d = t$, applies. The upper constant 8192 follows from (309) and $\log 4 \leq \log k$. $\square$

This is genuinely compressed noncommuting geometry. At $L = 2$, choose roots with inner product $a \neq 0$, $|a| = \sqrt{2/k}$. If their columns are $k^{-1/2}\mathbf{1}$ and $k^{-1/2}\tau$, orthonormalization gives coordinate leverage

$$\ell_i = \frac{2}{k} \frac{1 - a\tau_i}{1 - a^2}.$$

Both signs occur, so leverages differ. The two projected root vectors are neither orthogonal nor parallel, and their rank-one matrices do not commute. Arbitrary amplitudes in the space can also produce unequal coordinate magnitudes and cancellation.

The reusable identity is (310); cubic geometry supplies its root coherence, root probabilities and off-root energy bound. The coefficient dimension grows as $\sqrt{k/\log k}$, but the result does

not cover arbitrary signed-flat vectors or all k coefficient directions. A deterministic upgrade from modulation tests is ruled out by Proposition 19.8. Section 20 now proves a probabilistic enlargement for a specified isotropic family and closes genuinely noncommuting projection interactions in a common block algebra. Arbitrary coefficient spaces and unrestricted support selection remain open.

20 Closing cubic peaks and their coefficient interactions

The arbitrary-root theorem above controls coefficient dimension of order $\sqrt{k/\log k}$. Its inverse root Gram is essential: already at $\sqrt{2k} - 1$ independent roots, an omitted original modulation vector can lie in their span. We first identify this obstruction and repair the peak set. Retaining the signed Gram of a larger isotropic family then gives coefficient dimension of order $k/(\log k)^2$ at the same sampling scale. Finally, different repaired spaces admit genuinely noncommuting interactions inside a quantitative common block algebra. These are restricted geometric and computational theorems; they do not cover all Walsh supports.

Throughout this section r is odd, $k = 2^r$, and

$$A = \{(x, x^3) : x \in \mathbb{F}_k\}, \quad v_{\alpha, \beta}(x) = k^{-1/2}(-1)^{\text{Tr}(\alpha x + \beta x^3)}, \quad X_A(y) = \sqrt{k}v_y.$$

The trace pairing is an invertible relabeling of the original uniform Walsh row space, of cardinality k^2 . All logarithms without a subscript are natural. Write $q(x) = \text{Tr}(x^3)$ and $B(x, u) = q(x + u) + q(x) + q(u)$, an alternating binary form. Its radical is $\mathcal{R} = \{0, 1\}$. Indeed its radical equation is $u^{2^{r-1}} + u^2 = 0$, equivalently $u^4 = u$, and cubing is bijective on $\mathbb{F}_k^*$ for odd r . A symplectic basis of the quotient by $\mathcal{R}$ gives isotropic spaces of each dimension $1 \leq s \leq (r + 1)/2$ containing $\mathcal{R}$. On any such space U , the restriction of q is linear; choose α_U representing it under the trace pairing.

20.1 A reused direction that diagonal Gram data miss

Proposition 20.1 (Hidden peak and exact innovation). *Let $r \geq 3$ be odd and $D = 2^{(r-1)/2}$. There are disjoint original root families H_0, H_1 of size D , with $0 \in H_0$, whose normalized Gram is*

$$G_H = \begin{pmatrix} I_D & D^{-1}\mathbf{1}\mathbf{1}^T \\ D^{-1}\mathbf{1}\mathbf{1}^T & I_D \end{pmatrix}. \quad (311)$$

Deleting root zero leaves an independent family R_0 of size $2D - 1$, with

$$v_0 = \sum_{h \in H_1} v_h - \sum_{h \in H_0 \setminus \{0\}} v_h, \quad \lambda_{\min}(G_{R_0}) = 1 - \sqrt{1 - 1/D}. \quad (312)$$

In particular $\|\Pi_{\text{span } R_0} X_A(0)\|^2 = k$ although zero is absent from the selected roots and $\|G_{R_0}\| < 2$.

For a selected roots in H_0 and b in H_1 , with $ab < D^2$, the squared innovation on appending an unused root in the respective family is exactly

$$s_0(a, b) = \frac{1 - (a + 1)b/D^2}{1 - ab/D^2}, \quad s_1(a, b) = \frac{1 - a(b + 1)/D^2}{1 - ab/D^2}. \quad (313)$$

At $(a, b) = (D - 1, D)$ the first innovation is zero, whereas replacing the inverse Gram by its diagonal predicts $1 - 1/D$.

Proof. Choose a maximal isotropic U containing $\mathcal{R}$, put $W = U^{\perp_{\text{Tr}}}$, and take

$$H_0 = \{(\ell, 0) : \ell \in W\}, \quad H_1 = \{(\alpha_U + \ell, 1) : \ell \in W\}.$$

Here $|W| = D$ and $U^{\perp_B} = U$. Within each family, linear character orthogonality gives the identity Gram. For a cross correlation, partition the Walsh sum of $q(x) + \text{Tr}((\alpha_U + \ell)x)$ into U -cosets. The sum on $x + U$ vanishes unless $B(x, U) = 0$; only the coset U survives, with value $|U| = 2D$. Dividing by $k = 2D^2$ proves (311) with its positive signs. The two complete family sums agree coordinatewise: the factor $\sum_{\ell \in W} (-1)^{\text{Tr}(\ell x)}$ vanishes off U , and $1 - (-1)^{q(x) + \text{Tr}(\alpha_U x)}$ vanishes on U . This proves the dependence in (312).

On the two constant-family directions the Gram has eigenvalues 0, 2; all other eigenvalues equal one. Deleting one first-family root replaces the off-diagonal entry in this two-dimensional compression by $\sqrt{D(D-1)}/D$, proving independence and the stated minimum eigenvalue. The dependence places $\sqrt{kv_0}$ in its span.

More generally, the only nontrivial eigenvalues for sizes a, b are $1 \pm \sqrt{ab}/D$. For a new first-family root, its correlation vector is $(0_a, D^{-1}\mathbf{1}_b)$. Inverting the two-dimensional constant block gives its inverse-Gram energy $(b/D^2)/(1 - ab/D^2)$, whose complement is s_0 ; the other formula is symmetric. For the last missing root, the squared Euclidean correlation norm is $1/D$, but its inverse-Gram energy is one. $\square$

For arbitrary independent selected unit roots, with Gram G , smallest eigenvalue $\gamma > 0$, and coherence at most μ , an original row y outside the selected roots satisfies

$$\|\Pi_E X_A(y)\|^2 = k g_y^* G^{-1} g_y \leq \min\{k, k\mu^2 |R_0|/\gamma\}, \quad g_y = V^* v_y. \quad (314)$$

Thus a bounded upper Gram norm and small pair correlations cannot alone justify an $O(|R_0|)$ residual cap. In Proposition 20.1, the ratio to $k\mu^2 |R_0|$ is $D^2/(2D-1)$, at least $1/[4(1 - \sqrt{1 - 1/D})]$. The inverse-Gram dependence has the required order of magnitude.

There is a useful restricted repair. Adjoining the hidden root changes neither E nor the input matrix, but the redundant Gram has positive spectrum $\{1, 2\}$. Its Moore–Penrose inverse therefore gives

$$\|\Pi_E X_A(y)\|^2 \leq 4D \quad (y \notin H), \quad \left\| \sum_{h \in H} (\Pi_E X_A(h))(\Pi_E X_A(h))^* \right\| = 2k.$$

For $r \geq 5$, $4D < k$, so H contains exactly all original modulation roots whose vectors lie in E . No efficient general theorem for closing arbitrary root sets is inferred. Within the two fixed families, however, (313) is an exact extension rule with $O(r)$ -bit rational state parameters and $O(r^2)$ schoolbook bit work per update. Named roots must also be retained to detect repeats, using $O(|R_0|r)$ bits. Actual row correlations are additional input or computation, not a constant-cost oracle.

20.2 Signed isotropic channels at almost linear dimension

Now allow any isotropic $U \supseteq \mathcal{R}$ of dimension s . Define

$$\begin{aligned} W &= U^{\perp_{\text{Tr}}}, \\ H_U &= \{(\ell, 0) : \ell \in W\} \cup \{(\alpha_U + \ell, 1) : \ell \in W\}, \\ E_U &= \text{span}\{v_h : h \in H_U\}, \\ M &= 2^{r-s}, \quad \rho = 2^{r+1-2s}, \quad n = 2M - \rho. \end{aligned} \quad (315)$$

Changing the extension α_U translates the second family by an element of W and does not change these objects.

Lemma 20.2 (Exact signed channels). *The actual root Gram and its positive spectrum are*

$$G_U = \begin{pmatrix} I_M & C_U \\ C_U & I_M \end{pmatrix}, \quad \text{spec}(G_U) = \{2^{[\rho]}, 1^{[2(M-\rho)]}, 0^{[\rho]}\}. \quad (316)$$

Brackets denote multiplicities. In particular $\dim E_U = n$ and $\|G_U^+\| \leq 1$. Writing $K_U = G_U - I$, one has

$$C_U^3 = C_U, \quad G_U^+ = I + \frac{1}{4}K_U - \frac{3}{4}K_U^2. \quad (317)$$

For any orthonormal basis Q of E_U , put $\xi(y) = Q^* X_A(y)$. Then

$$\left\| \sum_{h \in H_U} \xi(h) \xi(h)^* \right\| = 2k, \quad \|\xi(y)\|^2 \leq \min\{k, 4M\} \quad (y \notin H_U). \quad (318)$$

Proof. Put $q_U(x) = q(x) + \text{Tr}(\alpha_U x)$, which vanishes on U . The cross block is symmetric convolution on W with kernel $k^{-1} \sum_x (-1)^{q_U(x) + \text{Tr}(\ell x)}$. Fourier eigenvectors on W are indexed by $x + U$, and their eigenvalues are

$$\lambda_{x+U} = |U|^{-1} \sum_{u \in U} (-1)^{q_U(x+u)} = \begin{cases} (-1)^{q_U(x)}, & x \in U^{\perp_B}, \\ 0, & \text{otherwise.} \end{cases} \quad (319)$$

This follows from $q_U(x+u) = q_U(x) + B(x, u)$. The signs and condition are well defined on cosets. Since U contains the one-dimensional radical, $\dim U^{\perp_B} = r - s + 1$, so exactly ρ channels are nonzero. Each nonzero channel gives eigenvalues 0, 2 of the full Gram, and each zero channel gives two ones. Evaluating the polynomial in (317) at the eigenvalues $-1, 0, 1$ of K_U proves that formula, including its signs.

For completeness, distinct cubic rows have correlation at most $\sqrt{2/k}$. For a nonzero cubic coefficient, scale by its unique cube root and square the quadratic Walsh sum. Summing first over the difference variable leaves only the two radical elements, giving squared Walsh sum zero or $2k$. A nonzero linear character sums to zero. Thus, if V has columns v_h and $g_y = V^* v_y$, the actual projection identity is

$$\|\Pi_{E_U} X_A(y)\|^2 = k g_y^* G_U^+ g_y.$$

For $y \notin H_U$ the $2M$ squared correlations sum to at most $4M/k$; (317) proves the residual bound. The coherent peak operator has the nonzero spectrum of kG_U , proving its norm exactly. $\square$

If $s \geq 3$, then $4M < k$ and H_U is already closed under membership of original roots in its span. There is also a stable full-coset extension: along a descending isotropic flag, use one correction agreeing with q on the largest member. Passing from dimension s to $s - 1$ doubles W and updates

$$(M, \rho, n) \mapsto (2M, 4\rho, 4M - 4\rho), \quad E_U \subseteq E_{U'}. \quad (320)$$

Recompute the signs in (319) on the refined cosets. Positive Gram eigenvalues stay in $\{1, 2\}$, so this permitted extension accumulates no conditioning loss. Partial arbitrary root additions are not covered, as Proposition 20.1 demonstrates.

The bases of U, W and the correction use $O(r^2)$ bits; listing roots uses $O(Mr)$ bits. The M -channel table costs at most $O(Mr^4)$ binary work with a supplied field basis. Convolution and the pseudoinverse apply in $O(M \log M)$ rational operations by Walsh transforms. With correlations supplied over a common denominator of bit length B_0 , intermediate lengths are $O(B_0 + r + \log M)$; forming the original correlations directly costs at most $O(kMr^3)$ binary work. One flag step doubles the named states and costs $O(Mr^4)$ for the new channel table. These are costs for a supplied space, not an enumeration of the whole family below.

20.3 A matching common-event sampling theorem

For every invertible binary ambient map T , frequency translation a , row modulation u , and isotropic U of dimension s , consider the support $a + TA$ and the coherently transported space generated by roots $(T^*)^{-1}(H_U + u)$. Let $F_s(m)$ denote the event that some unit vector in any

such space has empirical squared-norm distortion strictly greater than $1/2$, using the same m iid original uniform rows for all spaces. Real orthonormal representations show that complexification gives the same event.

Theorem 20.3 (Almost linear cubic coefficient dimension). *For odd $r \geq 5$, every $1 \leq s \leq (r+1)/2$, and every positive integer m ,*

$$\mathbb{P}(F_s(m)) \leq k^2 e^{-m \log k / (64k)} + 2n \exp\left((5r^2 + 2r) \log 2 - \frac{m}{160M}\right). \quad (321)$$

For odd $r \geq 15$, choose $s = \lceil \log_2(r^2) \rceil$. This is permitted,

$$(1 - r^{-2}) \frac{k}{r^2} < n \leq \frac{2k}{r^2}, \quad (322)$$

and $\mathbb{P}(F_s(m)) \leq \eta$ follows from

$$m \geq 1024 \left[k + \frac{k}{\log k} \log \frac{4}{\eta} \right], \quad 0 < \eta < 1. \quad (323)$$

For the least positive integer $m_{s,*}(\eta)$ with failure probability at most η , the same family satisfies

$$\frac{1}{16} \left[k + \frac{k}{\log k} \log \frac{1}{\eta} \right] \leq m_{s,*}(\eta) \leq \left\lceil 2048 \left[k + \frac{k}{\log k} \log \frac{1}{\eta} \right] \right\rceil. \quad (324)$$

Proof. For a fixed space, let $A_i = \mathbf{1}_{Y_i \notin H_U} \xi(Y_i) \xi(Y_i)^*$ and $Z_i = A_i - \mathbb{E}A_i$. The actual Parseval mean is $\mathbb{E}\xi\xi^* = I$, so

$$0 \preceq A_i \preceq 4MI, \quad \mathbb{E}A_i \preceq I, \quad \|Z_i\| \leq 4M, \quad \mathbb{E}Z_i^2 = \mathbb{E}A_i^2 - (\mathbb{E}A_i)^2 \preceq 4MI.$$

The last is a matrix variance bound; a directional fourth-moment constant cannot be substituted for it. For independent centered Hermitian Z_i with $\|Z_i\| \leq R$, $\mathbb{E}Z_i^2 \preceq RI$, $R \geq 1$, the exponential series and $j! \geq 2 \cdot 3^{j-2}$ give

$$\mathbb{E}e^{\theta Z_i} \preceq \exp\left(\frac{\theta^2 R}{2(1 - \theta R/3)}\right) I \quad (0 < \theta < 3/R).$$

Condition successively and use Golden-Thompson with this scalar envelope. Exponential Markov, optimized at $\theta = t/[R(1 + t/3)]$, gives the standard matrix concentration estimate, in the framework of [1],

$$\mathbb{P}\left\{\left\|m^{-1} \sum_i Z_i\right\| > t\right\} \leq 2n \exp\left(-\frac{mt^2}{2R(1 + t/3)}\right).$$

At $R = 4M, t = 1/4$, the residual operator fails its $1/4$ bound with probability at most $2ne^{-3m/(416M)} \leq 2ne^{-m/(160M)}$. The prefactor is $2n$, replacing an exponential-size directional net.

There are at most 2^{4r^2} ambient maps, 2^{2r} modulations, and $2^{rs} \leq 2^{r^2}$ possible U (count their ordered bases). A frequency translation changes individual row vectors only by scalar signs, so it adds no event count. All these transports preserve the actual row Gram and residual cap. A union bound proves the second term of (321).

Let n_y be the occupancies of the same k^2 original sample points. On $k \max_y n_y / m \leq 1/8$, the coherent sum in (318) bounds every empirical peak operator by $I/4$. The population peak operator has norm $2/k$. If all residual deviations are at most $1/4$, positivity on the appropriate side gives

$$(3/4 - 2/k)I \preceq G_m|_E \preceq (3/2)I,$$

precluding strict half-distortion for $k \geq 32$. For one point the binomial exponential bound at $\log(k/8)$ gives

$$\mathbb{P}\{n_y > m/(8k)\} \leq \exp\left[-\frac{m}{8k}\{\log(k/8) - 1 + 8/k\}\right] \leq e^{-m \log k/(64k)}.$$

For the last inequality subtract $(\log k)/8$ from the braces, check the positive value at 32, and note derivative $(7k/8 - 8)/k^2 > 0$. No rounding alters the strict event. Union over original points proves (321).

It is enough that

$$\begin{aligned} m &\geq 128k + \frac{64k}{\log k} \log(2/\eta), \\ m &\geq 160M[(5r^2 + 2r) \log 2 + \log(4n/\eta)]. \end{aligned}$$

At $r = 15$ the chosen s is eight. The inequality $(r + 2)^2 \leq 2r^2$ propagates its admissibility through larger odd r . Since $r^2 \leq 2^s < 2r^2$ and $n = 2M(1 - M/k)$, (322) follows. Also $n \leq k$ and $(5r^2 + 2r) \log 2 + \log n < (21/5)r^2$. The second threshold is at most $672k + 160(k/r^2) \log(4/\eta)$, proving (323) and its rounded upper version in (324).

For the lower bound, zero belongs to H_U , and modulation lets this same family contain every unit vector v_y . If $m < 2k/3$, the direction of the first sampled point has empirical energy at least $k/m > 3/2$, so failure is deterministic. Fix a point y_0 and put $q_0 = \lfloor 3m/(2k) \rfloor + 1 \leq m$. The first q_0 observations all equal y_0 with probability k^{-2q_0} , and this gives energy strictly greater than $3/2$ in v_{y_0} . Once success with $\eta < 1$ is possible, $m \geq 2k/3$ and hence $k^{-2q_0} \geq e^{-6m \log k/k}$. Combining these two necessary bounds proves the conservative lower constant $1/16$. No monotonicity in integer m is used. $\square$

The dimension improvement is for all coefficients in each specified isotropic space, simultaneously over the stated orbit and all U at that level. It does not assert the same result for arbitrary spaces of that dimension. Noncommutation is retained: some cross-family inner products have magnitude $\sqrt{2/k}$ strictly between zero and one. The corresponding rank-one matrices do not commute; for distinct orthogonal first-family roots, the full centered row errors have trace product $-k$.

20.4 A common algebra for noncommuting repaired projections

A descending flag gives nested, commuting projections. The following result allows genuinely different isotropic spaces and specifies exactly what keeps their subsequent ordered interactions tractable. On $\mathbb{C}^{\mathbb{F}_k}$, let P_U be the orthogonal projection onto E_U , and supply isotropic spaces $U_1, \dots, U_g$ containing $\mathcal{R}$. Put

$$K = \bigcap_j U_j, \quad T = \sum_j U_j, \quad \kappa = \dim K, \quad \tau = \dim T, \quad h = \tau - \kappa. \quad (325)$$

Every U_j is orthogonal to K , so $T \subseteq K^{\perp_B}$ and $\tau \leq r - \kappa + 1$.

Theorem 20.4 (Common blocks and exact interaction closure). *All P_{U_j} vanish on $E_K^\perp$. They preserve the mutually orthogonal spaces*

$$\mathcal{H}_C = E_K \cap \{f : \text{supp } f \subseteq C\}, \quad C \in \mathbb{F}_k/T,$$

which sum to E_K . There are $J = 2^{r-\tau}$ blocks, of exact dimensions

$$b_C = \begin{cases} 2^h, & C \subseteq K^{\perp_B}, \\ 2^{h+1}, & C \cap K^{\perp_B} = \emptyset. \end{cases} \quad (326)$$

The first case occurs for $2^{r-\kappa+1-\tau}$ blocks. In an explicit common real orthonormal basis, every generating projection has dyadic rational blocks. The unital algebra on E_K generated by these projections is contained in $\bigoplus_C M_{b_C}(\mathbb{C})$, with $b_C \leq b = 2^{h+1}$. Ordered sums, products, adjoints, range projections and range intersections stay inside this same block algebra.

Proof. Set $f_U(x) = (-1)^{q(x)+\text{Tr}(\alpha_U x)}$. The first root family spans functions constant on each U -coset; the second spans their products with f_U . On a coset $x + U$, f_U is a sign times the character $u \mapsto (-1)^{B(x,u)}$, hence constant or orthogonal to the constant function. Thus

$$E_U|_{x+U} = \text{span}\{1, f_U|_{x+U}\}.$$

The actual orthogonal projection kernel vanishes unless $x + y \in U$, and otherwise equals

$$P_U(x, y) = |U|^{-1} \begin{cases} 1, & x \in U^{\perp_B}, \\ 1 + f_U(x)f_U(y), & x \notin U^{\perp_B}. \end{cases} \quad (327)$$

This is also $V_U G_U^+ V_U^*$ from the signed root Gram. Since $K \subset U$, the difference $\alpha_U + \alpha_K$ annihilates K , so f_U/f_K is constant on each K -coset. Consequently $E_U \subseteq E_K$; no common correction for all U_j is required. The support of (327) then proves preservation of each T -coset.

For every K -coset $D = x + K$, take $|K|^{-1/2}1_D$, and also $|K|^{-1/2}f_K 1_D$ when $x \notin K^{\perp_B}$. These vectors are orthonormal and span E_K . Each T -coset contains 2^h K -cosets and is either inside or disjoint from $K^{\perp_B}$, proving (326) and the counts. Although normalizing the basis may use square roots, every compressed kernel entry has denominator dividing $|K||U|$, hence is dyadic with $O(r)$ bits. Blockwise matrix operations prove closure under sums, products and adjoints. Range and intersection projections follow from exact nullspace elimination and the true Gram inverse on an independent basis, still within the same blocks. $\square$

This family is not a commuting flag in disguise. Choose isotropic K with $\kappa \leq (r-1)/2$ and $e, f \in K^{\perp_B}$ satisfying $B(e, f) = 1$. Take $U = K + \langle e \rangle$ and $V = K + \langle f \rangle$, so $h = 2$. On the zero T -coset, in the normalized indicator basis of $K, e + K, f + K, e + f + K$, the actual projections are

$$P_U = I_4 - aa^T, \quad P_V = I_4 - cc^T, \quad a = \frac{1}{\sqrt{2}}(1, -1, 0, 0)^T, \quad c = \frac{1}{\sqrt{2}}(1, 0, -1, 0)^T. \quad (328)$$

Here $\langle a, c \rangle = 1/2$, so

$$\| [P_U, P_V] \|_F^2 = 3/8, \quad \text{spec}(P_U P_V P_U) = \{1, 1, 1/4, 0\}.$$

The intersection has rank two, whereas the ordered product retains a further singular value $1/2$. Recording intersections alone therefore loses a channel needed by subsequent products.

Each binary sum costs $O(Jb^2)$ arithmetic and each product, range, or intersection costs $O(Jb^3)$. Range calculations may introduce nondyadic rational denominators: determinant bounds give output lengths $O(b[B_0 + \log(b+1)])$ for a block with common-denominator input length B_0 . Appending a new isotropic space replaces K by $K' = K \cap U$ and T by $T' = T + U$. If $a = \kappa - \dim K'$ and $c = \dim T' - \tau$, then $h' = h + a + c$, and the block bound grows by 2^{a+c} . Rebuild the common basis and retained expressions; keeping the old blocks would be false closure. Repeated operations on a fixed subspace family do not increase h .

A conservative geometric preprocessing bound, evaluating the projection kernels directly and compressing one T -coset at a time, is $O(gJb^2 2^{2\tau})$ rational operations plus $O(gk^2 r^3)$ binary field work. Initial block storage is $O(gJb^2 r)$ bits. These bounds are polynomial in the explicit coefficient-coordinate input, although not necessarily optimal. They enumerate neither random outcomes nor all isotropic spaces.

20.5 Polynomial programs and the unchanged Walsh sampling event

The block algebra gives a concrete coefficient class for Theorem 9.1. Supply independent original signs $z_1, \dots, z_D$ and a written sum of q_0 ordered product programs of length at most L , with deterministic coefficients in the algebra of Theorem 20.4. At stage j , declare the union V_j of the primitive signs used by the local factors, with $|V_j| \leq a$ and at most t actual character terms after coherent merging. Charge padding, the written expression size, and the live primitive boundary

$$w = \max_j |(V_1 \cup \dots \cup V_j) \cap (V_{j+1} \cup \dots \cup V_L)|.$$

Reduced polynomial degree is at most $\min(D, La)$; it does not replace these structural parameters. Direct sums and fixed readouts encode the q_0 products inside each geometric block, with internal size at most $\beta = q_0 b$. Define

$$\begin{aligned} H_p &= \sum_C \binom{b_C + p - 1}{p}, \quad s_p = \binom{\beta + p - 1}{p}, \\ T_p(z) &= \sum_C \|\text{Sym}^p Z_C(z)\|_F^2, \quad \mathcal{F}_{2p} = (\mathbb{E} T_p)^{1/(2p)}. \end{aligned} \quad (329)$$

Exact conditional positive-message updates on the original primitive signs then give, for every integer $p \geq 1$,

$$H_p^{-1/(2p)} \mathcal{F}_{2p} \leq (\mathbb{E} \|Z\|^{2p})^{1/(2p)} \leq \mathcal{F}_{2p}. \quad (330)$$

The same-input proof is pointwise: $\|Z\|^{2p} \leq T_p \leq H_p \|Z\|^{2p}$, since singular values of a symmetric power are degree- p monomials in those of the original block. The internal size β affects computation; the output size b affects the comparison. At $p \geq b + \log(J/\eta)$ the comparison to the Markov upper threshold is at most $\sqrt{2e}$.

After geometric preprocessing, the arithmetic cost is

$$O\left(J(L+1)2^{w+a}[t\beta^2 + \beta p s_p^2 + s_p^3]\right), \quad O(2^w s_p^2) \text{ working storage.} \quad (331)$$

Input and fixed-readout construction are additional. If separate products have different character supports, t is their union term count, at most q_0 times the individual count. With a common input denominator of length B_0 and direct projection entries, an explicit sufficient intermediate bit bound, up to a universal constant, is

$$p(L+2)[B_0 + 2r + \log_2(t + \beta + 2)] + D + p \log_2(p + \beta + 2) + \log_2(J + 2).$$

Charge larger actual coefficient lengths if projections were themselves built by long arithmetic expressions. Schoolbook bit work is (331) times the square of this bound, plus label scheduling. For fixed h, q_0, a, t , the lift size is polynomial in p ; with logarithmic w and $p = O(\log(k/\eta))$, costs are polynomial in the explicit parameters and $\log(1/\eta)$. Large spread or many distributed product terms are charged rather than hidden.

Adding programs changes q_0 to at most $q_1 + q_2$; multiplying them changes it to at most $q_1 q_2$ and adds word lengths, preserving order. Primitive identifications substitute the same sign everywhere, cancel repeated powers modulo two and rebuild the live boundary. Already eliminated signs cannot be reused without revisiting the retained program. Algebraic operations have zero approximation error, but their new program costs must be paid.

A separate lower-tail certificate follows by computing $N_p = \mathbb{E} T_p^2$ with paired programs sharing the same original signs. For $\mathbb{E} T_p > 0$ and $0 < \theta < 1$,

$$\mathbb{P}\{\|Z\| > (\theta \mathbb{E} T_p / H_p)^{1/(2p)}\} \geq (1 - \theta)^2 (\mathbb{E} T_p)^2 / N_p.$$

Computing the pair uses J^2 block pairs, storage $O(2^w s_p^4)$, and replaces the cubic lift term in (331) by s_p^6 . This is an explicit confidence range, which can be small; it is not matching extreme quantiles merely from moment comparison.

There is a stronger, distinct confidence consequence for original Walsh sampling. If G_m is its actual sample Gram, the event of Theorem 20.3 for intersection dimension $\kappa = s$ implies simultaneously

$$\frac{1}{2}P^*P \preceq P^*G_mP \preceq \frac{3}{2}P^*P \quad (332)$$

for every operator whose range lies in any of the retained E_K , hence for every ordered sum, product, range and intersection built from its isotropic projection algebra. Insert $Pv \in E_K$ into the common quadratic-form event to prove (332). No entropy for word length is added; the event already covers every vector, including choices made after the sample. It gives $\sqrt{1/2}\|P\| \leq \|m^{-1/2}X_mP\| \leq \sqrt{3/2}\|P\|$, with a maximizing direction of this same P as lower witness. The family includes $P = P_K$, so its full common-event sample threshold has the same two-sided constants (324). This is a lower statement for the whole family, not a necessary sample count for every fixed operator.

For odd $r \geq 19$, one can have $\kappa = \lceil \log_2(r^2) \rceil$ and the spread-two noncommuting example (328) simultaneously. The retained coefficient rank is then of order k/r^2 , while common blocks have size at most eight. This separates large coefficient dimension from small irreducible interaction size. The actual sampled Gram generally couples these blocks; it is not automatically an input to the block moment calculus. Controlling those couplings or allowing arbitrary root spaces is still an additional mathematical problem.

21 Actual Walsh couplings beyond the cubic block algebra

The small blocks in Theorem 20.4 are invariant for the specified deterministic coefficient projections. The original sampled Walsh Gram need not preserve them. The result below quantifies these actual couplings: they have full cross covariance, and in a stated sample regime a very low-rank approximation misses their natural fluctuation scale with constant probability. A separate exact occupancy construction makes the entire centered Gram a flat cross coupling. These are obstructions to specified state reductions, not to every dense norm calculus.

21.1 Original cross covariance and whole-sample contractions

Let $S = \{s_1, \dots, s_k\} \subseteq \mathbb{F}_2^d$ consist of distinct labels with no four distinct labels summing to zero. Thus S is a Sidon Walsh support; the odd-dimensional cubic support is one example. Write $x(y)_i = (-1)^{s_i \cdot y}$ for the original uniform row. Supply real orthonormal coefficient matrices $Q_L \in \mathbb{R}^{k \times a}$ and $Q_R \in \mathbb{R}^{k \times b}$ with *disjoint coordinate supports*. Orthogonality alone is not the stated hypothesis. Set

$$u = Q_L^T x, \quad v = Q_R^T x, \quad Z = uv^T, \quad C_m = \frac{1}{m} \sum_{i=1}^m Z(Y_i) = Q_L^T (G_m(S) - I) Q_R, \quad (333)$$

where Y_i are independent observations from the same original uniform Walsh row law. In the cubic decomposition, the two matrices can use columns supported on disjoint collections of T -cosets, including the noncommuting geometries inside each collection.

Lemma 21.1 (Full original cross covariance). *One has*

$$\mathbb{E}Z = 0, \quad \mathbb{E}Z_{ij}Z_{kl} = \delta_{ik}\delta_{jl}, \quad \mathbb{E}ZZ^T = bI_a, \quad \mathbb{E}Z^TZ = aI_b. \quad (334)$$

Consequently, with $D_0 = ab$,

$$\mathbb{E} \text{vec}(C_m) \text{vec}(C_m)^T = m^{-1} I_{D_0}, \quad \mathbb{E} \|C_m\|_F^2 = D_0/m. \quad (335)$$

Every predetermined linear map $\mathcal{L} : \mathbb{R}^{a \times b} \rightarrow \mathbb{R}^{a \times b}$ of linear rank at most R satisfies

$$\mathbb{E} \|C_m - \mathcal{L}(C_m)\|_F^2 = m^{-1} \|I - \mathcal{L}\|_{\text{HS}}^2 \geq (D_0 - R)/m. \quad (336)$$

Equality in the last bound holds for orthogonal projection onto any fixed R -dimensional linear channel space.

Proof. Distinctness gives $\mathbb{E}xx^T = I$, even when zero is a selected frequency. The exact Sidon fourth identity, for real coefficient vectors c_j , is

$$\begin{aligned} \mathbb{E} \prod_{j=1}^4 \langle x, c_j \rangle &= \langle c_1, c_2 \rangle \langle c_3, c_4 \rangle + \langle c_1, c_3 \rangle \langle c_2, c_4 \rangle \\ &\quad + \langle c_1, c_4 \rangle \langle c_2, c_3 \rangle - 2 \sum_i \prod_{j=1}^4 (c_j)_i. \end{aligned} \quad (337)$$

Expand original characters: only even index multiplicities survive, and the correction removes two excess counts of the all-equal pattern. Apply this to two left and two right columns. Disjoint coordinate supports kill the diagonal correction, and orthogonality kills both cross pairings. The remaining pairing is $\delta_{ik}\delta_{jl}$. This proves (334); independence across sample names gives (335). Applying its scalar covariance to the linear operator $I - \mathcal{L}$ proves the identity in (336). An orthonormal basis of $\ker \mathcal{L}$, of dimension at least $D_0 - R$, contributes that much squared Hilbert–Schmidt norm; a fixed orthogonal projection attains equality. $\square$

Every individual observation in (333) has matrix rank at most one, but its linear span is the full ab -dimensional rectangular space by (334). Fixed linear-state dimension and the matrix rank of an adaptive approximation are different parameters. The next identities allow a separate statement about the latter.

Put $\mu_4 = \mathbb{E} \|Z\|_F^4 = \mathbb{E} \|u\|^4 \|v\|^4$. The complete original sample sum obeys

$$\mathbb{E} \|C_m\|_F^4 = \frac{m\mu_4 + m(m-1)(D_0^2 + 2D_0)}{m^4}, \quad (338)$$

$$\mathbb{E} \text{tr}((C_m C_m^T)^2) = \frac{m\mu_4 + m(m-1)ab(a+b+1)}{m^4}. \quad (339)$$

To prove them, expand the four sample names. A singleton name vanishes because $\mathbb{E}Z = 0$. The all-equal pattern contributes μ_4 in both expressions, since Z has rank at most one. For two paired names, the three Frobenius contractions are D_0^2, D_0, D_0 ; the three ordered trace contractions, for an independent original copy W , are

$$\mathbb{E} \text{tr}(ZZ^T WW^T) = ab^2, \quad \mathbb{E} \text{tr}(ZW^T WZ^T) = a^2b, \quad \mathbb{E} \text{tr}(ZW^T ZW^T) = ab.$$

The first two follow from the matrix second moments in (334); entrywise use of that same covariance twice proves the third. Each pairing has $m(m-1)$ assignments, proving the formulas. No positivity of each individual ordered trace word is assumed.

The only extra row contraction has the uniform bound

$$\|u\|^2 + \|v\|^2 \leq k, \quad \mu_4 \leq \frac{k^2}{4} \mathbb{E} \|Z\|_F^2 = \frac{k^2 ab}{4}. \quad (340)$$

The first inequality is the disjoint orthogonal projection bound, and the product of the two squared norms is at most $k^2/4$. An exact μ_4 can be retained if available; the displayed envelope does not require its full original law.

21.2 A constant-probability obstruction to adaptive low rank

Theorem 21.2 (Spectral mass beyond a small adaptive rank). *Assume $a = b = q \geq 2$ and integer $m \geq k^2/(4q)$. With probability at least $1/16$, simultaneously for every matrix B of rank at most any integer $0 \leq R \leq q/1024$,*

$$\|C_m - B\|_F > \frac{q}{2\sqrt{m}}, \quad \|C_m - B\| > \frac{1}{2}\sqrt{q/m}. \quad (341)$$

The approximation may depend on the entire observed sample. The statement allows a nonzero approximation rank when $q \geq 1024$; for $2 \leq q < 1024$ it is only the $B = 0$ norm certificate.

Proof. Let $F = \|C_m\|_F^2$ and $T = \text{tr}((C_m C_m^T)^2)$. The exact contractions, (340), and the sample hypothesis give

$$\mathbb{E}F = q^2/m, \quad \mathbb{E}F^2 \leq \frac{q^4}{m^2}(1 + 2/q^2 + 1/q) \leq \frac{2q^4}{m^2}, \quad \mathbb{E}T \leq \frac{4q^3}{m^2}.$$

Strict Paley–Zygmund yields $\mathbb{P}\{F > q^2/(2m)\} \geq 1/8$; Markov yields $\mathbb{P}\{T > 64q^3/m^2\} \leq 1/16$. Their intersection therefore has probability at least $1/16$. On it, the energy in the top R squared singular values is at most

$$\sqrt{RT} \leq \frac{8\sqrt{Rq^3}}{m} \leq \frac{q^2}{4m}.$$

Hence the remaining squared singular-value sum is strictly greater than $q^2/(4m)$. Singular-value truncation minimizes Frobenius error among rank- R matrices, proving the first inequality. For completeness, any rank- R perturbation has a kernel intersecting the span of the top $R + 1$ right singular vectors, so its operator error is at least σ_{R+1} . That singular value is at least the square root of the remaining energy divided by $q - R$, and is therefore greater than $\sqrt{q/m}/2$. The event was defined using only F, T , so the conclusion is simultaneous for all adaptive choices of B and all permitted R . $\square$

Thus rank $o(q)$ cannot achieve error below this scale with arbitrarily high confidence in the stated regime. The result is for one cut fixed before sampling; it does not assert failure on every adaptive partition, and it is not an impossibility theorem for algorithms that estimate only the norm.

21.3 An exact flat cross coupling in the original sample law

There is a separate deterministic realization whose diagonal component does not conceal the cross error. Let $a = b = q$, let O be a real orthogonal $q \times q$ matrix, and set

$$f(y) = u(y)^T O v(y), \quad S_0 = Q_L O Q_R^T + Q_R O^T Q_L^T, \quad \rho(y) = 1 + f(y)/k.$$

Then the same original Sidon law satisfies

$$\mathbb{E}f = 0, \quad \mathbb{E}xx^T f = S_0, \quad |f| \leq k/2, \quad \mathbb{E}[\rho(Y)x(Y)x(Y)^T] = I + k^{-1}S_0. \quad (342)$$

Indeed apply (337) to two coordinate vectors and the left and right columns. The trace pairing vanishes by orthogonality; the all-equal correction vanishes by disjoint supports; the other two pairings are exactly S_0 . Cauchy–Schwarz and (340) give $|f| \leq k/2$. Thus ρ is a probability density between $1/2$ and $3/2$ on the unchanged original row space. On range $Q_L \oplus \text{range } Q_R$, S_0 is $\begin{pmatrix} 0 & O \\ O^T & 0 \end{pmatrix}$. Its nonzero eigenvalues are $+1, -1$, each with multiplicity q . The centered Gram in (342) has norm exactly $1/k$, and all q singular values of its cross block are $1/k$. A rank- R approximation with $R < q$ therefore has cross operator error at least $1/k$, the norm of the entire centered Gram.

For the explicit cubic common basis, every column has entries zero or $\pm|K|^{-1/2}$. If O is rational over positive denominator D_O , then $|K|D_O f(y)$ is integral. With $N = k^2$, choose the positive integer original point occupancies

$$n_y = k|K|D_O + |K|D_O f(y), \quad m = Nk|K|D_O. \quad (343)$$

They sum to m because $\mathbb{E}f = 0$ and are positive because $|f| \leq k/2$. Their empirical weights are $\rho(y)/N$, so their actual empirical Gram is exactly (342). This multiset has original iid uniform probability $m!/(N^m \prod_y n_y!) > 0$. No replacement sampling law is used in this construction. The probability can be very small, so it is not a substitute for the constant-probability theorem above. Taking $O = I$ gives the rank example; for $q = 4^j$ a normalized Walsh matrix with denominator 2^j makes the coupling dense while preserving its flat singular spectrum.

21.4 Consequences for the cubic descriptor and explicit costs

In the spread-two geometry of Section 20, take odd $r \geq 19$, $k = 2^r$, $\kappa = \lceil \log_2(r^2) \rceil$, and $\tau = \kappa + 2$. The common blocks have size four or eight, and $J = 2^{r-\tau} \geq 2$. Split these blocks into two groups of $J/2$ and keep the same number q of orthonormal columns from each side, taking q to be the smaller side dimension. Their coordinate supports remain disjoint, and for $n = \dim E_K$,

$$n/4 \leq q \leq n/2, \quad q \asymp k/r^2. \quad (344)$$

Each side originally has dimension between $2J$ and $4J$, which proves these bounds. Thus small deterministic blocks coexist with growing cross rank. Theorem 21.2 applies for $m \geq k^2/(4q)$, a threshold of order kr^2 ; nonzero-rank exclusion also requires $q \geq 1024$. No union bound or cut-selection entropy is used. The fixed-linear bound charges q^2 linear channels, while the adaptive bound charges matrix rank of order q for the stated accuracy; these must not be identified.

There is nonetheless an exact update of the uncompressed state:

$$C_{m+1} = \frac{m}{m+1}C_m + \frac{1}{m+1}u(Y_{m+1})v(Y_{m+1})^T.$$

The scalar second and fourth moment formulas update by replacing m by $m+1$, with zero algebraic error. An insertion AZB transforms entry covariance into the tensor product of the actual matrices AA^T and B^TB ; it does not preserve the isotropic formulas unchanged unless the appropriate isometries hold. The coefficient sums must therefore remain in the state.

Given the coefficient pieces and a supplied or bounded μ_4 , evaluating (335), (338), and (339) takes constant arithmetic. The dimension-based envelopes have $O(\log(k+m+a+b))$ -bit rational parameters. An exact rational μ_4 carries its additional numerator/denominator bit length; arbitrary real inputs do not supply a rational oracle. Original coefficient projection preprocessing keeps its previously stated cost. Computing a new projected row directly costs $O(k(a+b))$ arithmetic plus character evaluation; a dense update costs $O(ab)$. A factored representation grows by at most one rank per observation. The obstruction rules out a proposed small-rank approximation at its specified tolerance, not a faster different representation. Listing all k^2 occupancies in (343) is an explicit verification construction, not an efficient general norm law.

The saved exact suite checks 21,008 original cross covariance entries, 266,304 original assignments in finite-sample expansions, all three ordered pair contractions, positive integer original occupancies, dense orthogonal couplings, and the probability/rank constants. For $r = 3, q = 2, m = 8$, an exact convolution aggregates $64^8 = 2^{48}$ original samples into 37,761 sum states and finds the spectral-mass event probability $369709833231/549755813888 > 1/16$. At this small q only rank zero is permitted; nonzero-rank constants are checked separately starting at $q = 1024$. These checks supplement the all-size proofs and do not establish them by extrapolation. Sections 23 and 24 now give two controlled successors: a supplied signed Fourier core propagates

one explicit remainder through higher interactions, and the quadratic HDH continuation has a two-sided same-input theorem. They do not produce a small exact cross-block state for every polynomial or simultaneously pool all Walsh supports. Full FR-10 remains open.

22 Dense closure after symmetry breaking and joint Grams

The grouped-sector theorem keeps every coefficient in a supplied permutation centralizer. Two later constructions enlarge that dense regime in different directions. The first absorbs finitely many arbitrary low-rank interactions into one exact cyclic space. The second allows signed, mixed-degree Clifford polynomials driven by several overlapping pencils, provided their complete attained joint Gram is available. Neither construction estimates the old and new pieces separately, so cancellation between them remains visible.

22.1 Cyclic seed spaces are stable under every later product

Supply an exact star-algebra representation

$$\mathcal{H} = \bigoplus_{\lambda \in \Lambda} (\mathbb{C}^{d_\lambda} \otimes \mathbb{C}^{\mu_\lambda}), \quad \mathcal{A} = \bigoplus_{\lambda \in \Lambda} M_{d_\lambda}(\mathbb{C}) \otimes I_{\mu_\lambda}, \quad (345)$$

and fixed seed vectors $v_1, \dots, v_r \in \mathcal{H}$. A rank- s operator is included by listing bases for both its range and its adjoint range, using at most $2s$ seeds. Put

$$N_{\mathcal{A}} = \sum_{\lambda} d_\lambda^2, \quad D_0 = \sum_{\lambda} d_\lambda.$$

Writing

$$v_a^{(\lambda)} = \sum_{i=1}^{d_\lambda} e_i \otimes r_{a,\lambda,i}, \quad M_\lambda = \text{span}\{r_{a,\lambda,i} : a, i\},$$

defines the cyclic active space

$$W = \mathcal{A} \text{span}\{v_a : a \leq r\} = \bigoplus_{\lambda} \mathbb{C}^{d_\lambda} \otimes M_\lambda. \quad (346)$$

Theorem 22.1 (Extension-stable cyclic seed reduction). *The space W reduces every base coefficient and every seed outer product $v_a v_b^*$. If $t_\lambda = \dim M_\lambda$, then*

$$t_\lambda \leq \min\{\mu_\lambda, r d_\lambda\}, \quad N_W := \dim W = \sum_{\lambda} d_\lambda t_\lambda \leq r N_{\mathcal{A}}. \quad (347)$$

For every ordered polynomial or evaluation circuit P in the base algebra, the seed outer products, and scalar random scores,

$$P \simeq P_W \oplus \bigoplus_{\lambda: \mu_\lambda > t_\lambda} P_\lambda^{(0)} \otimes I_{\mu_\lambda - t_\lambda}, \quad (348)$$

where $P_\lambda^{(0)}$ is obtained by setting every seed operator to zero on the untouched copy. The decomposition remains exact after arbitrary sums, ordered products, adjoints, and later base-algebra insertions.

Proof. The matrix unit E_{ij} in sector λ maps v_a to $e_i \otimes r_{a,\lambda,j}$ and kills every other sector. These vectors span the right side of (346); this proves the equality and (347). The space is reducing for $\mathcal{A}$. Every seed belongs to it, so $v_a v_b^*$ acts inside W and vanishes on $W^\perp$. The complement is exactly the sum of the remaining multiplicity copies, which proves (348). Restriction to a common reducing space is an algebra homomorphism and commutes with the adjoint, proving all future update statements. $\square$

The state can be built without choosing coordinates in an exponentially large multiplicity space. Retain the actual coefficient Gram entries

$$f_{ab,\lambda,ij} = \langle v_a, E_{\lambda,ij} v_b \rangle.$$

For a diagonal positive sector metric with entries $g_{\lambda,i}$, the matrix

$$H_\lambda[(a,i), (b,j)] = f_{ab,\lambda,ij}/g_{\lambda,i} = \langle r_{a,\lambda,i}, r_{b,\lambda,j} \rangle \quad (349)$$

determines t_λ , an exact active metric, every seed coordinate, and therefore every active outer product. All complex phases and cross-sector couplings are retained. Separate norms or separate seed Gram spectra do not determine (349).

For a finite original shared-score law and integer $h \geq 1$, let T_h be the sum of the h th powers of the retained squared singular values, with one copy of each nonempty untouched sector. If

$$D_{\text{keep}} = N_W + \sum_{\lambda: \mu_\lambda > t_\lambda} d_\lambda,$$

then the same physical matrix satisfies pointwise

$$\|P\|^{2h} \leq T_h \leq D_{\text{keep}} \|P\|^{2h}. \quad (350)$$

Thus its exact score law yields every positive norm moment and every strict upper-tail quantile within $D_{\text{keep}}^{1/(2h)}$. A maximizing active or untouched-sector vector lifts to the original matrix and supplies the lower witness. The factor is calculated once from the final state, rather than multiplied after each interaction.

For the collective-spin algebra on L sites, $N_{\mathcal{A}} = \binom{L+3}{3}$. If the seeds are supplied as open-boundary matrix-product states of bond at most χ , their complete coefficient Grams follow from the four-variable transfer

$$F_{ab}(g) = \langle v_a, g^{\otimes L} v_b \rangle.$$

There are $N_{\mathcal{A}}$ homogeneous coefficients, and all seed pairs cost $O(r^2 L \chi^4 N_{\mathcal{A}})$ arithmetic. Forming the common change-of-basis data costs $O(N_{\mathcal{A}} L^2)$, a conservative dense factorization costs $O(N_{\mathcal{A}}^3)$, and forming the seed right sides costs $O(r^2 N_{\mathcal{A}}^2)$, followed by $O(\sum_\lambda (rd_\lambda)^3)$ for exact Gram ranks and coordinates. These costs are polynomial for fixed bond and polynomial seed count, but they are not small for arbitrary physical seed arrays.

The dimension bound is sharp in the relevant sense. On $\mathbb{C}^n \otimes \mathbb{C}^n$, the algebra $M_n \otimes I_n$ together with the single maximally entangled seed $\sum_i e_i \otimes e_i$ generates the whole n^2 -dimensional active space and the full matrix algebra there. Low-rank input generators therefore do not imply low-rank outputs. Conversely, a spin-zero sector can have exponential multiplicity, while r seeds activate at most r of its multiplicity directions; the untouched scalar copy is precisely what keeps the exact representation economical.

22.2 Complete attained Grams close mixed-degree Clifford words

Let $A_1(z), \dots, A_q(z)$ be named Hermitian Clifford pencils on one even faithful ambient representation, with

$$A_i A_j + A_j A_i = 2Q_{ij}(z)I. \quad (351)$$

The retained datum is the complete positive semidefinite joint Gram $Q(z)$, including all cross-pencils. Let

$$Z(z) = \sum_{\alpha} C_{\alpha} \otimes A_{\alpha_1}(z) \cdots A_{\alpha_{r_{\alpha}}}(z), \quad C_{\alpha} \in M_b(\mathbb{C}),$$

where degrees may differ, coefficients may have either sign, and products remain in their written order.

Theorem 22.2 (Actual joint-Gram Clifford calculus). *For every attained $Q \succeq 0$ of rank $r(Q)$ there is an exact physical metric representation $\hat{Z}(Q)$ of dimension $b2^{r(Q)}$, with rational Clifford generators and metric when Q is rational, such that*

$$\|Z(z)\| = \|\hat{Z}(Q(z))\|_{\text{physical}}. \quad (352)$$

If $r_{\max}$ is the largest attained rank and $D_{\text{eff}} = b2^{r_{\max}}$, then

$$\tau_h(Q) = \text{tr}[(\hat{Z}(Q)^\# \hat{Z}(Q))^h],$$

where $\#$ is the adjoint in the physical metric of the representation, satisfies pointwise

$$\|Z\| \leq \tau_h(Q)^{1/(2h)} \leq D_{\text{eff}}^{1/(2h)} \|Z\|. \quad (353)$$

Consequently an exact finite law of the actual Gram gives all positive norm moments and all strict upper-tail quantiles within this factor, including atoms at rank changes.

Proof. Exact positive elimination selects an independent reached-vector basis with nonsingular principal Gram and expresses every named vector in that basis. Exterior creation and annihilation operators give a faithful Clifford representation in the induced positive metric. Relation (351) therefore extends to an algebra homomorphism on every ordered word, proving (352). The trace readout is the sum of the $2h$ th powers of at most D_{eff} singular values of this same matrix, which proves (353). Integration gives the moment statement; pointwise stochastic order gives the strict-quantile statement separately. $\square$

Addition joins term lists, multiplication concatenates words, and the adjoint reverses them and adjoints the external coefficients. Adding a pencil requires every new cross entry of Q , and identifying primitive signs substitutes the same name throughout the complete score program. A fresh affine condition is inserted into that actual program. Equal marginal radii, separate covariance matrices, or separate Gram tables cannot supply the missing cross terms.

If the Gram entries are determined by c bounded integer scores, each a sum of m local maps taking values in $\{-K, \dots, K\}$, of arity at most a with live primitive boundary w , their joint law is computed exactly in at most $2^w(2mK + 1)^c$ states. With $s_* = b2^q$, a conservative arithmetic bound is

$$O\left(mc2^{w+a}(2mK + 1)^c + (2mK + 1)^c \left[q^3 4^q + (G + \log(h + 1))s_*^3\right]\right), \quad (354)$$

where G is the actual circuit evaluation length. Exact pivot and coefficient bit lengths are charged through the corresponding rational minors. The useful regime has controlled q, c, a, w , numerical score range, circuit length, and coefficient bits. An arbitrary dense Parseval family need not admit such a score program.

For three cyclic shifts of $N = 32$ signs, the complete Gram is determined by two overlapping local scores and has 130 attained score pairs. The original physical Clifford matrix has dimension 65,536, while the reached regular matrices have dimension at most eight. At $h = 20$, order-40 moments and the strict confidence level 2^{-20} are bracketed within $8^{1/40} < 1.1$. The mixed polynomial

$$I + A_1 + \frac{1}{4}(A_2 A_3 - A_3 A_2)$$

has an exact two-score norm formula, so this benchmark checks the complete signed multiplication rather than only its comparison bound.

The even faithful ambient hypothesis records a real obstruction. For the irreducible Pauli triples (X, Y, Z) and $(X, Y, -Z)$ the positive Gram is I_3 in both cases, but

$$I + iXYZ = 0, \quad I + iXY(-Z) = 2I.$$

An irreducible Pauli representation of an odd-dimensional Clifford coordinate space must therefore retain the orientation of its central volume element. A doubled faithful representation contains both orientations and is covered by Theorem 22.2. This example also explains why a complete positive Gram can close one coefficient class while failing after an unrecorded change of representation.

22.3 What these dense extensions establish

The cyclic construction is a reusable update rule for supplied low-rank symmetry breaking: future interactions remain inside one exact coupled block, and all state growth is charged through new seed directions. The Clifford construction is a reusable rule for arbitrary signed mixed-degree words in a fixed faithful Clifford geometry: future interactions remain inside the algebra determined by the complete joint Gram. Both retain same-input lower sectors and direct strict quantiles when the original finite score law is tractable. They do not yield an affordable descriptor for arbitrary dense coefficients, arbitrary entangling ambient operators, or an unknown score law. Those input-construction questions remain model-specific.

23 Extension-stable signed conditional cores

This section gives a restricted affirmative answer to the central closure question. The retained state is an exact orthogonal projection in the *actual conditional Fourier Gram*; it keeps affine phases, named primitive reuse, and matrix multiplication order. Operations supported on the retained characters are exact bimodule operations. General products can return two discarded modes to the retained sector, so the update carries an additive remainder certificate through the whole continuation. This last point is essential: after such a product, the represented core is a certified approximation and is not silently relabelled as the exact projection.

23.1 The conditional projector and its exact algebra

Let the original binary primitives be uniform on the nonempty affine space

$$E = \{y \in \mathbb{F}_2^D : Cy = b\},$$

and retain the full equation system, its phases, and one origin $y_0 \in E$. Characters on $\ker C$ are indexed by $Q = \mathbb{F}_2^D / \text{row}(C)$. If $F = \sum_t A_t \chi_t$ is a matrix polynomial, define its actual conditional coefficients by

$$B_\gamma = \sum_{t + \text{row}(C) = \gamma} (-1)^{t \cdot y_0} A_t, \quad \gamma \in Q. \quad (355)$$

The origin phase occurs exactly once in this formula. The quotient characters, evaluated on $y - y_0 \in \ker C$, are orthonormal in $L^2(E)$. Thus all coefficient merging in (355) is a signed matrix sum performed before norms are taken.

Choose a subgroup $H \leq Q$ of rank h and set

$$A = P_H F = \sum_{\gamma \in H} B_\gamma \chi_\gamma, \quad R = F - A. \quad (356)$$

Then P_H is conditional expectation onto the sigma-algebra generated by the h retained characters. In particular it is an entrywise orthogonal projection, and the discarded energies are exactly

$$M_R^{\text{right}} = \mathbb{E}_E R^* R = \sum_{\gamma \notin H} B_\gamma^* B_\gamma, \quad M_R^{\text{left}} = \mathbb{E}_E R R^* = \sum_{\gamma \notin H} B_\gamma B_\gamma^*, \quad e_2^2 = \sum_{\gamma \notin H} \|B_\gamma\|_F^2. \quad (357)$$

These matrices are part of the certificate; replacing them by coordinatewise coefficient bounds would discard cancellations and directions needed by later interactions.

Lemma 23.1 (Exact bimodule rule and returning modes). *If L and B are H -supported matrix polynomials of compatible sizes, then*

$$P_H(LFB) = L(P_H F)B, \quad P_H(LRB) = 0. \quad (358)$$

The ordered coefficient convolution inside the core is

$$(AB)_\tau = \sum_{\gamma+\rho=\tau} A_\gamma B_\rho.$$

For arbitrary decompositions $F = A + R$ and $G = B + S$ one instead has the exact identity

$$\boxed{P_H(FG) = AB + P_H(RS).} \quad (359)$$

The last term need not vanish and is not determined by the two energy matrices separately.

Proof. Adding an element of H leaves every coset of H invariant. Convolution therefore proves (358). In the expansion of FG , the mixed terms AS and RB remain outside H , whereas two off-core frequencies can sum to an element of H . This gives (359) and also shows why a scalar residual radius cannot reconstruct the returned signed matrix coefficient. $\square$

If a fresh affine equation has quotient frequency $c \in H$, its two branches identify $\chi_c = \eta$ with the actual phase η . The new core is the image of H in $Q/\langle c \rangle$; an off-core coset cannot merge with a core coset. If s independent consistent equations of this kind are added, the elementary inequality $(U + \eta V)^*(U + \eta V) \leq 2(U^*U + V^*V)$ and its left analogue give

$$M_{R,\text{new}}^{\text{right}} \preceq 2^s M_R^{\text{right}}, \quad M_{R,\text{new}}^{\text{left}} \preceq 2^s M_R^{\text{left}}. \quad (360)$$

A reused equation has rank cost zero; an inconsistent phase is a zero state. An equation outside H can merge a discarded coset with a retained one. Restoring an exact orthogonal projection then requires promotion of that direction from the source dictionary, or recomputation. The energy certificate alone contains too little signed information to perform that promotion.

23.2 High moments and continuation through a circuit

The L^2 certificate becomes useful at confidence-scale moments when the discarded Fourier degree in the independent coordinates of the conditional law is controlled.

Lemma 23.2 (Weighted conditional Fourier energy). *Let $R = \sum_{S \subseteq [w]} B_S \varepsilon_S$ in independent free signs of E . For every real $q \geq 2$,*

$$\boxed{(\mathbb{E}_E \|R\|_F^q)^{1/q} \leq \left(\sum_S (q-1)^{|S|} \|B_S\|_F^2 \right)^{1/2}.} \quad (361)$$

In particular, free-coordinate degree at most a gives the bound $(q-1)^{a/2} e_2$.

Proof. The usual two-point hypercontractive induction remains valid for a Hilbert-valued function, taking the Frobenius matrices as the Hilbert space $\mathbf{K}$. It gives $\|T_\rho f\|_{L^q(\mathbf{K})} \leq \|f\|_{L^2(\mathbf{K})}$ for $\rho = (q-1)^{-1/2}$. Apply it to the polynomial whose coefficient at S is $\rho^{-|S|} B_S$ and use the exact Hilbert-space Parseval identity. One may first condition on the retained characters and charge only degree in the remaining free coordinates. $\square$

For each circuit input F_i , suppose that a common retained subgroup has produced $A_i = P_H F_i$ and certified, at exponent P ,

$$\| \|F_i - A_i\|_F \|_{L^P} \leq e_i, \quad \| \|A_i\| \|_{L^P} \leq a_i, \quad \| \|F_i\| \|_{L^P} \leq n_i \leq a_i + e_i. \quad (362)$$

The circuit may use constants, addition, adjoint, and ordered multiplication. Give a random input leaf weight one, a constant weight zero, the maximum weight at a sum, and the sum of weights at a product. Assume that the circuit contains a random input, so its final weight satisfies $\ell \geq 1$, and set $P = p\ell$ for a target $p \geq 2$. Constant-only circuits are evaluated exactly. A gate of weight $t > 0$ is estimated at exponent P/t .

Theorem 23.3 (Continuation-safe signed core). *Replace every random input by its core and evaluate the supplied circuit with the original coefficient order, obtaining A_{out} . Propagate norm and error radii through sums and adjoints in the evident way, and at an ordered product use*

$$a_{FG} \leq a_F a_G, \quad n_{FG} \leq n_F n_G, \quad \boxed{e_{FG} \leq e_F n_G + a_F e_G}. \quad (363)$$

Then the output of the same original circuit satisfies

$$\| \|F_{\text{out}} - A_{\text{out}}\|_F \|_{L^P} \leq e_{\text{out}}.$$

No resampling is made when a primitive or an input leaf is reused. If a written polynomial has word lengths at most ℓ , total norm of coefficient products at most C , and common leaf bounds (a, e) , then

$$\boxed{e_{\text{out}} \leq C\ell e \max\{1, a + e\}^{\ell-1}}. \quad (364)$$

After any final consistent affine restriction of new rank s , the valid error is at most $2^{s/p} e_{\text{out}}$.

Proof. At a product gate with positive child weights t_F, t_G , the assigned exponents satisfy $1/(P/(t_F + t_G)) = 1/(P/t_F) + 1/(P/t_G)$. Apply Hölder's inequality to

$$FG - A_F A_G = (F - A_F)G + A_F(G - A_G),$$

using $\|UV\|_F \leq \|U\|_F \|V\|$ and $\|UV\|_F \leq \|U\| \|V\|_F$. This proves (363). When one child has weight zero, use its exact L^∞ bound and omit its reciprocal exponent. Induction proves the circuit claim. Telescoping each word gives (364). Finally, if $E' \subseteq E$ has conditional probability 2^{-s} , then for every frozen matrix error R and $q > 0$,

$$(\mathbb{E}_{E'} \|R\|_F^q)^{1/q} \leq 2^{s/q} (\mathbb{E}_E \|R\|_F^q)^{1/q}.$$

Compiling all restrictions at the end gives one factor $2^{s/p}$ rather than multiplying a rank penalty at every historical gate. $\square$

The resulting lower estimate is also for the same matrix. If $a_p = \| \|A_{\text{out}}\| \|_{L^p}$ and the final error is e , then

$$\boxed{\max\{0, a_p - e\} \leq \| \|F_{\text{out}}\| \|_{L^p} \leq a_p + e}. \quad (365)$$

At even $p = 2r$, exact finite-core arithmetic may instead use

$$T_p = (\mathbb{E} \text{tr}((A_{\text{out}}^* A_{\text{out}})^r))^{1/p}, \quad d^{-1/p} T_p - e \leq \| \|F_{\text{out}}\| \|_{L^p} \leq T_p + e,$$

with the lower endpoint truncated at zero. The factor $d^{1/p}$ is constant when $p \gtrsim \log d$. Strict quantiles require a separate coupling step: if $\mathbb{P}\{\|F_{\text{out}} - A_{\text{out}}\| > u\} \leq \beta$ and $0 < \beta < \min(\eta, 1 - \eta)$, then

$$\max\{0, q_{\|A_{\text{out}}\|}(\eta + \beta) - u\} \leq q_{\|F_{\text{out}}\|}(\eta) \leq q_{\|A_{\text{out}}\|}(\eta - \beta) + u. \quad (366)$$

This does not turn moment comparison alone into matching extreme quantiles; the finite-core law must also be stable across the shifted confidence levels.

23.3 A noncommuting Parseval benchmark with a constant-size core

The calculus gives an unbounded improvement over a wordwise envelope on an original compressed Walsh example. Let

$$(\varepsilon_1, \dots, \varepsilon_6) = (t_1, t_2, t_3, t_4, t_5, \sigma t_1 t_2 t_3 t_4 t_5), \quad \sigma \in \{1, -1\}, \quad (367)$$

with independent t_i , and put

$$V = \frac{1}{9} \begin{pmatrix} 7 & -4 \\ -4 & 1 \\ -4 & -8 \end{pmatrix}, \quad V^T V = I_2.$$

Let W_L, W_R be arbitrary real Parseval matrices with two columns, whose fresh sign rows z_L, z_R are independent of each other and of (367). Define

$$u = V^T \varepsilon_{1:3}, \quad v = V^T \varepsilon_{4:6}, \quad g = W_L^T z_L, \quad h = W_R^T z_R,$$

choose $\alpha, \beta > 0$ with $\alpha^2 + \beta^2 = 1$, and write $\lambda = \beta/\alpha$. The normalized cross block of the original block-diagonal Parseval frame is

$$X = \alpha^{-2}(\alpha u + \beta g)(\alpha v + \beta h)^T = uv^T + \lambda(uh^T + gv^T) + \lambda^2 gh^T. \quad (368)$$

The full five-dimensional character space in (367) is the retained subgroup. Hence $A = P_H X = uv^T$ has at most 32 characters, independent of the number of tail rows. The geometry is noncommuting: for the first two rows r_1, r_2 of V , $[r_1 r_1^T, r_2 r_2^T]_{12} = 32/729$. Conditional Parseval contraction and Lemma 23.2 give, for every $P \geq 2$,

$$\boxed{\|X - A\|_F \leq e_P, \quad e_P^2 = 12(P-1)\lambda^2 + 4(P-1)^2\lambda^4, \quad \|A\|_\infty \leq 3.} \quad (369)$$

The bound contains no tail-row count. One may therefore choose the tail matrices W_L, W_R as arbitrarily many duplicated, rescaled copies of V . Their maximum tail-row leverage then tends to zero while the six retained core rows and this certificate remain fixed.

To audit signed continuation, let

$$p_3(x, y) = \frac{16550x^3 - 46641x^2y - 56694xy^2 - 6895y^3}{22680} = \sum_{r=0}^3 c_r x^{3-r} y^r.$$

This satisfies $p_3(V^T e) = e_1 e_2 e_3$ for every $e \in \{\pm 1\}^3$. If $i(r)$ is the length-three list with $3-r$ entries equal to 1 and r entries equal to 2, define the legal ordered cubic continuation

$$\mathcal{Q}(B) = \sum_{r,s=0}^3 c_r c_s E_{1,i(r)_1} B E_{i(s)_1, i(r)_2} B E_{i(s)_2, i(r)_3} B E_{i(s)_3, 1} - E_{11}. \quad (370)$$

The exact retained outputs are

$$\boxed{\mathcal{Q}(A) = 0 \quad (\sigma = 1), \quad \mathcal{Q}(A) = -2E_{11} \quad (\sigma = -1).} \quad (371)$$

The two phase laws have identical complete entry covariance, all deterministic two-occurrence contractions, and every even entry tensor. At every sample count their cross-block sample matrices are negatives of one another in law, so their entire singular-value laws also agree. The cubic continuation (370) distinguishes precisely the omitted odd signed channel. Since $\sum_r |c_r| < 6$, Theorem 23.3 at $P = 3p$ gives

$$\boxed{\|\mathcal{Q}(X) - \mathcal{Q}(A)\|_{L^p} \leq 108e_{3p}(3 + e_{3p})^2.} \quad (372)$$

Every word in (370) is a multiple of E_{11} , so the fixed vector e_1 witnesses the norm of the same noisy matrix. If $p \geq 2$, $0 < \epsilon \leq 1$, and

$$\lambda \leq \frac{\epsilon}{32768\sqrt{3p}}, \quad (373)$$

then the error in (372) is at most $\epsilon/3$. Thus the plus-phase moment is at most $\epsilon/3$, while the minus-phase moment is within $\epsilon/3$ of 2. In contrast, applying the triangle inequality to the sixteen signed words and the constant in (370) gives a valid certificate at least 1 even at $\lambda = 0$. The signed-core upper certificate is $O(\lambda\sqrt{p})$ when $\lambda\sqrt{p} \leq 1$, so the improvement over that specified wordwise certificate is unbounded as $\lambda \downarrow 0$.

The same retained phase also recovers the established fourth- and sixth-order corrections. Put

$$\xi = (\alpha u + \beta g) \oplus (\alpha v + \beta h) \in \mathbb{R}^4.$$

For rows r_i of the full d -dimensional Parseval frame ($d = 4$ here), put $\ell_i = \|r_i\|^2$, $S_1 = \sum_i \ell_i r_i r_i^T$, $S_2 = \sum_i \ell_i^2 r_i r_i^T$, $q_{\text{lev}} = \sum_i \ell_i^2$, and $Y = \xi \xi^T - I_d$. For arbitrary $D \in \mathbb{R}^{d \times d}$, one has

$$\mathbb{E}[YDY] = \text{tr}(D)I + D^T - 2 \sum_i (r_i^T D r_i) r_i r_i^T, \quad (374)$$

$$\mathbb{E}Y^3 = (d^2 + 3d + 4 - 2q_{\text{lev}})I - (4d + 18)S_1 + 16S_2 + C_6, \quad (375)$$

where the signed relation term is the ordered contraction

$$C_6 = \sigma \sum_{\pi \in S_6} r_{\pi(1)} r_{\pi(2)}^T \langle r_{\pi(3)}, r_{\pi(4)} \rangle \langle r_{\pi(5)}, r_{\pi(6)} \rangle. \quad (376)$$

Here $r_1, \dots, r_6$ in C_6 are the six core rows in their actual two-block positions. Individual trace words in this channel can have either sign; the coherent sum is the invariant retained by the core.

23.4 Costs and exact scope

The state stores the affine register, a basis of H , at most 2^h signed core matrices, a source dictionary or structural contraction certificate, and moment bounds for the remainder at the required horizon. Direct core convolution costs $O(4^h d^3)$ arithmetic; evaluating gates on the exact core-assignment table costs $O(2^h d^3)$ per gate, with $O(h 2^h d^2)$ for a Fourier transform between the two forms. Dense binary elimination costs $O(D^3)$ per affine recomputation. A T -term source dictionary can be projected and assigned weighted energy in polynomial work in T, D, d ; the input-reading and coefficient-bit costs remain part of the descriptor. The benchmark has $h = 5$ and fixed word count even when the primitive tail rank diverges.

This is an extension-stable *core plus certified remainder* calculus, not a theorem that every dense input has a small accurate core. A large discarded energy, growing free-coordinate degree, a large continuation coefficient mass, or adversarial conditioning can make the displayed error useless. Returning modes cannot be promoted from a scalar radius, and general products are exact projections only when their source coefficients are recomputed. The result therefore identifies both a useful closure mechanism and the precise information required when closure is refreshed.

24 Finite signed-polynomial transfer and a two-sided quadratic calculus

The previous section controls a small signed core against a diffuse remainder. A complementary module handles a sum of independent matrix tuples by comparing the complete ordered polynomial to the jointly Gaussian tuple with the same full entry covariance. A nilpotent path linearization

retains every coefficient and every reuse of an input name. For general polynomials the comparison is additive and the Gaussian readout can still be difficult. For the noncommuting quadratic class HDH , however, exact fourth-order identities and a same-input random-vector witness reduce the functional to $(\nu n/m)\|D\|$ with dimension-independent constants. Linearization and Gaussian transfer for noncommutative polynomials are established tools; Brailovskaya and van Handel use classical linearization in their asymptotic polynomial transfer, Section 9.4.1 of [12]. The module below contributes explicit finite grid, covariance, truncation, and arithmetic bookkeeping for the written circuit rather than a new universality principle.

24.1 Exact ordered linearization and finite spectral transfer

Let $X = (X_1, \dots, X_g)$ be a Hermitian $n \times n$ tuple with

$$X_j = M_j + \sum_{a=1}^m Z_{aj}, \quad (377)$$

where the tuples $(Z_{a1}, \dots, Z_{ag})$ are independent across a and centered. Dependencies within one tuple are retained. Let G be a jointly Gaussian Hermitian tuple with the same means and the complete real entry covariance of X . Supply a written ordered polynomial

$$P(X) = D_0 + \sum_{\alpha=1}^T D_{\alpha 0} X_{j_{\alpha 1}} D_{\alpha 1} \cdots X_{j_{\alpha r_{\alpha}}} D_{\alpha r_{\alpha}}, \quad 1 \leq r_{\alpha} \leq R. \quad (378)$$

Assume $T \geq 1$. A constant-only polynomial ($T = 0$) is evaluated exactly and requires no path space. Repeated names in (378) refer to the same random matrix. No coefficient is replaced by its norm.

Construct a disjoint directed path of $r_{\alpha} + 1$ vertices for every nonconstant word and set $q = \sum_{\alpha} (r_{\alpha} + 1)$. On the direct sum of the qn vertex spaces, put

$$T(X)_{(\alpha, \ell-1), (\alpha, \ell)} = X_{j_{\alpha \ell}} D_{\alpha \ell}, \quad K(X) = I - T(X).$$

Let the block row B contain $D_{\alpha 0}$ at every initial vertex, and let the block column C contain I_n at every final vertex. Then

$$T^{R+1} = 0, \quad P(X) = D_0 + BK(X)^{-1}C, \quad K^{-1} = \sum_{j=0}^R T^j. \quad (379)$$

There is exactly one complete path from each initial vertex to its matching final vertex, so (379) reproduces the written matrix order without a commutation step.

Let $H(A) = \begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}$, form

$$Q(X) = \begin{pmatrix} 0 & K(X) \\ K(X)^* & 0 \end{pmatrix}, \quad U_1 = \begin{pmatrix} B \\ 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 \\ C^* \end{pmatrix}, \quad W = (U_2 \ U_1),$$

and define the affine Hermitian pencil

$$L_X(z) = \begin{pmatrix} zI_{2n} - H(D_0) & W \\ W^* & Q(X) \end{pmatrix}. \quad (380)$$

Its dimension is $d_L = 2n + 2qn$, and the Schur complement of $Q(X)$ is $zI_{2n} - H(P(X))$. Consequently

$$0 \in \text{sp } L_X(z) \iff z \in \text{sp } H(P(X)). \quad (381)$$

Suppose an envelope event satisfies

$$\|K(X)^{-1}\|, \|K(G)^{-1}\| \leq \kappa, \quad \|W\| \leq \omega, \quad \|P(X)\|, \|P(G)\| \leq B_P.$$

Put $A_0 = (1 + \omega\kappa)^2$. For a target additive error $u > 0$, take

$$h = \frac{\min\{u, \kappa^{-1}\}}{4A_0}, \quad J \leq 2 + \frac{2B_P}{h}. \quad (382)$$

At one grid point, the matrix spectrum comparison of Brailovskaya and van Handel supplies a universal $C_* \geq 1$ and the error

$$e_L(t) = C_* \left[\sigma_{*,L} \sqrt{t} + R_L^{1/3} \sigma_L^{2/3} t^{2/3} + R_L t \right], \quad (383)$$

where R_L is the maximum component cap and $\sigma_L, \sigma_{*,L}$ are the full variance and bilinear variance parameters of the linearized sum. Fix one joint coupling of (X, G) ; the primary theorem applies to that same coupled pair at every grid point [12].

Theorem 24.1 (Finite ordered-polynomial transfer). *For $t > 0$, if the envelope event has probability at least $1 - \beta$ and $e_L(t) \leq h$, with h, J as in (382), then*

$$\boxed{\mathbb{P}\{ \left| \|P(X)\| - \|P(G)\| \right| > u \} \leq \beta + J d_L e^{-t}.} \quad (384)$$

For the unbalanced path construction, define

$$R_0 = \max_{a,j} \text{ess sup} \|Z_{aj}\|, \quad \sigma_0 = \max_j \|\mathbb{E}(X_j - M_j)^2\|^{1/2},$$

$$\sigma_{*,0} = \max_j \sup_{\|v\|=\|w\|=1} (\mathbb{E}|v^*(X_j - M_j)w|^2)^{1/2}.$$

If $a = \max_{\alpha, \ell \geq 1} \|D_{\alpha\ell}\|$ and each input component has parameters $R_0, \sigma_0, \sigma_{*,0}$, then

$$R_L \leq aR_0, \quad \sigma_L \leq a\sigma_0, \quad \sigma_{*,L} \leq a\sigma_{*,0}.$$

These bounds remain valid when the matrices inside each tuple are dependent, provided their complete joint covariance is used for G .

Proof sketch. Block Gaussian elimination factors $L_X(z)$ through the direct sum of its Schur complement and $Q(X)$. The two triangular factors and their inverses have norm at most $1 + \omega\kappa$. Hence a point of $\text{sp } \mathbf{H}(P(X))$ within h of a grid point forces an eigenvalue of $L_X(z)$ within h of zero, while spectral comparison and the factorization place the Gaussian polynomial spectrum within u . Repeat in the other direction and union bound over the J scalar grid points. The path random block is a weighted shift, so its norm is the largest edge norm; Cauchy–Schwarz over disjoint edge endpoints gives the bilinear variance bound without breaking within-tuple dependencies. $\square$

This linearization is reusable under addition, product, and adjoint by updating the written word list. Product concatenates ordered paths and has at most $(T_1 + 1)(T_2 + 1) - 1$ nonconstant words and degree at most $R_1 + R_2$. The explicit path dimension, inverse condition κ , scalar grid, and coefficient-bit growth are charged. The theorem compares the final polynomial directly, so comparison losses are not multiplied after every algebraic update.

24.2 The exact quadratic energy identity

The quadratic class admits a stronger readout that does not require a high-order word expansion.

Lemma 24.2 (Global quadratic energy with component reuse). *Let Y_a be independent centered matrix tuples and let G_a be independent centered jointly Gaussian tuples with the respective complete entry covariances. Assume that all fourth-order moments displayed below are finite. Put $S = \sum_a Y_a$ and $G = \sum_a G_a$. For every homogeneous ordered quadratic matrix polynomial Q , including arbitrary deterministic insertions and rectangular output, one has the matrix identity*

$$\mathbb{E}Q(S)^*Q(S) = \mathbb{E}Q(G)^*Q(G) - \sum_a \mathbb{E}Q(G_a)^*Q(G_a) + \sum_a \mathbb{E}Q(Y_a)^*Q(Y_a). \quad (385)$$

The corresponding left-energy identity also holds. For iid Y_a and a weighted sum $S = \sum_a w_a Y_a$ with real weights w_a , set $v = \sum_a w_a^2 > 0$ and $\rho = \sum_a w_a^4 / v^2$. If G_0 matches one component, then

$$\mathbb{E}Q(S/\sqrt{v})^*Q(S/\sqrt{v}) = (1 - \rho)\mathbb{E}Q(G_0)^*Q(G_0) + \rho\mathbb{E}Q(Y_1)^*Q(Y_1). \quad (386)$$

Proof. Expand the four random factors in their original order. Any component name appearing once integrates to zero. The remaining name patterns are two components used twice each, which depend only on their two full covariances, and one component used four times. The first class agrees with the Gaussian expansion. Subtracting its one-component Gaussian terms and adding the actual one-component terms gives (385). Under iid weighting, Gaussian homogeneity supplies v^2 and the one-component energies supply $\sum_a w_a^4$, proving (386). Aliased observations must first have their weights combined; treating them as fresh would give the wrong ρ . $\square$

The last two terms in (385) need not have a positive difference for heterogeneous components. The iid formula (386) is a positive mixture. Both identities cover noncommuting words and negative individual trace-word coefficients; their positivity occurs only after the complete ordered energy is formed.

24.3 A tractable two-sided theorem for HDH

The sandwich form is not itself a new random-matrix object. In particular, Chenakkod, Dereziński, and Dong form $X_i = S_i U$ and $Y_i = X_{2i-1}^T X_{2i}$ from independent copies of a fully independent unscaled OSNAP. Their Lemma 5.1 bounds $\|Y_1^T V^T V Y_1\|_{2q}$ by diagonal terms and an independent-copy successor [10]. This is a recursive upper estimate with a positive-semidefinite insertion, rather than the theorem below verbatim. The Gaussian covariance comparison used below is likewise an application of the Brailovskaya–van Handel framework, not a new universality principle. The statement here should be read as a restricted model theorem: it combines the imported upper-moment estimate with an exact fourth-order map and a same-realization lower certificate for arbitrary symmetric insertions. Its oversampled SparseStack regime is not competitive with the near-optimal embedding dimension proved in that work.

Fix a real Parseval frame $U \in \mathbb{R}^{k \times n}$, $U^T U = I_n$, with rows u_i^T , and $n \geq 64$. Let x be an original row on a fixed Sidon Walsh support and put $\xi = U^T x$. The centered one-row Gram error is $\xi \xi^T - I_n$. Its matched unnormalized Gaussian matrix G_U has the exact covariance map

$$\begin{aligned} \Psi(B) &= \mathbb{E}G_U B G_U = \text{tr}(B)I + B^T - 2 \sum_i (u_i^T B u_i) u_i u_i^T, \\ V &= \mathbb{E}G_U^2 = (n+1)I - 2 \sum_i \|u_i\|^2 u_i u_i^T. \end{aligned} \quad (387)$$

In particular $(n-1)I \preceq V \preceq (n+1)I$.

The lower witness depends on the signed crossing map rather than on a rank condition for D . Define

$$S_U = \sum_i \|u_i\|^2 u_i u_i^T, \quad \mathcal{T}(B) = \sum_i (u_i^T B u_i) u_i u_i^T, \quad \mathcal{L}(B) = U^T [(U U^T)^{\circ 2} \circ (U B U^T)] U.$$

These are positive maps of operator norm at most one. A complete Wick contraction gives, for every real symmetric D ,

$$\begin{aligned} \Omega_U(D) &:= \mathbb{E} G_U^2 D G_U^2 = V D V + \Psi(\Psi(D)) + \mathcal{K}(D), \\ \mathcal{K}(D) &= (n+3)D - 2(DS_U + S_U D) - 4\mathcal{T}(D) + 4\mathcal{L}(D), \end{aligned} \quad (388)$$

and hence

$$\left\| n^{-2} \Omega_U(D) - \left[D + n^{-1} \operatorname{tr}(D) I \right] \right\| \leq \left(\frac{7}{n} + \frac{25}{n^2} \right) \|D\|. \quad (389)$$

The term $\mathcal{K}$ is the ordered crossing pairing. Formula (388) retains it even though individual crossing traces can be negative.

Let C_0 denote the universal constant in the Brailovskaya–van Handel moment-universality inequality [12]. The following theorem is the strongest explicit two-sided matrix class presently obtained from this module.

Theorem 24.3 (Covariance-shaped HDH calculus). *Let $Y_1, \dots, Y_m$ be iid centered real symmetric matrices satisfying*

$$\|Y_a\| \leq K \quad \text{almost surely}, \quad K \geq n, \quad \operatorname{Cov}(Y_a) = \nu \operatorname{Cov}(G_U), \quad 0 < \nu \leq 1, \quad (390)$$

where covariance means the complete real entry covariance. Put $H = m^{-1} \sum_{a=1}^m Y_a$. For every fixed real symmetric D and every integer p satisfying

$$\max\{2, \lceil \log n \rceil\} \leq p \leq n, \quad m \geq \max \left\{ \frac{24K^2}{\nu n}, \frac{4C_0^2 K^2 p^4}{\nu(n-1)} \right\}, \quad (391)$$

the same original matrix obeys

$$\boxed{\frac{\nu n}{4m} \|D\| \leq (\mathbb{E} \|HDH\|^p)^{1/p} \leq 576 \frac{\nu n}{m} \|D\|.} \quad (392)$$

The comparison has no rank, definiteness, spectral-mass, or commutation hypothesis on D . The frame U and insertion D are fixed before sampling.

Same-input lower certificate and upper bound. Normalize $X = (m\nu n)^{-1/2} \sum_a Y_a$, so that its covariance equals that of $G_U/\sqrt{n}$. Write $\tau(D) = n^{-1} \operatorname{tr} D$ and $B_D = D + \tau(D)I$. Since $D = B_D - \frac{1}{2}\tau(B_D)I$, one has $\|B_D\| \geq 2\|D\|/3$. Choose a deterministic extreme eigenvector v of D for which $|v^T B_D v| = \|B_D\|$.

The exact four-position component expansion, together with (389), gives

$$\mathbb{E} X^2 D X^2 = (1 - m^{-1}) n^{-2} \Omega_U(D) + \frac{1}{m\nu^2 n^2} \mathbb{E} Y_1^2 D Y_1^2. \quad (393)$$

Both terms are signed linear maps in D , while their operator errors are controlled by their values on the identity. The cap and covariance give $\|\mathbb{E} Y_1^2 D Y_1^2\| \leq K^2 \nu(n+1) \|D\|$. Under the first sample condition in (391), the rational estimate at $n = 64$ and its monotone continuation yield

$$\|\mathbb{E} X^2 D X^2 - B_D\| \leq \frac{1}{6} \|D\|, \quad |v^T \mathbb{E} X^2 D X^2 v| \geq \frac{1}{2} \|D\|. \quad (394)$$

The same four-position calculation gives $\mathbb{E}\|Xv\|^4 \leq 4$. Pointwise,

$$|(Xv)^T(XDX)(Xv)| \leq \|XDX\| \|Xv\|^2.$$

Cauchy–Schwarz and (394) therefore imply $\|XDX\|_{L^2} \geq \|D\|/4$, and the same lower bound holds for every $p \geq 2$. This vector belongs to the same realized matrix; no different Gaussian model supplies the lower witness.

For the upper bound, the imported moment-universality inequality and the second sample condition in (391) give $\|X\|_{L^{2p}} \leq 24$. Thus $\|XDX\|_{L^p} \leq 24^2\|D\| = 576\|D\|$. Finally $HDH = (\nu n/m)XDX$, which proves (392). $\square$

24.4 Three distinct input laws covered by the same theorem

1. **Fixed-Sidon Walsh rows.** For independent observations $Y_a = \xi_a \xi_a^T - I_n$ on any fixed Sidon support, $\nu = 1$ and $K = k$. The feature signs inside one row need not be jointly independent: their actual fourth character moments give (387). Put

$$H_m = \frac{1}{m} \sum_{a=1}^m (\xi_a \xi_a^T - I_n).$$

Hence

$$\frac{n}{4m} \|D\| \leq (\mathbb{E}\|H_m D H_m\|^p)^{1/p} \leq 576 \frac{n}{m} \|D\|, \quad m \geq \max \left\{ \frac{24k^2}{n}, \frac{4C_0^2 k^2 p^4}{n-1} \right\}.$$

2. **Independent Rademacher rows.** Take an independent sign vector in $\mathbb{R}^k$ and compress it by any real Parseval U . The complete covariance is again (387), so the same formula and sample condition hold. This is a dense, generally noncommuting coefficient benchmark rather than a coordinate-frame special case.
3. **Actual SparseStack layers.** For each layer let $S_a \in \mathbb{R}^{b \times k}$ have column i equal to $\varepsilon_{ai} e_{h_{ai}}$, with independent signs and uniform hashes $h_{ai} \in [b]$. Set

$$Y_a = U^T S_a^T S_a U - I_n, \quad S = m^{-1/2} \begin{bmatrix} S_1 \\ \vdots \\ S_m \end{bmatrix}, \quad H = U^T S^T S U - I_n.$$

Then $K = k$ and $\text{Cov}(Y_a) = b^{-1} \text{Cov}(G_U)$. Therefore

$$\boxed{\frac{n}{4mb} \|D\| \leq (\mathbb{E}\|HDH\|^p)^{1/p} \leq 576 \frac{n}{mb} \|D\|,} \quad (395)$$

provided

$$m \geq \max \left\{ \frac{24bk^2}{n}, \frac{4C_0^2 bk^2 p^4}{n-1} \right\}. \quad (396)$$

Indeed the actual layer expands as

$$Y_a = \sum_{i < j} \varepsilon_{ai} \varepsilon_{aj} \mathbf{1}_{\{h_{ai}=h_{aj}\}} (u_i u_j^T + u_j u_i^T).$$

Only an identical unordered edge survives sign expectation in a product of two entries, and its collision indicator has expectation $1/b$. This proves the complete covariance scaling. It does not declare collision events independent: a reused triangle has probability b^{-2} , not b^{-3} , and its higher interaction stays in the actual one-layer fourth-order term of (393).

Thus one descriptor and one functional work in three genuinely different models while retaining noncommuting coefficient geometry. The maps in (387) and (388) can be formed in polynomial arithmetic: $\mathcal{T}(D)$ costs $O(kn^2)$ and $\mathcal{L}(D)$ costs $O(k^2n + kn^2)$ from the frame Gram. Computing $\|D\|$ and its extreme eigenvector supplies the functional and lower witness. The moment order enters the explicit sample condition; no $2p$ -word enumeration is required for this quadratic class.

24.5 Limitations and the remaining bridge

The general linearization theorem is a comparison module, not by itself a tractable two-sided calculus for every ordered polynomial. A generic Gaussian polynomial norm can remain hard to compute; path lists and coefficient bits can grow exponentially under repeated products; and poor inverse conditioning makes the additive spectral transfer useless. An affine core may be conditioned first, but the remaining component groups must actually be independent on every retained atom. Constraints that couple all groups can leave one large component and destroy the useful comparison regime.

The HDH theorem is correspondingly precise about its range. It assumes $n \geq 64$, logarithmic-to-linear integer moments, bounded independent components, and the oversampling condition (391). The universal constant C_0 is inherited from the primary moment theorem and is not assigned a numerical value here. The SparseStack application controls an oversampled layer regime and does not give a near-optimal joint bucket/sparsity tradeoff. The result gives matching norm moments for fixed U, D ; it does not give simultaneous postselection over Walsh supports or matching extreme quantiles. Full FR-10 still requires a uniform original-sample argument over the support family, including the unresolved pooled all-Sidon residual process. These limits distinguish the completed two-sided class from the larger general target.

25 Exact aggregate-moment boundary calculus

The boundary calculi developed earlier retain primitive labels or complete operator-valued messages. A different exact regime becomes available when the random matrix is a polynomial in a supplied, small list of additive statistics. The statistics may share primitive signs, their joint law may have exponentially many atoms, and their matrix coefficients may generate the full matrix algebra. The useful parameters are the number of aggregate coordinates and the live boundary of a supplied elimination schedule. This section gives an exact, extension-stable moment calculation for that class. It is a substantive realization of the general target, but it does not assert that arbitrary Walsh or SparseStack inputs admit a representation with small parameters.

25.1 The model and its live boundary

Let $\xi_1, \dots, \xi_M$ be independent uniform signs. For $j = 1, \dots, L$, supply a set $V_j \subseteq [M]$, with $|V_j| \leq a$, a rational vector table

$$s_j : \{\pm 1\}^{V_j} \rightarrow \mathbb{Q}^w,$$

and a zero-one constraint table g_j on the same variables. Put

$$T = \sum_{j=1}^L s_j(\xi_{V_j}), \quad \mathcal{E} = \left\{ \prod_{j=1}^L g_j(\xi_{V_j}) = 1 \right\}, \quad \rho = \mathbb{P}(\mathcal{E}). \quad (397)$$

We condition on $\mathcal{E}$ and require $\rho > 0$. Repeated occurrences of a primitive sign are the same named variable. Thus repeated equalities, overlapping parity triangles and implied constraints remain actual shared events rather than independent probability penalties.

Let

$$P(t) = \sum_{|\alpha| \leq R} A_\alpha t^\alpha, \quad Z = P(T), \quad A_\alpha \in M_d(\mathbb{Q}(i)). \quad (398)$$

The scalar aggregate coordinates commute, while the coefficient matrices do not. A written ordered product of affine matrix pencils, with fixed matrices inserted at specified locations, is included by multiplying its coefficients in their original order and collecting equal scalar monomials.

For the supplied factor order define

$$B_j = \left(\bigcup_{i \leq j} V_i \right) \cap \left(\bigcup_{i > j} V_i \right), \quad b = \max_j |B_j|, \quad E_j = B_{j-1} \cup V_j. \quad (399)$$

The descriptor consists of the local tables, the original variable names, the schedule and live boundaries, the aggregate names, and the signed ordered coefficient polynomial. Finding a low-width schedule or a low-dimensional aggregate representation is part of the model-specific work and is not charged as a free preprocessing step.

25.2 An exact truncated exponential transfer

Fix an integer $p \geq 1$, set $D = 2Rp$, and work in

$$\mathcal{R}_D = \mathbb{Q}[z_1, \dots, z_w] / \langle z^\alpha : |\alpha| > D \rangle, \quad H_D = \binom{w+D}{w}. \quad (400)$$

Define the local truncated exponential

$$K_j(u; z) = g_j(u) \sum_{|\alpha| \leq D} \frac{s_j(u)^\alpha}{\alpha!} z^\alpha.$$

Starting with $F_0(\emptyset; z) = 1$, update the boundary message by

$$F_j(v; z) = 2^{-|E_j \setminus B_j|} \sum_{\substack{u \in \{\pm 1\}^{E_j} \\ u|_{B_j} = v}} F_{j-1}(u|_{B_{j-1}}; z) K_j(u|_{V_j}; z) \pmod{\deg > D}. \quad (401)$$

Lemma 25.1 (Exact aggregate moments and constraint reuse). *For every multi-index α with $|\alpha| \leq D$,*

$$[z^\alpha] F_L(\emptyset; z) = \frac{\mathbb{E}[\mathbf{1}_\mathcal{E} T^\alpha]}{\alpha!}. \quad (402)$$

In particular the constant term is ρ , and multiplication by $\alpha!/\rho$ gives every required conditional moment. The transfer has zero approximation error.

Proof. In $\mathcal{R}_D$, the product of two truncated exponentials of scalar linear forms is the truncated exponential of their sum. Terms above degree D cannot affect a coefficient of degree at most D . For each complete sign assignment, the product of the K_j is therefore $\mathbf{1}_\mathcal{E} \exp(\langle z, T \rangle)$ through degree D . After factor j , every primitive omitted from B_j has no future occurrence, so averaging it in (401) is exact under the original product sign law. Induction identifies F_j with the appropriate partial expectation; at the empty terminal boundary, the exponential coefficient gives (402). Because $g^2 = g$ for a zero-one constraint and logical implications hold pointwise before averaging, repeated or implied constraints acquire no guessed fresh probability factor. $\square$

25.3 Two-sided norm moments and a separate strict-tail certificate

Form the ordered matrix polynomial

$$Q_p(t) = (P(t)^* P(t))^p = \sum_{|\alpha| \leq D} C_{p,\alpha} t^\alpha$$

and use Lemma 25.1 to compute

$$M_p = \sum_{|\alpha| \leq D} \text{tr}(C_{p,\alpha}) \mathbb{E}[T^\alpha \mid \mathcal{E}], \quad \mathcal{F}_{2p} = M_p^{1/(2p)}. \quad (403)$$

Individual ordered trace terms may be negative or nonreal. They are summed before any absolute value is taken; the final number is real and nonnegative.

Theorem 25.2 (Same-input aggregate calculus). *For every input (397)–(399) with $\rho > 0$,*

$$M_p = \mathbb{E}[\text{tr}((Z^* Z)^p) \mid \mathcal{E}], \quad \boxed{d^{-1/(2p)} \mathcal{F}_{2p} \leq (\mathbb{E}[\|Z\|^{2p} \mid \mathcal{E}])^{1/(2p)} \leq \mathcal{F}_{2p}}. \quad (404)$$

The ratio of the two certificates is at most $\sqrt{e}$ when $p \geq \log d$.

Proof. Lemma 25.1 evaluates the expectation of every scalar monomial in the ordered expansion of Q_p . For the singular values of this same realization of Z ,

$$\|Z\|^{2p} \leq \text{tr}((Z^* Z)^p) \leq d \|Z\|^{2p}.$$

Conditional expectation and a $2p$ th root prove the result. $\square$

Thus, for $0 < \eta < 1$,

$$\mathbb{P}\{\|Z\| > \eta^{-1/(2p)} \mathcal{F}_{2p} \mid \mathcal{E}\} \leq \eta. \quad (405)$$

Matching moments do not by themselves imply matching quantiles. A separate strict lower certificate is nevertheless computable. Run the transfer to degree $4Rp$ and let

$$N_p = \mathbb{E}\left[\left(\text{tr}((Z^* Z)^p)\right)^2 \mid \mathcal{E}\right].$$

If $M_p > 0$ and $0 < \theta < 1$, strict Paley–Zygmund gives

$$\mathbb{P}\{\|Z\| > (\theta M_p / d)^{1/(2p)} \mid \mathcal{E}\} \geq (1 - \theta)^2 \frac{M_p^2}{N_p}. \quad (406)$$

The displayed success probability is part of the conclusion and may be small. If $M_p = 0$, then $Z = 0$ almost surely under the conditional law.

25.4 Reusable operations and explicit costs

The descriptor is closed under the following operations, all exactly.

- Products and word extension use ordered coefficient convolution $C_\gamma = \sum_{\alpha+\beta=\gamma} A_\alpha B_\beta$; degrees add and future insertions see the resulting signed matrix sums.
- Addition is performed before norms are taken. The adjoint conjugates and transposes coefficient matrices and reverses every ordered matrix product.
- Different aggregate lists are concatenated, with zeros inserted in unused coordinates and every original primitive identification retained. Coincident statistics are merged only after their equality has been supplied or proved.

- A new local interaction is appended to the source tables and the transfer is recomputed at the required horizon. A late interaction involving a forgotten primitive forces recomputation with the larger boundary; a terminal moment table alone is not an extension-stable descriptor.
- Actual equalities or parity relations are imposed by identifying named variables or adjoining their local zero-one indicators, followed by recomputation of (a, b, L) and the normalizing constant ρ .

With naive truncated convolution, the transfer and dense matrix readout use at most

$$O(L2^{b+a}(w+1)H_D^2), \quad O(d^3H_D^2 \log(p+1)) \quad (407)$$

rational arithmetic operations, respectively. Their storage is $O(2^b H_D)$ scalar coefficients and $O(d^2 H_D)$ matrix coefficients. Put $H_R = \binom{w+R}{w}$. An explicit sum of J ordered words of length at most R in affine pencils can be collected in $O(JR(w+1)d^3 H_R)$ arithmetic operations. At fixed w, a, b, R and $p = O(\log(d/\eta))$, these costs are polynomial in the dense matrix dimension and input length; the power of the logarithmic moment scale coming from H_D^2 is at most $2w$.

The magnitude of rational data is also visible. Let S bound the total binary input length, including all local tables, let $B \leq S$ bound an individual numerator or denominator, and recall $M \leq La$ after unused signs are deleted. A conservative bound on every intermediate numerator or denominator length is

$$O((D+2p+1)S + D \log((L+1)(D+1)) + p \log((d+1)(H_R+1)) + M + \log(H_D+1)). \quad (408)$$

Multiplying (407) by a polynomial in (408) gives an explicit schoolbook bit bound. In particular an aggregate with a very large range does not enlarge the state count, although its rational bit length is still charged.

25.5 Applications, verification, and the boundary of the theorem

Three model classes illustrate distinct uses of the same transfer.

- (i) For independent rational weighted rows, $T_\ell = \sum_j a_{\ell j} \xi_j$, one has $a = 1$ and $b = 0$. The value range may still be exponential. In the exact benchmark $T = \sum_{j=0}^{79} 2^j \xi_j$ has 2^{80} values, but every moment through degree 12 is carried by only 13 scalar message coefficients.
- (ii) The overlapping path coordinates

$$T_1 = \sum_j a_j \xi_j, \quad T_2 = \sum_j b_j \xi_j \xi_{j+1}, \quad T_3 = \sum_j c_j \xi_j \xi_{j+2}$$

have $w = 3$, $a \leq 3$ and a schedule with $b \leq 2$. Their dependence is retained rather than replaced by independent aggregate coordinates.

- (iii) Encode a bucket in $B = 2^h$ locations by its h original bits and put the vertex sign beside those bits. On a graph of vertex-frontier width t , weighted collision statistics have boundary at most $(h+1)t$ and local arity at most $2(h+1)$. Reused edges and triangle closures are actual constraints; in particular an actual triangle collision has probability B^{-2} , not three independent edge penalties. Any polynomial in a fixed number of these statistics can carry dense, irreducible, noncommuting coefficient matrices.

For example, a diagonal matrix with distinct entries together with a symmetric path adjacency generates every matrix unit, so the last claim does not hide a simultaneous diagonalization. What

the theorem requires is that all random entries factor through a small supplied aggregate vector. An arbitrary dense compressed SparseStack matrix need not have small w or small frontier width, and no such representation is inferred here.

The companion exact suite checked original assignments, arbitrary rational weights, ordered noncommuting continuation, negative individual trace coefficients, reused and contradictory constraints, and actual triangle collisions. It passed 591 exact assertions, including an irreducible three-dimensional coefficient algebra and the 80-sign benchmark above. These checks supplement the proof; they do not certify a uniform Walsh or SparseStack sampling theorem.

26 Signed Walsh pooling: what is closed and what remains open

The Walsh problem exposes a different frontier. The clipped part of the quadratic process can be pooled over every sparse support and amplitude profile with the desired entropy. Several large-observation families admit exact same-input signed reductions, and finite witnesses now control every strict Sidon tail count on the original sample with one remaining height logarithm. The sharp stage has two explicit formulations: a coherent signed excess estimate or a sharper tail-count contraction at the required levels.

26.1 The common process and a complete clipped contraction

Let $G = \mathbb{F}_2^d$, $N = 2^d$, and $\chi_v(y) = (-1)^{v \cdot y}$. For a deterministic list of original rows $Q = (y_1, \dots, y_m)$, repetitions allowed, let $\mathcal{C}_k$ be the real unit vectors with Sidon support of size at most k , and put

$$F_c(y) = \sum_v c_v \chi_v(y), \quad f_c(y) = F_c(y)^2, \quad R_Q^2 = \sup_{c \in \mathcal{C}_k} \sum_{i=1}^m f_c(y_i),$$

$$\mathcal{W}_Q(g) = \mathbb{E}_\varepsilon \sup_{c \in \mathcal{C}_k} \left| \sum_{i=1}^m \varepsilon_i g_c(y_i) \right|. \quad (409)$$

The signs ε_i are shared across the entire supremum. For $L > 0$ write

$$b_{c,L} = \min(f_c, L), \quad h_{c,L} = (f_c - L)_+.$$

Theorem 26.1 (Pooled clipped contraction). *For every Q , $1 \leq k \leq N$ and $L > 0$,*

$$\boxed{\mathcal{W}_Q(b_L) \leq 4\sqrt{2Lmk \log(2eN/k)} \leq 4\sqrt{2} R_Q \sqrt{Lk \log(2eN/k)}}. \quad (410)$$

The first inequality remains valid over all supports of size at most k ; Sidonicity is not used.

Proof. The scalar map $u \mapsto \min(u^2, L)$ is $2\sqrt{L}$ -Lipschitz and vanishes at zero. Rademacher contraction, including both signs for the absolute supremum, bounds the left side by $4\sqrt{L}$ times the sparse linear width. If $A_v = \sum_i \varepsilon_i \chi_v(y_i)$, that linear supremum is the Euclidean norm of the k largest $|A_v|$. The marginal tail $\mathbb{P}\{A_v^2 > u\} \leq 2e^{-u/(2m)}$ is enough; no independence among Walsh columns is needed. Integrating above $u = 2m \log(2N/k)$ gives

$$\mathbb{E} \sum_{j \leq k} |A|_{(j)}^2 \leq 2mk \log(2eN/k).$$

Jensen and $R_Q \geq \sqrt{m}$ finish the proof. □

At $L = \log(2k)$ this is already the desired scale for every support and every real amplitude profile. Moreover

$$|\mathcal{W}_Q(f) - \mathcal{W}_Q(h_L)| \leq \mathcal{W}_Q(b_L). \quad (411)$$

Consequently the all-Sidon step is equivalent, up to changing the universal constant in the target-scale bound, to controlling only the excess. When Q consists of iid uniform original Walsh rows, write $\mathbb{E}_Q$ for expectation over that row list. The exact averaged statement still missing is

$$\boxed{\mathbb{E}_Q \mathcal{W}_Q(h_{\log(2k)}) \leq C \sqrt{k \log(2k) \log(2N/k)} \mathbb{E}_Q R_Q.} \quad (412)$$

Equation (412) is a conjectural target, not a theorem in this report.

There is already a complete positive result under a checkable dense-sample condition. Let n_y be the multiplicity of row y in Q and put $n_{\max} = \max_y n_y$.

Proposition 26.2 (Balanced-sample pooling). *For $1 \leq k \leq N/2$,*

$$\mathcal{W}_Q(f) \leq 2\sqrt{6Nn_{\max}\{\log 2 + k \log(9eN/k)\}}.$$

Consequently, if

$$Nn_{\max} \leq BR_Q^2 \log(2k),$$

then the target-scale bound holds with a universal constant times $\sqrt{B}$.

Proof. For each Sidon support sphere of size $s \leq k$, choose a $1/4$ -net of size at most 9^s . The absolute quadratic supremum of a symmetric matrix is at most twice its maximum on this net. Across all supports there are at most $(9eN/k)^k$ net points. Moreover,

$$\sum_i f_c(y_i)^2 \leq n_{\max} \sum_y f_c(y)^2 \leq 3Nn_{\max},$$

so every resulting Rademacher sum has variance proxy at most $3Nn_{\max}$. The exponential-moment maximum bound, applied to both signs, gives the stated estimate. Substitution of the displayed sample condition finishes the proof. $\square$

If every row has the same positive multiplicity, then $R_Q^2 = m$ and $Nn_{\max} = m$; the proposition gives the stronger $O(\sqrt{mk \log(2N/k)})$ bound. This is a full-amplitude pooled theorem, but its dense-sample condition does not cover the sparse iid regime of FR-10.

For a fixed Sidon support the complete amplitude geometry can be lifted without loss:

$$\begin{aligned} q_c(0) &= 1, & q_c(u+v) &= 2c_u c_v \quad (u < v), & f_c(y) &= \sum_w q_c(w) \chi_w(y), \\ \|q_c\|_2^2 &= 3 - 2 \sum_v c_v^4 \leq 3, \\ \sum_i f_c(y_i)^2 &= q_c^T H_Q q_c, & H_Q[w, z] &= \sum_i \chi_{w+z}(y_i). \end{aligned} \quad (413)$$

This state retains coefficient signs and amplitudes and the orientation of the actual empirical Gram. It does not close the next interaction automatically. For instance, with $F_{(3)} = \sum_v c_v^3 \chi_v$ and $s_4 = \sum_v c_v^4$, the exact successor is

$$F_c^4 = 6F_c^2 - 3 - 8F_c F_{(3)} + 6s_4 + 24 \sum_{a < b < c < d} c_a c_b c_c c_d \chi_{a+b+c+d}. \quad (414)$$

The last signed four-distinct-label sum is the surviving interaction that a population fourth moment or an un-oriented spectrum cannot discard.

26.2 Finite witnesses on the original Sidon sample

The clipped estimate leaves a large-observation question. The following result attacks the original strict counts directly, without auxiliary row samples. Assume $Q = (Y_1, \dots, Y_m)$ consists of iid uniform rows of $\mathbb{F}_2^d$, $N = 2^d$, and $2 \leq k \leq N$. For $c \in \mathcal{C}_k$ write

$$n_c(t) = \#\{i : F_c(Y_i)^2 > t\}, \quad A_w = 2N + 1, \quad L_t = \left\lceil \frac{32k}{t} \log(2t^2) \right\rceil. \quad (415)$$

Theorem 26.3 (Simultaneous finite-witness count bound). *For every $t \geq 2$ and $u \geq 0$, with probability at least $1 - e^{-u}$ over the one original list Q , simultaneously for every $c \in \mathcal{C}_k$,*

$$n_c(t) \leq \frac{131m}{t^2} + 3L_t \log A_w + 3u. \quad (416)$$

Moreover,

$$\mathbb{E}_Q \sup_{c \in \mathcal{C}_k} n_c(t) \leq \frac{131m}{t^2} + 3L_t \log A_w + 3. \quad (417)$$

For $t \geq k$ the strict count is identically zero. For $2 \leq t < k$, (417) implies

$$\mathbb{E}_Q \sup_c n_c(t) \leq \frac{131m}{t^2} + \frac{96k \log(2t^2) \log(2N + 1)}{t} + 3 \log(2N + 1) + 3. \quad (418)$$

The probability version has $3u$ in place of the final 3.

Proof. Sidonicity and Walsh orthogonality give $\mathbb{E}F_c^4 = 3 - 2 \sum_v c_v^4 \leq 3$ and $\|c\|_1 \leq \sqrt{k}$. Represent F_c as the auxiliary expectation of a random function Z from the fixed alphabet

$$\{0\} \cup \{\sqrt{k}\chi_v, -\sqrt{k}\chi_v : 1 \leq v \leq N\}.$$

The probabilities are $|c_v|/\sqrt{k}$, with the remaining mass at zero. For $G = L_t^{-1} \sum_{a=1}^{L_t} Z_a$, Hoeffding's inequality gives, at every original row,

$$\mathbb{P}_Z\{|G - F_c| > \sqrt{t}/4\} \leq 2e^{-L_t t/(32k)} \leq t^{-2} \leq 1/4.$$

There are at most $A_w^{L_t}$ possible G . For $B_G = \{|G| > 3\sqrt{t}/4\}$, $\mu_G = \mathbb{P}(B_G)$, and $N_G = \sum_i \mathbf{1}_{B_G}(Y_i)$, the binomial exponential identity and a union bound give, on one event of probability at least $1 - e^{-u}$,

$$N_G \leq 2m\mu_G + 2[L_t \log A_w + u]$$

for every witness. This event is fixed before c is selected from Q . Averaging it under the coefficient-dependent auxiliary law gives

$$\mathbb{E}_Z N_G \geq (1 - t^{-2})n_c(t), \quad \mathbb{E}_Z \mu_G \leq \mathbb{P}\{|F_c| > \sqrt{t}/2\} + t^{-2} \leq 49t^{-2}.$$

Division by $1 - t^{-2} \geq 3/4$ yields (416); integration of the $3u$ excess gives (417). Finally, expanding the ceiling in L_t proves (418). $\square$

If a size- k Sidon support exists, then $k^2 \leq 4N$ and $\log(2N + 1) \leq 2 \log(2eN/k)$. With $D_0 = k \log(2eN/k)$, the complexity term in (418) is therefore $O(D_0 \log(2t)/t)$. If no size- k Sidon support exists, one retains the direct formula or replaces k by the largest possible support size.

Theorem 26.4 (Fixed-confidence consequence of finite witnesses). *Put $D_0 = k \log(2eN/k)$. Fix $T \geq 167680$, let*

$$J = \max\{0, \lceil \log_2(k/T) \rceil\}, \quad t_j = 2^j T \quad (0 \leq j < J),$$

and define, with empty sums equal to zero,

$$B_T = 3 \log(2N + 1) \sum_{j=0}^{J-1} t_j L_{t_j} + 3 \sum_{j=0}^{J-1} t_j.$$

If

$$m \geq \max\{32768TD_0, 128T^2 \log 20, 640B_T\}, \quad (419)$$

then, with probability at least $9/10$, every $c \in \mathcal{C}_k$ satisfies

$$\frac{7}{8} - \frac{3}{4T} \leq \frac{1}{m} \sum_{i=1}^m F_c(Y_i)^2 \leq \frac{19}{16}. \quad (420)$$

For fixed T , whenever the size- k Sidon class is nonempty, the displayed conditions have order $O(k \log^2(2k) \log(2N/k))$.

Proof. The strict dyadic inequality $(x - T)_+ \leq \sum_{j < J} t_j \mathbf{1}_{\{x > t_j\}}$ for $0 \leq x \leq k$ and (417) give

$$\mathbb{E}_Q \sup_c \frac{1}{m} \sum_i (f_c(Y_i) - T)_+ \leq \frac{262}{T} + \frac{B_T}{m}.$$

The last term is at most $1/640$ under (419), while $262/T \leq 1/640$; hence Markov bounds the excess by $1/16$ with probability at least $19/20$. The first two sample conditions are exactly those in the clipped-process concentration argument: with probability at least $19/20$ it bounds the empirical clipped energy simultaneously between $7/8 - 3/(4T)$ and $9/8$. Intersecting the events and adding the excess proves (420). Finally

$$B_T \leq 96kJ \log(2N + 1) \log(2k^2) + 6k \log(2N + 1) + 6k,$$

which gives the stated order. $\square$

The rate retains one height, hence one sparsity, logarithm beyond the target. It matches the known $O(k \log^2 k \log N)$ general Fourier order of Haviv and Regev [7]; no theorem from that paper is used above. The finite alphabet has A_w symbols and at most $A_w^{L_t}$ ordered words or $\binom{2N+L_t}{L_t}$ multiplicity histograms. Only the logarithm of this family enters the probability proof. Enumeration is generally exponential, so this is a finite probabilistic certificate with explicit entropy rather than a cheap algorithm for evaluating the supremum.

26.3 The noise floor of a positive witness

The logarithm in L_t is necessary for the particular positive population-witness step just used. Let Y be uniform on $\mathbb{F}_2^k$, use the independent-character Sidon support $S = \{e_1, \dots, e_k\}$, and put

$$F(Y) = k^{-1/2} \sum_{j=1}^k \chi_{e_j}(Y).$$

For iid uniform auxiliary labels $V_1, \dots, V_L \in [k]$, independent of Y , define

$$G_V(Y) = \frac{\sqrt{k}}{L} \sum_{a=1}^L \chi_{e_{V_a}}(Y), \quad \pi_{k,L}(t) = \mathbb{P}_{Y,V} \{|G_V(Y)| > 3\sqrt{t}/4\}.$$

Theorem 26.5 (Positive-witness population floor). *If*

$$L \geq 64, \quad 2 \leq t \leq k/64, \quad L(L-1) \leq k,$$

then

$$\pi_{k,L}(t) \geq \frac{1}{16} \exp(-18 - 6Lt/k). \quad (421)$$

For any row-independent law on ordered lists of L distinct labels, the coefficient improves to $1/8$ and the last hypothesis becomes $L \leq k$. Consequently a requirement $\pi_{k,L}(t) \leq A_0/t^2$ forces

$$L \geq \frac{k}{6t} [2 \log t - 18 - \log(16A_0)]_+. \quad (422)$$

If also $t/8 \geq 18 + 6Lt/k + \log 64$, then the iid witness has genuine false-positive mass at least

$$\mathbb{P}_{Y,V}\{|G_V(Y)| > 3\sqrt{t}/4, |F(Y)| \leq \sqrt{t}/2\} \geq \frac{1}{32} e^{-18-6Lt/k}.$$

Proof outline. Conditional on L distinct labels, $LG_V/\sqrt{k}$ is a sum of L independent symmetric signs. An elementary central-binomial strip bound gives

$$\mathbb{P}\{S_L > \frac{3}{4}\sqrt{Lq}\} \geq \frac{1}{8} e^{-18-6q} \quad (L \geq 64, 0 \leq q \leq L/64).$$

Take $q = Lt/k$. Iid labels are distinct with probability at least $1/2$ under $L(L-1) \leq k$, giving (421); a law supported on distinct lists removes that factor. Taking logarithms gives (422). Finally $\mathbb{P}\{|F| > \sqrt{t}/2\} \leq 2e^{-t/8}$ and the last displayed hypothesis give the false-positive claim by subtraction. The companion proof note contains the parity-safe strip calculation and endpoint checks. $\square$

This is a lower bound only for positive, row-independent Maurey witnesses and their distinct-label variants. It does not bound signed, correlated, or sample-dependent representations, the true all-Sidon count, the open signed excess, or FR-10.

26.4 Exact signed large-observation families

There are already nontrivial classes on which the excess closes exactly. Let $k = 2^r$ with odd $r \geq 3$, identify the row space with $\mathbb{F}_{2^r}^2$, and take the cubic Sidon support

$$S = \{(x, x^3) : x \in \mathbb{F}_{2^r}\}.$$

For the flat positive vector and its row translations, the complete squared observation law is

$$\begin{array}{ccccc} f(y) & k & 2 & 0 & \\ \# \{y\} & 1 & k(k-1)/2 & (k-1)(k+2)/2. & \end{array} \quad (423)$$

Thus, whenever $2 < t < k$, the translated residual is exactly $h_z(y) = (k-t)\mathbf{1}_{\{y=z\}}$. If $T_y = \sum_{i:y_i=y} \varepsilon_i$, then for every deterministic Q ,

$$\mathbb{E}_\varepsilon \sup_z \left| \sum_i \varepsilon_i h_z(y_i) \right| = (k-t) \mathbb{E}_\varepsilon \max_y |T_y|. \quad (424)$$

For independent uniform original rows and $1 \leq m \leq k^{4/3}$ this expectation lies between $(k-t)/2$ and $4(k-t)$. This is a sharp signed benchmark in the same sampled model, including the range $m \asymp k \log^2 k$ for large k .

A broader theorem allows nonflat amplitudes and exact signed pooling across all admissible profiles, embeddings, and modulations. For every real $\alpha = (\alpha_a)_{a \in \mathbb{F}_{2r}}$ satisfying

$$\sum_a \alpha_a^2 = 1, \quad 4\|\alpha\|_1^2 \leq \tau, \quad 4 \leq \tau < k,$$

define

$$c_x = k^{-1/2} \sum_a \alpha_a (-1)^{\text{Tr}(ax)} \quad \text{on } S.$$

Let $\mathcal{A}_{\text{cub}}(\tau)$ contain these vectors for every such profile α , every injective linear embedding of this support into $\mathbb{F}_2^d$, $d \geq 2r$, and every row modulation. Let $\mathcal{C}_{2r}$ be the collection of affine codimension- $2r$ cells in the ambient row space, $M = |\mathcal{C}_{2r}|$, $n_C = \#\{i : y_i \in C\}$ and $n_{\max} = \max_C n_C$. For this restricted family define

$$R_{\text{cub}}(Q)^2 = \sup_{c \in \mathcal{A}_{\text{cub}}(\tau)} \sum_i f_c(y_i), \quad \mathcal{W}_Q(h_{\text{cub}}) = \mathbb{E}_\varepsilon \sup_{c \in \mathcal{A}_{\text{cub}}(\tau)} \left| \sum_i \varepsilon_i (f_c(y_i) - \tau)_+ \right|.$$

Theorem 26.6 (Exact cubic excess descriptor). *For every original row list and every common sign vector,*

$$\sup_{c \in \mathcal{A}_{\text{cub}}(\tau)} \left| \sum_i \varepsilon_i (f_c(y_i) - \tau)_+ \right| = (k - \tau) \max_{C \in \mathcal{C}_{2r}} \left| \sum_{i: y_i \in C} \varepsilon_i \right|. \quad (425)$$

For the same input family,

$$\begin{aligned} R_{\text{cub}}(Q)^2 &\geq kn_{\max}, \\ \mathcal{W}_Q(h_{\text{cub}}) &\leq R_{\text{cub}}(Q) \sqrt{2k \log(2M)} \\ &\leq 4R_{\text{cub}}(Q) \sqrt{k \log(2k) \log(2N/k)}. \end{aligned} \quad (426)$$

Proof. For the chosen injective embedding ι , let $\pi = \iota^*$ be its surjective adjoint row map, and let y_0 denote the row shift induced by the modulation. The cubic Walsh estimate gives $F_c(a, 0) = \sqrt{k} \alpha_a$ and $|F_c(a, b)| \leq 2\|\alpha\|_1$ for $b \neq 0$. Hence the exact excess after embedding and modulation is a disjoint cell sum

$$h_c(y) = \sum_a w_a \mathbf{1}_{\{\pi(y+y_0)=(a,0)\}}, \quad w_a = (k\alpha_a^2 - \tau)_+, \quad \sum_a w_a \leq k - \tau.$$

This proves the upper half of (425). Conversely $\alpha = e_0$ realizes any affine codimension- $2r$ cell with weight $k - \tau$, proving equality and $R_{\text{cub}}^2 \geq kn_{\max}$. Each cell's signed count is subgaussian with variance proxy n_C ; maximize over the M cells and use the radius certificate. Counting ordered normals gives $M \leq N^{2r+1}$, which yields the last constant in (426). $\square$

In the base ambient dimension $d = 2r$, these cells are single rows. If n_y is the row multiplicity and

$$B_n(t) = 2^{-n} \sum_{j: |n-2j| \leq t} \binom{n}{j},$$

then, for $Z_Q = \max_y |\sum_{i: y_i=y} \varepsilon_i|$,

$$\mathbb{P}\{Z_Q \leq t\} = \prod_{y: n_y > 0} B_{n_y}(t). \quad (427)$$

The complete restricted excess supremum has law $(k - \tau)Z_Q$. Its exact moments, quantiles and strict lattice tails are therefore computable from the same bin counts in polynomial bit complexity when τ is rational, charging its supplied bit length. At $\tau = \log(2k)$ the scalar can

remain symbolic; numerical quantiles instead charge a certified precision. This is an exact signed-process law, not by itself a confidence theorem for the full random matrix.

When $k \geq 256$, exact long-list consistency also works for a structured four-quadrant sign profile on the same cubic support. Let σ_V denote the coefficient sign at V . For an auxiliary list $T = (V_1, \dots, V_\ell)$ drawn uniformly with replacement from the signed support, define its agreement cell

$$\mathcal{C}_T = \{y : \chi_{V_j+V_1}(y) = \sigma_{V_j}\sigma_{V_1}, 2 \leq j \leq \ell\}.$$

Its consistency probability satisfies

$$p_\ell \leq \mathbb{P}\{\mathcal{C}_T \neq \emptyset\} \leq p_\ell + 2(k^2 - 4)(5/8)^\ell, \quad p_\ell = 4(3/4)^\ell - 6(1/2)^\ell + 4(1/4)^\ell,$$

and hence, with $b_\ell = (3/4)^\ell + (1/4)^\ell$,

$$4 - 6(2/3)^\ell \leq \frac{\mathbb{P}\{\mathcal{C}_T \neq \emptyset\}}{b_\ell} \leq 4 + 2k^2(5/6)^\ell. \quad (428)$$

A full-rank normalized kernel converges uniformly to the four genuine peak rows with explicit errors of the same exponential orders. Thus actual phase consistency can cancel an apparently exponential normalization, but only because the coefficient Fourier geometry identifies the correct peak sector.

26.5 Pooling that is useful but does not cover all supports

There is also a useful common-event theorem for a supplied union of master coefficient spaces. Let $T_1, \dots, T_J$ be actual frequency sets of size K , and let $\mathcal{F}$ consist of the unit vectors supported on at most k coordinates of one master. With $q = \lceil K/k \rceil$ and with the radius computed over that same family,

$$\mathcal{W}_{Q,\mathcal{F}} \leq 2R_{Q,\mathcal{F}} \sqrt{qK \log(2KJ)}. \quad (429)$$

All masters share the original signs; a block-diagonal matrix-series readout is used only after that dependence has been retained. When each master is Sidon, the fixed-master operator event pools all of its child supports and all deterministic Parseval compressions at once. For one cubic master of size $2k$, this gives a two-sided confidence scale

$$m \asymp k + \frac{k}{\log k} \log(1/\eta)$$

for a family containing exponentially many affine-inequivalent child supports, with a lower witness drawn from that same family. The comparison constants include the numerically unspecified universal constant in the imported fixed-master covariance theorem.

26.6 Earlier reductions that the exact examples rule out

The remaining obstruction is not simply a shortage of fourth moments. The following counterexamples specify information that a successful descriptor must retain.

- (i) *A uniform empirical fourth-energy cap fails even on a good sample.* For every nonempty Q and every available Sidon support of size k , an amplitude vector aligned with one sampled row gives

$$\sup_c \sum_i f_c(y_i)^2 \geq k^2.$$

If the same Q already has ordinary k -RIP constant at most $1/2$, then $R_Q^2 \leq 3m/2$, so the ratio to $R_Q^2 \log(2k)$ is at least $2k^2/[3m \log(2k)]$ and is unbounded along any such sample sequence with $m = o(k^2/\log k)$. Thus the population identity $\mathbb{E}_y f_c^2 \leq 3$ cannot be promoted by taking the supremum after sampling.

- (ii) *Exact small-master covers have prohibitive global entropy.* If $k^3 \leq N$ and real spaces $W_1, \dots, W_J$ of dimension at most K contain every unit vector on every Sidon k -support, even when each vector may select its own space, then

$$J \geq \left(\frac{N}{2K} \right)^k. \quad (430)$$

For $K = O(k)$, inserting this into (429) loses a factor of order $k/\log k$ in the key parameter. This is an obstruction to the stated finite exact cover method, not a sampling lower bound.

- (iii) *A global positive list normalization misallocates amplitude.* For a one-bit lift of the cubic Sidon support with $k = 2^a$, $a \geq 8$, a $3/4$ – $1/4$ coefficient sign split, and admitted agreement-list lengths $\ell \geq 2a - 11$, every auxiliary agreement cell is consistent. Nevertheless any single globally scaled, high-rank agreement-kernel majorant of the $\log(2k)$ excess has signed width at least

$$c_0 k^{1+\theta}, \quad \theta = 2 \log_2(4/3) > \frac{1}{2},$$

while the true modulation family obeys $\mathcal{W}_Q(h_{\text{mod}}) \leq R_Q \sqrt{2k \log(2N')}$ with $N' = 2k^2$. The same power loss survives nonnegative mixtures of all admitted list lengths, exact consistency and occupancies, and subtraction of the exact scalar population mean. Their multipliers may depend on the modulation z and the realized Q , but are fixed independently of the fresh symmetrization signs ε . The loss comes from using one multiplier for an intermediate row and for a movable true peak.

- (iv) *Scalar centering does not cancel a nonconstant successor sector.* For the three-value law (423), a centered majority detector still distinguishes the $f = 0$ and $f = 2$ sectors. Write q_j for the unscaled majority-detector probability on the sector $f = j$. Under $2 < t \leq k/4$ and $16k/t \leq \ell \leq k/4$ its detector gap satisfies

$$q_2 - q_0 \geq \frac{\ell}{k} e^{-2\ell t/k}.$$

If $m \geq (32/7)(k/\ell)^2 e^{4\ell t/k}$, the centered surrogate width is at least $\ell \sqrt{m} e^{-2\ell t/k}/8$. Taking $t = \log(2k)$, $\ell \asymp k/t$ and $m \asymp kt^2$ makes its ratio to the target diverge as $\sqrt{k}/t^2$. In the same original-row model, the averaged true excess width remains of order k . The exact correction

$$h_z(y) = \frac{k-t}{k(k-2)} [f_z(y)^2 - 2f_z(y)] \quad (431)$$

cancels the $f = 2$ sector, showing why a signed nonconstant correction is qualitatively different from scalar centering.

- (v) *The empirical pair-Gram spectrum loses its orientation.* For every integer $s \geq 2$, put $t_s = 2^s$, $r_s = 3s$, $M_s = 2^{r_s}$, $m = t_s M_s$, $d = t_s + r_s$, and $N = 2^d$. Already for the complete two-sparse coefficient family, there are explicit full-affine-rank deterministic Walsh row lists Q_H, Q_L with the same number of distinct rows, the same complete empirical pair-Gram eigenvalues and the same exact family radius $R_Q^2 = 2m - 2$, but

$$\frac{\mathcal{W}_{Q_H}}{\mathcal{W}_{Q_L}} \geq \frac{2^{s/2}}{10\sqrt{s}} \rightarrow \infty. \quad (432)$$

The same separation holds for the Gaussian width. The oriented form $q_c^T H_Q q_c$ in (413), rather than only the spectrum of H_Q , is therefore necessary for a bounded-factor two-sided empirical descriptor.

None of these examples disproves (412). They rule out specified reductions while identifying what was lost: empirical orientation, amplitude-dependent allocation, nonconstant signed corrections, or an economical way to share events across supports.

26.7 Peak orientation defeats an all-sample smooth contraction

The finite-witness theorem leaves one height logarithm. A natural attempt to remove it was to contract a smooth tail indicator using only its empirical mass. Let

$$\mathcal{F}_k = \{0\} \cup \{F_c : c \in \mathcal{C}_k\}$$

and define the exact ramp

$$\phi_t(x) = \begin{cases} 0, & |x|/\sqrt{t} \leq 1/2, \\ 3v^2 - 2v^3, & v = 2|x|/\sqrt{t} - 1 \in (0, 1), \\ 1, & |x|/\sqrt{t} \geq 1. \end{cases}$$

For the original iid list Q , put

$$r_t(Q) = \sup_{F \in \mathcal{F}_k} \frac{1}{m} \sum_i \phi_t(F(Y_i)),$$

$$\widehat{R}_Q(t) = \frac{1}{m} \mathbb{E}_\varepsilon \sup_{F \in \mathcal{F}_k} \left| \sum_i \varepsilon_i \phi_t(F(Y_i)) \right|, \quad D_0 = k \log(2eN/k).$$

The signs are shared across the supremum. The proposed uniform contraction was

$$\mathbb{E}_Q \widehat{R}_Q(t) \stackrel{?}{\leq} C \sqrt{\frac{D_0}{mt}} \sqrt{\mathbb{E}_Q r_t(Q)}. \quad (433)$$

Theorem 26.7 (No absolute all-sample smooth-contraction constant). *For every odd integer $r \geq 9$, set*

$$k = 2^r, \quad d = 4r, \quad N = k^4, \quad t = k/2, \quad m = 2^{30} k^6 r^2.$$

For the full all-Sidon class above,

$$\mathbb{E}_Q r_t(Q) \leq \frac{256}{k^2}, \quad \mathbb{E}_Q \widehat{R}_Q(t) \geq \frac{1}{32} \sqrt{\frac{4r^2 \log 2}{mk^2}},$$

and hence

$$\frac{\mathbb{E}_Q \widehat{R}_Q(t)}{\sqrt{D_0/(mt)} \sqrt{\mathbb{E}_Q r_t(Q)}} \geq \frac{\sqrt{r}}{1024}. \quad (434)$$

Thus no absolute C makes (433) valid uniformly over all sample sizes. The obstruction also holds with fewer samples than rows: for odd $r \geq 11$, take $d = 10r$, $N = k^{10}$, and $m = 2^{32} k^6 r^2 < N$; the ratio is at least $\sqrt{8/5} \sqrt{r}/1024$.

Proof outline. Embed the cubic Sidon graph in each binary graph subspace. Its flat squared Walsh profile has values $0, 2, k$ and a unique peak, so at $t = k/2$ the ramp is exactly the indicator of a graph cell. The 2^{4r^2} cell orientations have population mass k^{-2} and a packing of logarithmic size $4r^2 \log 2$. The finite sign-packing lower bound yields the displayed lower width. A finite net over the full all-Sidon coefficient class, followed by the actual iid-row concentration estimate, gives $\mathbb{E}_Q r_t(Q) \leq 256/k^2$. Substitution and $D_0/t \leq 8r$ prove (434). The $m < N$ construction uses $2r$ -by- $8r$ graph maps and the same argument. The source map identifies the complete packing, separation, and net calculations. $\square$

Both constructions are strongly oversampled. They refute only the absolute-constant assertion (433) for every m ; they do not refute a target-sample-only hypothesis, the strict count theorem, (412), or FR-10.

There is a quantitatively compatible conditional repair.

Proposition 26.8 (A summable conditional orientation correction). *Fix $T \geq 184320$, put*

$$J = \max\{0, \lceil \log_2(k/T) \rceil\}, \quad t_j = 2^j T \quad (0 \leq j < J).$$

Suppose that, at the sample size under consideration and only at these levels, constants $C_0, C_1 \geq 0$ satisfy

$$\mathbb{E}_Q \hat{R}_Q(t_j) \leq C_0 \sqrt{\frac{D_0}{mt_j}} \sqrt{\mathbb{E}_Q r_{t_j}(Q)} + C_1 \sqrt{\frac{D_0 \log(2t_j)}{mt_j^3}}. \quad (435)$$

If

$$m \geq \max \left\{ 32768TD_0, 128T^2 \log 20, 3840C_0^2 D_0 J, 23040^2 C_1^2 \frac{D_0 \log(2T)}{T} \right\}, \quad (436)$$

then, with probability at least $9/10$, every $c \in \mathcal{C}_k$ satisfies

$$\frac{7}{8} - \frac{3}{4T} \leq \frac{1}{m} \sum_i F_c(Y_i)^2 \leq \frac{19}{16}.$$

For fixed C_0, C_1, T , the displayed conditions have order $O(D_0 \log(2k))$.

Only the implication in this proposition is proved. Its assumed corrected contraction (435) remains open. The additive term matches the peak-orientation floor and is summable over the required dyadic levels.

26.8 Exact cell aggregation still loses cross-cell coherence

The next example permits arbitrary signed coefficients after aggregating all identical high-rank agreement cells. Let $k = 2^a$, $a \geq 12$, put $d = k$, $N = 2^k$, and use $S = \{e_1, \dots, e_k\} \subset \mathbb{F}_2^k$ with flat coefficients. Define

$$F(y) = k^{-1/2} \sum_{j=1}^k (-1)^{y_j}, \quad f = F^2, \quad \tau = \log(2k), \quad h = (f - \tau)_+, \quad R_{\text{cell}} = 2a - 12.$$

For nonempty $T \subset [k]$, let

$$A_T(y) = \mathbf{1}_{\{(-1)^{y_j} : j \in T\} \text{ are all equal}}, \quad \mu_T = \mathbb{E}_y A_T(y) = 2^{1-|T|}.$$

After observing the original iid sample Q , but before drawing the fresh signs, permit arbitrary real choices in

$$g_Q(y) = b_Q + \sum_{|T| \geq R_{\text{cell}}+1} a_T(Q) [A_T(y) - \beta_T(Q)]$$

subject only to

$$\max_i |g_Q(Y_i) - h(Y_i)| \leq \epsilon, \quad 0 \leq \epsilon \leq 1.$$

The constant b_Q is free. Define

$$\mathfrak{T}_Q(a, \beta) = \sum_{|T| \geq R_{\text{cell}}+1} |a_T(Q)| \mathbb{E}_\epsilon \left| \sum_i \epsilon_i [A_T(Y_i) - \beta_T(Q)] \right|$$

and let $\mathfrak{T}_*(Q)$ be its infimum over all feasible choices. Here the comparison uses the full ambient all-Sidon radius

$$\mathcal{R}_Q^2 = \sup_{c \in \mathcal{C}_k} \sum_i F_c(Y_i)^2.$$

Theorem 26.9 (Cellwise triangle obstruction). *If $m \geq 8N \log(4N)$, then*

$$\mathbb{E}_Q \mathfrak{T}_*(Q) \geq \frac{\sqrt{m} k \tau}{3072}, \quad \mathbb{E}_Q \mathcal{R}_Q \leq \sqrt{2m},$$

and consequently

$$\frac{\mathbb{E}_Q \mathfrak{T}_*(Q)}{(\mathbb{E}_Q \mathcal{R}_Q) \sqrt{k \log(2k) \log(2N/k)}} \geq \frac{\sqrt{\log(2k)}}{3072\sqrt{2}}. \quad (437)$$

Proof outline. A constant-annihilating functional compares uniform measure with a Hamming level where $h \geq \tau$. Its action on every cell is at most $4 \cdot 2^{-R_{\text{cell}}/2} \sqrt{\mu_T}$. On the event that every Walsh row occurs between half and twice its mean multiplicity, the sampled-row approximation is uniform over the row space, so the functional forces $\sum_T |a_T| \sqrt{\mu_T} \geq k\tau/512$. On this same event, fresh-sign Khintchine interpolation gives each centered cell a width at least $\sqrt{m\mu_T}/3$. The event has probability at least $1/2$ under the sample hypothesis, yielding the first inequality. Complete Walsh orthogonality and the maximum row multiplicity give the full radius bound. The details retain arbitrary centers and coefficients chosen from Q before the fresh signs. $\square$

The true translated residual does not suffer this loss. For every deterministic Q , with $h_z(y) = h(y + z)$,

$$\mathbb{E}_\varepsilon \max_z \left| \sum_i \varepsilon_i h_z(Y_i) \right| \leq \mathcal{R}_Q \sqrt{2k \log(2N)}. \quad (438)$$

Exact high-rank representations exist; a symmetric one has $k/2 + 1$ coefficients and takes $O(k^2)$ rational operations, with coefficient bit bound $O(B + k^2)$ for B -bit rational levels. The noncommuting matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + h(y) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

inherits the obstruction by same-input scalar compression. Thus the failure is the triangle operation across different cells, after all within-cell signed aggregation and scalar centering. The construction is strongly oversampled and does not refute the coherent residual, target-scale sampling, matching quantiles, or FR-10.

26.9 The exact remaining routes

For the signed-width route, the remaining all-Sidon target is (412). It is sufficient but no longer the only known bridge: a sharp bound for the actual strict tail counts on the required dyadic levels and sample sizes would also complete the fixed-confidence all-Sidon stage. Theorem 26.3 proves that route with one extra $\log(2t)$ factor and therefore one extra sparsity logarithm after integration. The clipped process, flat translated peaks, restricted inverse-Fourier amplitudes, finite master families, and near-peak cells already satisfy the intended scale. What remains is to remove that last factor while retaining the oriented pair lift and all shared signs through the supremum.

Even a proof of the all-Sidon residual estimate would leave a second bridge: arbitrary supports may contain repeated additive pair sums and higher relations. A general FR-10 argument must stratify or contract those collisions without paying for every support separately and without assigning independence to reused relations. The existing density filtration and master-space theorems settle important subfamilies, but do not yet supply this global support-selection rule.

The most plausible retained state suggested by the positive and negative results is multiscale and oriented. It should keep the coefficient-dependent pair lift, the action of the actual empirical Gram on that lift, and signed amplitude sectors that permit identities such as (431). Any compression of that state must be tested under the successor (414); preserving a single local

Gram or a scalar energy is not extension stability. The viable next routes are a contraction restricted to the necessary sample regime, a proof of the summable correction (435), or a direct coherent estimate retaining cancellation across high-rank cells. Matching matrix quantiles remain a separate task after the width and moment comparison has been proved.

27 What a unified descriptor cannot discard

The positive results have three different kinds of closure. The Gram algebra closes pointwise products of its represented inputs. A live-label register closes prescribed label-dependent extensions. An analytic upper potential closes an inequality while potentially losing exact spectral information. They should not be identified. The following obstructions isolate this distinction without requiring a catalogue of special models.

27.1 Coherent cancellation is not permission to retire a live label

Consider the eight rows

$$(1, 0), (0, 1), (1, 1), (-1, -1), (1, -1), (1, -1), (2, 0), (0, 2),$$

all divided by 3. Their outer products sum to I_2 . Name the first four rows a, b, v, w and put

$$A = A_{av} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = A_{vb} = \frac{1}{9} \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then $A_{aw} = -A$, $A_{wb} = -B$, and

$$A_{av} + A_{aw} = 0, \quad A_{av}A_{vb} + A_{aw}A_{wb} = \frac{2}{81} \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \neq 0. \quad (439)$$

The coefficients do not commute: $[A, B] = \frac{4}{81} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Thus the coherent-sum encoding $(C_v, C_w) \mapsto C_v + C_w$ fails the kernel test (21) for the incident right-multiplication update.

This is realized on actual hash and sign inputs. Condition on all four hashes being equal and $\sigma_v = \sigma_w$, an event of probability $1/(2b^3)$. The path sum $z_{av}A_{av} + z_{aw}A_{aw}$ is zero pointwise on this event, while its incident extension is $\sigma_a\sigma_b$ times the nonzero matrix in (439). Its conditional squared Frobenius norm is $8/729$. The current scalar words agree and the extended scalar words agree; both conditional scalar Grams are the two-by-two all-ones matrix. What is missing is the association of each summand with its future coefficient. This is an obstruction to that encoding, not a lower bound for the full sketch from a selected positive sector.

27.2 Even all pairwise positive data can miss a signed cycle

The next obstruction keeps fixed dimension, fixed polynomial degree and only three primitive signs. It separates exact overlap information from signed coefficient alignment.

Theorem 27.1 (A fixed-size polynomial separation). *There are two triples of real symmetric 8×8 matrices $A_i^\pm$ with the following identical retained data: the law of the three independent signs, every individual coefficient spectrum, $\sum_i (A_i^\pm)^2 = I_8$, each positive operator $[A_i^\pm, A_j^\pm]^* [A_i^\pm, A_j^\pm]$, every real-weighted one- or two-term signed-sum norm law, and the normalized second and fourth trace moments of $X_\pm = \sum_i \varepsilon_i A_i^\pm$. Nevertheless, for the same polynomial $p_\tau(x) = (x^2 - 2/3)(x^2 - 2) + \tau$, $0 < \tau \leq 2/9$,*

$$\|p_\tau(X_+)\| = \tau, \quad \|p_\tau(X_-)\| \geq 2/9 \quad \text{on every original sign input.} \quad (440)$$

Consequently no constant-factor estimator of all these polynomial outputs can depend only on the listed retained data and the specified polynomial.

Proof. Let $I = I_2$, $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Define

$$\Gamma_1 = D \otimes I \otimes I, \quad \Gamma_2 = J \otimes D \otimes I, \quad \Gamma_3 = J \otimes J \otimes D, \quad \Gamma_4 = J \otimes J \otimes J.$$

They are traceless symmetric matrices, square to I_8 , and anticommute in distinct indices, since $JD = -DJ$ at the first differing tensor position. Thus for $\Gamma(v) = \sum_j v_j \Gamma_j$,

$$\Gamma(v)^2 = \|v\|^2 I_8, \quad \Gamma(v)\Gamma(w) + \Gamma(w)\Gamma(v) = 2\langle v, w \rangle I_8. \quad (441)$$

Choose

$$\begin{aligned} v_1^+ &= (1, 0, 0, 0), & v_2^+ &= (1, 1, 1, 1)/2, & v_3^+ &= (1, 1, 1, -1)/2, \\ v_1^- &= (1, 0, 0, 0), & v_2^- &= (-1, 1, 1, 1)/2, & v_3^- &= (-1, -1, -1, -1)/2, \end{aligned}$$

and set $A_i^\pm = \Gamma(v_i^\pm)/\sqrt{3}$. These vectors have unit norm and pairwise inner products all $+1/2$ or all $-1/2$, respectively. Therefore $(A_i^\pm)^2 = I_8/3$, and each spectrum is $\pm 1/\sqrt{3}$ with multiplicity four. Equation (441) implies for unit v, w of inner product c that $[\Gamma(v), \Gamma(w)]^*[\Gamma(v), \Gamma(w)] = 4(1 - c^2)I_8$. Every retained pairwise positive operator is consequently $I_8/3$. For any real weights a, b ,

$$\|a\varepsilon_i A_i^\pm + b\varepsilon_j A_j^\pm\|^2 = \frac{a^2 + b^2 \pm ab\varepsilon_i \varepsilon_j}{3},$$

whose law does not depend on the sign of the pair alignment.

On the same three primitive signs put $T = \varepsilon_1 \varepsilon_2 + \varepsilon_1 \varepsilon_3 + \varepsilon_2 \varepsilon_3$. Then $T = 3$ with probability $1/4$ and $T = -1$ with probability $3/4$, and

$$X_\pm^2 = (1 \pm T/3)I_8.$$

Both normalized second moments are 1 and both normalized fourth moments are $4/3$, but the normalized sixth moments are $20/9$ and $16/9$. The missing signed invariant is the triangle product of the three pair inner products, $+1/8$ versus $-1/8$. A coefficient sign change alters two of its edges and cannot change this product. Finally $T^2 = 3 + 2T$ gives

$$p_0(X_+) = 0, \quad p_0(X_-) = (4T/9)I_8.$$

Adding $\tau \in (0, 2/9]$ proves (440). Since the output is scalar, the same coordinate vector certifies its lower bound on every input. $\square$

The L^p norm ratios and every strict quantile ratio in this example are at least $2/(9\tau)$; raw p th-moment ratios are at least $(2/(9\tau))^p$. The functional degree is four, and after reduction by $\varepsilon_i^2 = 1$ the displayed output has Boolean degree at most two. This rules out the specified positive pairwise state, not the complete signed coefficient array, the actual commutator matrices, or vectorized covariance. Those stronger data distinguish the models. It also does not refute the scalar-column Walsh RIP conjecture.

27.3 Positive return kernels do not give a Loewner recurrence

Even with all odd first returns zero, replacing the trace comparison in Theorem 2.2 by $K_{2q} \preceq V(V + c^2 I)^{q-1}$, $c = \|C\|$, is invalid. Take

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad D = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = V - D \succeq 0.$$

On a rest space $(\mathbb{R}^2 \oplus \mathbb{R}^2) \oplus \mathbb{R}^2$, put $Bx = (A^{1/2}x, D^{1/2}x; 0)$ and $C(y, z; w) = (w, 0; y)$. Then $B^*B = V$, $\|C\| = 1$, odd first returns vanish, and $F_2 = V$, $F_{2j} = A$ for $j \geq 2$. Renewal gives $K_6 = V^3 + VA + AV + A$. The proposed upper matrix minus this exact value is

$$V(V + I)^2 - K_6 = D + VD + DV = \frac{1}{4} \begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix},$$

which has determinant $-1/16$. This is an obstruction at the general finite block-operator level; no realization of this particular block example as a SparseStack input is asserted or needed. Complete positivity of each sandwich map does not permit substitution inside a noncommuting polynomial. Trace Hölder, block row/column contractions and pointwise Gram multiplication remain legitimate; they establish different statements.

28 The unified result and the remaining frontier

The strongest completed methodology is modular. The exact random-function representation identifies the relevant Gram. The descent criterion tests whether a proposed reduction is meaningful under later interactions. The Gram algebra gives true pointwise circuit closure when the coefficient list is retained. Joint feature aggregation or a collision budget supplies a computable probability law on specified structural classes. Conditional bucket estimates supply a different form of contraction when a large diffuse population would make exact enumeration expensive. Coupled perturbation bounds combine exact and analytic pieces on the same input. Walsh sampling then shows that even a successful fixed-support contraction must separately pay for simultaneous selection of the support. The later results sharpen both sides of this boundary: live parity messages provide computational closure, localized relation codes preserve complete compressed moments, and coherent peak operators control growing coefficient spheres on shared events. The sparse-block theorem now combines all requested operations with useful moment and strict-quantile comparisons on one explicit class. The dense grouped-sector theorem removes the coordinate-sparsity requirement by retaining the full algebra associated with supplied permutation groups and marked sites. Low-rank and homogeneous Clifford calculi add different dense geometries. Even-innovation junctions and cubic interaction blocks extend the analytic and geometric interfaces; the actual cross-block obstruction identifies a limit of the latter. These achievements still have different structural hypotheses.

These results should be read as one set of obligations with several proved ways to discharge them. They do not assert that every model splits into “bulk plus extremes,” that a scalar Gram spectrum is sufficient, or that positive retained energy provides a pointwise norm lower bound for the complete matrix. The counterexamples identify the precise steps at which such assertions would fail.

28.1 A concrete general target

For an explicit class of polynomial matrices Z of degree r and dimension d , seek a descriptor $\mathcal{D}(Z)$ and a functional $\mathcal{F}_p$ such that

$$c(r, w, p)\mathcal{F}_p(\mathcal{D}(Z)) \leq (\mathbb{E}\|Z\|^p)^{1/p} \leq C(r, w, p)\mathcal{F}_p(\mathcal{D}(Z)), \quad (442)$$

where w is a stated structural parameter and the arithmetic and bit costs of the descriptor are explicit. Achieving this target for a substantial, clearly defined class would constitute a general calculus for that class; it need not encompass every polynomial random matrix. Every claimed lower bound in (442) must be witnessed in that same Z .

Three further requirements make this a specification of a reusable calculus.

1. **Reusable update rules.** The descriptor must support extending a word, multiplying represented expressions, and identifying shared random constraints. Each permitted operation must have a stated update rule with quantitative control of the resulting state size and accumulated approximation error. Closure must survive subsequent interactions. This is the additional structure that turns a two-sided characterization into a calculus.
2. **Useful costs and comparison factors.** Explicit dependence alone is insufficient: enumerating the entire random law has an explicit cost, and a comparison factor such as $\exp(p^2)$ is explicit too. State regimes in which computation is affordable and the ratio $C(r, w, p)/c(r, w, p)$ remains useful. Track arithmetic and bit costs, state growth, and accumulated losses throughout the required moment range, especially when p is comparable to $\log(d/\delta)$. No dependence on growing degree, structural width, or moment order may be hidden.
3. **Separate moment sharpness from quantile sharpness.** Equation (442) gives matching norm moments and hence upper-tail bounds by Markov's inequality. Matching quantiles require an additional argument about the distribution of the same matrix, for example a suitable lower-tail certificate or a direct finite-law calculation. Moment comparison alone must not be advertised as matching confidence bounds. Strict thresholds and the intended confidence range remain part of the statement.

The central requirement can therefore be stated as follows:

The descriptor admits reusable rules for the specified interactions, with quantitative control of state size, computational cost, and accumulated comparison loss throughout the required moment range.

Theorem 10.2 now realizes this specification for a clearly defined sparse-block/local-randomness class. Its descriptor retains actual matrix coefficients and named constraints; full addition, ordered multiplication, adjoint and primitive identification have exact rules. The final structural factor is $\sqrt{b\Delta_r\Delta_c}$, not a product of losses over preceding operations. A local finite-law transfer computes the envelope distribution, and the pointwise compression witness gives matching quantiles of the same matrix at every strict confidence endpoint. Fixed local parameters, logarithmic boundary and polynomial actual coefficient lengths give useful costs at p comparable to $\log(d/\eta)$. This is a restricted achievement of the general target, not a claim of new historical priority for elementary block bounds or variable elimination.

Theorem 11.3 is a broader realization of the same target. It retains all full, noncommuting sectors of the algebra commuting with permutations within supplied groups, together with arbitrary marked-site coefficients. Only proved identity multiplicities are discarded. A physical sector embeds in the same original matrix, so its singular vector supplies the lower witness. The exact joint distribution of overlapping integer scores gives strict quantiles separately from moments. The comparison can be made arbitrarily close to one by increasing the explicitly charged readout order h .

Its useful regime has fixed group count, score count and local arity, logarithmically many marked sites and live random names, polynomial score range, and polynomial actual circuit/coefficient lengths. The cost is polynomial in numerical site count and score range, not just their binary encodings. Site marking and common partition refinement preserve the same physical operator before the next interaction, while increasing these parameters explicitly. Dense invertible examples show that this achievement does not rely on bounded coordinate-block sparsity. Invoking an abstract symmetry decomposition alone would not supply the required construction or update algorithm; those are part of the theorem.

Section 12 gives two complementary routes. A pointwise rank certificate makes a Frobenius readout useful and its scalar transfer admits ordinary symmetric powers. Identity shifts can

have full physical rank, but the residual-rank restriction is essential for their same-input lower witness. Homogeneous Clifford expressions have an exact signed radial normal form, even when the ambient coefficient algebra is irreducible. Different degrees require the retained radial law and affine phases. These moment/PZ results do not inherit the sector theorem’s all-confidence quantile conclusion.

Section 9 distinguishes three earlier computational regimes. Symmetric-power messages are polynomial in moment order at fixed internal size. Pure tensor factors permit growing irreducible coefficient dimension with compact-input efficiency. General matrix-product operators admit full sums and interacting local factors, but their cyclic replica bond is exponential in moment order; polynomial dependence on d/η does not imply polynomial dependence on compact input or confidence bits. The explicit legal-continuation obstruction shows why ordinary replica symmetrization cannot silently remove this cost.

The all-name theorem in Section 17 resolves independent-name integration in every order, with one second-moment gain per name and constant one under amplification. Actual affine registers distinguish fresh, reused and contradictory constraints, including coefficient phase. Even-innovation junction trees provide a tractable analytic contraction with no depth loss and arbitrary predetermined signed real Parseval compression. These results do not control cancellations after summing arbitrarily many nonempty relation channels.

Section 20 proves a separate geometric closure: actual isotropic cubic projections have common blocks controlled by the spread between the sum and intersection of their subspaces. Ordered sums, products, adjoints, ranges and intersections remain in that algebra. The result includes noncommutation at spread two and large coefficient rank; random program width is still charged separately. The original sampling event controls every generated deterministic operator at once, but the sample Gram generally couples the blocks and is not thereby an input to the small-block moment theorem. Section 21 now quantifies that distinction: the actual cross block has full vector covariance, and a constant-probability spectral-mass event defeats every very low-rank approximation at its natural scale, even when chosen after sampling. Small deterministic blocks and rank-one observations alone therefore do not give bounded-rank closure.

Sections 23–25 add three modules that were absent at the dense-calculus checkpoint. The signed-core state is closed under every core-supported insertion and charges the only returning remainder after a product. Its noncommuting example proves a quantitative improvement over wordwise estimation. The nilpotent linearization keeps arbitrary written polynomial order and random-name reuse, while the quadratic specialization gives a same-input two-sided theorem for fixed-Sidon Walsh rows, independent Rademacher rows under fixed compression, and actual SparseStack layers in the stated oversampled fixed- (U, D) regime. The aggregate-moment state is a different exact calculus: when the output depends polynomially on a few additive statistics, it contracts their truncated exponential messages rather than enumerating their exponentially large joint range. These are substantial classes with reusable update rules and explicit costs; they do not imply a compact state for every polynomial random matrix.

Section 22 extends the dense results by proving closure under supplied low-rank symmetry-breaking seeds and under mixed-degree Clifford interactions determined by a complete joint Gram. The affine-phase example in Section 23, together with the pooled pair-lift and high-rank agreement-cell examples, sharpens the boundary. Complete covariance, singular values, or Gram eigenvalues can omit the signed orientation needed by a legal continuation. Even exact within-cell aggregation, arbitrary signed cell coefficients, and scalar centering can lose an unbounded factor if a triangle inequality is then taken across cells. The same-input 2×2 compression makes this a noncommuting obstruction. A useful general descriptor must retain the relevant orientation and cross-cell cancellation or restrict its permitted future interactions precisely.

28.2 The current calculus frontier and the full FR-10 pooling problem

The immediate Walsh bottleneck is now more precise than an unspecified fixed-support moment remainder. Section 26 proves the clipped quadratic-process estimate for all sparse supports, the surviving excess estimate on a broad cubic profile class, and the simultaneous original-sample count

$$\mathbb{E}_Q \sup_c n_c(t) \leq \frac{131m}{t^2} + 3L_t \log(2N + 1) + 3.$$

Whenever a size- k Sidon support exists, dyadic integration of this theorem gives a fixed-confidence all-Sidon result with sample order

$$O\left(k \log^2(2k) \log(2N/k)\right).$$

It removes two earlier logarithmic losses but retains one height logarithm. At the target sampling order, the signed-width route requires the following coherent same-sample estimate.

$$\mathbb{E}_Q \mathbb{E}_\varepsilon \sup_{c \in \mathcal{C}_k} \left| \sum_{i=1}^m \varepsilon_i (f_c(y_i) - \log(2k))_+ \right| \leq C \sqrt{k \log(2k) \log(2N/k)} \mathbb{E}_Q R_Q. \quad (443)$$

It must use the same realized row list Q and its actual radius R_Q . The specified reductions based only on positive agreement kernels, majority lists, scalar centering, empirical Gram spectra, finite exact master-space covers, or cellwise triangle aggregation fail as standalone general routes to (443). The formerly proposed log-free smooth contraction is also false if required uniformly over all sample sizes. These failures do not disprove (443), a target-sample-only contraction, the strict tail-count target, or FR-10. If no size- k Sidon support exists, all count-rate statements use the largest feasible Sidon size instead.

The live all-Sidon routes are therefore: prove (443); remove the last height logarithm directly from the strict counts; prove the summable orientation correction (435) at only the needed sample sizes and levels; or estimate the retained high-rank family coherently across cells. Completing any one of these routes would still leave passage from Sidon to arbitrary k -supports in the unrestricted range $d \ll k \ll N$. Such supports contain repeated pair sums and higher relations. The core-coset, bounded-flat-union, and cubic theorems show how to pool structured families, but no current state supplies the arbitrary-support bridge without enumerating all named supports.

Three additional methodological directions remain open. First, the signed core must currently be supplied or discovered model by model; an algorithm that finds a small useful core and certifies a small remainder for a broader class would turn the analytic rule into a more automatic calculus. Second, the aggregate theorem needs a small statistic dimension and a bounded-width primitive schedule supplied with the input; arbitrary dense compressed SparseStack or Walsh support selection need not have either. Third, matching moments yield upper tails, but matching strict quantiles still require a same-input lower-tail witness or an exact finite-law transfer. The sector, sparse-block, Clifford-score, and cyclic-seed calculi have such transfers; the aggregate and universality theorems generally do not.

For exact-constant tails, a separate milestone remains a uniform local-limit or saddle-point theorem for the actual joint lattice score, including moving thresholds and boundary atoms. The fixed-dimensional benchmark of Section 7 establishes the finite-law mechanism but does not settle that asymptotic problem.

28.3 Provenance and verification

The starting material consisted of the supplied manuscripts on graded multiplication, SparseStack, graph matrices, and structured transforms. This report develops the branch concerning vacuum weights, random-constraint overlaps, retained coefficient states, and the subsequent Walsh

investigation. It does not reassert every theorem in those original manuscripts, nor claim solutions to the other queued Fourier, tensor, or graph-theta problems. General matrix-chaos theory already gives powerful coefficient-based inequalities; see Bandeira, Lucca, Nizić-Nikolac, and van Handel [11]. The present synthesis specifies which further overlap and extension information is used in the restricted results here.

The current verification index records 94 saved suite statuses, 77 proof-review records, 134 authored documents, and 1,078 checked local links. The latest checkpoint added four suites. Its index performed 764 hash comparisons and found zero mismatches and zero missing links; it also records 252 indexed-source attestations, 111 repeated/dependency comparisons, and 90 prior frozen-suite records. The historical suites were not all rerun for this report edit, and 17 legacy source-attestation omissions remain flagged. This distinction is retained rather than converting an indexed artifact count into a stronger verification claim.

The source map now continues from the dense general-calculus checkpoint through the dense-interaction, signed-core, finite polynomial-transfer, aggregate-moment, signed-correction, tail-profile, and signed-pooling notes. The newest branch contains

- `notes/finite_witness_and_coherence_checkpoint.md` and `results/finite_witness_and_coherence_verification.json`;
- `problems/FR-10_Walsh_RIP/finite_witness_tail_calculus.md`;
- `problems/FR-10_Walsh_RIP/sidon_witness_noise_floor.md`;
- `problems/FR-10_Walsh_RIP/smooth_contraction_diagnostic.md`;
- `problems/FR-10_Walsh_RIP/high_rank_signed_pooling_bridge.md`;
- the four corresponding pairs of `_checks.py` and `_checks.json` companions.

Research sources and prior report versions are preserved. Separate report reviews check transcription, hypotheses, constants, costs, applications and scope; production checks cover references, bibliography, compilation and rendered layout. These exact finite tests and agent reviews are not formal certification, external human peer review, or a novelty certificate.

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